

Question 1

T5

Gina correctly started to answer the following question. Complete her solution.

Question: Solve the following equation for all real values of θ .
Express your answer in radians correct to 3 decimal places.

$$3\sin^2 \theta - 14\sin \theta - 5 = 0$$

Gina's solution: $3\sin^2 \theta - 14\sin \theta - 5 = 0$
 $(3\sin \theta + 1)(\sin \theta - 5) = 0$

Solution

Method 1

$$3\sin^2 \theta - 14\sin \theta - 5 = 0$$

$$(3\sin \theta + 1)(\sin \theta - 5) = 0$$

$$\sin \theta = -\frac{1}{3} \qquad \sin \theta = 5$$

no solution

$\frac{1}{2}$ mark for $\sin \theta = 5$
 $\frac{1}{2}$ mark for no solution

$$\theta_r = 0.339\ 837$$

$$\theta = 3.481\ 429, 5.943\ 348$$

1 mark ($\frac{1}{2}$ mark for each value of θ)

$$\theta = 3.481 + 2k\pi, k \in \mathbb{I}$$

$$\theta = 5.943 + 2k\pi, k \in \mathbb{I}$$

1 mark for general solution

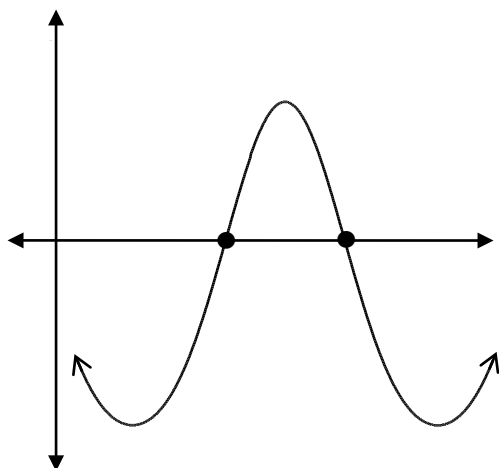
3 marks

Note(s):

- give maximum of 2 marks for solution in degrees: $\theta = 199.471^\circ + 360^\circ k, k \in \mathbb{I}$
 $\theta = 340.529^\circ + 360^\circ k, k \in \mathbb{I}$

Method 2

$$y = 3 \sin^2 \theta - 14 \sin \theta - 5$$

 $\frac{1}{2}$ mark for equation**or** $\frac{1}{2}$ mark for justification

Find all zeros over the reals.

$$\theta = 3.481 + 2k\pi, k \in \mathbb{I}$$

1 mark for solutions

$$\theta = 5.943 + 2k\pi, k \in \mathbb{I}$$

1 mark for general solution

3 marks

Find and simplify the 6th term in the binomial expansion of $\left(3x^4 - \frac{1}{x^3}\right)^9$.

Solution

$$\begin{aligned}t_6 &= {}_9C_5 \left(3x^4\right)^4 \left(-\frac{1}{x^3}\right)^5 \\&= (126)(81x^{16})\left(-\frac{1}{x^{15}}\right) \\&= -10\,206x\end{aligned}$$

2 marks (1 mark for ${}_9C_5$, $\frac{1}{2}$ mark for each consistent factor)

1 mark for simplification ($\frac{1}{2}$ mark for evaluating coefficient, $\frac{1}{2}$ mark for simplifying variable)

3 marks

Question 3

R10

The number of times a website is visited can be modeled by the function:

$$A = 800(e)^{rt}$$

where A = the total number of visitors at time t

t = the time in days ($t \geq 0$)

r = the rate of growth

After 5 days, 40 000 people have visited the site.

Determine the number of visitors expected after 9 days.

Express your answer as a whole number.

Solution

Method 1

$$40\,000 = 800e^{r5}$$

½ mark for substitution

$$\frac{40\,000}{800} = e^{5r}$$

$$\ln 50 = \ln e^{5r}$$

½ mark for applying logs

$$\ln 50 = 5r$$

1 mark for log theorem

$$\frac{\ln 50}{5} = r$$

$$r = 0.782\,404\,601$$

$$A = 800e^{(0.782\,404\,601)(9)}$$

½ mark for substitution

$$A = 914\,610.103$$

½ mark for calculations with base e

$$A = 914\,610$$

3 marks

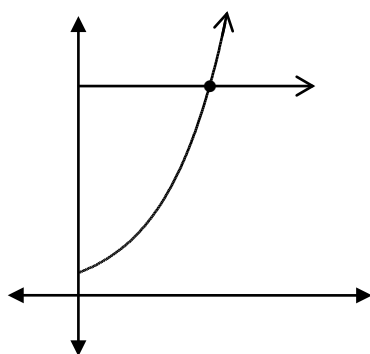
Method 2

Use calculator to find the value of r .

$$y = 40\,000$$

$\frac{1}{2}$ mark for equations

$$y = 800e^{5x}$$



or

$\frac{1}{2}$ mark for justification

Find the point of intersection of these two functions.

$$x = 0.782\,404\,601$$

1 mark for finding the value of x at the intersection point

Substitute x for r into formula.

$$\therefore A = 800e^{(0.782\,404\,601)(9)}$$

$$A = 914\,610.103$$

$$A = 914\,610$$

1 mark for value of A

3 marks

Question 4

R10

Solve algebraically:

$$10^{3x} = 7^{x+5}$$

Express your answer correct to 3 decimal places.

Solution

Method 1

$$10^{3x} = 7^{x+5}$$

$$\log 10^{3x} = \log 7^{x+5}$$

$$3x(\log 10) = (x+5)\log 7$$

$$3x \log 10 = x \log 7 + 5 \log 7$$

$$3x \log 10 - x \log 7 = 5 \log 7$$

$$x = \frac{5 \log 7}{3 \log 10 - \log 7}$$

$$x = 1.960\ 873$$

$$x = 1.961$$

$\frac{1}{2}$ mark for applying logs

1 mark for power rule

$\frac{1}{2}$ mark for collecting terms with x

$\frac{1}{2}$ mark for solving for x

$\frac{1}{2}$ mark for evaluating quotient of logarithms

Method 2

$$10^{3x} = 7^{x+5}$$

$$\log 10^{3x} = \log 7^{x+5}$$

$$3x(\log 10) = (x+5)\log 7$$

$$3x = x \log 7 + 5 \log 7$$

$$3x - x \log 7 = 5 \log 7$$

$$x = \frac{5 \log 7}{3 - \log 7}$$

$$x = 1.960\ 873$$

$$x = 1.961$$

3 marks

Question 5

P2

A word contains two Ms, two Es, two Ns, and no other repeated letters.

Suppose one of the Ns is replaced by an M.

Will this replacement result in greater or fewer permutations?

Justify your reasoning.

Solution

Method 1

$$2 \text{ Ms, } 2 \text{ Es, } 2 \text{ Ns: } \frac{(\text{total number of letters})!}{2!2!2!}$$

1 mark for dividing the total number of letters by $2!2!2!$

$$\frac{(\text{total number of letters})!}{8}$$

$$3 \text{ Ms, } 2 \text{ Es, } 1 \text{ N: } \frac{(\text{total number of letters})!}{3!2!1!}$$

1 mark for dividing the total number of letters by $3!2!1!$

$$\frac{(\text{total number of letters})!}{12}$$

2 marks

If one of the Ns is changed to an M, there would be fewer permutations.

Method 2

If one of the Ns is changed to an M, there would be fewer permutations because you would be dividing the total number of letters by a larger number.

2 marks for justification

2 marks

Question 6

P3

There is a group of 16 boys and 12 girls. How many ways can a committee of 3 people be formed if there must be at least 2 girls on the committee?

Express your answer as a whole number.

Solution

Method 1

Case 1: 2 girls, 1 boy

$${}_{12}C_2 \cdot {}_{16}C_1 = 1056$$

1 mark for case 1 ($\frac{1}{2}$ mark for each factor shown in a product)

Case 2: 3 girls

$${}_{12}C_3 = 220$$

1 mark for case 2

$$1056 + 220 = 1276$$

1 mark for addition of cases

3 marks

Method 2

Case 1: 2 boys, 1 girl

$${}_{16}C_2 \cdot {}_{12}C_1 = 1440$$

1 mark for case 1 ($\frac{1}{2}$ mark for each factor shown in a product)

Case 2: 3 boys

$${}_{16}C_3 = 560$$

1 mark for case 2

All possibilities

$${}_{28}C_3 = 3276$$

$\frac{1}{2}$ mark for all possible cases

$$3276 - 1440 - 560 = 1276$$

$\frac{1}{2}$ mark for subtraction from total

3 marks

Question 7

T1

A student is using the formula $s = \theta r$ to find an arc length of a circle. Given a central angle measure of 35° and a radius of 6 cm, the student's solution is as follows:

$$s = (35)(6)$$

$$s = 210 \text{ cm}$$

Explain why this solution is incorrect.

Write the correct solution.

Solution

When using the formula $s = \theta r$, the angle must be in radians. $\frac{1}{2}$ mark for explanation

$$35^\circ \left(\frac{\pi}{180^\circ} \right) = \frac{35\pi}{180} \quad \text{or} \quad \frac{7\pi}{36}$$

1 mark for conversion to radians

$$\therefore s = \theta r$$

$$= \frac{7\pi}{36}(6)$$

$$= \frac{42\pi}{36} \text{ cm} \quad \text{or} \quad \frac{7\pi}{6} \text{ cm} \quad \text{or} \quad 3.665 \text{ cm}$$

$\frac{1}{2}$ mark for simplification

2 marks

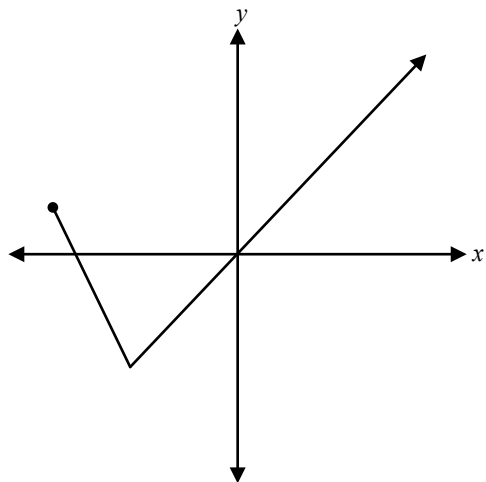
Note(s):

- award 1 mark for correct final answer with no work

Question 8

R1

Given the graph of $f(x)$ below, explain how you would sketch the graph of $y = |f(x)|$.



Solution

The negative y values are reflected over the x -axis.

1 mark for explanation

1 mark

Question 9

R10

Claire correctly solves the following equation:

$$\log_2(6-x) + \log_2(3-x) = 2$$

She finds two possible values of x : $x = 2$ and $x = 7$.

Identify which one of these values is unacceptable and explain why.

Solution

If x is greater than 3, you have a negative argument $\therefore x = 2$ but $x \neq 7$.

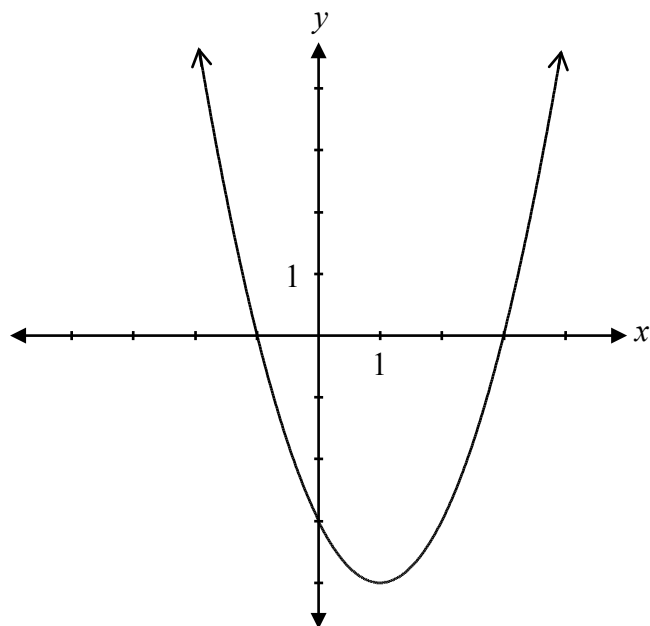
or

1 mark for explanation

The domain is restricted to values of $x < 3 \therefore x = 2$.

1 mark

Given the graph of the function $f(x)$ below, state the domain of $y = \sqrt{f(x)}$.



Solution

The domain is $\{x \in \mathbb{R} \mid x \leq -1 \text{ or } x \geq 3\}$.

or

The domain is $(-\infty, -1] \cup [3, \infty)$.

1 mark for domain

1 mark

Question 11

P1

A school offers 4 different Science courses, 3 different Mathematics courses, and 2 different English courses.

Julie must select 1 Science course, 1 Mathematics course, and 1 English course. She thinks this creates 9 options for her timetable.

Show why Julie is incorrect.

Solution

Julie should have multiplied the number of options.

1 mark for justification

or

1 mark

$4 \times 3 \times 2 = 24$ different options

Explain how Pascal's triangle can be used to determine the coefficients in the binomial expansion of $(x + y)^n$.

Solution

The $(n + 1)^{\text{st}}$ row of Pascal's triangle corresponds to the coefficients of the terms in the expansion of the binomial $(x + y)^n$.

1 mark for explanation

1 mark

Prove the identity below for all permissible values of x :

$$\frac{1 + \cos 2x}{\sin 2x} = \cot x$$

Solution

Method 1

$$\begin{aligned} \text{LHS} &= \frac{1 + 2\cos^2 x - 1}{2\sin x \cos x} && \begin{array}{l} 1 \text{ mark for identity} \\ \frac{1}{2} \text{ mark for identity} \end{array} \\ &= \frac{2\cos^2 x}{2\sin x \cos x} \\ &= \frac{\cos x}{\sin x} && 1 \text{ mark for simplification} \\ &= \cot x && \frac{1}{2} \text{ mark for identity} \\ &= \text{RHS} \end{aligned}$$

3 marks

Method 2

$$\begin{aligned} \text{LHS} &= \frac{1 + 1 - 2\sin^2 x}{2\sin x \cos x} && \begin{array}{l} \frac{1}{2} \text{ mark for identity} \\ \frac{1}{2} \text{ mark for identity} \end{array} \\ &= \frac{2 - 2\sin^2 x}{2\sin x \cos x} \\ &= \frac{1 - \sin^2 x}{\sin x \cos x} && \frac{1}{2} \text{ mark for simplification} \\ &= \frac{\cos^2 x}{\sin x \cos x} && \frac{1}{2} \text{ mark for identity} \\ &= \frac{\cos x}{\sin x} && \frac{1}{2} \text{ mark for simplification} \\ &= \cot x && \frac{1}{2} \text{ mark for identity} \\ &= \text{RHS} \end{aligned}$$

3 marks

Solution**Method 3**

$$\text{LHS} = \frac{1 + \cos^2 x - \sin^2 x}{2 \sin x \cos x}$$

½ mark for identity

½ mark for identity

$$= \frac{2 \cos^2 x}{2 \sin x \cos x}$$

½ mark for identity $(1 - \sin^2 x = \cos^2 x)$

$$= \frac{\cos x}{\sin x}$$

1 mark for simplification

$$= \cot x$$

½ mark for identity

$$= \text{RHS}$$

3 marks

Your classmate, Leo, was absent for one of his math lessons.

Explain to Leo how to determine the cosecant ratio for an angle in standard position given that $P(-3, -4)$ is a point on the terminal arm of the angle.

Solution

Method 1

- Locate the point $(-3, -4)$ on the coordinate plane.
- Draw a right-angle triangle by connecting the point $(-3, -4)$ to the x -axis and to the origin.
- The angle created with the x -axis and the line connecting the point with the origin (the terminal arm) is θ . ½ mark
- Determine the length of the terminal arm by using the Pythagorean Theorem. ½ mark
- Once you have the lengths of all three sides of the triangle, find $\sin \theta$. ½ mark
- Invert the $\sin \theta$ ratio to find $\csc \theta$. ½ mark

2 marks

Method 2

- Determine the value of r using the Pythagorean Theorem. ½ mark
- Determine the value of $\sin \theta \left(\sin \theta = \frac{y}{r} \right)$. ½ mark
- Identify the correct quadrant. ½ mark
- Invert the $\sin \theta$ ratio to find $\csc \theta$. ½ mark

2 marks

Note(s):

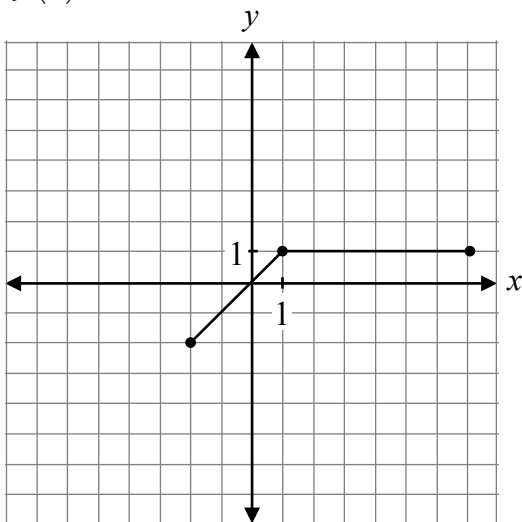
- award a maximum of 1 mark for correct work shown without an explanation in words

Question 15

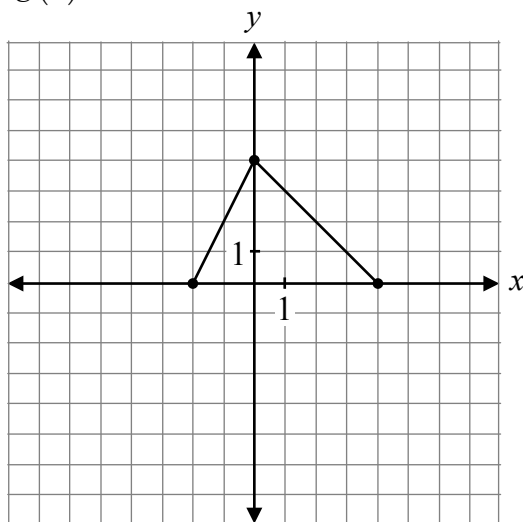
R1

Given the following graphs:

$f(x)$

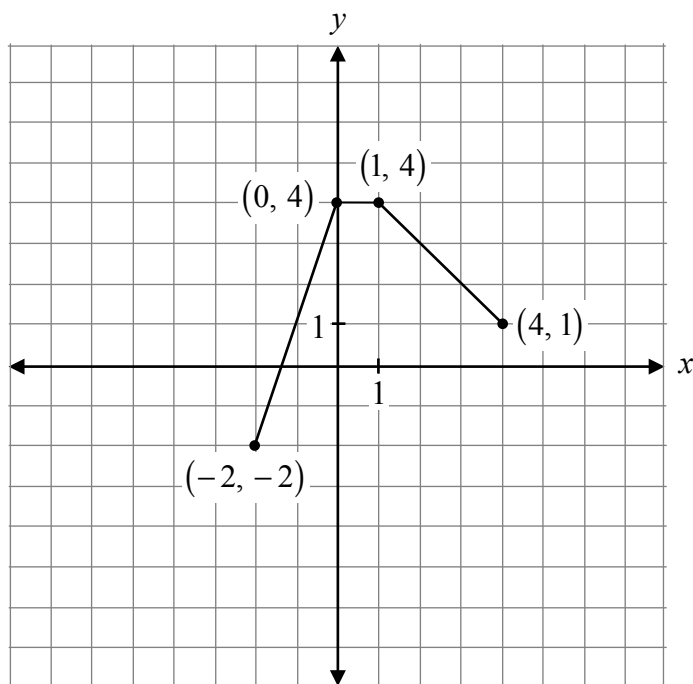


$g(x)$



Sketch the graph of $f(x) + g(x)$.

Solution



2 marks ($\frac{1}{2}$ mark for each point where the graph changes direction)

$[(-2, -2); (0, 4); (1, 4); (4, 1)]$

2 marks

Note(s):

- deduct a maximum of 1 mark for incorrect domain

Question 25

R7

Given $\log_b a = 3$, give one example of possible values for a and b that make this equation true.

Solution

Answers will vary but $b^3 = a$.

Some possible solutions are: $a = 8$ $b = 2$

or

$a = 27$ $b = 3$

or

$a = 64$ $b = 4$

1 mark

Question 26

R5

The range of the graph of $y = f(x)$ is $[-3, 2]$.

Explain why there is no effect on the range of the graph that is a result of the transformation $y = f(-x)$.

Solution

$y = f(-x)$ is a reflection over the y -axis.

The domain is affected, but the range remains the same. 1 mark for explanation

1 mark

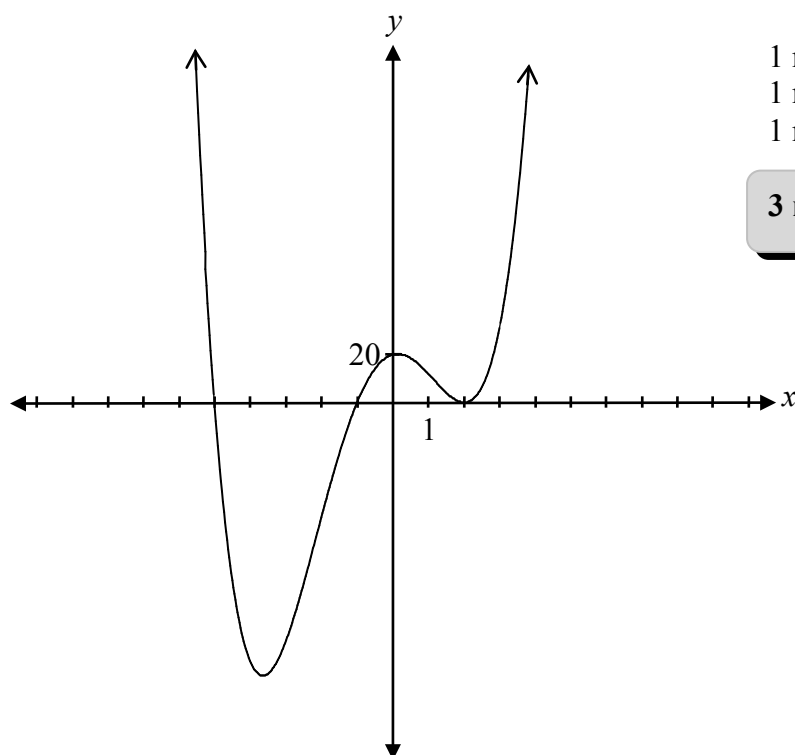
Sketch the graph of $y = (x + 1)(x - 2)^2(x + 5)$.

Identify the x -intercepts and y -intercept.

Solution

x -intercepts: -5 , -1 , and 2

y -intercept: 20



1 mark for x -intercepts

1 mark for y -intercept

1 mark for multiplicity of a zero at $x = 2$

3 marks

Note(s):

- if no graph is shown, give $\frac{1}{2}$ mark for x -intercepts (-5 , -1 , and 2) and $\frac{1}{2}$ mark for y -intercept (20)
- relative maximum and minimum are not required

Question 28

T4

The graph of the function $y = \sin x$ has been transformed to create a new graph.

The range of this new graph is $[-4, 4]$ and the zeros are $x = k\frac{\pi}{2}$, where k is an integer.

Write the equation that corresponds to this new graph.

Solution

$$\text{Amplitude} = 4$$

1 mark for correct amplitude

$$\text{Period} = \pi$$

$\frac{1}{2}$ mark for correct period

$$\therefore b = \frac{2\pi}{\pi} = 2$$

$\frac{1}{2}$ mark for consistent value of b

2 marks

$$y = 4 \sin(2x)$$

or

$$y = -4 \sin(2x)$$

Note(s):

- deduct $\frac{1}{2}$ mark for a translation that results in an incorrect range and/or zeros

Question 29

R1

Given the functions $f(x) = x^2 - 1$ and $g(x) = x + 1$, state the domain of $\frac{g(x)}{f(x)}$.

Solution

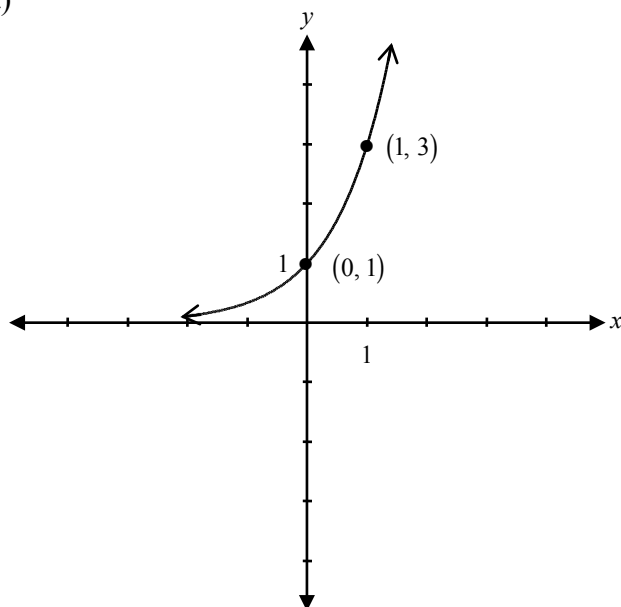
Domain: $x \in \mathbb{R}$ where $x \neq 1$ and $x \neq -1$ 1 mark ($\frac{1}{2}$ mark for $x \neq 1$, $\frac{1}{2}$ mark for $x \neq -1$)

1 mark

- a) Sketch the graph of $y = 3^x$.
- b) Explain how the graph of $y = 3^x$ can be used to sketch the graph of $y = \log_3 x$.

Solutions

a)



½ mark for increasing exponential function
½ mark for y-intercept at $(0, 1)$
½ mark for consistent point on exponential function
½ mark for asymptotic behaviour

2 marks

b)

To graph $y = \log_3 x$, you can reflect the graph of $y = 3^x$ over the line $y = x$.

or

You can switch the x and y coordinates of $y = 3^x$ to get the graph of $y = \log_3 x$.

1 mark for explanation

1 mark

Note(s):

- in (b), give ½ mark for having only stated they are inverse functions of each other

Question 31

R11, R12

A box in the shape of a rectangular prism has side lengths x , $x + 2$, and $x + 10$.

Write a function, $V(x)$, to express the volume of the box in terms of x .

Find all possible values of x , given that the volume of the box is 96 cm^3 .

State the dimensions of the box.

Solution

$$V(x) = (x)(x+2)(x+10)$$

$\frac{1}{2}$ mark for expressing volume in terms of x

$$96 = (x)(x+2)(x+10)$$

$$96 = x^3 + 12x^2 + 20x$$

$$0 = x^3 + 12x^2 + 20x - 96$$

$\frac{1}{2}$ mark for equating to zero

when $x = 2$:

$$0 = (2)^3 + 12(2)^2 + 20(2) - 96$$

$$0 = 8 + 48 + 40 - 96$$

$$0 = 0$$

$\therefore x = 2$ is a possible value.

1 mark for identifying one possible value of x

$$\begin{array}{r|rrrr} 2 & 1 & 12 & 20 & -96 \\ & & 2 & 28 & 96 \\ \hline & 1 & 14 & 48 & 0 \end{array}$$

1 mark for division

$$(x-2)(x^2 + 14x + 48) = 0$$

$$(x-2)(x+8)(x+6) = 0$$

$$x = -8, -6, 2$$

1 mark for identifying all possible values of x

$x \neq -8$ and $x \neq -6$ because dimensions cannot be negative values.

$\therefore x = 2$ is the only solution.

$\frac{1}{2}$ mark for rejecting extraneous roots

The dimensions of the box are $2 \text{ cm} \times 4 \text{ cm} \times 12 \text{ cm}$.

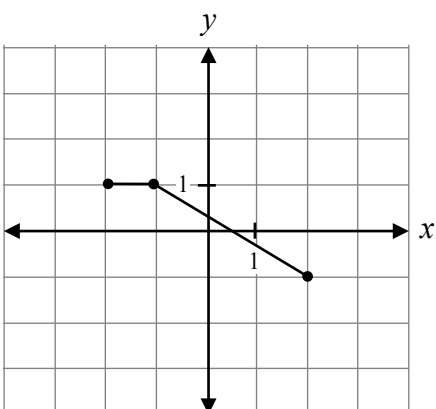
$\frac{1}{2}$ mark for stating the dimensions of the box

5 marks

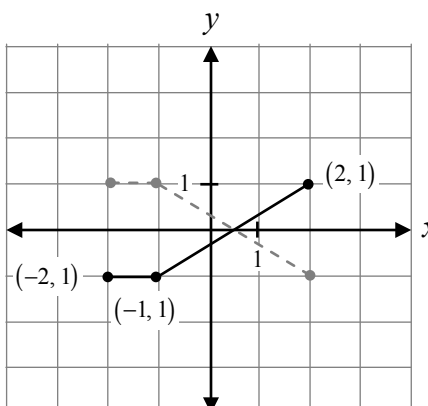
Question 32

R5

Given the graph of $f(x)$ below, sketch the graph of $y = -f(x)$.



Solution



1 mark

Question 33

T2

Determine the coordinates of a point (x, y) on the unit circle if you are given $\theta = 30^\circ$ where θ is in standard position.

Solution

$$P(\theta) = (\cos \theta, \sin \theta)$$

If $\theta = 30^\circ$, then the coordinates of $P(\theta)$ would be $(\cos 30^\circ, \sin 30^\circ)$.

or

$$P(30^\circ) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right)$$

1 mark for answer stated
as coordinate point

1 mark

Given the following sinusoidal equation:

$$P(t) = 3000 \sin\left[\frac{\pi}{10}(t - 2010)\right] + 10\,000$$

Determine the maximum value of $P(t)$ and a value of t at which this maximum occurs.

Solution

Method 1

$$\begin{aligned}\text{Maximum value} &= 10\,000 + 3000 \\ &= 13\,000\end{aligned}$$

1 mark for maximum value

$$13\,000 = 3000 \sin\left[\frac{\pi}{10}(t - 2010)\right] + 10\,000$$

$$3000 = 3000 \sin\left[\frac{\pi}{10}(t - 2010)\right]$$

$$1 = \sin\left[\frac{\pi}{10}(t - 2010)\right]$$

$\frac{1}{2}$ mark for simplifying

$$\frac{\pi}{2} = \frac{\pi}{10}(t - 2010)$$

1 mark for exact value

$$5 = t - 2010$$

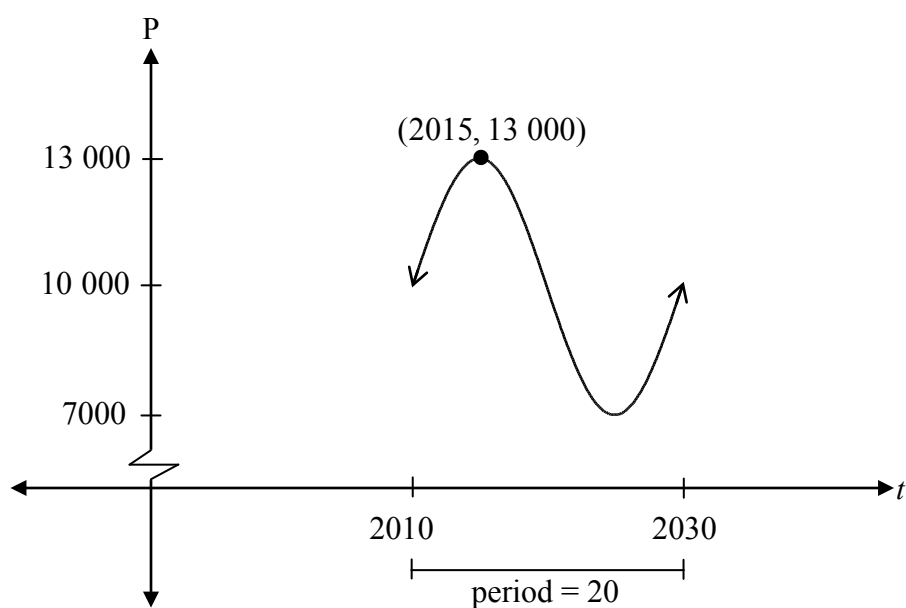
$$t = 2015$$

$\frac{1}{2}$ mark for solving for t

3 marks

Note(s):

- the period of the function is 20 $\therefore$ other acceptable answers are: $t = 2015 \pm 20$

Solution**Method 2**

1 mark for period

$$P(t) = 13\,000$$

1 mark for maximum value

$$t = 2015$$

1 mark for solving for t

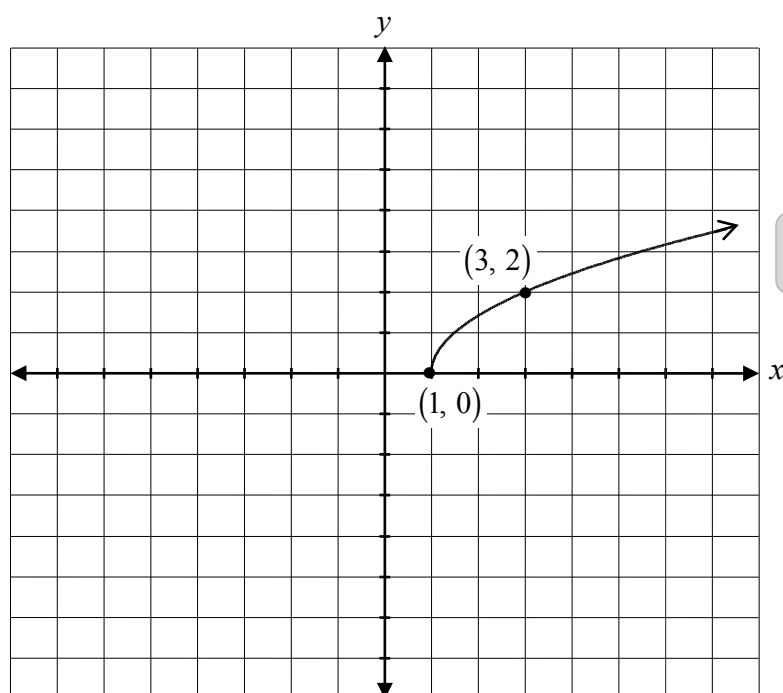
$\therefore$ the maximum value is 13 000 when $t = 2015$.

3 marks

Sketch the graph of $y = \sqrt{2x - 2}$.

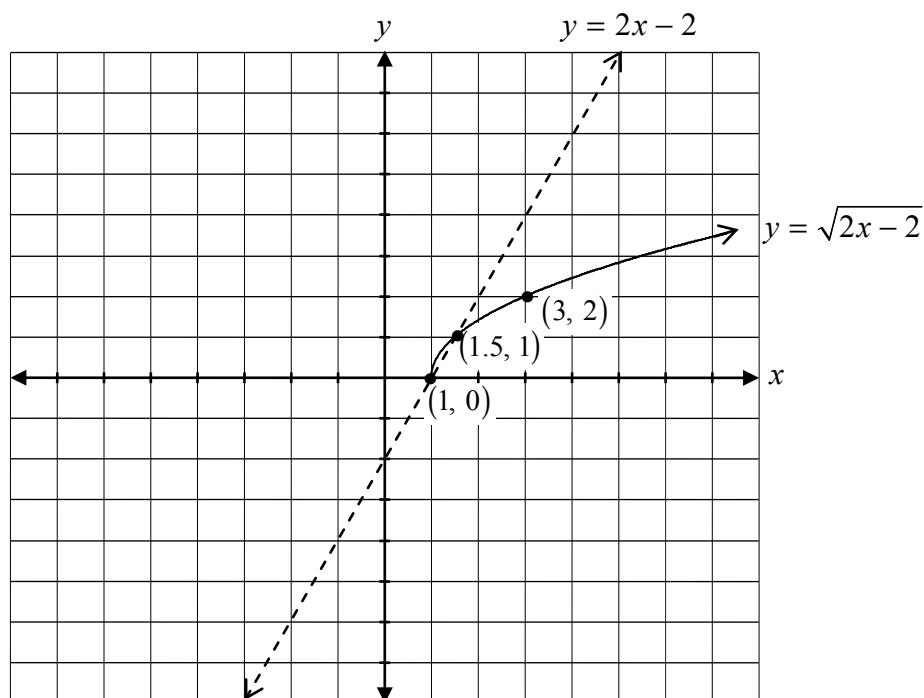
Solution**Method 1**

$$\begin{aligned} y &= \sqrt{2x - 2} \\ &= \sqrt{2(x - 1)} \end{aligned}$$



- 1 mark for domain: $[1, \infty)$
- 1 mark for shape (graph of a radical function)
- 1 mark for horizontal compression

3 marks

Method 2

1 mark for domain of $y = \sqrt{2x - 2} : [1, \infty)$

1 mark for invariant points where $y = 0$ and $y = 1$ ($\frac{1}{2}$ mark for each point)

$\frac{1}{2}$ mark for graph of $y = \sqrt{2x - 2}$ drawn above the graph of $y = 2x - 2$ between the invariant points

$\frac{1}{2}$ mark for graph of $y = \sqrt{2x - 2}$ drawn below the graph of $y = 2x - 2$ after the invariant point where $y = 1$

3 marks

Question 36

R6

Given $f(x) = 2x - 6$, write the equation of $f^{-1}(x)$.

Solution

Switch the x and y values.

$$x = 2y - 6$$

1 mark for switching x and y values

$$x + 6 = 2y$$

$$\frac{x + 6}{2} = y$$

$\frac{1}{2}$ mark for solving for y

$$\therefore f^{-1}(x) = \frac{x + 6}{2}$$

$\frac{1}{2}$ mark for writing equation of $f^{-1}(x)$

2 marks

Question 37

R8

Frank tried to expand a logarithmic expression using the laws of logarithms. He made one error.

$$\text{Frank's solution: } \log_a \frac{(x+2)}{zw} = \log_a x + \log_a 2 - \log_a z - \log_a w$$

Write the correct solution.

Solution

$$\text{Correct solution: } \log_a \frac{(x+2)}{zw} = \log_a (x+2) - \log_a z - \log_a w$$

1 mark for correct solution

1 mark

Question 38

T6

Determine all non-permissible values of θ over the interval $[0, 2\pi]$.

$$\frac{\sin \theta}{1 + \cos \theta} + \csc \theta + \cot \theta$$

Explain your reasoning.

Solution

To determine non-permissible values, the denominator needs to equal zero.

1 mark for explanation

The denominators of this expression are “ $1 + \cos \theta$ ” and “ $\sin \theta$ ” $\left(\text{since } \csc \theta = \frac{1}{\sin \theta} \text{ and } \cot \theta = \frac{\cos \theta}{\sin \theta} \right)$.

1 mark ($\frac{1}{2}$ mark for identifying each restriction)

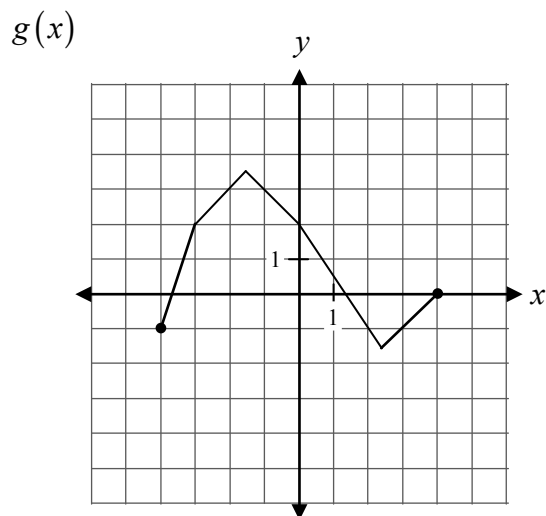
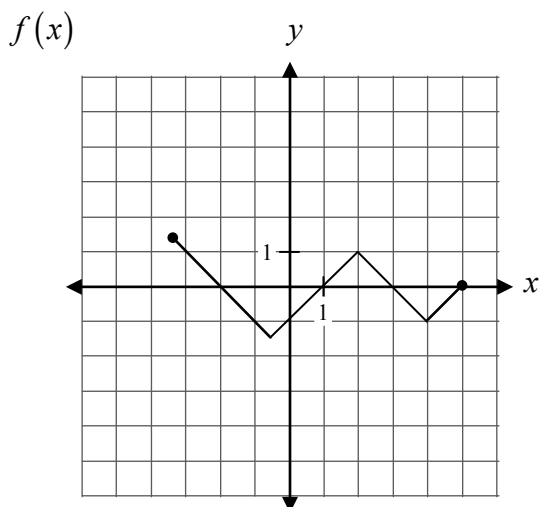
$$\begin{aligned} \therefore 1 + \cos \theta &= 0 & \sin \theta &= 0 \\ \cos \theta &= -1 & \theta &= 0, \pi, 2\pi \\ \theta &= \pi \end{aligned}$$

1 mark for all non-permissible values of θ ($\frac{1}{2}$ mark for each equation)

$\therefore$ the non-permissible values of θ over $[0, 2\pi]$ are $0, \pi$, and 2π .

3 marks

Given the following graphs:



- Determine the value of $[f \cdot g](0)$.
- Determine the value of $g(f(4))$.
- Determine a value for k where $f(k)=1$.

Solutions

$$\begin{aligned} \text{a) } f(0) &= -1 \\ g(0) &= 2 \end{aligned}$$

$$\begin{aligned} [f \cdot g](0) &= (-1)(2) \\ &= -2 \end{aligned}$$

1 mark for the value of $[f \cdot g](0)$

1 mark

$$\begin{aligned} \text{b) } f(4) &= -1 \\ g(-1) &= 3 \end{aligned}$$

$\frac{1}{2}$ mark for $f(4)$

$\frac{1}{2}$ mark for $g(f(4))$ consistent with $f(4)$ value

1 mark

$$\text{c) } k = 2 \quad \text{or} \quad k = -3$$

1 mark for a value for k

1 mark

Question 40

R1

Given that $h(x) = 2x^2 + 5x - 3$ and that $h(x) = f(x) \cdot g(x)$, determine $f(x)$ and $g(x)$.

Solution

$$f(x) = 2x - 1$$

$$g(x) = x + 3$$

1 mark for two correct factors of $h(x)$

1 mark

Other answers are possible.

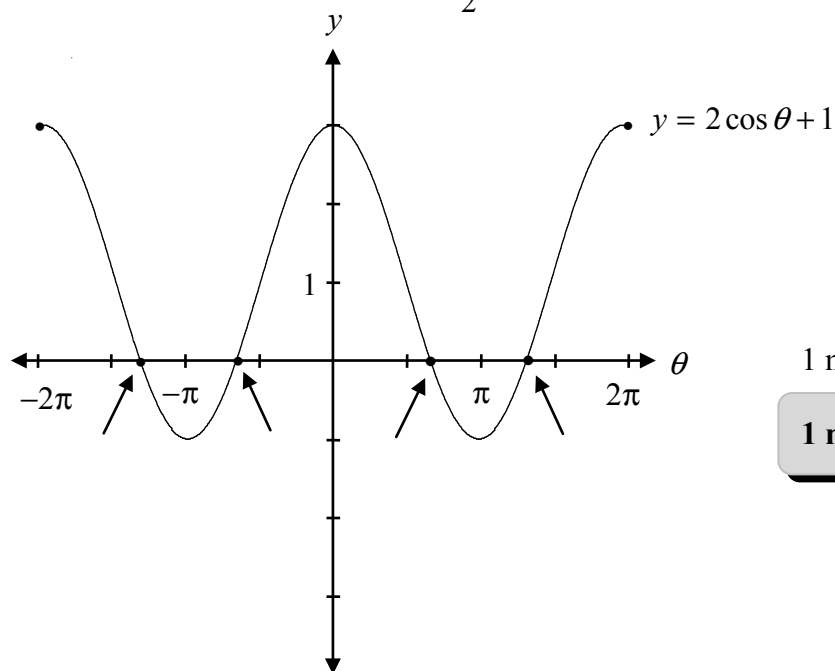
Question 41

T5

The graph of $y = 2 \cos \theta + 1$ below can be used to solve the equation $\cos \theta = -\frac{1}{2}$ over the interval $[-2\pi, 2\pi]$. Indicate on the graph where to find the solutions to the equation $\cos \theta = -\frac{1}{2}$.

Solution

The solution to the equation $\cos \theta = -\frac{1}{2}$ is found where the graph of $y = 2 \cos \theta + 1$ crosses the x-axis.



1 mark for indicating solutions on graph

1 mark

Note(s):

- give $\frac{1}{2}$ mark for indicating 1, 2, or 3 of the solutions

Question 42

R1

The function $f(x)$ is transformed.

A new function, $y = \frac{1}{f(x)}$, is created that does not have any vertical asymptotes.

What can you conclude about the original function $f(x)$?

Solution

If $f(x)$ does not have any x -intercepts, then the transformation would not have any vertical asymptotes.

or

$f(x)$ cannot equal zero.

1 mark for conclusion

1 mark

or

$f \neq 0$

Note(s):

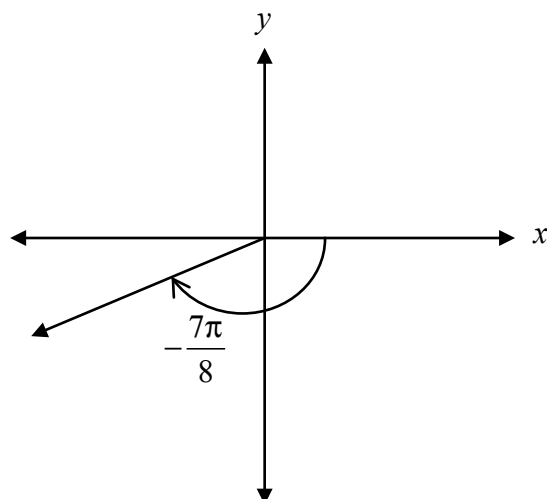
- award $\frac{1}{2}$ mark for a correct example without a given conclusion

Question 43

T1

Draw the angle $-\frac{7\pi}{8}$ in standard position.

Solution



1 mark for angle drawn in Quadrant III

1 mark

Determine the exact value of:

$$4 \cos\left(\frac{11\pi}{12}\right)$$

Solution

Method 1

$$\begin{aligned} 4 \cos\left(\frac{11\pi}{12}\right) &= 4 \cos\left(\frac{\pi}{4} + \frac{2\pi}{3}\right) \\ &= 4 \left[\cos \frac{\pi}{4} \cos \frac{2\pi}{3} - \sin \frac{\pi}{4} \sin \frac{2\pi}{3} \right] \\ &= 4 \left[\left(\frac{\sqrt{2}}{2}\right) \left(-\frac{1}{2}\right) - \left(\frac{\sqrt{2}}{2}\right) \left(\frac{\sqrt{3}}{2}\right) \right] \\ &= 4 \left[\frac{-\sqrt{2} - \sqrt{6}}{4} \right] \\ &= -\sqrt{2} - \sqrt{6} \end{aligned}$$

1 mark for combination

2 marks ($\frac{1}{2}$ mark for each exact value)

3 marks

Solution**Method 2**

$\frac{11\pi}{12}$ has a reference angle of $\frac{\pi}{12}$.

½ mark for correct reference angle

$$\begin{aligned}\cos\left(\frac{\pi}{12}\right) &= \cos\left(\frac{\pi}{4} - \frac{\pi}{6}\right) \\ &= \cos\frac{\pi}{4}\cos\frac{\pi}{6} + \sin\frac{\pi}{4}\sin\frac{\pi}{6} \\ &= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) \\ &= \frac{\sqrt{6} + \sqrt{2}}{4}\end{aligned}$$

½ mark for substitution into correct identity

1 mark for exact values

Since $\frac{11\pi}{12}$ is in Quadrant II, cosine is negative.

$$\begin{aligned}\therefore 4\cos\left(\frac{11\pi}{12}\right) &= 4\left(\frac{-\sqrt{6} - \sqrt{2}}{4}\right) \\ &= -\sqrt{6} - \sqrt{2}\end{aligned}$$

1 mark for the concept that $\cos\left(\frac{11\pi}{12}\right)$ is < 0

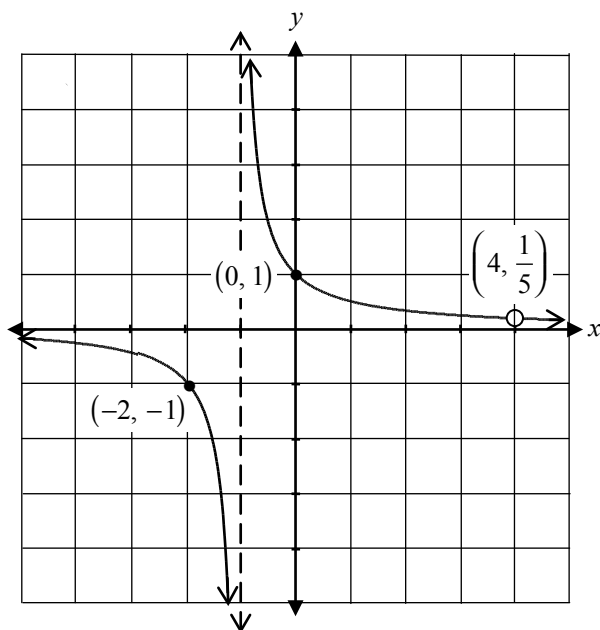
3 marks

Note(s):

- $\frac{-2 - 2\sqrt{3}}{\sqrt{2}}$ is an acceptable way to write the solution
- in Method 1, another possible combination is $\left(\frac{\pi}{6} + \frac{3\pi}{4}\right)$
- deduct ½ mark if answer is not expressed as a single fraction
- in Method 2, the reference angle need not be explicitly stated in order to obtain full marks

Sketch the graph of $f(x) = \frac{x-4}{x^2-3x-4}$.

Solution



1 mark for vertical asymptote at $x = -1$
 $\frac{1}{2}$ mark for graph left of vertical asymptote
 $\frac{1}{2}$ mark for graph right of vertical asymptote
 1 mark for point of discontinuity at $x = 4$

3 marks

$$\begin{aligned} f(x) &= \frac{x-4}{x^2-3x-4} \\ &= \frac{x-4}{(x-4)(x+1)} \\ &= \frac{1}{x+1} \text{ with a point of discontinuity at } x = 4 \end{aligned}$$

point of discontinuity: $f(4) = \frac{1}{5}$

$\therefore$ there is a point of discontinuity at $\left(4, \frac{1}{5}\right)$

$$\begin{aligned} \text{y-intercept: } f(0) &= \frac{0-4}{(0)^2-3(0)-4} \\ &= \frac{-4}{-4} \\ &= 1 \end{aligned}$$

Question 46

R7

Estimate the value of $\log_5 35$.

Justify your answer.

Solution

$$5^2 = 25$$

$$5^3 = 125$$

$\frac{1}{2}$ mark for justification

The value of $\log_5 35$ is more than 2 but less than 2.5.

$\frac{1}{2}$ mark for estimated answer

1 mark

Question 47

R11

If $p(x) = x^5 - 12x + 1$, determine the remainder when $p(x)$ is divided by $(x + 2)$.

Solution**Method 1**

$$p(x) = x^5 - 12x + 1$$

$$\begin{aligned} p(-2) &= (-2)^5 - 12(-2) + 1 \\ &= -32 + 24 + 1 \\ &= -7 \end{aligned}$$

1 mark for
substitution

1 mark

Method 2

$$-2 \overline{) 1 \ 0 \ 0 \ 0 \ -12 \ 1}$$

1 mark for correct
set-up of synthetic
division

$$\begin{array}{r} -2 \overline{) 1 \ 0 \ 0 \ 0 \ -12 \ 1} \\ \downarrow -2 \ 4 \ -8 \ 16 \ -8 \\ \hline 1 \ -2 \ 4 \ -8 \ 4 \ -7 \end{array}$$

1 mark

The remainder is -7 .

Describe the effects on the graph of $y = f(x)$ when asked for the graph of $y = f(x - 3) + 5$.

Solution

Shift right 3 and up 5

$\frac{1}{2}$ mark for horizontal translation

$\frac{1}{2}$ mark for vertical translation

or

$(x + 3, y + 5)$

1 mark

Find the exact value of the following expression:

$$\sin\left(\frac{11\pi}{3}\right) \cdot \sec\left(\frac{4\pi}{3}\right) \cdot \tan\left(-\frac{5\pi}{6}\right)$$

Solution

$$= \left(-\frac{\sqrt{3}}{2}\right)(-2)\left(\frac{1}{\sqrt{3}}\right)$$

$$= 1$$

1 mark for $\sin\left(\frac{11\pi}{3}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

1 mark for $\sec\left(\frac{4\pi}{3}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

1 mark for $\tan\left(-\frac{5\pi}{6}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

3 marks

Find the coterminal angle to $\frac{27\pi}{5}$ over the interval $[-360^\circ, 0^\circ)$.

Solution**Method 1**

$$\begin{aligned}\frac{27\pi}{5} \left(\frac{180^\circ}{\pi} \right) &= 27(36^\circ) \\ &= 972^\circ\end{aligned}$$

1 mark for conversion to degrees

$$972^\circ - (360^\circ)(3) = -108^\circ$$

$\frac{1}{2}$ mark for coterminal angle

$\frac{1}{2}$ mark for correct domain

2 marks

Method 2

$$-\frac{3\pi}{5} \text{ is a coterminal angle to } \frac{27\pi}{5}.$$

$\frac{1}{2}$ mark for coterminal angle

$\frac{1}{2}$ mark for correct domain

$$-\frac{3\pi}{5} \cdot \frac{180^\circ}{\pi} = -108^\circ$$

1 mark for conversion to degrees

2 marks

Solve the following equation over the interval $0 \leq \theta < 2\pi$.

$$(\tan \theta - 3)(\tan \theta + 1) = 0$$

Solution

$$(\tan \theta - 3)(\tan \theta + 1) = 0$$

$$\tan \theta = 3$$

$$\tan \theta = -1$$

1 mark (½ mark for each branch)

$$\theta_r = 1.249\ 046$$

$$\theta_r = \frac{\pi}{4}$$

2 marks (½ mark for each value of θ)

$$\theta = 1.249\ 046$$

$$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$\theta = 4.390\ 639$$

or

$$\theta = 1.249, 4.391$$

$$\theta = 2.356\ 194$$

$$\theta = 5.497\ 787$$

3 marks

$$\therefore \theta = 1.249, \frac{3\pi}{4}, 4.391, \frac{7\pi}{4}$$

or

$$\theta = 1.249, 2.356, 4.391, 5.498$$

Question 3

R10

An earthquake in Vancouver had a magnitude of 6.3 on the Richter scale. An earthquake in Japan had a magnitude of 8.9 on the Richter scale.

How many times more intense was the Japan earthquake than the Vancouver earthquake?

You may use the formula below:

$$M = \log \left(\frac{A}{A_0} \right)$$

where M is the magnitude of the earthquake on the Richter scale

A is the intensity of the earthquake

A_0 is the intensity of a standard earthquake

Express your answer as a whole number.

Solution

Method 1

Vancouver: substitute $M = 6.3$

$$6.3 = \log \left(\frac{A}{A_0} \right)$$

$$10^{6.3} = \frac{A}{A_0}$$

$$A = 10^{6.3} A_0$$

½ mark for exponential form

Japan: substitute $M = 8.9$

$$8.9 = \log \left(\frac{A}{A_0} \right)$$

$$10^{8.9} = \frac{A}{A_0}$$

$$A = 10^{8.9} A_0$$

½ mark for exponential form

To compare the two earthquakes divide their intensities.

$$\begin{aligned} \frac{\text{the intensity of Japan}}{\text{the intensity of Vancouver}} &= \frac{10^{8.9} A_0}{10^{6.3} A_0} \\ &= 398.107 \\ &= 398 \end{aligned}$$

1 mark for comparison

2 marks

Method 2

$$\frac{I_J}{I_V} = \frac{10^{8.9}}{10^{6.3}}$$
$$= 398$$

1 mark for comparison

 $\frac{1}{2}$ mark for exponential form $\frac{1}{2}$ mark for exponential form**2 marks**

Find and simplify the last term in the expansion of $(2y - 3x)^7$.

Solution

$$\begin{aligned}t_8 &= {}_7C_7 (2y)^0 (-3x)^7 \\ &= -2187x^7\end{aligned}$$

½ mark for ${}_7C_7$

½ mark for $(2y)^0$

1 mark for $(-3x)^7$ which is equal to $-2187x^7$

2 marks

Note(s):

- no deduction if ${}_7C_7$ and $(2y)^0$ are not shown

Given $\log_a 9 = 1.129$ and $\log_a 4 = 0.712$, find the value of $\log_a 12$.

Solution

Method 1

$$\log_a 9 = 1.129$$

$$\log_a 3^2 = 1.129$$

$$2 \log_a 3 = 1.129$$

$$\log_a 3 = 0.5645$$

1 mark for power rule

$$\log_a 12 = \log_a (4 \cdot 3)$$

1 mark for writing 12 as a product

$$= \log_a 4 + \log_a 3$$

1 mark for product rule

$$= 0.712 + 0.5645$$

$$= 1.2765$$

$$= 1.277$$

3 marks

Method 2

$$\log_a 12 = \log_a (\sqrt{9} \cdot 4)$$

1 mark for writing 12 as a product

$$= \frac{1}{2} \log_a 9 + \log_a 4$$

1 mark for power rule

$$= \frac{1}{2}(1.129) + 0.712$$

1 mark for product rule

$$= 1.2765$$

$$= 1.277$$

3 marks

Method 3

$$\log_a 9 = 1.129$$

$$a^{1.129} = 9$$

1 mark for exponential form

$$a = 9^{\frac{1}{1.129}}$$

$$a = 7$$

1 mark for solving for a

$$\log_7 12 = \frac{\log 12}{\log 7}$$

$$= 1.276\ 989$$

$$= 1.277$$

1 mark for value of $\log_a 12$

3 marks

Question 6

P2

How many different ways can 4 girls and 4 boys be arranged in a row if the girls and the boys must alternate?

Solution

Method 1

$$\begin{array}{ccccccccccc}
 & & 8 & \cdot & 4 & \cdot & 3 & \cdot & 3 & \cdot & 2 & \cdot & 2 & \cdot & 1 & \cdot & 1 \\
 \nearrow & & \uparrow & & \underbrace{\hspace{1.5cm}} & & & & & & & & & & & & \\
 \text{choose} & & \text{choose} & & \text{alternate the rest} \\
 \text{one} & & \text{the} & & & & & & & & & & & & & & \\
 \text{person} & & \text{other} & & & & & & & & & & & & & & \\
 & & \text{gender} & & & & & & & & & & & & & &
 \end{array}
 = 1152 \text{ ways}$$

1 mark for beginning with either gender
1 mark for alternating genders

2 marks

Method 2

$$\text{Case 1: } \frac{4}{B} \cdot \frac{4}{G} \cdot \frac{3}{B} \cdot \frac{3}{G} \cdot \frac{2}{B} \cdot \frac{2}{G} \cdot \frac{1}{B} \cdot \frac{1}{G} = 576$$

1 mark for arrangement of alternating genders

$$\text{Case 2: } \frac{4}{G} \cdot \frac{4}{B} \cdot \frac{3}{G} \cdot \frac{3}{B} \cdot \frac{2}{G} \cdot \frac{2}{B} \cdot \frac{1}{G} \cdot \frac{1}{B} = 576$$

½ mark for two cases

$$\text{Total number of ways: } 576 + 576 = 1152 \text{ ways}$$

½ mark for addition of cases

2 marks

Question 7

T3, T5, T6

Solve the following equation over the interval $[0, 2\pi]$.

$$2 \cos 2\theta - 1 = 0$$

Solution

Method 1

$$2 \cos 2\theta - 1 = 0$$

$$2(2 \cos^2 \theta - 1) - 1 = 0$$

$$4 \cos^2 \theta - 3 = 0$$

$$\cos^2 \theta = \frac{3}{4}$$

$$\cos \theta = \pm \frac{\sqrt{3}}{2}$$

$$\theta_r = \frac{\pi}{6}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

1 mark for identity

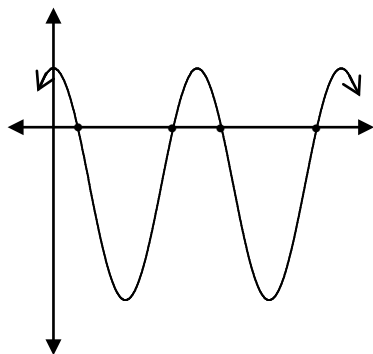
1 mark for solving for $\cos \theta$

2 marks for solutions ($\frac{1}{2}$ mark for each consistent solution)

4 marks

Method 2

$$y = 2 \cos 2\theta - 1$$



1 mark for equation

1 mark for the graph with zeroes indicated

Find all zeros over the interval $[0, 2\pi]$.

$$\theta = 0.524, 2.618, 3.665, 5.760$$

2 marks for solutions

4 marks

Solution**Method 3**

$$2 \cos 2\theta - 1 = 0$$

$$\cos 2\theta = \frac{1}{2}$$

½ mark for solving for $\cos 2\theta$

$$\text{let } x = 2\theta$$

$$\therefore \cos x = \frac{1}{2}$$

$$x_r = \frac{\pi}{3}$$

½ mark for reference angle

$$x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

1 mark for correct angles

$$\therefore 2\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

1 mark for coterminal angles

1 mark for all solutions over the interval $[0, 2\pi]$.

4 marks

Note(s):

- in Method 1, deduct a maximum of 1 mark if missing $\cos \theta = -\frac{\sqrt{3}}{2}$

Question 8

R4

Alex incorrectly explains to Rashid that the graph of $y = 2f(x) + 5$ means you first move the graph of $y = f(x)$ up 5 units and then multiply the y values by 2.

Explain to Rashid the correct way to transform the graph.

Solution

Alex explains the transformations correctly, but not in the correct order.

First multiply the y -values by 2, then move the graph up 5 units.

1 mark for explanation

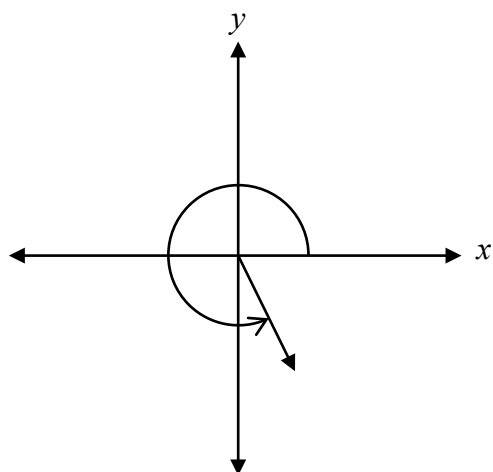
1 mark

Question 9

T1

Sketch the angle of 5 radians in standard position.

Solution



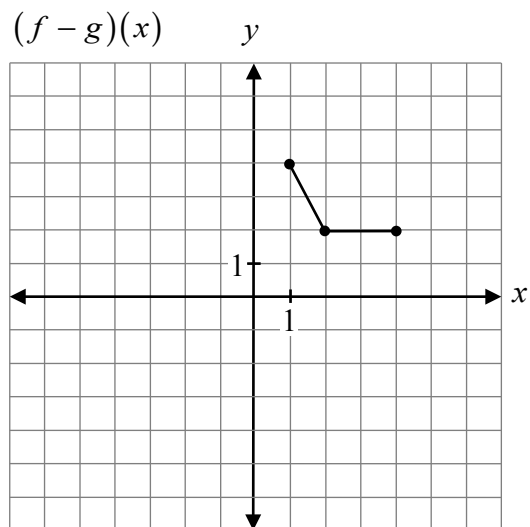
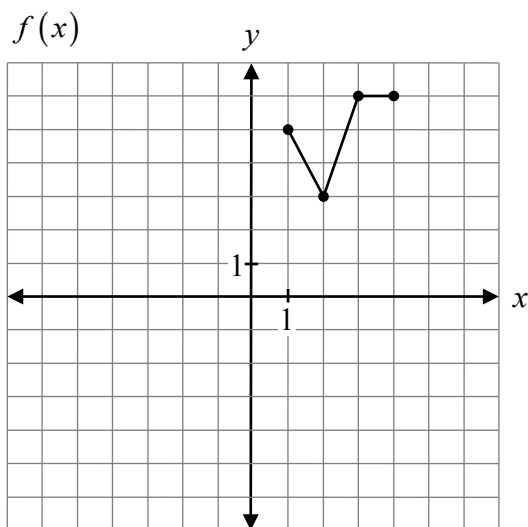
1 mark for angle drawn in Quadrant IV

1 mark

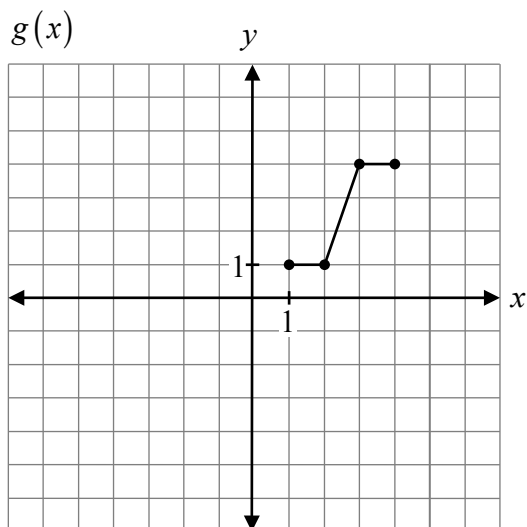
Question 10

R1

Given the graphs of $f(x)$ and $(f - g)(x)$, sketch the graph of $g(x)$.



Solution



1 mark for subtraction of $f(x) - (f - g)(x)$
 1 mark for shape representing the operation given

2 marks

A particular math class has a large number of students. From this class, you are to create a committee of 4 students that has at least 1 girl.

Without actually solving the problem, explain the strategy you would use to find the total number of ways to select this committee.

Solution

Method 1

Find all ways to form the committee and subtract the number of ways with no girls on the committee.

1 mark for identifying cases

1 mark for operation on these cases

2 marks

Method 2

Find all scenarios with girls on the committee: 1 girl on the committee, 2 girls on the committee, 3 girls on the committee, and a committee with all girls. Then add all scenarios together to get the total number of ways to create a committee with at least 1 girl.

1 mark for identifying cases

1 mark for operation on these cases

2 marks

Note(s):

- in Method 2, deduct $\frac{1}{2}$ mark if one of the cases is missing

Question 12

a) T6 b) T5, T6

a) Prove the identity below for all permissible values of θ .

$$\frac{1 + 2\cos^2 \theta}{\cos^2 \theta} = \tan^2 \theta + 3$$

b) Determine all the non-permissible values for θ .

Solution

Method 1

a)	LHS	RHS
	$\frac{1}{\cos^2 \theta} + \frac{2\cos^2 \theta}{\cos^2 \theta}$	$\tan^2 \theta + 3$
	$\sec^2 \theta + 2$	
	$\tan^2 \theta + 1 + 2$	
	$\tan^2 \theta + 3$	

$\therefore \text{LHS} = \text{RHS}$

1 mark for appropriate algebraic strategy
1 mark for appropriate identity substitutions

2 marks

b) $\cos^2 \theta = 0$
 $\cos \theta = 0$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$\therefore$ the non-permissible values of θ

are $\frac{\pi}{2} + k\pi, k \in \mathbb{I}$

or

$90^\circ + 180k, k \in \mathbb{I}$.

$\frac{1}{2}$ mark for $\cos^2 \theta = 0$

$\frac{1}{2}$ mark for any non-permissible value of θ

1 mark for all non-permissible values of θ

2 marks

Solution**Method 2**

a)	LHS	RHS
	$\frac{1 + 2\cos^2 \theta}{\cos^2 \theta}$	$\tan^2 \theta + 3$
		$\sec^2 \theta - 1 + 3$
		$\sec^2 \theta + 2$
		$\frac{1}{\cos^2 \theta} + 2$
		$\frac{1}{\cos^2 \theta} + \frac{2\cos^2 \theta}{\cos^2 \theta}$
		$\frac{1 + 2\cos^2 \theta}{\cos^2 \theta}$

1 mark for appropriate identity substitutions
 1 mark for appropriate algebraic strategy

2 marks

$\therefore \text{LHS} = \text{RHS}$

b) $\cos^2 \theta = 0$

$\cos \theta = 0$

$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$

$\therefore$ the non-permissible values of θ

are $\frac{\pi}{2} + k\pi, k \in \mathbb{I}$

or

$90^\circ + 180k, k \in \mathbb{I}.$

$\frac{1}{2}$ mark for $\cos^2 \theta = 0$

$\frac{1}{2}$ mark for any non-permissible value of θ

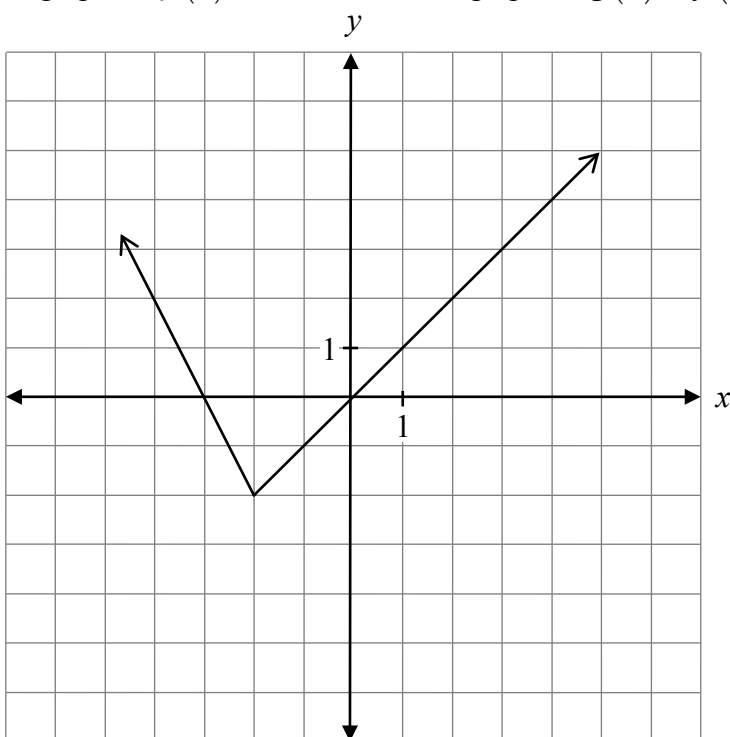
1 mark for all non-permissible values of θ

2 marks

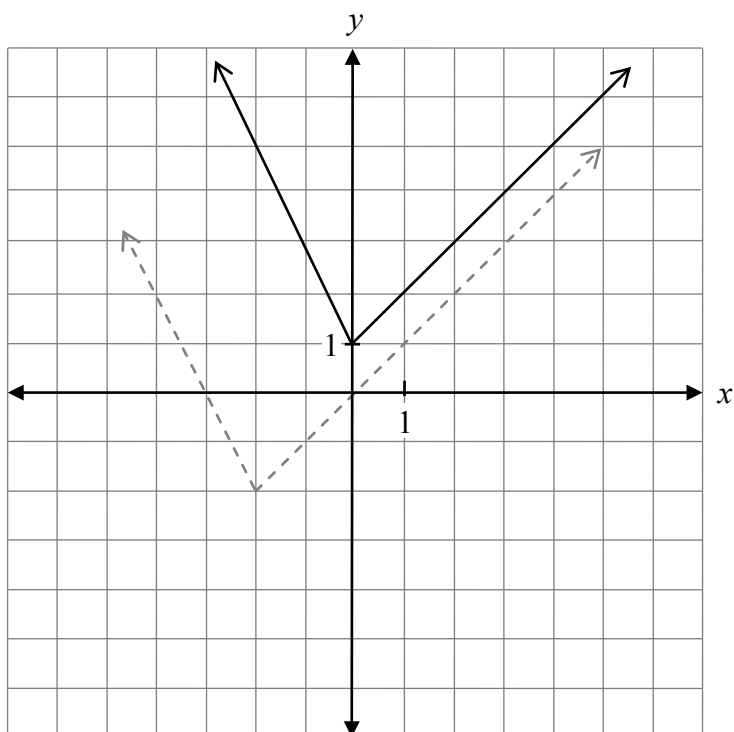
Question 13

R2

Given the graph of $f(x)$ below, sketch the graph of $g(x) = f(x - 2) + 3$.



Solution



1 mark for horizontal shift
1 mark for vertical shift

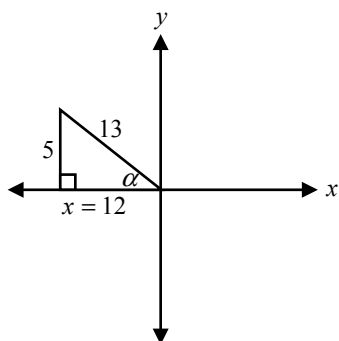
2 marks

Given that $\sin \alpha = \frac{5}{13}$, where α is in Quadrant II, and $\cos \beta = \frac{2}{5}$, where β is in Quadrant IV, find the exact value of:

- a) $\cos(\alpha + \beta)$ b) $\sin 2\alpha$

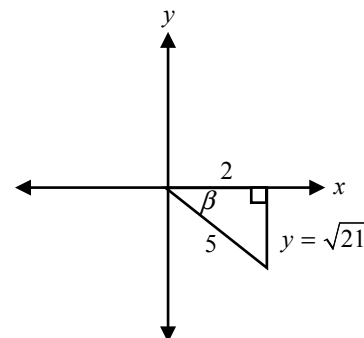
Solution

a)



$$\begin{aligned}x^2 + y^2 &= r^2 \\x^2 + 25 &= 169 \\x^2 &= 144 \\x &= \pm 12 \\x &= -12\end{aligned}$$

$$\begin{aligned}x^2 + y^2 &= r^2 \\4 + y^2 &= 25 \\y^2 &= 21 \\y &= \pm\sqrt{21} \\y &= -\sqrt{21}\end{aligned}$$



$\frac{1}{2}$ mark for value of x
 $\frac{1}{2}$ mark for value of y

$$\begin{aligned}\cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\&= \left(-\frac{12}{13}\right)\left(\frac{2}{5}\right) - \left(\frac{5}{13}\right)\left(-\frac{\sqrt{21}}{5}\right) \\&= -\frac{24}{65} + \frac{5\sqrt{21}}{65} \\&= \frac{5\sqrt{21} - 24}{65}\end{aligned}$$

$\frac{1}{2}$ mark for $\cos \alpha$
 $\frac{1}{2}$ mark for $\sin \beta$

1 mark for substitution into correct identity

3 marks

b) $\sin 2\alpha = 2 \sin \alpha \cos \alpha$

$$\begin{aligned}&= 2\left(\frac{5}{13}\right)\left(-\frac{12}{13}\right) \\&= -\frac{120}{169}\end{aligned}$$

1 mark for substitution into correct identity

1 mark

Note(s):

- accept any of the following values for x : $x = \pm 12$, $x = -12$ or $x = 12$
- accept any of the following values for y : $y = \pm\sqrt{21}$, $y = -\sqrt{21}$ or $y = \sqrt{21}$

If $f(x) = x^3$ and $g(x) = 2x - 3$, what is the value of $\left(\frac{f}{g}\right)(-1)$?

Solution

$$\begin{aligned} f(-1) &= (-1)^3 \\ &= -1 \end{aligned}$$

½ mark for substituting into $f(x)$ and $g(x)$

$$\begin{aligned} g(-1) &= 2(-1) - 3 \\ &= -5 \end{aligned}$$

$$\begin{aligned} \left(\frac{f}{g}\right)(-1) &= \frac{-1}{-5} \\ &= \frac{1}{5} \end{aligned}$$

½ mark for evaluating $\left(\frac{f}{g}\right)(-1)$

1 mark

One of the factors of $P(x) = x^3 - kx^2 - 7x + 10$ is $(x - 2)$.

Find the value of k .

Solution

Method 1

$$x = 2$$

$\frac{1}{2}$ mark for $x = 2$

$$0 = (2)^3 - k(2)^2 - 7(2) + 10$$

1 mark for remainder theorem

$$0 = 8 - 4k - 14 + 10$$

$$0 = 4 - 4k$$

$$4k = 4$$

$$k = 1$$

$\frac{1}{2}$ mark for solving for k

2 marks

Method 2

$$x = 2$$

$\frac{1}{2}$ mark for $x = 2$

2	1	$-k$	-7	10
		2	$-2k + 4$	$-4k - 6$
	1	$-k + 2$	$-2k - 3$	$-4k + 4$

$\frac{1}{2}$ mark for synthetic division

$$-4k + 4 = 0$$

$\frac{1}{2}$ mark for equating the remainder to zero

$$4k = 4$$

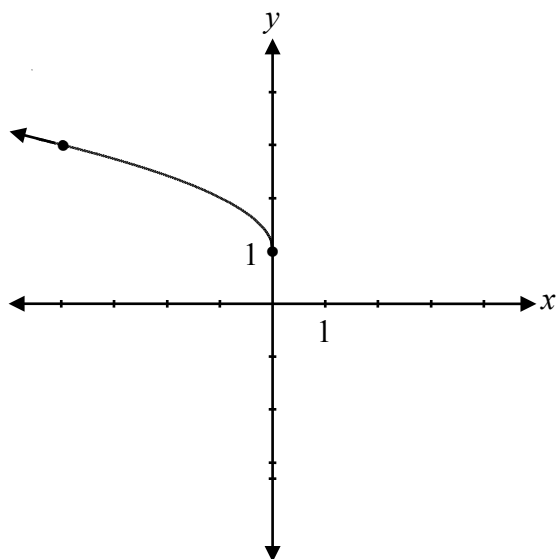
$$k = 1$$

$\frac{1}{2}$ mark for solving for k

2 marks

a) Sketch the graph of the function $y = \sqrt{-x} + 1$.

Solution

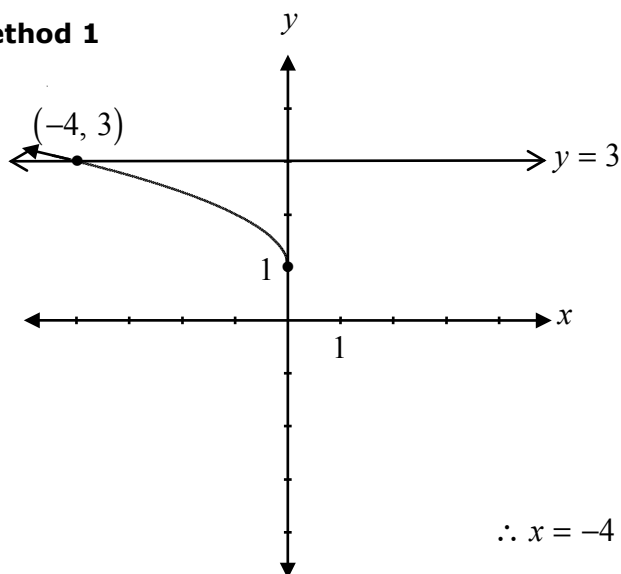


1 mark for general shape
1 mark for horizontal reflection
1 mark for vertical shift

3 marks

b) Determine the value of x when $y = 3$.

Method 1



$$\therefore x = -4$$

1 mark for consistent value of x

1 mark

Method 2

$$y = \sqrt{-x} + 1$$

$$3 = \sqrt{-x} + 1$$

$$2 = \sqrt{-x}$$

$$4 = -x$$

$$x = -4$$

1 mark for consistent value of x

1 mark

Solve the following equation:

$${}_nP_2 = {}_nC_3$$

Solution

Method 1

$$\frac{n!}{(n-2)!} = \frac{n!}{(n-3)!3!}$$

½ mark for substituting for ${}_nP_2$

$$\cancel{n!} (n-3)!3! = \cancel{n!} (n-2)!$$

½ mark for substituting for ${}_nC_3$

$$6 = \frac{(n-2)!}{(n-3)!}$$

1 mark for simplification

$$6 = \frac{(n-2) \cancel{(n-3)!}}{\cancel{(n-3)!}}$$

1 mark for expansion of $(n-2)!$

$$6 = n - 2$$

$$8 = n$$

3 marks

Method 2

$${}_nP_2 = {}_nC_3; \text{ we know } n \geq 3$$

$$\frac{n!}{(n-2)!} = \frac{n!}{(n-3)!3!}$$

½ mark for substituting for ${}_nP_2$

½ mark for substituting for ${}_nC_3$

$$\frac{n(n-1) \cancel{(n-2)!}}{\cancel{(n-2)!}} = \frac{n(n-1)(n-2) \cancel{(n-3)!}}{\cancel{(n-3)!} 6}$$

½ mark for expansion of ${}_nP_2$

½ mark for expansion of ${}_nC_3$

$$6 = \frac{\cancel{n} \cancel{(n-1)} (n-2)}{\cancel{n} \cancel{(n-1)}}$$

we can divide by n
and $(n-1)$ since
we know $n \geq 3$

$$6 = n - 2$$

$$8 = n$$

1 mark for simplification

3 marks

Solution**Method 3**

$${}_nP_2 = {}_nC_3$$

$$\frac{n!}{(n-2)!} = \frac{n!}{(n-3)!3!}$$

$$\frac{n(n-1)\cancel{(n-2)!}}{\cancel{(n-2)!}} = \frac{n(n-1)(n-2)\cancel{(n-3)!}}{\cancel{(n-3)!}3!}$$

$$6n(n-1) = n(n-1)(n-2)$$

$$6n(n-1) - n(n-1)(n-2) = 0$$

$$n(n-1)(6 - (n-2)) = 0$$

$$n(n-1)(8-n) = 0$$

$$\cancel{n=0} \quad \cancel{n=1} \quad n=8$$

½ mark for substituting for ${}_nP_2$

½ mark for substituting for ${}_nC_3$

½ mark for expansion of ${}_nP_2$

½ mark for expansion of ${}_nC_3$

1 mark for simplification

3 marks

Note(s):

- deduct ½ mark for not rejecting non-permissible values

Given $f(x) = x^2 - 1$ and $g(x) = \sqrt{x+1}$, sketch the graph of $y = f(g(x))$ and state its domain.

Solution**Method 1**

$$f(x) = x^2 - 1$$

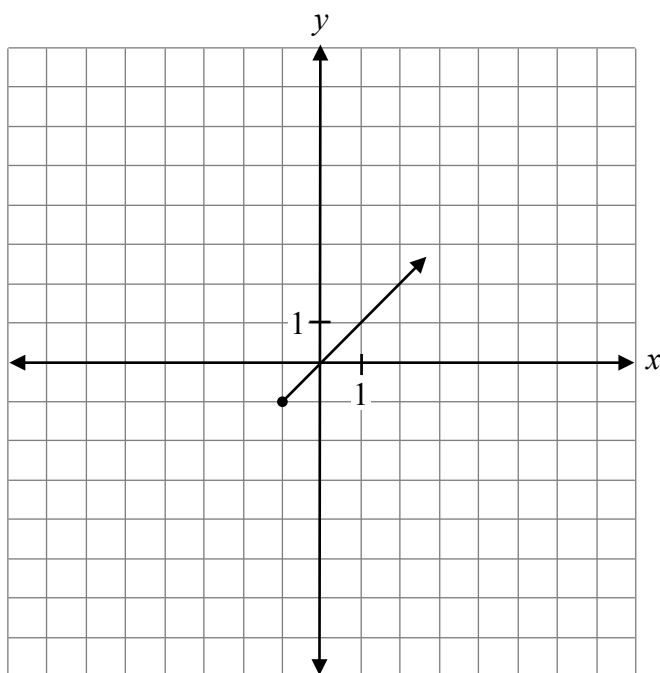
Domain: $x \in \mathbb{R}$

$$g(x) = \sqrt{x+1}$$

Domain: $[-1, \infty)$

$$\begin{aligned} f(g(x)) &= (\sqrt{x+1})^2 - 1 \\ &= x + 1 - 1 \\ &= x \end{aligned}$$

1 mark for determining the function $f(g(x))$



1 mark for graph of composite function

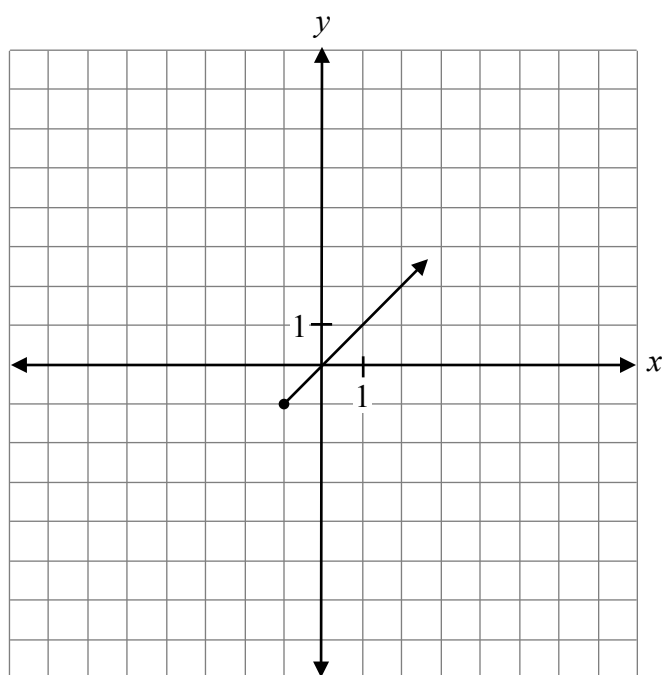
Domain of $f(g(x)) : [-1, \infty)$ **or** $\{x | x \geq -1, x \in \mathbb{R}\}$ 1 mark for restricted domain

3 marks

Solution**Method 2**

x	$g(x)$	$f(g(x))$
-2		
-1	0	-1
0	1	0
1		1
2		2
3	2	3

1 mark for table of values



1 mark for graph of composite function

Domain of $f(g(x)) : [-1, \infty)$ or $\{x | x \geq -1, x \in \mathbb{R}\}$

1 mark for restricted domain

3 marks

Write the equation of the horizontal asymptote for the function $f(x) = \frac{x-3}{x-2}$.

Solution

$$y = 1$$

1 mark for equation of horizontal asymptote

1 mark

The x -intercept of $f(x)$ is 4 and the x -intercept of $g(x)$ is 4.

Benjamin concludes that the x -intercept of $f(x) + g(x)$ is 8.

Do you agree with Benjamin? Justify your answer.

Solution

No, I do not agree with Benjamin.

Benjamin added the x -values instead of adding the y -values.

If the x -intercept of $f(x)$ is 4, then $y = 0$.

If the x -intercept of $g(x)$ is 4, then $y = 0$.

$\therefore$ the x -intercept of $f(x) + g(x)$ is 4.

1 mark for justification

1 mark

Solve the following equation:

$$2 \log_4 x - \log_4 (x + 3) = 1$$

Solution

$$2 \log_4 x - \log_4 (x + 3) = 1$$

$$\log_4 \left(\frac{x^2}{x+3} \right) = 1$$

$$4^1 = \left(\frac{x^2}{x+3} \right)$$

$$4(x+3) = x^2$$

$$x^2 - 4x - 12 = 0$$

$$(x-6)(x+2) = 0$$

$$x = 6 \quad \cancel{x = -2}$$

1 mark for power rule

1 mark for quotient rule

1 mark for exponential form

$\frac{1}{2}$ mark for solving for x

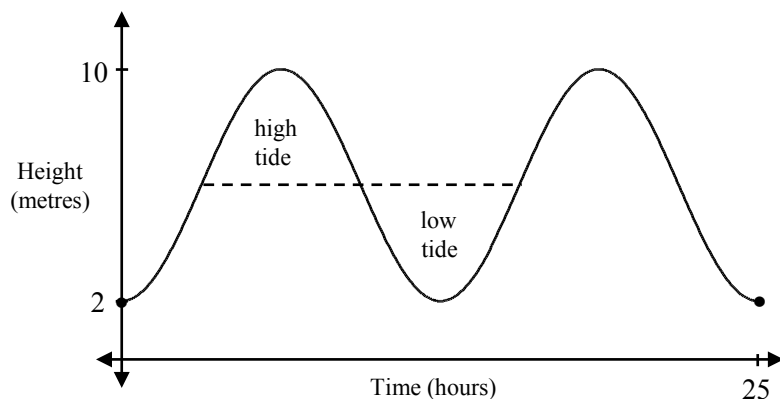
$\frac{1}{2}$ mark for rejecting extraneous root

4 marks

Question 33

T4

The following graph represents tidal levels in the Bay of Fundy over a 25-hour period.



- What is the average height of the water?
- What is the period of the graph above?

Explain what the period represents in this situation.

Solution

- 6 metres

1 mark

$$\begin{aligned} \text{b) Period} &= \frac{25}{2} \\ &= 12.5 \text{ hours} \end{aligned}$$

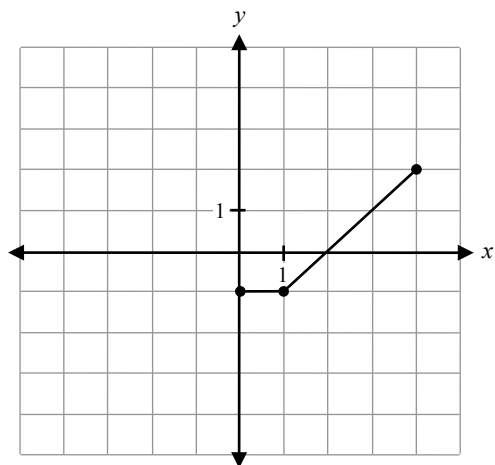
1 mark for period

The period represents the time to complete one cycle of tidal levels in the Bay of Fundy.

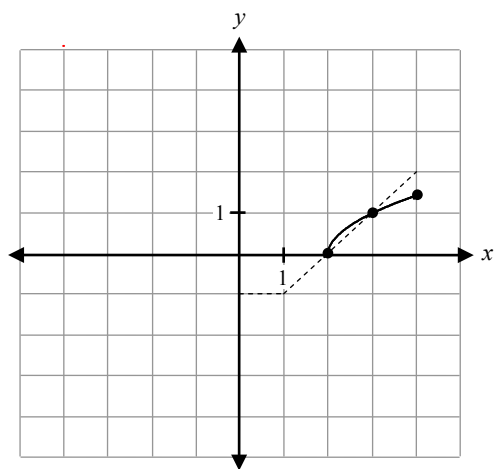
1 mark for explanation

2 marks

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



Solution



1 mark for restricting the domain

$\frac{1}{2}$ mark for graph above $y = f(x)$ over the range $[0, 1]$

$\frac{1}{2}$ mark for graph below $y = f(x)$ over the range $[1, 2]$

2 marks

When $P(x)$ is divided by $x - 3$, it has a quotient of $2x^2 + x - 6$ and a remainder of 4.

Determine $P(x)$.

Solution

$$P(x) = (x - 3)(2x^2 + x - 6) + 4$$

or

1 mark for polynomial $P(x)$

1 mark

$$P(x) = 2x^3 - 5x^2 - 9x + 22$$

Identify the domain and range of the following function:

$$f(x) = \frac{3}{x^2 + 1}$$

Solution

Domain: $\{x \in \mathbb{R}\}$

1 mark for domain

or

$(-\infty, \infty)$

Range: $\{y \in \mathbb{R} \mid 0 < y \leq 3\}$

1 mark for range

or

$(0, 3]$

2 marks

Evaluate:

$$\csc\left(\frac{11\pi}{6}\right) + \sin^2\left(-\frac{3\pi}{4}\right) + \cos\left(\frac{23\pi}{3}\right)$$

Solution

$$= (-2) + \left(-\frac{\sqrt{2}}{2}\right)^2 + \frac{1}{2}$$

$$= -2 + \frac{1}{2} + \frac{1}{2}$$

$$= -1$$

1 mark for $\csc\left(\frac{11\pi}{6}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

1 mark for $\sin^2\left(-\frac{3\pi}{4}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

1 mark for $\cos\left(\frac{23\pi}{3}\right)$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

3 marks

Evaluate the coefficient of the term containing x^3 in the expansion of $(1+x)^7$.

Justify your answer.

Solution

Method 1

				1				
			1		1			
		1		2		1		
	1		3		3		1	
	1	4		6		4		1
1		5	10		10	5		1
1	6		15	20		15	6	1
1	7	21	(35)	35	21	7	1	

1 mark for justification

The coefficient of x^3 is 35.

1 mark for identifying coefficient

2 marks

Method 2

$$t_4 = {}_7C_3 (1)^4 (x)^3$$

$${}_7C_3 = \frac{7!}{3!4!}$$

$$= \frac{7 \cdot \cancel{6} \cdot 5 \cdot \cancel{4}!}{3! \cancel{4}!}$$

$$= 35$$

1 mark for justification

1 mark for evaluating coefficient

2 marks

The coefficient of x^3 is 35.

Which of the following equations could be solved without the use of logarithms?

Without actually solving the problem, explain your choice.

$$4^x = 10^{3x+1} \quad \text{or} \quad \left(\frac{1}{3}\right)^{2x+1} = 27^{4x-1}$$

Solution

$\left(\frac{1}{3}\right)^{2x+1} = 27^{4x-7}$ can be solved without the use of logarithms because $\frac{1}{3}$ and 27 can both be changed to a base of 3.

1 mark for explanation

1 mark

Sketch the graph of $y = x^3 + x^2 - 5x + 3$ given that one of the x -intercepts is 1.

Identify the x -intercepts and y -intercept.

Solution

$$x = 1$$

1	1	-5	3
	1	2	-3
1	2	-3	0

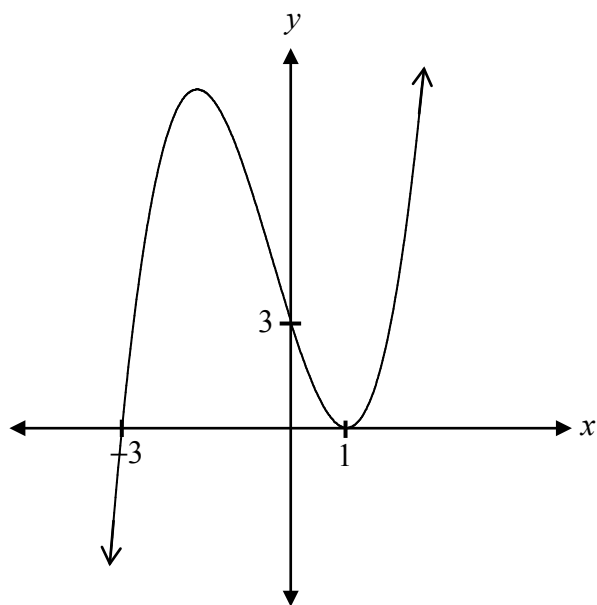
1 mark for synthetic division

$$y = (x - 1)(x^2 + 2x - 3)$$

$$y = (x - 1)(x + 3)(x - 1)$$

$$y = (x + 3)(x - 1)^2$$

1 mark for identifying the factors



2 marks for graph ($\frac{1}{2}$ mark for x -intercepts, $\frac{1}{2}$ mark for multiplicity, $\frac{1}{2}$ mark for y -intercept, $\frac{1}{2}$ mark for end behaviour)

4 marks

If $f(x) = \frac{1}{x-2}$ and $g(x) = x-2$, what is the domain of $f(x) \cdot g(x)$?

Solution

$$f(x) = \frac{1}{x-2} \qquad g(x) = x-2$$

Domain: $x \in \mathbb{R}, x \neq 2$ Domain: $x \in \mathbb{R}$

Domain of $f(x) \cdot g(x)$: $\{x \in \mathbb{R} \mid x \neq 2\}$

1 mark for domain of $f(x) \cdot g(x)$

1 mark

Given $f(x) = (x+1)^2$ for $x \leq -1$, write the equation of $y = f^{-1}(x)$.

Solution

Method 1

$$y = (x+1)^2$$

$$x = (y+1)^2$$

$$y = \pm\sqrt{x} - 1$$

1 mark for inverse

½ mark for solving for y

Since the domain of $f(x)$ is $x \leq -1$,
the range of the inverse is $y \leq -1$.

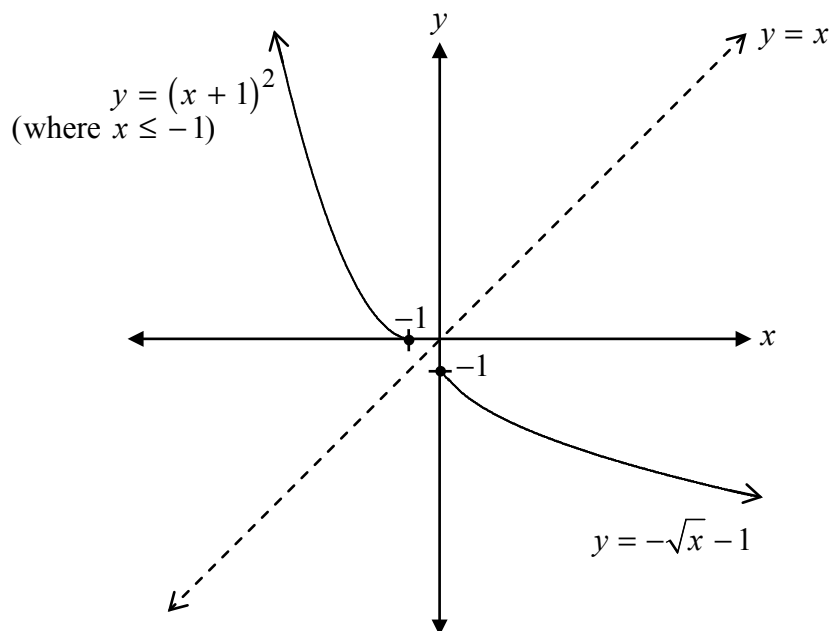
$$\therefore y = -\sqrt{x} - 1$$

$$f^{-1}(x) = -\sqrt{x} - 1$$

½ mark for rejecting $y = \sqrt{x}$

2 marks

Method 2



1 mark for reflection over the line $y = x$

1 mark for correct equation

2 marks

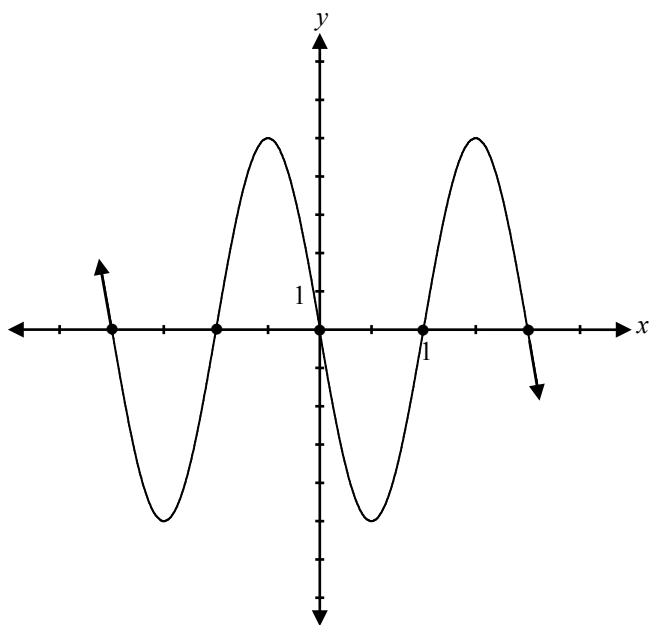
Sketch a graph of at least one period of the function $y = 5 \sin[\pi(x+1)]$.

Clearly indicate the x -intercepts.

Solution

$$b = \pi$$

$$\therefore \text{period} = \frac{2\pi}{\pi} = 2$$



1 mark for amplitude
1 mark for horizontal shift
1 mark for period
1 mark for clearly indicating at least two x -intercepts consistent with graph

4 marks

Sketch the graph of the following function:

$$f(x) = \frac{x-2}{(2x-3)(x-2)}$$

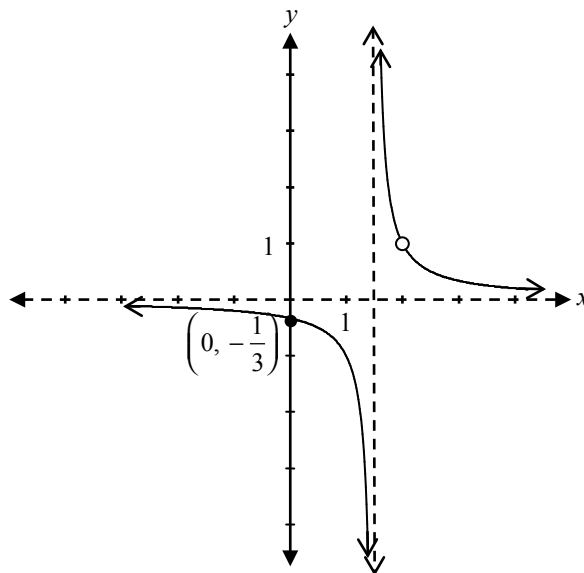
Solution

$$\begin{aligned} f(x) &= \frac{x-2}{(2x-3)(x-2)} \\ &= \frac{1}{2x-3} \text{ with a point of discontinuity at } x=2 \end{aligned}$$

point of discontinuity: $f(2) = 1$

$\therefore$ there is a point of discontinuity at $(2, 1)$

$$\begin{aligned} \text{y-intercept: } f(0) &= \frac{0-2}{(2(0)-3)(0-2)} \\ &= -\frac{2}{6} \\ &= -\frac{1}{3} \end{aligned}$$



1 mark for horizontal asymptote at $y = 0$

1 mark for vertical asymptote at $x = \frac{3}{2}$

$\frac{1}{2}$ mark for graph left of vertical asymptote

$\frac{1}{2}$ mark for graph right of vertical asymptote

1 mark for point of discontinuity at $(2, 1)$; ($\frac{1}{2}$ mark for $x = 2$, $\frac{1}{2}$ mark for $y = 1$)

4 marks

Convert $-\frac{13\pi}{5}$ to degrees.

Solution

$$-\frac{13\pi}{5} \times \frac{180^\circ}{\pi}$$
$$-468^\circ$$

1 mark

Question 2

P2, P3

- a) From a group of 9 people, in how many ways can you select a committee of 4 members?
- b) From a group of 9 people, in how many ways can you select a president, a vice president, a secretary, and a treasurer?
- c) Explain why the answers in a) and b) are different.

Solution

a) ${}_9C_4 = 126$ ways

1 mark for ${}_9C_4$

1 mark

b) ${}_9P_4 = 3024$ ways

1 mark for ${}_9P_4$

1 mark

- c) Part a) is a combination because the order does not matter; part b) is a permutation because committee members have specific roles.

1 mark

Question 3

R10

A population of 500 bacteria will triple in 20 hours.

Using the formula given below,

$$A = Pe^{rt}$$

A = population after t hours

P = initial population

r = rate of growth

t = time in hours

- a) Determine the rate of growth, r .
- b) Determine how many hours it will take for the initial population to double with the same rate of growth.

Solution

a)

$$1500 = 500e^{r(20)}$$

½ mark for substitution

$$3 = e^{20r}$$

$$\ln 3 = \ln e^{20r}$$

½ mark for applying logarithms

$$\ln 3 = 20r \cdot \ln e$$

½ mark for power rule

$$r = \frac{\ln 3}{20}$$

$$r = 0.054\ 930\ 614$$

½ mark for evaluating quotient of logarithms

2 marks

b)

$$1000 = 500e^{0.054\ 930\ 614t}$$

½ mark for substitution

$$2 = e^{0.054\ 930\ 614t}$$

$$\ln 2 = \ln e^{0.054\ 930\ 614t}$$

½ mark for applying logarithms

$$\ln 2 = 0.054\ 930\ 614t \cdot \ln e$$

½ mark for power rule

$$t = \frac{\ln 2}{0.054\ 930\ 614}$$

$$t = 12.619 \text{ hours}$$

½ mark for evaluating quotient of logarithms

2 marks

Question 4

T5

Talla incorrectly solved the following trigonometric equation:

Solve: $2 \sec x - 5 = 0$; $0^\circ \leq x \leq 360^\circ$.

Talla's work:

$$2 \sec x - 5 = 0$$

$$\sec x = \frac{5}{2}$$

No solution, $\sec x$ cannot be greater than 1.

- Explain her error.
- Determine the correct solution.

Solution

- Talla incorrectly stated that $\sec x$ cannot be greater than 1.
The value of $\cos x$ cannot be greater than 1.

1 mark

$$\text{b) } \sec x = \frac{5}{2}$$

$$\cos x = \frac{2}{5}$$

$$x_r = 66.421\ 821$$

$$x = 66.422^\circ$$

$$x = 293.578^\circ$$

1 mark for reciprocal

1 mark for solving for x ($\frac{1}{2}$ mark for each value of x)

2 marks

Question 5

P4

Simplify the 6th term in the expansion of:

$$\left(2x - \frac{3}{x^2}\right)^{10}$$

Solution

$$t_6 = {}_{10}C_5 (2x)^5 \left(-\frac{3}{x^2}\right)^5$$

2 marks (1 mark for ${}_{10}C_5$, $\frac{1}{2}$ mark for each consistent factor)

$$= 252 \left(32x^5\right) \left(-\frac{243}{x^{10}}\right)$$

$$= -1\,959\,552\,x^{-5}$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark for exponent)

3 marks

Question 6

T1

Determine the arc length subtended by a central angle if the diameter is 19 cm and the central angle is 1.6 radians.

Solution

$$s = \theta r$$

$$s = (1.6)(9.5)$$

$$s = 15.2 \text{ cm}$$

1 mark

Question 7

T3, T5

Solve the following equation algebraically for x , where $0 \leq x \leq 2\pi$.

$$2 \cos^2 x = -3 \sin x$$

Solution

$$2(1 - \sin^2 x) = -3 \sin x$$

1 mark for identity

$$2 - 2 \sin^2 x = -3 \sin x$$

$$0 = 2 \sin^2 x - 3 \sin x - 2$$

$$0 = (2 \sin x + 1)(\sin x - 2)$$

$$\sin x = -\frac{1}{2}$$

~~$$\sin x = 2$$~~

1 mark for solving for $\sin x$

No Solution

1 mark for indicating no solution

$$x = \frac{7\pi}{6}$$

1 mark for solving for x ($\frac{1}{2}$ mark for each value)

$$x = \frac{11\pi}{6}$$

4 marks

In how many different ways can you arrange the letters in the word VOLLEYBALL?

State your answer as a factorial.

Solution

$$\frac{10!}{4!}$$

1 mark

Note(s):

- award full marks for 151 200

Question 9

R11

Is $(x - 2)$ a factor of the polynomial $p(x) = -x^4 - 3x^3 + 11x^2 + 3x - 10$?

Justify your response.

Solution

Method 1

$$\begin{aligned} p(2) &= -(2)^4 - 3(2)^3 + 11(2)^2 + 3(2) - 10 \\ &= -16 - 24 + 44 + 6 - 10 \\ &= 0 \end{aligned}$$

The remainder is zero, so $(x - 2)$ is a factor.

$\frac{1}{2}$ mark for $p(2)$

1 mark for the remainder theorem

$\frac{1}{2}$ mark for justification

2 marks

Method 2

2	-1	-3	11	3	-10
	↓	-2	-10	2	10
	-1	-5	1	5	<u>0</u>

The remainder is zero, so $(x - 2)$ is a factor.

$\frac{1}{2}$ mark for $x = 2$

1 mark for synthetic division
(or for any equivalent strategy)

$\frac{1}{2}$ mark for justification

2 marks

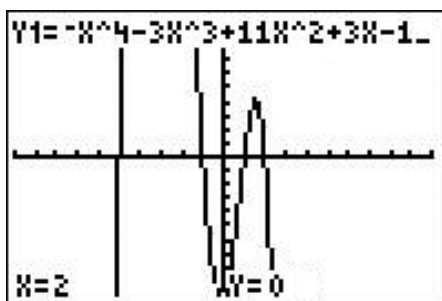
Method 3

I entered $y = -x^4 - 3x^3 + 11x^2 + 3x - 10$ into my calculator and located the zeroes.

$x = 2$ was a zero, which means $(x - 2)$ is a factor.

1 mark for graphing calculator method

1 mark for relating the zeroes to the factors



2 marks

Determine the period of the sinusoidal function $y = \frac{1}{2} \sin\left(\frac{1}{3}x\right)$.

State your answer in radians.

Solution

$$p = \frac{2\pi}{|b|}$$

$$p = \frac{2\pi}{\left|\frac{1}{3}\right|}$$

$$p = 6\pi \quad \text{or} \quad p = 18.850$$

$\frac{1}{2}$ mark for correct value of b

$\frac{1}{2}$ mark for period consistent with b

1 mark

Question 11

R1

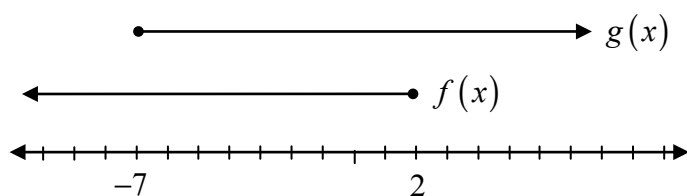
The domain of $f(x)$ is $x \leq 2$. The domain of $g(x)$ is $x \geq -7$.

State the domain of $f(x) + g(x)$.

Justify your answer.

Solution

Both $f(x)$ and $g(x)$ have restricted domains, so both domains need to be considered.



The solution is the intersection of the two domains.

1 mark for justification

$$\{x | x \in \mathbb{R}, -7 \leq x \leq 2\} \quad \text{or} \quad [-7, 2]$$

1 mark for domain

2 marks

Question 12

T6

Prove the identity below for all permissible values of θ .

$$\frac{1}{1 + \cos \theta} = \csc^2 \theta - \frac{\cot \theta}{\sin \theta}$$

Solution

Method 1

Left-Hand Side	Right-Hand Side	
$\frac{1}{1 + \cos \theta}$	$\csc^2 \theta - \frac{\cot \theta}{\sin \theta}$	
	$\frac{1}{\sin^2 \theta} - \frac{\frac{\cos \theta}{\sin \theta}}{\sin \theta}$	1 mark for correct substitution of identities
	$\frac{1}{\sin^2 \theta} - \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta}$	
	$\frac{1}{\sin^2 \theta} - \frac{\cos \theta}{\sin^2 \theta}$	1 mark for algebraic strategies
	$\frac{1 - \cos \theta}{\sin^2 \theta}$	
	$\frac{1 - \cos \theta}{1 - \cos^2 \theta}$	1 mark for logical process to prove an identity
	$\frac{1 - \cancel{\cos \theta}}{(1 - \cancel{\cos \theta})(1 + \cos \theta)}$	
	$\frac{1}{1 + \cos \theta}$	3 marks

Solution**Method 2**

Left-Hand Side	Right-Hand Side	
$\frac{1}{1 + \cos \theta} (1 - \cos \theta)$	$\csc^2 \theta - \frac{\cot \theta}{\sin \theta}$	1 mark for algebraic strategies
$\frac{1 - \cos \theta}{1 - \cos^2 \theta}$		
$\frac{1 - \cos \theta}{\sin^2 \theta}$		1 mark for correct substitution of identities
$\frac{1}{\sin^2 \theta} - \frac{\cos \theta}{\sin^2 \theta}$		
$\csc^2 \theta - \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta}$		1 mark for logical process to prove an identity
$\csc^2 \theta - \frac{\cot \theta}{\sin \theta}$		3 marks

Explain how the end behaviours of the graphs of polynomial functions with an even degree and with an odd degree are different.

Solution

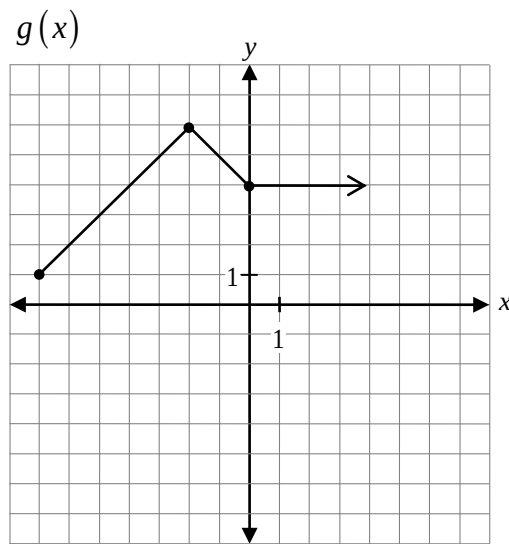
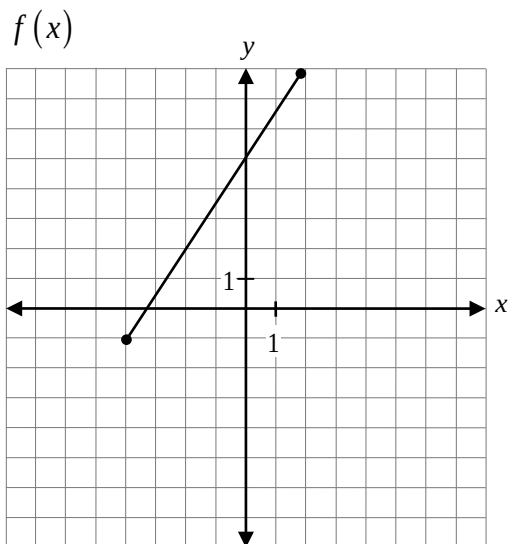
When the degree is odd, the end behaviour is in opposite directions.
When the degree is even, the end behaviour is in the same direction.

1 mark

Question 14

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $g(x) - f(x)$.



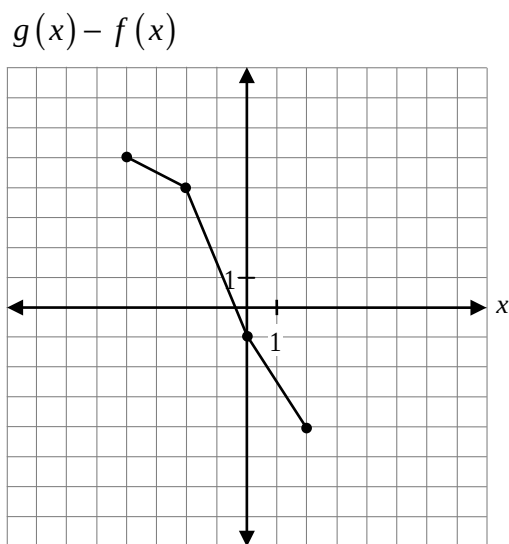
Solution

x	$g(x)$	$f(x)$	$(g - f)(x)$
-4	4	-1	5
-2	6	2	4
0	4	5	-1
2	4	8	-4

1 mark for subtraction of $g(x) - f(x)$

1 mark for restricting domain on graph

2 marks



Given $f(x) = -3x + 7$, evaluate $f^{-1}(-2)$.

Solution

$$\text{Let } y = f(x)$$

$$f(x) = -3x + 7$$

$$y = -3x + 7$$

$$x = -3y + 7$$

$$x - 7 = -3y$$

$$y = \frac{x - 7}{-3}$$

$$f^{-1}(x) = \frac{x - 7}{-3}$$

$$f^{-1}(-2) = \frac{-2 - 7}{-3}$$

$$f^{-1}(-2) = 3$$

1 mark for switching x and y

$\frac{1}{2}$ mark for $f^{-1}(x)$

$\frac{1}{2}$ mark for $f^{-1}(-2)$

2 marks

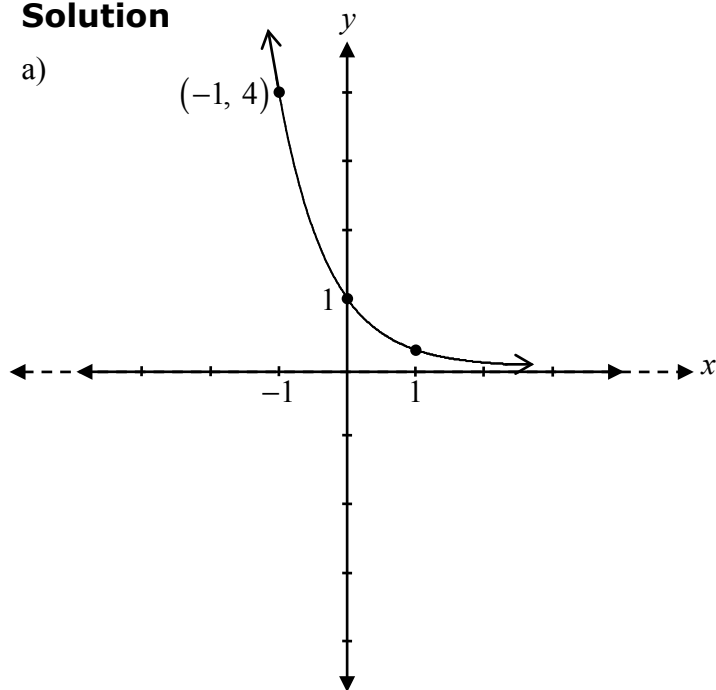
Sketch the graphs of:

a) $y = \left(\frac{1}{4}\right)^x$

b) $y = 2\left(\frac{1}{4}\right)^x$

Solution

a)



$\frac{1}{2}$ mark for decreasing exponential function

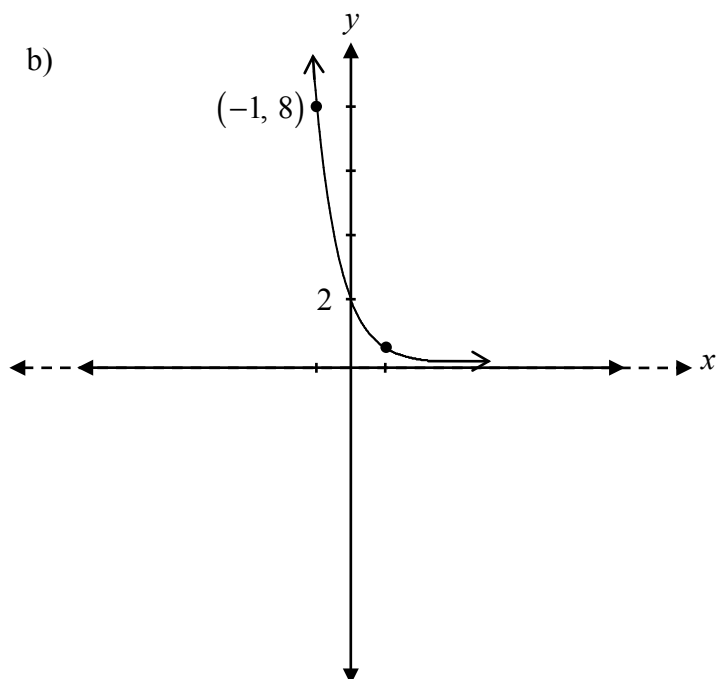
$\frac{1}{2}$ mark for y-intercept $(0, 1)$

$\frac{1}{2}$ mark for consistent point of an exponential function

$\frac{1}{2}$ mark for horizontal asymptote at $y = 0$

2 marks

b)



1 mark for a vertical stretch by a factor of 2 of the graph consistent with a)

1 mark

Determine all of the zeroes of the function $p(x) = x^3 - 5x^2 - 2x + 24$, given one of the factors of $p(x)$ is $(x - 3)$.

Solution

$$0 = x^3 - 5x^2 - 2x + 24$$

3	1	-5	-2	24
	↓	3	-6	-24
	1	-2	-8	0

½ mark for $x = 3$

1 mark for synthetic division (or for any equivalent strategy)

$$x^2 - 2x - 8 = 0$$

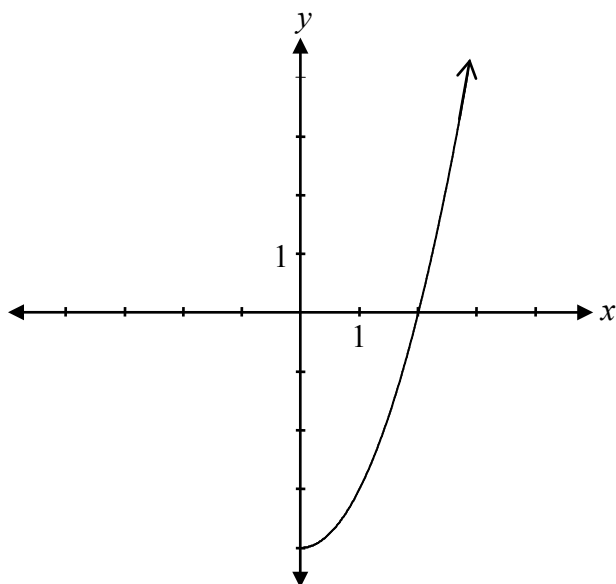
$$(x - 4)(x + 2) = 0$$

zeroes: 3, 4, -2

½ mark for consistent zeroes

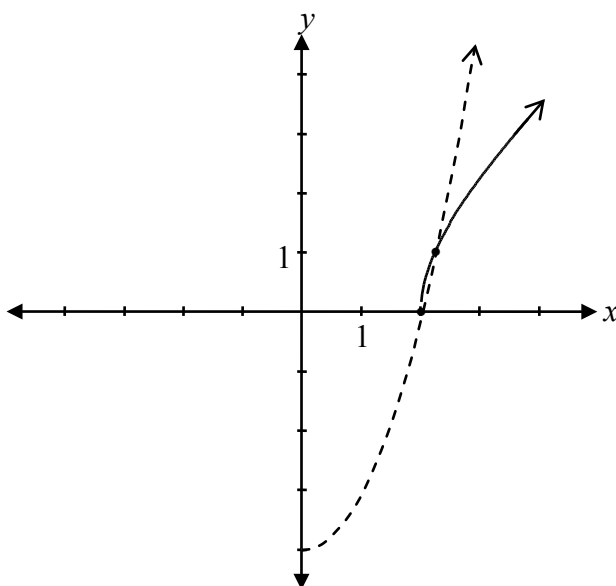
2 marks

Given the graph of $f(x)$,



sketch the graph of $y = \sqrt{f(x)}$.

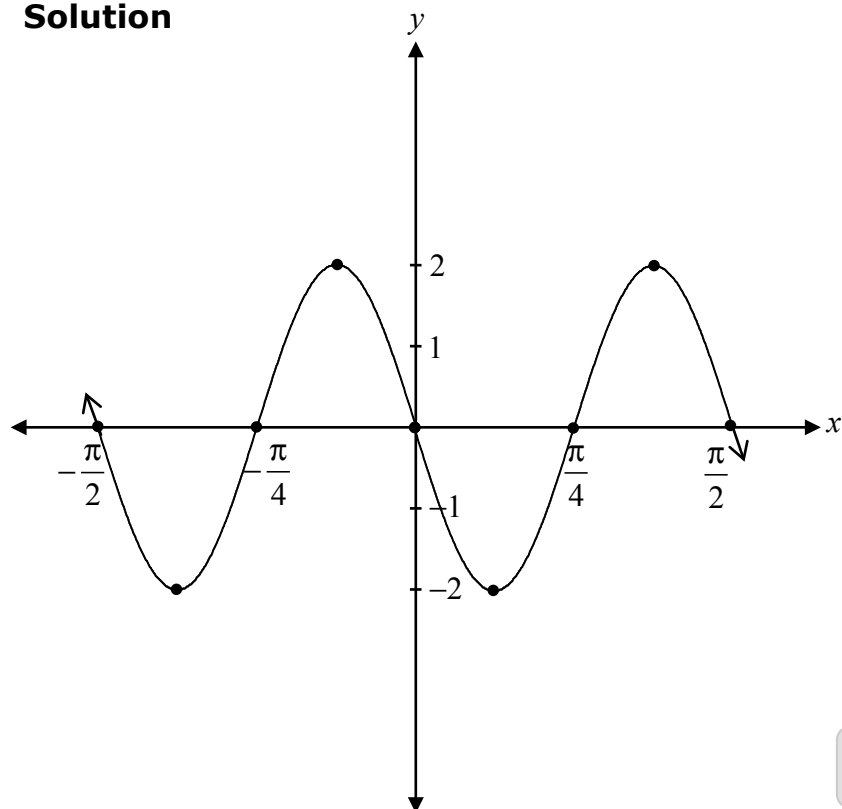
Solution



1 mark for restricting domain
 $\frac{1}{2}$ mark for shape between invariant points
 $\frac{1}{2}$ mark for shape to the right of the invariant points

2 marks

Sketch the graph of at least one period of the function $y = -2\sin(4x)$.

Solution

1 mark for amplitude
1 mark for period
1 mark for reflection in the x -axis

3 marks

Evaluate:

$$\frac{1}{2} \log_3 144 - \log_3 4 + 2 \log_3 3$$

Solution

$$\log_3 (144)^{\frac{1}{2}} - \log_3 4 + \log_3 (3)^2$$

1 mark for power rule

$$\log_3 12 - \log_3 4 + \log_3 9$$

$$\log_3 \left(\frac{12 \cdot 9}{4} \right)$$

$\frac{1}{2}$ mark for product rule

$\frac{1}{2}$ mark for quotient rule

$$\log_3 27$$

$$3$$

1 mark for evaluating a logarithm

3 marks

Match each function with its correct description.

- a) The graph of this function has a vertical asymptote at $x = -1$.
- b) The graph of this function has a point of discontinuity (hole) at $x = 3$.
- c) The graph of this function has a horizontal asymptote at $y = 4$.
- d) The domain of this function is $x \in \mathbb{R}$.

Place the appropriate letter in this column.

Solution

$$f(x) = \frac{4}{x^2 + 1} \quad \underline{\quad \text{d)} \quad}$$

$$g(x) = \frac{4x}{x + 3} \quad \underline{\quad \text{c)} \quad}$$

$$h(x) = \frac{4(x - 3)(x + 2)}{(x - 3)} \quad \underline{\quad \text{b)} \quad}$$

$$k(x) = \frac{4(x - 3)}{(x + 3)(x + 1)} \quad \underline{\quad \text{a)} \quad}$$

$\frac{1}{2}$ mark for each correct answer

2 marks

The point $(-3, 4)$ is on the graph of $y = \frac{1}{2}f(3x)$.

State the coordinates of the corresponding point on the graph of $y = f(x)$.

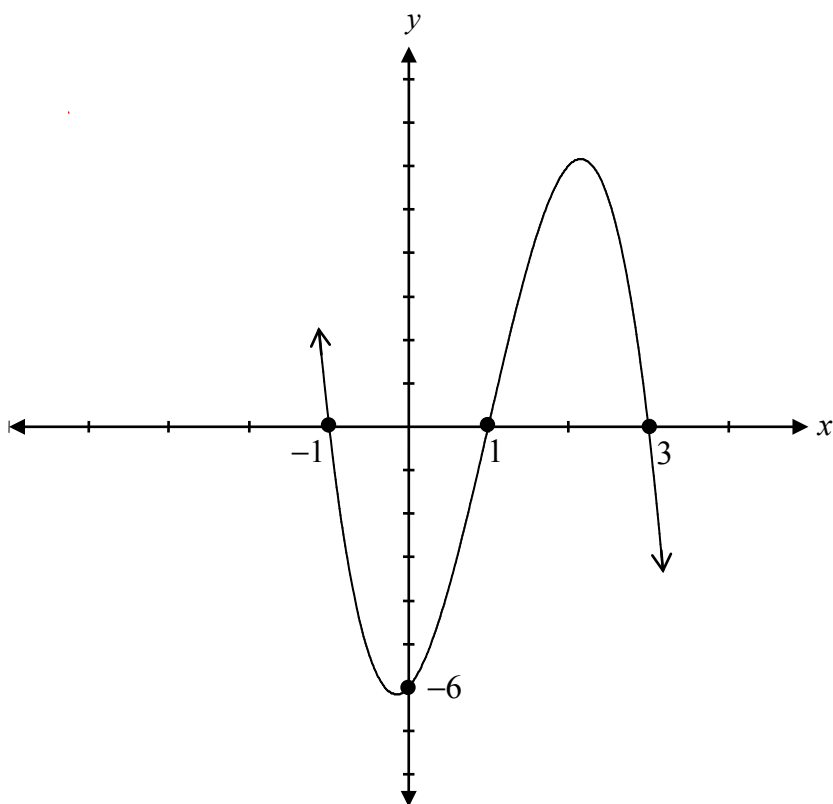
Solution

$(-9, 8)$

$\frac{1}{2}$ mark for each coordinate

1 mark

Sketch the graph of $y = -2(x-1)(x-3)(x+1)$.

Solution

1 mark for x -intercepts
1 mark for y -intercept
1 mark for end behaviour

3 marks

- a) Verify that the equation $\frac{1 - \sin^2 x}{\cos x} = \frac{\sin 2x}{2 \sin x}$ is true for $x = \frac{\pi}{3}$.
- b) Explain why verifying the equation for $x = \frac{\pi}{3}$ is insufficient to conclude that the equation is an identity.

Solution

$$\begin{array}{l|l}
 \text{a)} \quad \frac{1 - \sin^2\left(\frac{\pi}{3}\right)}{\cos \frac{\pi}{3}} & \frac{\sin 2\left(\frac{\pi}{3}\right)}{2 \sin \frac{\pi}{3}} \\
 \\
 \frac{1 - \left(\frac{\sqrt{3}}{2}\right)^2}{\frac{1}{2}} & \frac{\frac{\sqrt{3}}{2}}{2\left(\frac{\sqrt{3}}{2}\right)} \\
 \\
 \frac{1 - \frac{3}{4}}{\frac{1}{2}} & \frac{\frac{\sqrt{3}}{2}}{\frac{2}{\sqrt{3}}} \\
 \\
 \frac{\frac{1}{4}}{\frac{1}{2}} & \frac{1}{2} \\
 \frac{1}{2} &
 \end{array}$$

LHS = RHS

1 mark for exact values ($\frac{1}{2}$ mark for $\sin \frac{\pi}{3}$, $\frac{1}{2}$ mark for $\cos \frac{\pi}{3}$)1 mark for simplification ($\frac{1}{2}$ mark for LHS, $\frac{1}{2}$ mark for RHS)**2 marks**

- b) Proving that it is true for one value does not mean that it is true for all values.

1 mark

Evaluate:

$$\frac{{}_7P_2}{{}_7P_5}$$

Solution

$$\frac{\frac{7!}{(7-2)!}}{\frac{7!}{(7-5)!}}$$

½ mark for substitution

$$\frac{\frac{7!}{5!}}{2!}$$

$$\frac{2!}{5!}$$

½ mark for simplification

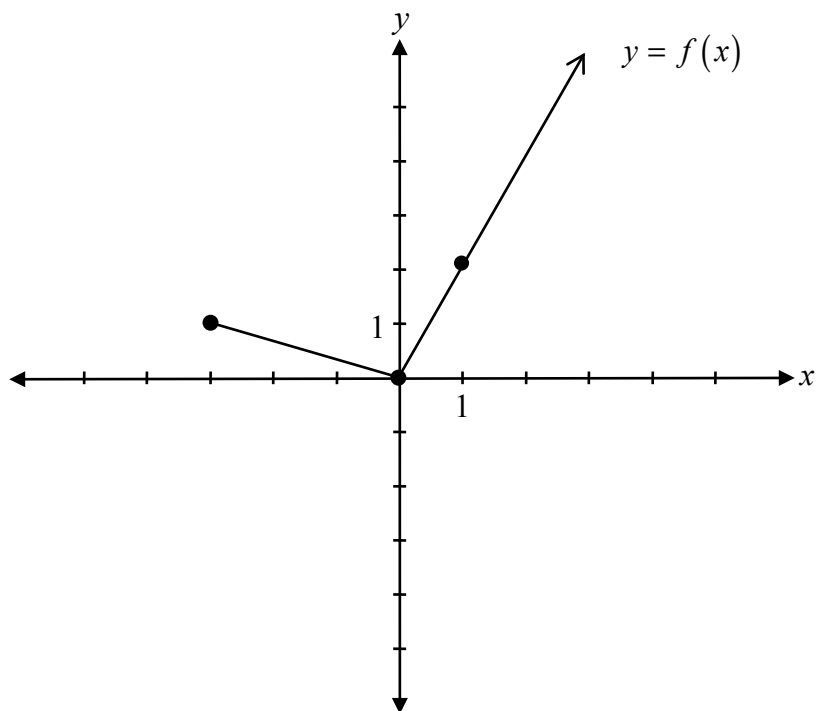
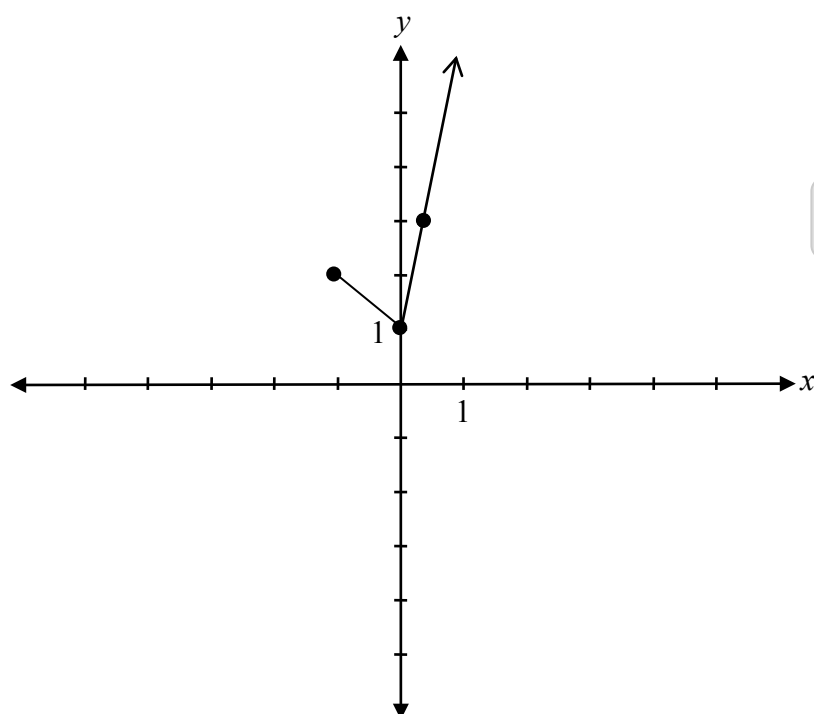
$$\frac{2 \times 1}{5 \times 4 \times 3 \times 2 \times 1}$$

1 mark for expanding factorials

$$\frac{1}{60}$$

2 marks

Use the graph of $y = f(x)$ to sketch the graph of $y = f(3x) + 1$.

**Solution**

1 mark for horizontal compression
1 mark for vertical translation

2 marks

Solve the following equation:

$$\log_4(x+2) + \log_4 3 = \log_4 x$$

Solution

Method 1

$$\log_4(x+2) + \log_4 3 = \log_4 x$$

$$\log_4[(x+2)3] = \log_4 x$$

$$3(x+2) = x$$

$$3x + 6 = x$$

$$x = -3$$

No solution

1 mark for product rule

1 mark for equating arguments

$\frac{1}{2}$ mark for solving for x

$\frac{1}{2}$ mark for rejecting extraneous root

3 marks

Method 2

$$\log_4(x+2) + \log_4 3 = \log_4 x$$

$$\log_4(x+2) + \log_4 3 - \log_4 x = 0$$

$$\log_4 \left[\frac{3(x+2)}{x} \right] = 0$$

$$4^0 = \frac{3x+6}{x}$$

$$x = -3$$

~~$$x = -3$$~~

1 mark for logarithmic rules ($\frac{1}{2}$ mark for product rule; $\frac{1}{2}$ mark for quotient rule)

1 mark for exponential form

$\frac{1}{2}$ mark for solving for x

$\frac{1}{2}$ mark for rejecting extraneous root

3 marks

Determine the coordinates of the point of discontinuity (hole) for the graph of the function

$$y = \frac{(2-x)(x-3)}{(x-2)}.$$

Solution

$$x \neq 2$$

$$y = -(x-3)$$

$$y = -(2-3)$$

$$y = 1$$

$$(2, 1)$$

1 mark for point of discontinuity (hole) at $(2, 1)$

($\frac{1}{2}$ mark for $x = 2$, $\frac{1}{2}$ mark for consistent y -coordinate)

1 mark

Evaluate and simplify $\sec\left(\frac{5\pi}{6}\right) \cdot \tan\left(-\frac{\pi}{6}\right)$.

Solution

$$\left(-\frac{2}{\sqrt{3}}\right) \cdot \left(-\frac{1}{\sqrt{3}}\right)$$

$$\frac{2}{3}$$

1 mark for $\sec\left(\frac{5\pi}{6}\right)$ ($\frac{1}{2}$ mark for value, $\frac{1}{2}$ mark for quadrant)

1 mark for $\tan\left(-\frac{\pi}{6}\right)$ ($\frac{1}{2}$ mark for value, $\frac{1}{2}$ mark for quadrant)

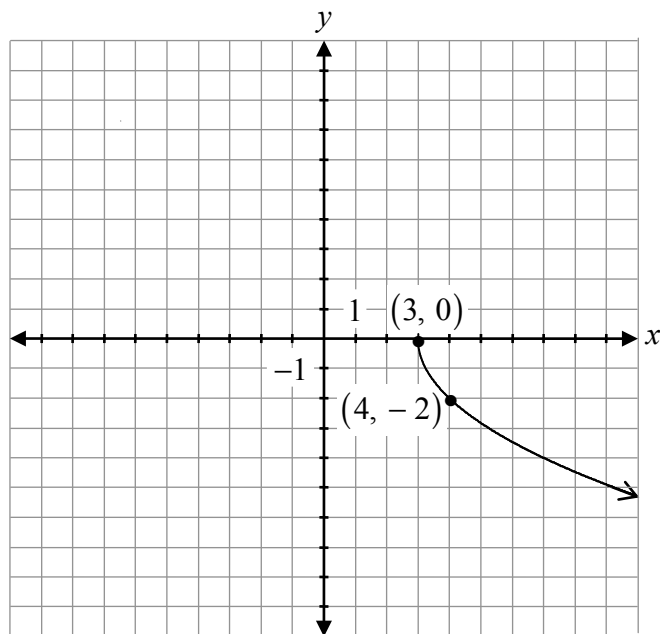
2 marks

Sketch the graph of the following function:

$$y = -2\sqrt{x-3}$$

Solution

Method 1



1 mark for shape (graph of a radical function)

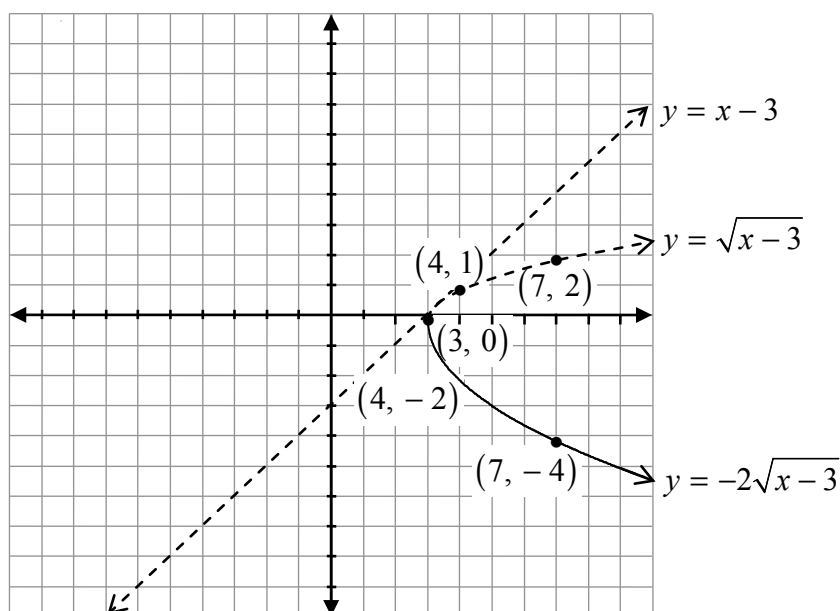
1 mark for vertical reflection

1 mark for horizontal shift

1 mark for vertical stretch

4 marks

Method 2



1 mark for invariant points where $y = 0$ and $y = 1$ ($\frac{1}{2}$ mark for each point)

1 mark for domain of $[3, \infty)$

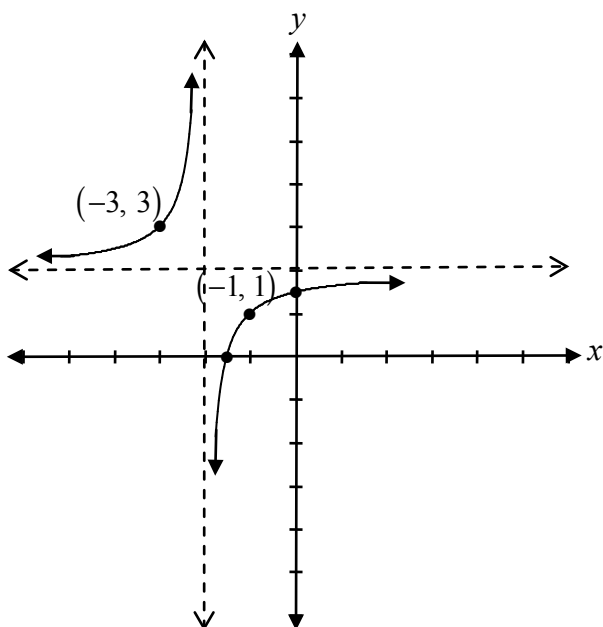
$\frac{1}{2}$ mark for shape between invariant points

$\frac{1}{2}$ mark for shape to the right of the invariant points

1 mark for applying transformations (vertical stretch, vertical reflection)

4 marks

Sketch the graph of $f(x) = \frac{2x+3}{x+2}$.

Solution

1 mark for vertical asymptote

1 mark for horizontal asymptote

$\frac{1}{2}$ mark for graph left of the vertical asymptote

$\frac{1}{2}$ mark for graph right of the vertical asymptote

3 marks

- a) Given the functions $f(x) = \sqrt{4+x}$ and $g(x) = |3x-6|$, evaluate $f(g(-5))$.
b) Is it possible to evaluate $g(f(-5))$?

Justify your answer.

Solution

a) $g(-5) = |3(-5) - 6|$

$$g(-5) = 21$$

1 mark for $g(-5)$

$$f(21) = \sqrt{4+21}$$

$$= \sqrt{25}$$

$$= 5$$

1 mark for consistent value of $f(g(-5))$

2 marks

- b) No, because $f(x)$ is undefined when $x = -5$

1 mark for justification

or

$$f(-5) = \sqrt{4+(-5)}$$

$$f(-5) = \sqrt{-1}$$

1 mark

$f(-5)$ is undefined because you cannot evaluate the square root of a negative number.

Identify which of these values is greater. Justify your answer.

$$\log_5 80 \quad \text{or} \quad \log_3 30$$

Solution

$$\begin{aligned} 5^2 &= 25 \\ 5^3 &= 125 \end{aligned} \quad \log_5 80 \text{ is less than } 3$$

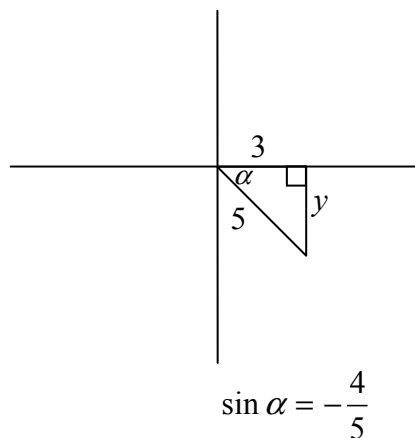
$$\begin{aligned} 3^3 &= 27 \\ 3^4 &= 81 \end{aligned} \quad \log_3 30 \text{ is more than } 3$$

$\therefore \log_3 30$ is greater

1 mark for justification

1 mark

Given $\cos \alpha = \frac{3}{5}$, where α is in quadrant IV, and $\cos \beta = -\frac{2}{3}$, where β is in quadrant II, determine the exact value of $\sin(\alpha - \beta)$.

Solution

$$x^2 + y^2 = r^2$$

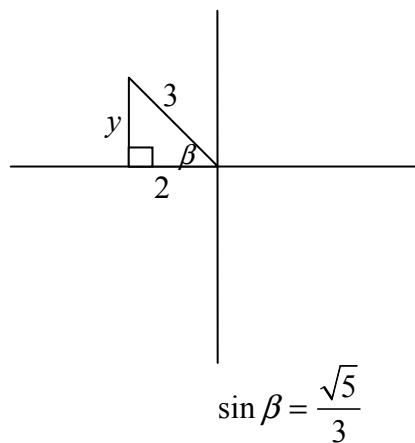
$$9 + y^2 = 25$$

$$y^2 = 16$$

$$y = \pm 4$$

$$y = -4$$

½ mark for value of y



$$x^2 + y^2 = r^2$$

$$4 + y^2 = 9$$

$$y^2 = 5$$

$$y = \pm\sqrt{5}$$

$$y = \sqrt{5}$$

½ mark for value of y

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= \left(-\frac{4}{5}\right)\left(-\frac{2}{3}\right) - \left(\frac{3}{5}\right)\left(\frac{\sqrt{5}}{3}\right)$$

$$= \frac{8 - 3\sqrt{5}}{15}$$

½ mark for $\sin \alpha$

½ mark for $\sin \beta$

1 mark for substitution into correct identity

3 marks

Determine the number of possible sandwiches from the following menu.

MENU

Select one item from each column:

<u>Bread</u>	<u>Sauce</u>	<u>Meat</u>	<u>Vegetable</u>
White	Mayo	Turkey	Tomato
Rye	Mustard	Ham	Onion
Brown		Roast Beef	Lettuce
		Chicken	

Solution

$$3 \times 2 \times 4 \times 3$$

72 sandwiches

1 mark

Question 1

T1

A pizza with a diameter of 15 inches is cut into equal slices, each with a central angle of 36° . Determine the length of the crust on the outer edge of one slice of pizza.

Solution

$$\frac{36^\circ}{180^\circ} = \frac{\theta}{\pi}$$

$$\theta = \frac{\pi}{5}$$

1 mark for conversion

$$s = \theta r$$

$$s = \left(\frac{\pi}{5}\right)\left(\frac{15}{2}\right)$$

1 mark for substitution

$$s = \frac{3\pi}{2} \text{ inches}$$

2 marks

or

$$s = 4.712 \text{ inches}$$

Question 2

P3

There are 9 girls and 7 boys in a math class from which a committee of 5 is to be chosen.

- a) How many different committees of 5 can be formed if one of the boys, William, must be on the committee?
- b) How many different committees of 5 can be formed if there must be 2 girls and 3 boys on the committee?

Solution

a) ${}_1C_1 \cdot {}_{15}C_4 = 1365$

1 mark

b) ${}_9C_2 \cdot {}_7C_3 = 1260$

$\frac{1}{2}$ mark for ${}_9C_2$

$\frac{1}{2}$ mark for ${}_7C_3$

1 mark for the product of combinations

2 marks

Note(s):

- ${}_1C_1$ does not need to be shown

Question 3

T5

Solve the following equation over the interval $[0, 2\pi]$:

$$\sin^2 \theta + 6 \sin \theta - 2 = 0$$

Solution

$$\sin \theta = \frac{-6 \pm \sqrt{(6)^2 - 4(1)(-2)}}{2(1)}$$

$$\sin \theta = \frac{-6 \pm \sqrt{36 + 8}}{2}$$

$$\sin \theta = \frac{-6 \pm \sqrt{44}}{2}$$

$$\sin \theta = 0.316\ 624\dots$$

$$\theta_r = 0.322\ 169$$

$$\theta = 0.322$$

$$\theta = 2.819$$

$$\sin \theta = -6.316\ 624\dots$$

no solution

1 mark for solving for $\sin \theta$

2 marks for solving for θ ($\frac{1}{2}$ mark for each value, 1 mark for indicating no solution)

3 marks

Question 4

R10

Solve:

$$6(5)^{3x+2} = 9^{2-x}$$

Solution

$$\log \left[6(5)^{3x+2} \right] = \log 9^{2-x}$$

½ mark for applying logarithms

$$\log 6 + \log 5^{3x+2} = \log 9^{2-x}$$

1 mark for product law

$$\log 6 + (3x + 2) \log 5 = (2 - x) \log 9$$

1 mark for power law

$$\log 6 + 3x \log 5 + 2 \log 5 = 2 \log 9 - x \log 9$$

$$3x \log 5 + x \log 9 = 2 \log 9 - 2 \log 5 - \log 6$$

½ mark for collecting terms with x

$$x(3 \log 5 + \log 9) = 2 \log 9 - 2 \log 5 - \log 6$$

$$x = \frac{2 \log 9 - 2 \log 5 - \log 6}{3 \log 5 + \log 9}$$

½ mark for solving for x

$$x = -0.087\ 707$$

½ mark for evaluating quotient of logarithms

$$= -0.088$$

4 marks

Solve $(2 \sin \theta - 1)(\sin \theta + 1) = 0$ where $\theta \in \mathbb{R}$.

Solution

$$2 \sin \theta - 1 = 0$$

$$\sin \theta = \frac{1}{2}$$

$$\theta_r = \frac{\pi}{6}$$

$$\theta = \frac{\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

$$\theta = \frac{5\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

or

$$\theta_r = 30^\circ$$

$$\theta = 30^\circ + 360^\circ k, k \in \mathbb{Z}$$

$$\theta = 150^\circ + 360^\circ k, k \in \mathbb{Z}$$

$$\sin \theta + 1 = 0$$

$$\sin \theta = -1$$

$$\theta_r = \frac{3\pi}{2}$$

$$\theta = \frac{3\pi}{2} + 2k\pi, k \in \mathbb{Z}$$

or

$$\theta_r = 270^\circ$$

$$\theta = 270^\circ + 360^\circ k, k \in \mathbb{Z}$$

1 mark for solving for $\sin \theta$

2 marks for solving for θ (1 mark for each branch)

1 mark for general solution

4 marks

The roots of the polynomial equation $3(x - 2)(x + 1)^2 = 0$ are $x = 2$ and $x = -1$.

Explain what these roots represent on the graph of $p(x) = 3(x - 2)(x + 1)^2$.

Solution

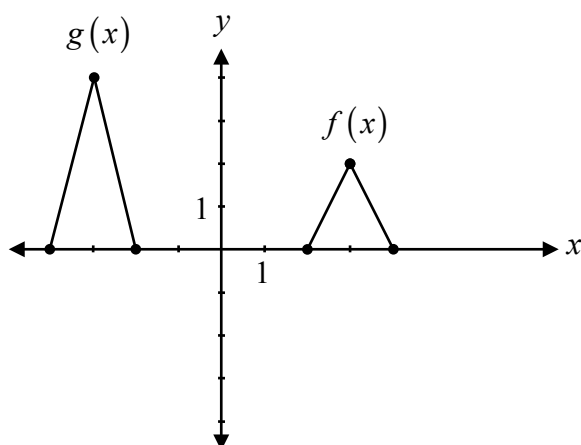
They are the x -intercepts of the graph of $p(x)$.

1 mark

Question 7

R2, R3

Determine an equation for $g(x)$ as a transformation of $f(x)$.



Solution

$$g(x) = \underline{2f(x+6)}$$

1 mark for vertical stretch

1 mark for horizontal translation

2 marks

or

$$g(x) = \underline{2f(-x)}$$

1 mark for vertical stretch

1 mark for horizontal reflection

2 marks

Question 8

R11

A student must determine the factors of $5x^4 - 2x^3 + 4x - 1$. He used 5, -2 , 4, and -1 as the coefficients of the polynomial when using synthetic division.

Explain the student's error.

Solution

The student did not write the coefficient of 0 for the x^2 term.

1 mark

Describe the transformations of $y = f(x)$ when asked to sketch the graph of $y = -f(x - 4)$.

Solution

$f(x)$ is reflected over the x -axis and translated 4 units to the right.

1 mark for vertical reflection

1 mark for horizontal translation

2 marks

Prove the identity below for all permissible values of θ :

$$\sin \theta + \frac{\cos \theta}{\tan \theta} = \frac{1}{\cos \theta \tan \theta}$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\sin \theta + \frac{\cos \theta}{\tan \theta}$	$\frac{1}{\cos \theta \tan \theta}$
$\sin \theta + \frac{\cos \theta}{\frac{\sin \theta}{\cos \theta}}$	$\frac{1}{\cos \theta \frac{\sin \theta}{\cos \theta}}$
$\sin \theta + \frac{\cos^2 \theta}{\sin \theta}$	$\frac{1}{\sin \theta}$
$\frac{\sin^2 \theta}{\sin \theta} + \frac{\cos^2 \theta}{\sin \theta}$	
$\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta}$	
$\frac{1}{\sin \theta}$	

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove an identity

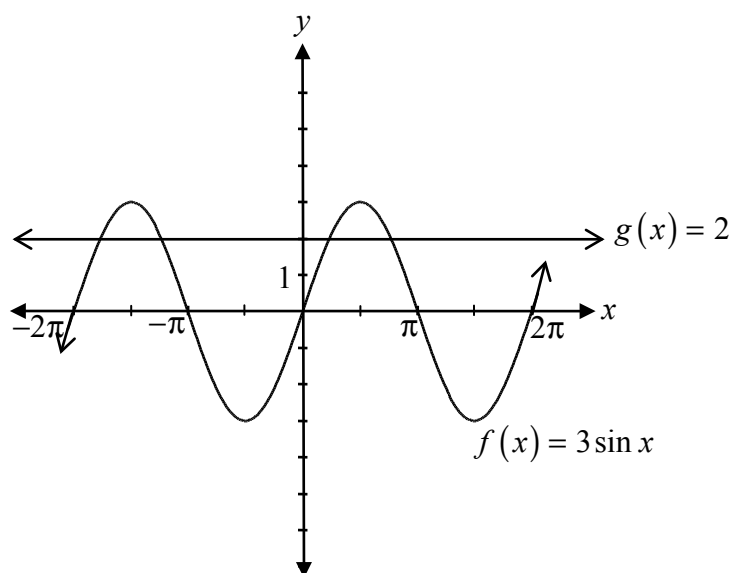
3 marks

Method 2

Left-Hand Side	Right-Hand Side	
$\sin \theta + \frac{\cos \theta}{\tan \theta}$	$\frac{1}{\cos \theta \tan \theta}$	
	$\frac{1}{\cos \theta \frac{\sin \theta}{\cos \theta}}$	1 mark for correct substitution of identities
	$\frac{1}{\sin \theta}$	
	$\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta}$	
	$\frac{\sin^2 \theta}{\sin \theta} + \frac{\cos^2 \theta}{\sin \theta}$	
	$\sin \theta + \cos \theta \frac{\cos \theta}{\sin \theta}$	1 mark for algebraic strategies
	$\sin \theta + \cos \theta \cot \theta$	
	$\sin \theta + \frac{\cos \theta}{\tan \theta}$	1 mark for logical process to prove an identity

3 marks

Describe how to use the graphs of $f(x) = 3 \sin x$ and $g(x) = 2$ to solve the equation $3 \sin x = 2$.

**Solution**

The solution will be the x -values where the two graphs intersect.

1 mark

Question 12

P1

A hockey arena has 5 doors.

Determine the number of ways that you can enter through one door but exit through a different door.

Solution

$$\underline{5} \cdot \underline{4} = 20 \text{ ways}$$

1 mark

Given that $(x + 3)$ is one of the factors, express $2x^3 + 7x^2 + 2x - 3$ as a product of factors.

Solution

$$\begin{array}{r|rrrr}
 -3 & 2 & 7 & 2 & -3 \\
 & \downarrow & & & \\
 \hline
 & 2 & 1 & -1 & 0
 \end{array}$$

$\frac{1}{2}$ mark for $x = -3$

1 mark for synthetic division (or for any other equivalent strategy)

$$(x + 3)(2x^2 + x - 1)$$

$\frac{1}{2}$ mark for consistent product of factors

or

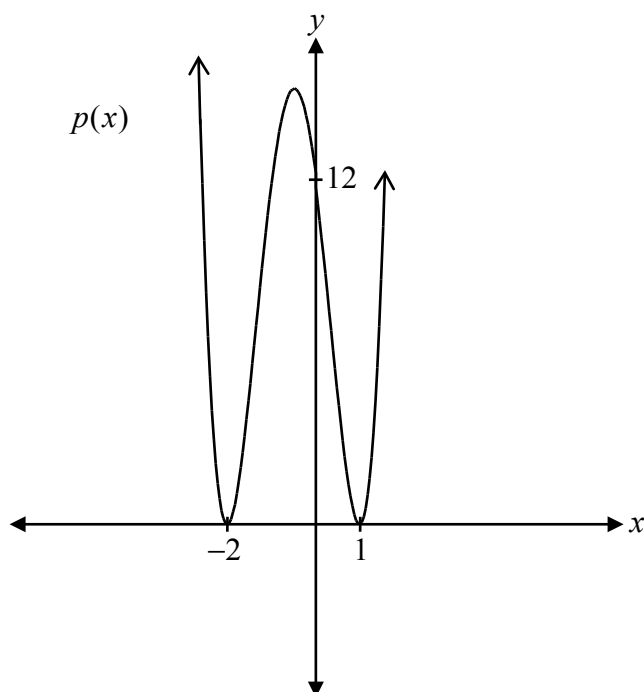
$$(x + 3)(2x - 1)(x + 1)$$

2 marks

Question 22

R12

Determine the equation of the polynomial function, $p(x)$, represented by the graph.



Solution

$$p(x) = \underline{3(x+2)^2(x-1)^2}$$

1 mark for factors

1 mark for multiplicity of 2 (½ mark for each)

1 mark for correct value of a (consistent with factors and multiplicity)

3 marks

Evaluate:

$$\log_4 2$$

Solution

$$\frac{1}{2}$$

1 mark

Evaluate:

$$\left(\cos \frac{11\pi}{3}\right)\left(\csc \frac{11\pi}{6}\right)$$

Solution

$$\left(\frac{1}{2}\right)(-2)$$

1 mark for $\cos \frac{11\pi}{3}$ ($\frac{1}{2}$ mark for the quadrant, $\frac{1}{2}$ mark for the value)

1 mark for $\csc \frac{11\pi}{6}$ ($\frac{1}{2}$ mark for the quadrant, $\frac{1}{2}$ mark for the value)

-1

2 marks

Estimate the value of $\log_2 5$.

Justify your answer.

Solution

$$\left. \begin{array}{l} \log_2 4 = 2 \\ \log_2 8 = 3 \end{array} \right\} \quad \frac{1}{2} \text{ mark for justification}$$
$$\therefore \log_2 5 \approx 2.3 \quad \frac{1}{2} \text{ mark for estimated value in the interval } [2.1, 2.4]$$

1 mark

If θ terminates in quadrant III and $\cos \theta = -\frac{6}{7}$, determine the exact value of $\tan \theta$.

Solution

$$\cos \theta = \frac{x}{r}$$

$$x^2 + y^2 = r^2$$

$$y^2 = (7)^2 - (-6)^2 \quad \frac{1}{2} \text{ mark for substitution of } x = -6 \text{ and } r = 7$$

$$y^2 = 13$$

$$y = \pm\sqrt{13}$$

$\frac{1}{2}$ mark for solving for y

$$\tan \theta = \frac{\sqrt{13}}{6}$$

1 mark for the value of $\tan \theta$ ($\frac{1}{2}$ mark for the quadrant, $\frac{1}{2}$ mark for the value)

2 marks

Given $f(x) = x^2 + x - 4$ and $g(x) = \sqrt{x+5}$, Taz was asked to find $f(g(x))$.

Taz's solution:

$$\begin{aligned} f(g(x)) &= (\sqrt{x+5})^2 + x - 4 \\ &= x + 5 + x - 4 \\ &= 2x + 1, \quad x \geq -5 \end{aligned}$$

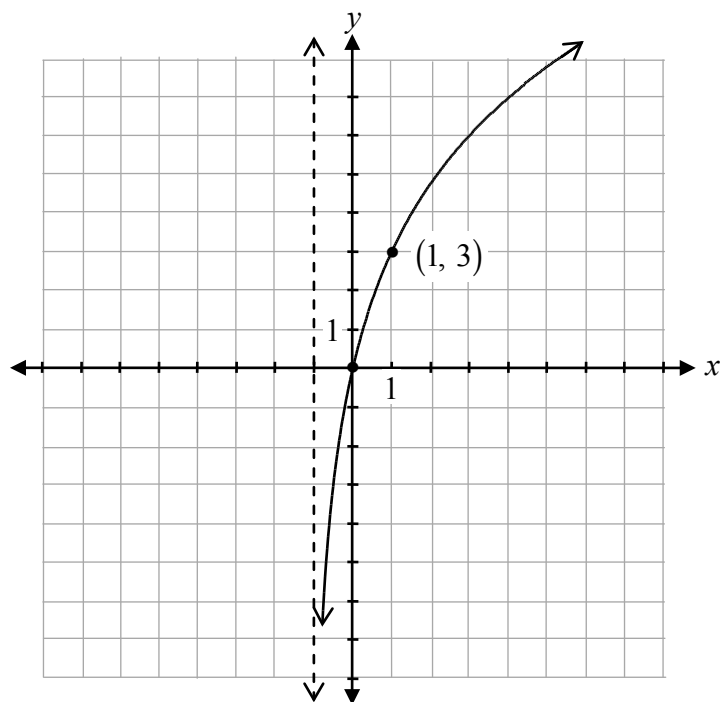
Describe the error in Taz's solution.

Solution

Taz must substitute $g(x)$ in both terms containing x in $f(x)$ and then simplify.

1 mark

Sketch the graph of the function $f(x) = 3\log_2(x+1)$.

Solution

1 mark for increasing logarithmic function
1 mark for vertical stretch
1 mark for asymptotic behaviour at $x = -1$

3 marks

Write an equation of a rational function that would not have any vertical asymptotes.

Solution

Various equations, such as the following, are possible:

$$y = \frac{(x-2)(x+1)}{(x-2)}$$

or

$$y = \frac{4}{x^2 + 4}$$

1 mark

Determine the exact value of $\tan 75^\circ$.

Solution

$$\tan 75^\circ = \tan(30^\circ + 45^\circ)$$

1 mark for combination

$$= \frac{\tan 30^\circ + \tan 45^\circ}{1 - \tan 30^\circ \tan 45^\circ}$$

$$= \frac{\frac{1}{\sqrt{3}} + 1}{1 - \frac{1}{\sqrt{3}}(1)}$$

1 mark for exact values ($\frac{1}{2}$ mark for each)

2 marks

$$= \frac{1 + \sqrt{3}}{\frac{\sqrt{3}}{\sqrt{3} - 1}}$$

$$= \frac{1 + \sqrt{3}}{\sqrt{3} - 1}$$

or

$$= \frac{\sqrt{3} + 3}{3 - \sqrt{3}}$$

Note(s):

- Other combinations are possible.

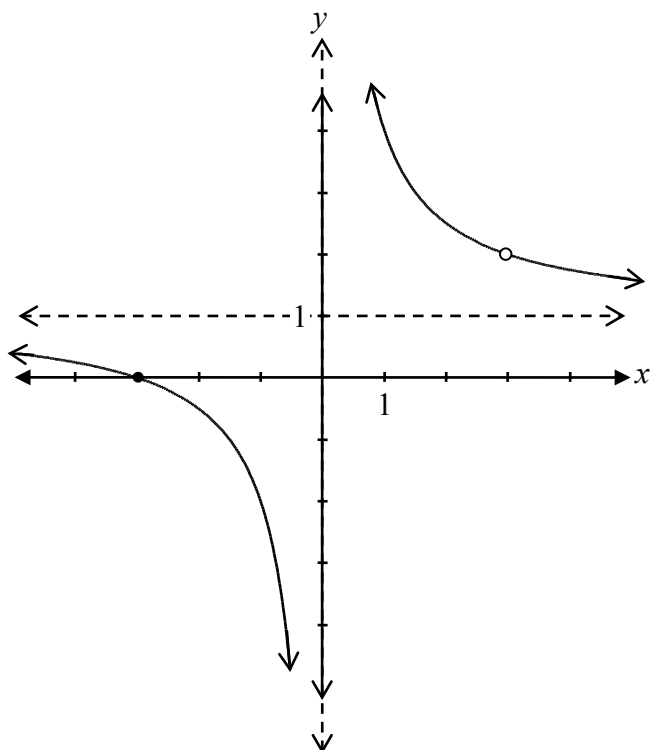
Sketch the graph of the following function:

$$f(x) = \frac{(x+3)(x-3)}{x(x-3)}$$

Solution

$$\begin{aligned} f(x) &= \frac{(x+3)(\cancel{x-3})}{x(\cancel{x-3})} \\ &= \frac{x+3}{x}, x \neq 3 \end{aligned}$$

$\therefore$ there is a point of discontinuity (hole) at $(3, 2)$



1 mark for asymptotic behaviour at $y = 1$

1 mark for asymptotic behaviour at $x = 0$

1 mark for point of discontinuity (hole) at $(3, 2)$

($\frac{1}{2}$ mark for $x = 3$, $\frac{1}{2}$ mark for $y = 2$)

$\frac{1}{2}$ mark for graph left of vertical asymptote

$\frac{1}{2}$ mark for graph right of vertical asymptote

4 marks

In the binomial expansion of $\left(\frac{1}{x^3} - 2x^2\right)^9$, determine which term contains x^3 .

Solution

Method 1

$$x^3 = (x^{-3})^{9-k} (x^2)^k$$

½ mark for substitution

$$x^3 = x^{-27+3k+2k}$$

$$3 = -27 + 3k + 2k$$

$$30 = 5k$$

$$6 = k$$

½ mark for solving for k

∴ term 7 would contain x^3

1 mark for identifying the 7th term
(or consistent term with the value of k)

2 marks

Method 2

$$\left(\frac{1}{x^3}\right)^9, \left(\frac{1}{x^3}\right)^8 (x^2), \left(\frac{1}{x^3}\right)^7 (x^2)^2$$

$$x^{-27}, x^{-22}, x^{-17}$$

1 mark for determining the pattern

∴ term 7 would contain x^3

1 mark for identifying the 7th term
(or consistent term with the pattern)

2 marks

Question 33

T4

José and Dana get on a Ferris wheel, which is 1 metre off the ground. The diameter of the Ferris wheel is 16 metres. Their ride lasts for 4 minutes, in which time the Ferris wheel makes one revolution.

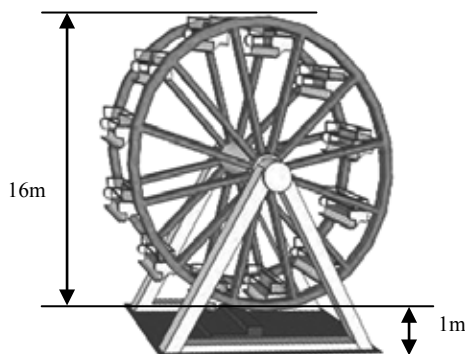
Determine the values of A , B , C , and D , if the sinusoidal function that models the situation is $h(t) = A\cos[B(t - C)] + D$, where h is the height at which José and Dana are located on the Ferris wheel, from the ground, in metres, and t is the time, in minutes.

$A =$ _____

$B =$ _____

$C =$ _____

$D =$ _____



Solution

$$A = \underline{\quad 8 \quad}$$

or

$$A = \underline{\quad -8 \quad}$$

1 mark for A

$$B = \underline{\quad \frac{\pi}{2} \quad}$$

$$B = \underline{\quad \frac{\pi}{2} \quad}$$

1 mark for B

$$C = \underline{\quad 2 \quad}$$

$$C = \underline{\quad 0 \quad}$$

1 mark for C

$$D = \underline{\quad 9 \quad}$$

$$D = \underline{\quad 9 \quad}$$

1 mark for D

4 marks

Note(s)

- Other answers are possible.

Solve algebraically:

$${}_nP_3 = 4!(n-1)$$

Solution

$$\frac{n!}{(n-3)!} = 4!(n-1)$$

$$\frac{n(\cancel{n-1})(n-2)(\cancel{n-3})!}{(\cancel{n-3})!} = 4!(\cancel{n-1})$$

$$n(n-2) = 24$$

$$n^2 - 2n - 24 = 0$$

$$(n-6)(n+4) = 0$$

$$n = 6 \quad \cancel{n = -4}$$

½ mark for substitution

1 mark for factorial expansion

½ mark for simplification of factorials

½ mark for rejecting extraneous root

½ mark for the value of n

3 marks

Given $f(x) = \frac{2}{x-1}$, determine the equation of the inverse, $f^{-1}(x)$.

Solution**Method 1**

$$\text{let } y = f(x)$$

$$f(x) = \frac{2}{x-1}$$

$$y = \frac{2}{x-1}$$

$$x = \frac{2}{y-1}$$

1 mark for switching x and y values

$$y-1 = \frac{2}{x}$$

$$y = \frac{2}{x} + 1$$

$\frac{1}{2}$ mark for solving for y

$$f^{-1}(x) = \frac{2}{x} + 1$$

$\frac{1}{2}$ mark for writing equation of $f^{-1}(x)$

2 marks

Method 2

$$\text{let } y = f(x)$$

$$f(x) = \frac{2}{x-1}$$

$$y = \frac{2}{x-1}$$

$$x = \frac{2}{y-1}$$

1 mark for switching x and y values

$$x(y-1) = 2$$

$$xy - x = 2$$

$$xy = 2 + x$$

$$y = \frac{2+x}{x}$$

$\frac{1}{2}$ mark for solving for y

$$f^{-1}(x) = \frac{2+x}{x}$$

$\frac{1}{2}$ mark for writing equation of $f^{-1}(x)$

2 marks

Solve:

$$4 \log_3 2 - \frac{1}{3} \log_3 8 = \log_3 a$$

Solution**Method 1**

$$\log_3 2^4 - \log_3 8^{\frac{1}{3}} = \log_3 a$$

$$\log_3 16 - \log_3 2 = \log_3 a$$

$$\log_3 \left(\frac{16}{2} \right) = \log_3 a$$

$$\log_3 8 = \log_3 a$$

$$a = 8$$

1 mark for power law (½ mark for each)

1 mark for quotient law

1 mark for equating arguments

3 marks**Method 2**

$$\log_3 2^4 - \log_3 8^{\frac{1}{3}} - \log_3 a = 0$$

$$\log_3 16 - \log_3 2 - \log_3 a = 0$$

$$\log_3 \left(\frac{16}{2a} \right) = 0$$

$$\log_3 \left(\frac{8}{a} \right) = 0$$

$$3^0 = \frac{8}{a}$$

$$1 = \frac{8}{a}$$

$$a = 8$$

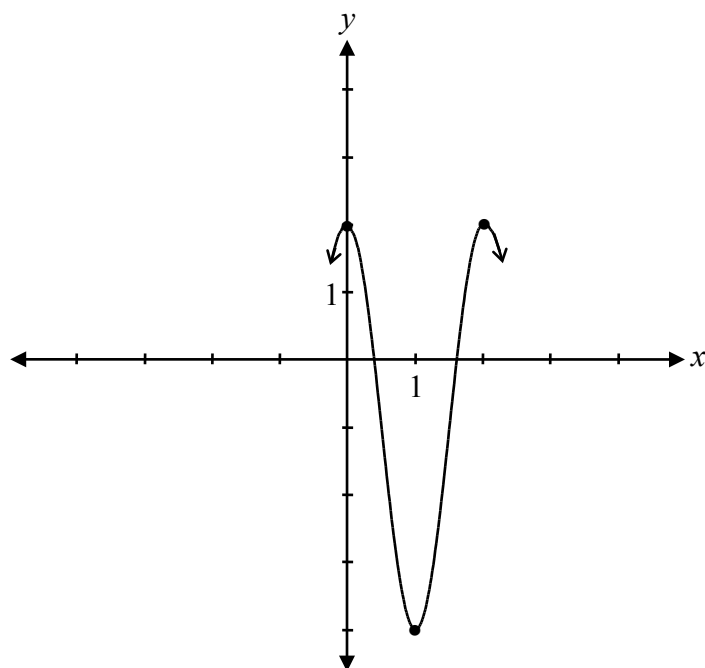
1 mark for power law (½ mark for each)

1 mark for quotient law

1 mark for exponential form

3 marks

Sketch the graph of at least one period of the function $y = 3\cos(\pi x) - 1$.

Solution

$$\begin{aligned}\text{period} &= \frac{2\pi}{\pi} \\ &= 2\end{aligned}$$

1 mark for amplitude
1 mark for period
1 mark for vertical translation

3 marks

Using the laws of logarithms, fully expand the expression:

$$\log_a \left(\frac{x^3}{y\sqrt{z}} \right)$$

Solution

$$\log_a \left(\frac{x^3}{y\sqrt{z}} \right) = \log_a x^3 - (\log_a y + \log_a \sqrt{z})$$

1 mark for quotient law

1 mark for product law

$$= 3 \log_a x - \left(\log_a y + \frac{1}{2} \log_a z \right)$$

1 mark for power law (½ mark for each)

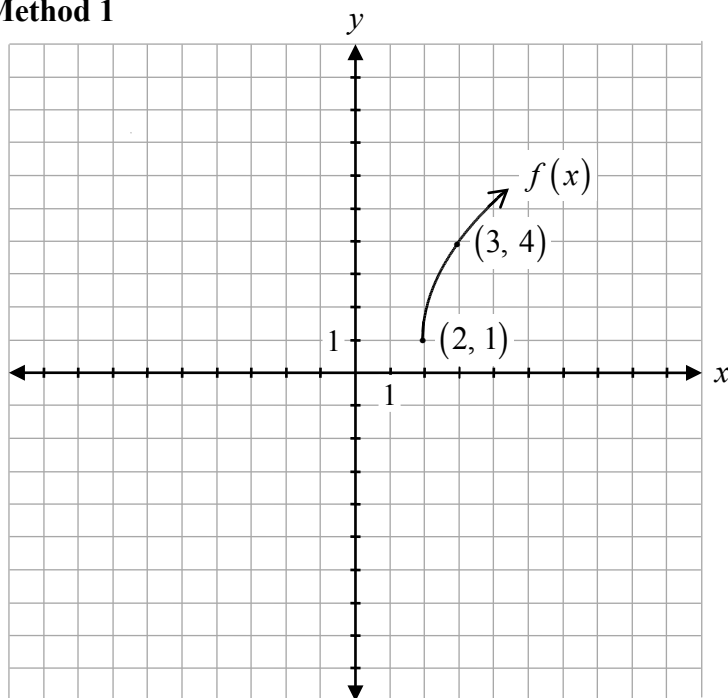
$$= 3 \log_a x - \log_a y - \frac{1}{2} \log_a z$$

3 marks

Sketch the graph of $f(x) = 3\sqrt{x-2} + 1$.

Solution

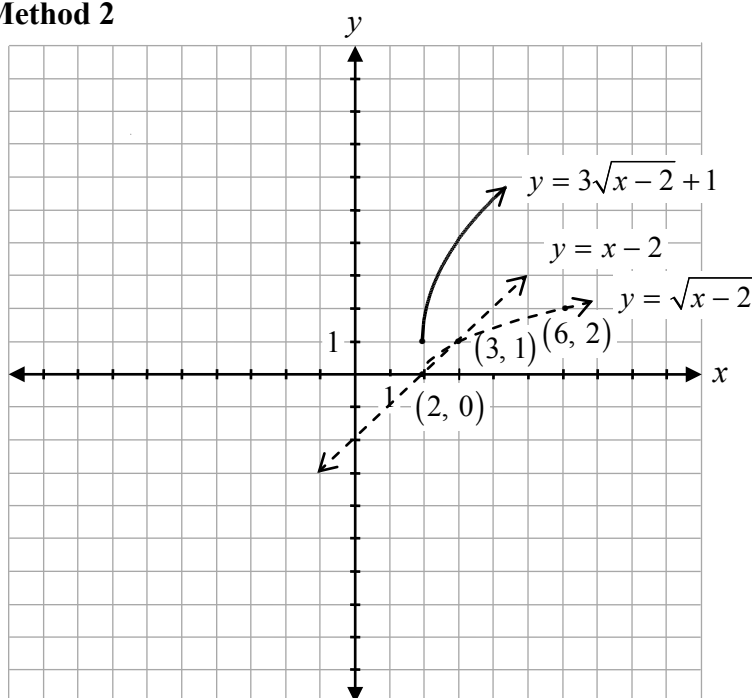
Method 1



- 1 mark for horizontal translation
- 1 mark for vertical translation
- 1 mark for shape of a radical function
- 1 mark for vertical stretch

4 marks

Method 2



- 1 mark for invariant points where $y = 0$ and $y = 1$ ($\frac{1}{2}$ mark for each point)
- 1 mark for domain $[2, \infty)$
- $\frac{1}{2}$ mark for shape between invariant points
- $\frac{1}{2}$ mark for shape to the right of the invariant points
- 1 mark for applying transformations ($\frac{1}{2}$ mark for vertical stretch, $\frac{1}{2}$ mark for vertical translation)

4 marks

- a) Determine the domain of the graph of the function $f(x) = \sqrt{x^2 - 4}$.
- b) Explain why the domain of $f(x) = \sqrt{x^2 - 4}$ is restricted.

Solution

a) $\{x \mid x \leq -2 \cup x \geq 2\}$

or

D: $(-\infty, -2] \cup [2, \infty)$

1 mark for domain ($\frac{1}{2}$ mark for $x \leq -2$, $\frac{1}{2}$ mark for $x \geq 2$)

1 mark

- b) The domain is restricted because you cannot take the square root of a negative number.

1 mark

Question 41

R1, R2, R5

Given the point $(-12, -18)$ on the graph of $f(x)$, determine the new points after the following transformations of $f(x)$.

a) $\frac{1}{f(x)}$

b) $f(-x) + 10$

Solution

a) $\left(-12, \frac{-1}{18}\right)$

1 mark

b) $(12, -8)$

1 mark ($\frac{1}{2}$ mark for x -value, $\frac{1}{2}$ mark for y -value)

1 mark

Explain why there is no solution for the equation $\csc \theta = -\frac{1}{2}$.

Solution

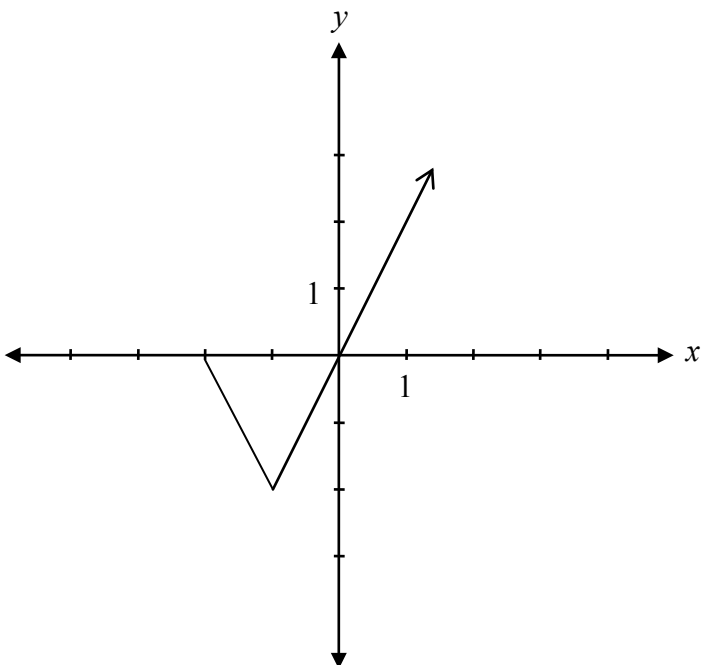
The value of $\csc \theta$ cannot be between -1 and 1 .

or

The value of $\sin \theta$ cannot be less than -1 .

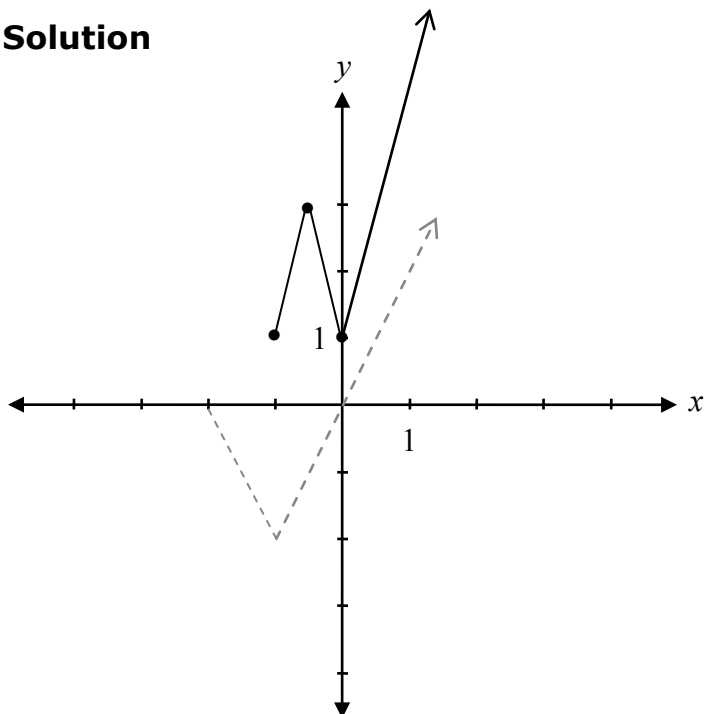
1 mark

Given the graph of $y = f(x)$,



sketch the graph of $y = |f(2x)| + 1$.

Solution



1 mark for absolute value
1 mark for horizontal compression
1 mark for vertical translation

3 marks

Given $f(x) = 2^x + 1$, state the equation of the horizontal asymptote.

Solution

$$y = 1$$

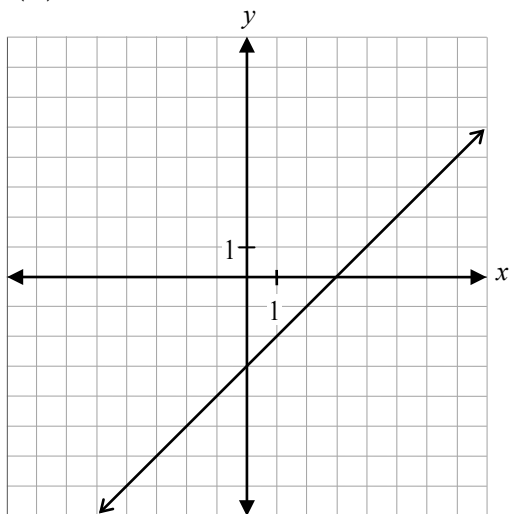
1 mark

Question 45

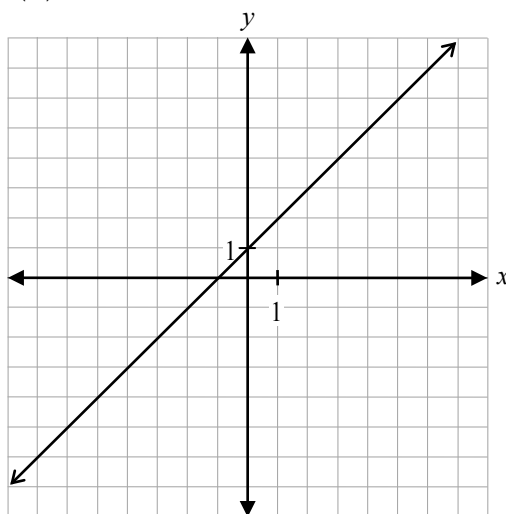
R1

Given the following graphs of $f(x) = x - 3$ and $g(x) = x + 1$,

$f(x)$

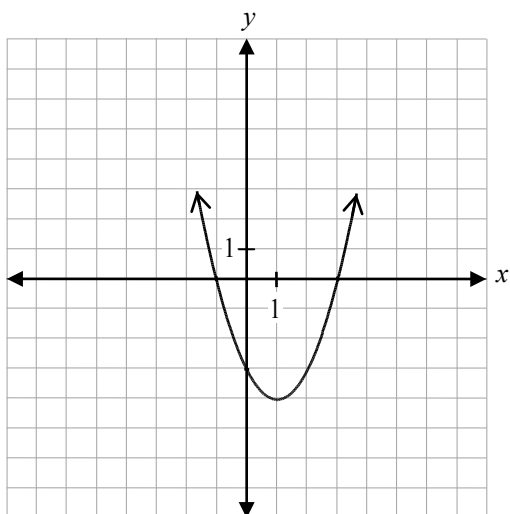


$g(x)$



sketch the graph of $h(x) = (f \cdot g)(x)$.

Solution



1 mark for operation of multiplication

1 mark for shape representing the given operation

2 marks

Question 1

P3

There are 24 different movies Kiandra can download to her computer. Determine the number of ways she can select 15 movies.

Solution

$${}_{24}C_{15} = 1\,307\,504 \text{ combinations}$$

1 mark

Question 2

T1

Given $\theta = 40^\circ$,

- a) convert θ to radians.
- b) determine the coterminal angles of θ where $\theta \in \mathbb{R}$.

Solution

a)

$$\theta = 40 \left(\frac{\pi}{180} \right)$$

$$\theta = \frac{2\pi}{9}$$

or

$$\theta = 0.698$$

1 mark

b)

$$\theta = \frac{2\pi}{9} + 2\pi k \in \mathbb{Z}$$

or

$$\theta = 0.698 + 2\pi k \in \mathbb{Z}$$

or

$$\theta = 40^\circ + 360^\circ k, k \in \mathbb{Z}$$

1 mark

Question 3

R10

Peter invests \$560 per month at an annual interest rate of 4.2%, compounded monthly. Determine how many monthly investments he will need to make to obtain at least \$500 000. Express your answer as a whole number.

Use the formula:

$$FV = \frac{R \left[(1+i)^n - 1 \right]}{i}$$

where FV = the future value

R = the investment amount each period

$i = \frac{\text{the annual interest rate}}{\text{the number of compounding periods per year}}$

n = the number of investments

Solution

$$500\,000 = \frac{560 \left[\left(1 + \frac{0.042}{12} \right)^n - 1 \right]}{\frac{0.042}{12}}$$

½ mark for substitution

$$500\,000 = \frac{560 \left[(1 + 0.0035)^n - 1 \right]}{0.0035}$$

$$500\,000 = 160\,000 (1.0035^n - 1)$$

$$3.125 = 1.0035^n - 1$$

$$4.125 = 1.0035^n$$

½ mark for simplification

$$\log 4.125 = \log 1.0035^n$$

½ mark for applying logarithms

$$\log 4.125 = n \log 1.0035$$

1 mark for power law

$$n = \frac{\log 4.125}{\log 1.0035}$$

$$n = 405.584$$

½ mark for solving for n

∴ 406 monthly investments are needed.

3 marks

Question 4

P2

Ishmael has 4 dogs, 5 cats, and 3 horses.

If he arranges all of them in a row, determine how many ways they can be arranged if each type of animal must be grouped together.

Solution

$$\frac{3!}{\text{types of animals}} \cdot \frac{4!}{\text{dogs}} \cdot \frac{5!}{\text{cats}} \cdot \frac{3!}{\text{horses}} = 103\,680 \text{ ways}$$

1 mark for arrangement of types of animals
1 mark for arrangement of dogs, cats, and horses

2 marks

Solve the following equation algebraically over the interval $0 \leq \theta \leq 2\pi$.

$$2 \cos^2 \theta + 9 \cos \theta - 5 = 0$$

Solution

$$2 \cos^2 \theta + 9 \cos \theta - 5 = 0$$

$$(2 \cos \theta - 1)(\cos \theta + 5) = 0$$

$$\cos \theta = \frac{1}{2}$$

$$\cos \theta = -5$$

1 mark for solving for $\cos \theta$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

no solution

2 marks for solving for θ

($\frac{1}{2}$ mark for each value, 1 mark for indicating no solution)

3 marks

Determine which term contains $\frac{1}{x^6}$ in the binomial expansion of $\left(\frac{2}{x^3} + 3x^2\right)^7$.

Solution

Method 1

$$\left(x^{-3}\right)^{7-k} \left(x^2\right)^k = x^{-6}$$

½ mark for substitution

$$x^{-21+3k} x^{2k} = x^{-6}$$

$$x^{-21+5k} = x^{-6}$$

$$-21 + 5k = -6$$

$$5k = 15$$

$$k = 3$$

½ mark for solving for k

∴ the fourth term contains $\frac{1}{x^6}$

1 mark for the 4th term (or a term consistent with the value of k)

2 marks

Method 2

$$\left(\frac{1}{x^3}\right)^7, \left(\frac{1}{x^3}\right)^6 (x^2), \left(\frac{1}{x^3}\right)^5 (x^2)^2, \dots$$

$$\frac{1}{x^{21}}, \frac{1}{x^{16}}, \frac{1}{x^{11}}, \dots$$

1 mark for determining a pattern

∴ the fourth term contains $\frac{1}{x^6}$

1 mark for the 4th term (or a term consistent with the pattern)

2 marks

Question 7

T1

Determine the radius of a circle which has an arc length of 5 cm with a central angle of 3 radians.

Solution

$$s = \theta r$$

$$5 = 3r$$

$$r = \frac{5}{3} \text{ cm}$$

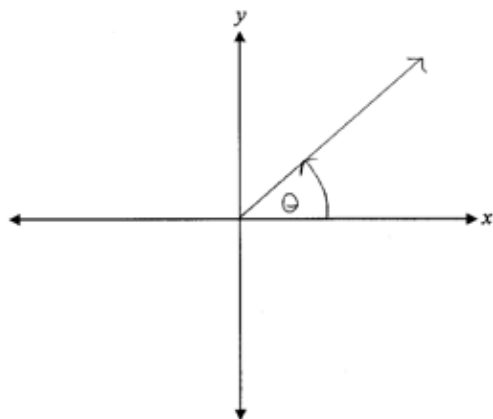
1 mark

Question 8

T1

Tyler incorrectly sketched the angle $\theta = -\frac{7\pi}{4}$ in standard position.

Describe his error.



Solution

Tyler incorrectly indicated the direction of the angle as positive.

or

Tyler sketched the reference angle, not $\theta = -\frac{7\pi}{4}$.

1 mark

Given the identity $\sec \theta + \cos \theta = \frac{2 - \sin^2 \theta}{\cos \theta}$,

- a) determine the non-permissible values of θ , over the interval $0 \leq \theta \leq 2\pi$.
- b) prove the identity for all permissible values of θ .

Solution

a) $\cos \theta = 0$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

1 mark for non-permissible values
($\frac{1}{2}$ mark for each value)

1 mark

Solution**Method 1**

b)

Left-Hand Side	Right-Hand Side
$\sec \theta + \cos \theta$	$\frac{2 - \sin^2 \theta}{\cos \theta}$
$\frac{1}{\cos \theta} + \cos \theta$	
$\frac{1 + \cos^2 \theta}{\cos \theta}$	1 mark for correct substitution of appropriate identities
$\frac{1 + 1 - \sin^2 \theta}{\cos \theta}$	1 mark for algebraic strategies
$\frac{2 - \sin^2 \theta}{\cos \theta}$	1 mark for logical process to prove the identity
	3 marks

Method 2

Left-Hand Side	Right-Hand Side
$\sec \theta + \cos \theta$	$\frac{2 - \sin^2 \theta}{\cos \theta}$
	$\frac{2 - (1 - \cos^2 \theta)}{\cos \theta}$
	1 mark for correct substitution of appropriate identities
	$\frac{1 + \cos^2 \theta}{\cos \theta}$
	1 mark for algebraic strategies
	$\frac{1}{\cos \theta} + \frac{\cos^2 \theta}{\cos \theta}$
	1 mark for logical process to prove the identity
	$\sec \theta + \cos \theta$
	3 marks

Question 10

R8

Expand using the laws of logarithms.

$$\log\left(\frac{a}{b^4}\right)$$

Solution

$$\log a - \log b^4$$

1 mark for quotient law

$$\log a - 4 \log b$$

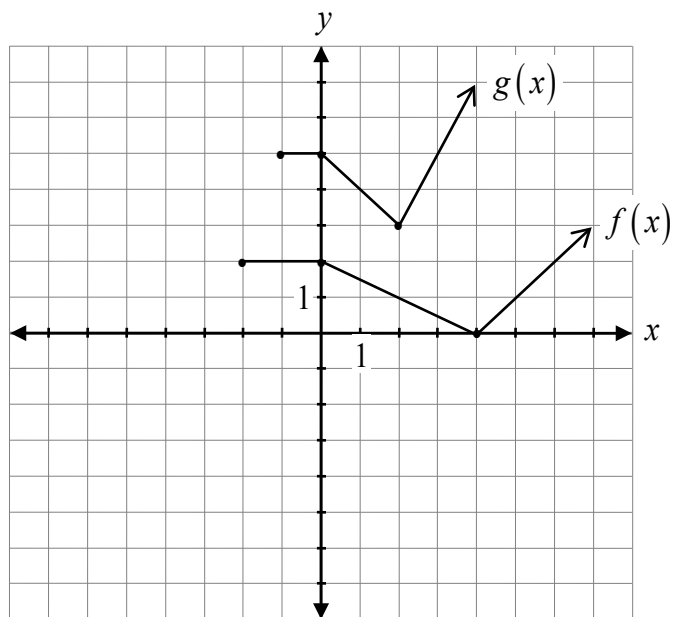
1 mark for power law

2 marks

Question 11

R4

State the equation of $g(x)$ in terms of $f(x)$.



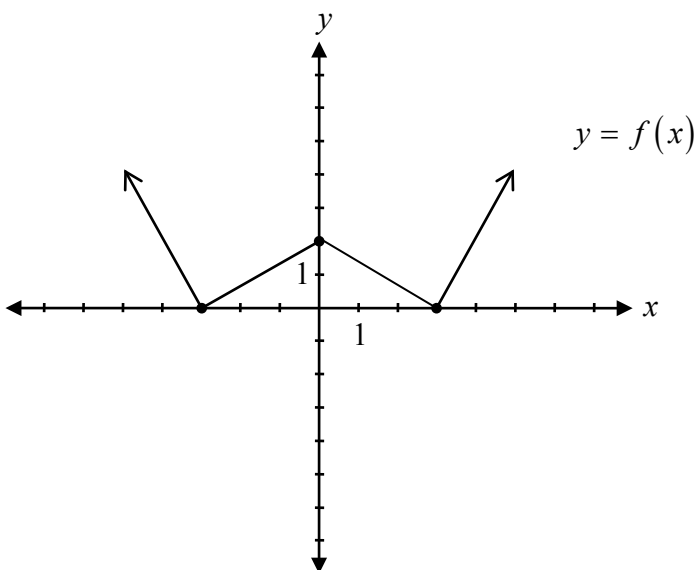
Solution

$$g(x) = f(2x) + 3$$

1 mark for horizontal compression
1 mark for vertical translation

2 marks

Explain why the inverse of the graph of $y = f(x)$ is not a function.

**Solution**

The domain of $f(x)$ was not restricted to ensure there is only one value of y for each x and one value of x for each y .

or

The graph of the inverse will not pass the vertical line test.

or

The graph of $f(x)$ does not pass the horizontal line test.

1 mark

Question 13

R10

Solve the following equation algebraically:

$$\log(x^2 + 5) - \log(x^2 + 1) = \log 3$$

Solution

$$\log\left(\frac{x^2 + 5}{x^2 + 1}\right) = \log 3$$

1 mark for quotient law

$$\frac{x^2 + 5}{x^2 + 1} = 3$$

½ mark for equating arguments

$$x^2 + 5 = 3(x^2 + 1)$$

$$x^2 + 5 = 3x^2 + 3$$

$$2 = 2x^2$$

$$1 = x^2$$

$$\pm 1 = x$$

½ mark for solving for x

2 marks

Describe the transformations used to obtain the graph of the function $y = 5f(x + 1)$ from the graph of $y = f(x)$.

Solution

Vertically stretch the graph by a factor of 5 and translate the graph one unit left.

1 mark for vertical stretch
1 mark for horizontal translation

2 marks

Question 15

P3

Solve algebraically:

$${}_nC_2 = 3n + 4$$

Solution

$$\frac{n!}{(n-2)!2!} = 3n + 4$$

½ mark for substitution into equation

$$\frac{n(n-1)\cancel{(n-2)!}}{\cancel{(n-2)!}} = 2!(3n + 4)$$

1 mark for factorial expansion

½ mark for simplification of factorials

$$n(n-1) = 2(3n + 4)$$

$$n^2 - n = 6n + 8$$

$$n^2 - 7n - 8 = 0$$

$$(n-8)(n+1) = 0$$

½ mark for rejecting the extraneous root

$$n = 8 \quad \cancel{n = -1}$$

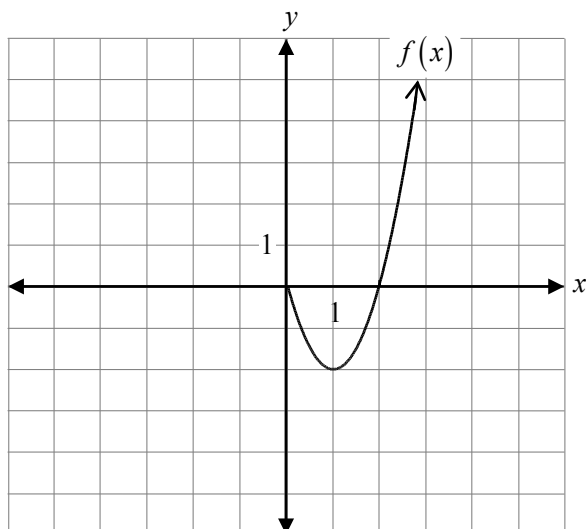
½ mark for the value of n

3 marks

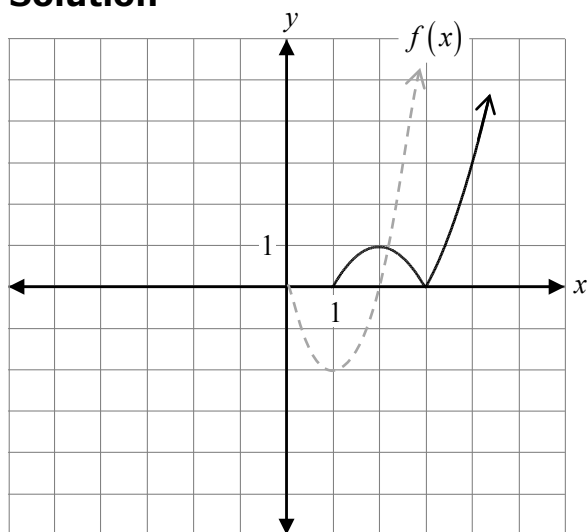
Question 16

R1, R2, R3

Given the graph of $f(x)$, sketch the graph of $y = \left| \frac{1}{2} f(x-1) \right|$.



Solution



1 mark for vertical stretch
1 mark for horizontal translation
1 mark for absolute value

3 marks

Explain why $f(x) = (x+2)^3(x-1)^{\frac{1}{2}}$ is not a polynomial function.

Solution

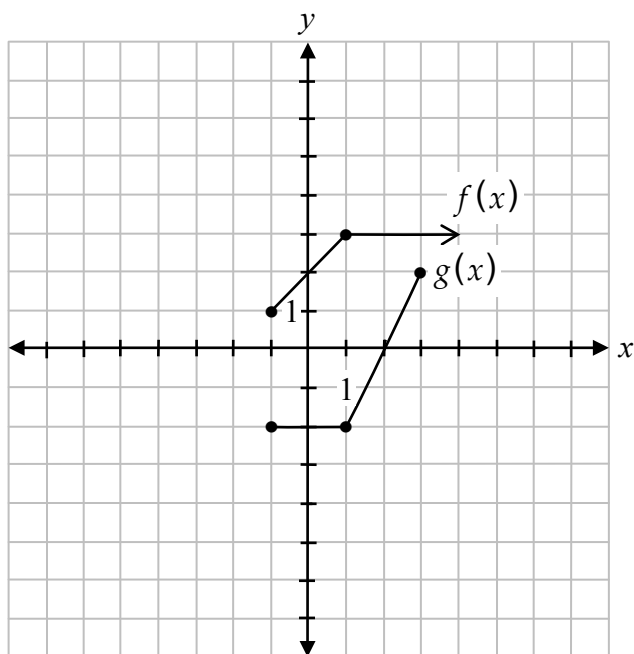
All factors in a polynomial function must have exponents that are whole numbers.

1 mark

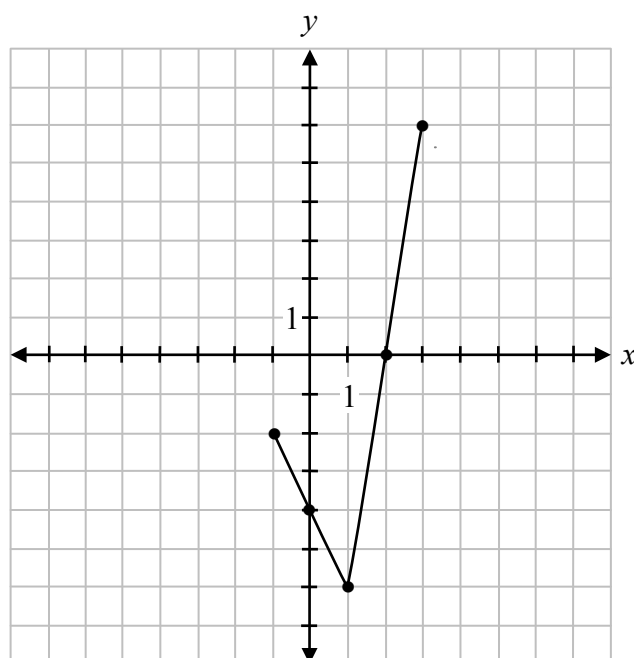
Question 18

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = (f \cdot g)(x)$.



Solution



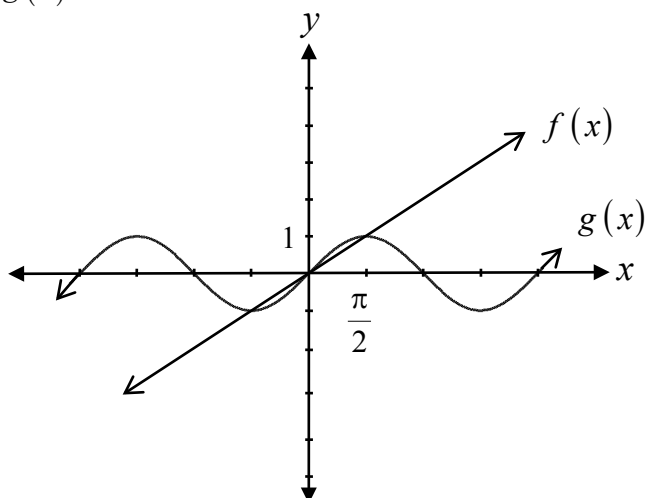
x	$f(x)$	$g(x)$	$(f \cdot g)(x)$
-1	1	-2	-2
1	3	-2	-6
3	3	2	6

1 mark for operation of multiplication

1 mark for restricted domain

2 marks

Given the graphs of $f(x)$ and $g(x)$, identify the choice with all of the solutions of the equation $f(x) = g(x)$.



a) $x = -2\pi, -\pi, 0, \pi, 2\pi$

b) $x = -\frac{\pi}{2}, 0, \frac{\pi}{2}$

c) $x = \frac{\pi}{2}$

d) $x = -1, 0, 1$

Match the following equations with their graphs:

Solution

Place the appropriate letter
in this column.

$$f(x) = (x-1)^3(x+1)(x-3) \quad \underline{\hspace{2cm} C \hspace{2cm}}$$

$$g(x) = (x+1)^2(x-1)(x+3) \quad \underline{\hspace{2cm} B \hspace{2cm}}$$

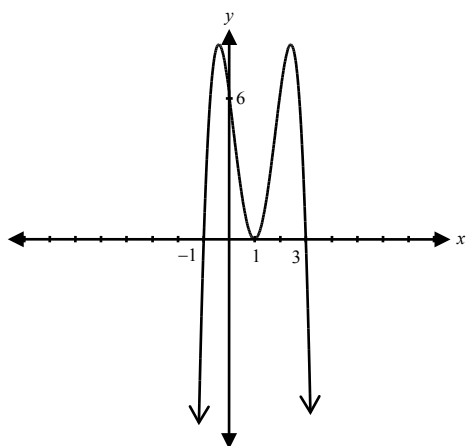
$$h(x) = -2(x-1)^2(x+1)(x-3) \quad \underline{\hspace{2cm} A \hspace{2cm}}$$

$$k(x) = 2(x+1)^2(x-1)(x+3) \quad \underline{\hspace{2cm} D \hspace{2cm}}$$

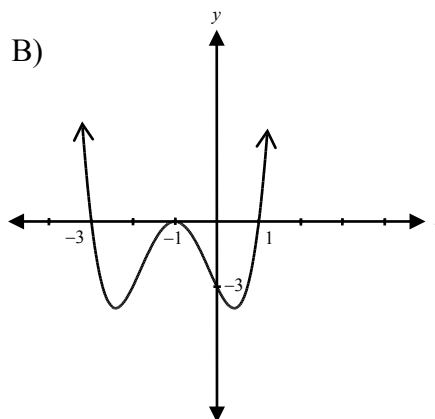
$\frac{1}{2}$ mark for each correct answer

2 marks

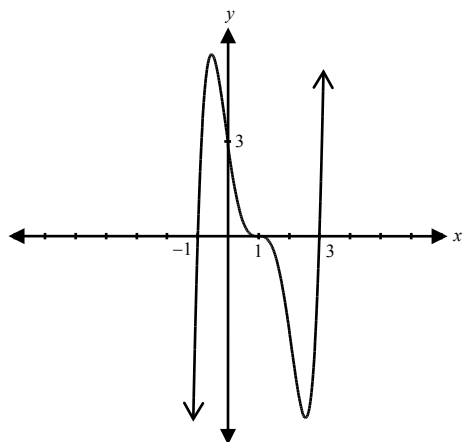
A)



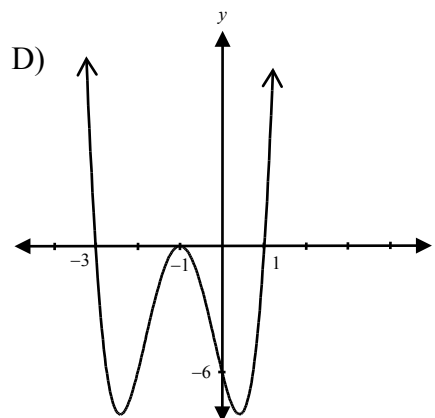
B)



C)



D)



If $\log 6 = p$, $\log 5 = r$ and $\log 2 = q$, express $\log 60$ in terms of p , q and r .

Solution

$$\log 60 = \log(6 \cdot 5 \cdot 2)$$

$\frac{1}{2}$ mark for combination

$$= \log 6 + \log 5 + \log 2$$

1 mark for product law

$$= p + r + q$$

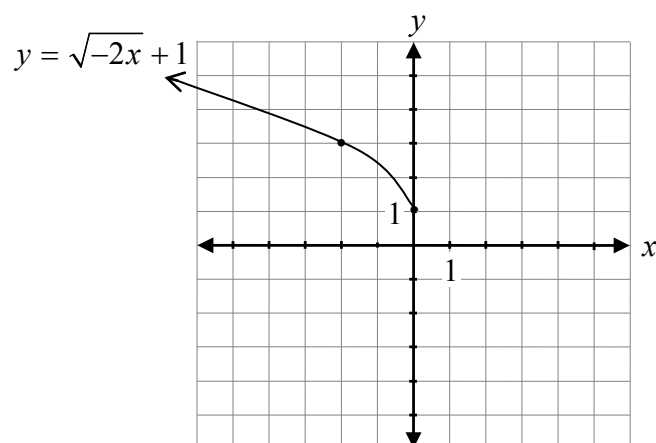
$\frac{1}{2}$ mark for substitution

2 marks

Sketch the graph of $y = \sqrt{-2x} + 1$.

Solution

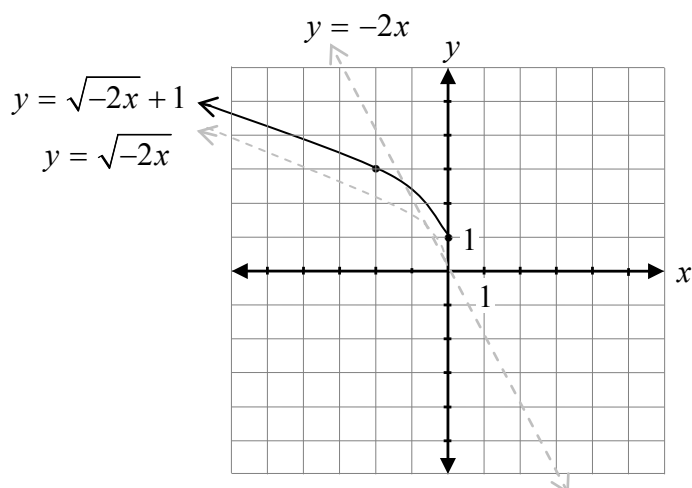
Method 1



- 1 mark for shape of a radical function
- 1 mark for horizontal compression
- 1 mark for vertical translation
- 1 mark for horizontal reflection

4 marks

Method 2



- ½ mark for shape between invariant points
- ½ mark for shape to the left of the invariant points
- 1 mark for invariant points where $y = 0$ and $y = 1$ (½ mark for each point)
- 1 mark for domain of $(-\infty, 0]$
- 1 mark for vertical translation

4 marks

Justify why the letters of the word FRANCE have a greater number of possible arrangements than the letters of the word CANADA.

Solution

CANADA has repeated letters that, when placed in different orders, do not change the arrangements of the letters.

$$\text{FRANCE: } 6! \qquad \text{CANADA: } \frac{6!}{3!}$$

1 mark

$\therefore$ France has a greater number of arrangements.

Question 31

R1, R14

Given $f(x) = \frac{1}{x-2}$ and $g(x) = x+5$,

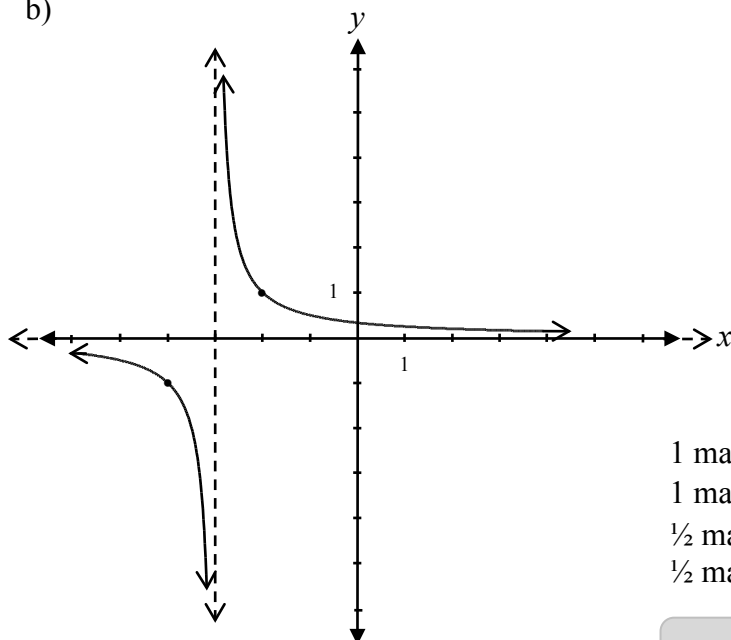
- determine the equation for $f(g(x))$.
- sketch the graph of $f(g(x))$.

Solution

$$\begin{aligned} \text{a) } f(g(x)) &= \frac{1}{(x+5)-2} \\ &= \frac{1}{x+3} \end{aligned}$$

1 mark

b)



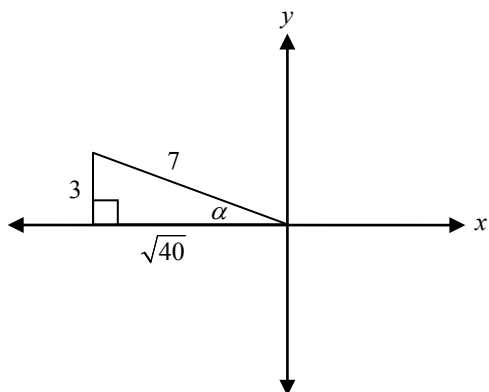
- 1 mark for asymptotic behaviour approaching $x = -3$
- 1 mark for asymptotic behaviour approaching $y = 0$
- $\frac{1}{2}$ mark for branch left of vertical asymptote
- $\frac{1}{2}$ mark for branch right of vertical asymptote

3 marks

Given that $\sin \alpha = \frac{3}{7}$, where α is in Quadrant II, and $\cos \beta = \frac{4}{5}$, where β is in Quadrant IV, determine the exact value of:

a) $\sin(\alpha - \beta)$

Solution

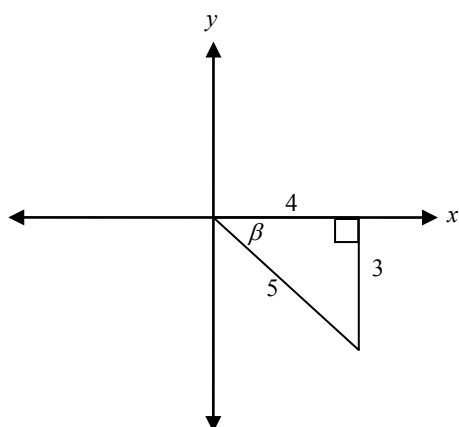


$$x^2 + y^2 = r^2$$

$$x^2 + 9 = 49$$

$$x^2 = 40$$

$$x = \pm\sqrt{40} \quad \frac{1}{2} \text{ mark for value of } x$$



$$x^2 + y^2 = r^2$$

$$16 + y^2 = 25$$

$$y^2 = 9$$

$$y = \pm 3 \quad \frac{1}{2} \text{ mark for value of } y$$

$$\begin{aligned} \sin(\alpha - \beta) &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\ &= \left(\frac{3}{7}\right)\left(\frac{4}{5}\right) - \left(\frac{-\sqrt{40}}{7}\right)\left(\frac{-3}{5}\right) \end{aligned}$$

$$= \frac{12}{35} - \frac{3\sqrt{40}}{35}$$

$$= \frac{12 - 3\sqrt{40}}{35} \quad \text{or} \quad \frac{12 - 6\sqrt{10}}{35}$$

$\frac{1}{2}$ mark for $\cos \alpha$

$\frac{1}{2}$ mark for $\sin \beta$

1 mark for substitution into correct identity

3 marks

Note(s):

- accept any of the following values for x : $x = \pm\sqrt{40}$, $x = -\sqrt{40}$ or $x = \sqrt{40}$
- accept any of the following values for y : $y = \pm 3$, $y = -3$ or $y = 3$

b) $\cos 2\alpha$ **Solution**

$$\begin{aligned}\cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha \\ &= \left(\frac{-\sqrt{40}}{7}\right)^2 - \left(\frac{3}{7}\right)^2 \\ &= \frac{40}{49} - \frac{9}{49} \\ &= \frac{31}{49}\end{aligned}$$

or

$$\begin{aligned}\cos 2\alpha &= 2\cos^2 \alpha - 1 \\ &= 2\left(\frac{-\sqrt{40}}{7}\right)^2 - 1 \\ &= 2\left(\frac{40}{49}\right) - 1 \\ &= \frac{80}{49} - 1 \\ &= \frac{31}{49}\end{aligned}$$

or

$$\begin{aligned}\cos 2\alpha &= 1 - 2\sin^2 \alpha \\ &= 1 - 2\left(\frac{3}{7}\right)^2 \\ &= 1 - 2\left(\frac{9}{49}\right) \\ &= 1 - \frac{18}{49} \\ &= \frac{31}{49}\end{aligned}$$

1 mark for substitution of an appropriate identity

1 mark

Determine the domain and range of $f(x) = \sqrt{x-5} - 1$.

Solution

Domain: $\{x \in \mathbb{R} \mid x \geq 5\}$ **or** $[5, \infty)$ 1 mark for domain

Range: $\{y \in \mathbb{R} \mid y \geq -1\}$ **or** $[-1, \infty)$ 1 mark for range

2 marks

Justify why 4.7 is a better estimate than 4.3 for the value of $\log_2 26$.

Solution

$$2^4 = 16 \quad 2^5 = 32$$

or

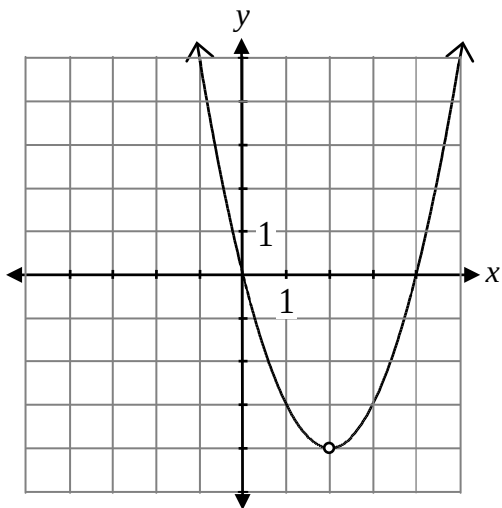
$$\log_2 16 = 4 \quad \log_2 32 = 5$$

26 is closer to 32 than 16; therefore $\log_2 26$ is closer to 5 than 4.

1 mark

Sketch the graph of the function:

$$f(x) = \frac{x(x-2)(x-4)}{(x-2)}$$

Solution

1 mark for point of discontinuity (hole) at (2, -4)
(½ mark for x value, ½ mark for y value)

½ mark for shape of a parabola

½ mark for end behaviour

2 marks

Evaluate:

$$\sec^2\left(\frac{\pi}{6}\right) + \tan\left(\frac{7\pi}{6}\right) \csc\left(-\frac{2\pi}{3}\right)$$

Solution

$$\left(\frac{2}{\sqrt{3}}\right)^2 + \left(\frac{\sqrt{3}}{3}\right)\left(-\frac{2}{\sqrt{3}}\right)$$

1 mark for $\sec\left(\frac{\pi}{6}\right)$ (½ mark for value, ½ mark for quadrant)

$$\left(\frac{4}{3}\right) - \left(\frac{2}{3}\right)$$

1 mark for $\tan\left(\frac{7\pi}{6}\right)$ (½ mark for value, ½ mark for quadrant)

$$\frac{2}{3}$$

1 mark for $\csc\left(-\frac{2\pi}{3}\right)$ (½ mark for value, ½ mark quadrant)

3 marks

The graph of $f(x) = 3x + 7$ is reflected over the y -axis.

Determine the equation of the new function.

Solution

$$y = -3x + 7$$

or

$$y = f(-x)$$

1 mark

Determine the equations of all of the asymptotes of the function:

$$y = \frac{2x + 1}{x - 3}$$

Solution

Horizontal asymptote at $y = 2$

1 mark for horizontal asymptote

Vertical asymptote at $x = 3$

1 mark for vertical asymptote

2 marks

Question 39

R11

One of the zeros of $p(x) = x^3 + 6x^2 - 32$ is $x = 2$. Determine all of the other zeros of $p(x)$.

Solution

2		1	6	0	-32
		↓	2	16	32
		1	8	16	0

1 mark for synthetic division (or any equivalent strategy)

$$0 = x^2 + 8x + 16$$

½ mark for the other factors

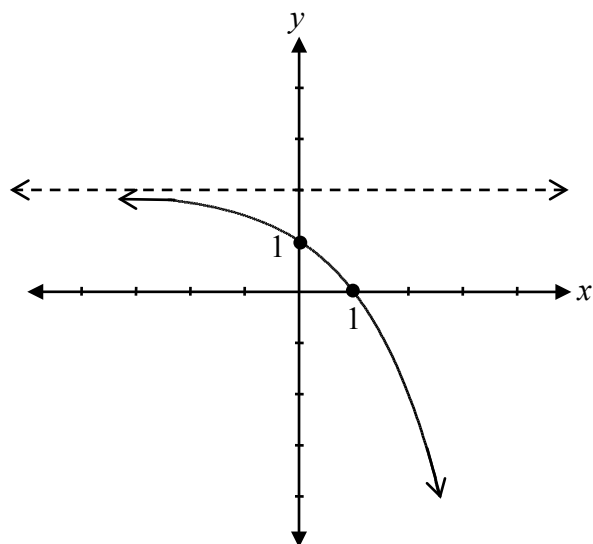
$$0 = (x + 4)(x + 4)$$

$$x = -4$$

½ mark for the other zeros

2 marks

Sketch the graph of $y = -2^x + 2$.

Solution

1 mark for shape of an exponential function
1 mark for vertical reflection
1 mark for asymptotic behaviour approaching $y = 2$

3 marks

Given the function $f(x) = \frac{2}{x} - 1$, justify why $f(f(2))$ is undefined.

Solution

$$f(2) = \frac{2}{2} - 1$$

$$= 1 - 1$$

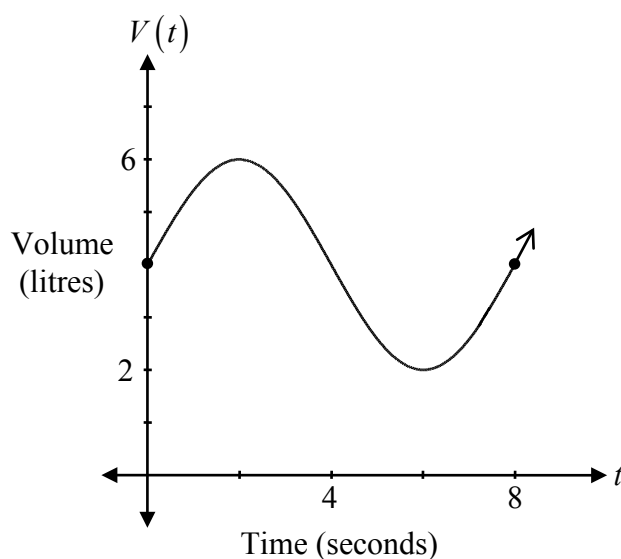
$$= 0$$

$f(f(2)) = \frac{2}{0} - 1$, which is undefined because the denominator cannot be zero.

1 mark for justification

1 mark

The following graph represents the volume of air in an adult's lungs. If $V(t)$ is the volume of air in litres and t is the time in seconds, determine an equation that represents this sinusoidal function.

**Solution**

$$V(t) = 2 \sin\left(\frac{\pi}{4}t\right) + 4$$

or

$$V(t) = 2 \cos\left[\frac{\pi}{4}(t - 2)\right] + 4$$

1 mark for amplitude

$\frac{1}{2}$ mark for period

$\frac{1}{2}$ mark for consistent value of b

1 mark for vertical translation

3 marks

Explain why the domain of $y = \log_2(x - 1)$ is $x > 1$.

Solution

The argument of a logarithmic function must be positive.

1 mark

The point $(-2, 7)$ is on the terminal arm of an angle in standard position.

Determine the coordinates of the corresponding point, $P(\theta)$, on the unit circle.

Solution

$$x^2 + y^2 = r^2$$

$$(-2)^2 + (7)^2 = r^2$$

$\frac{1}{2}$ mark for substitution of $x = \pm 2$ and $y = 7$

$$4 + 49 = r^2$$

$$53 = r^2$$

$\frac{1}{2}$ mark for solving for r

$$\sqrt{53} = r$$

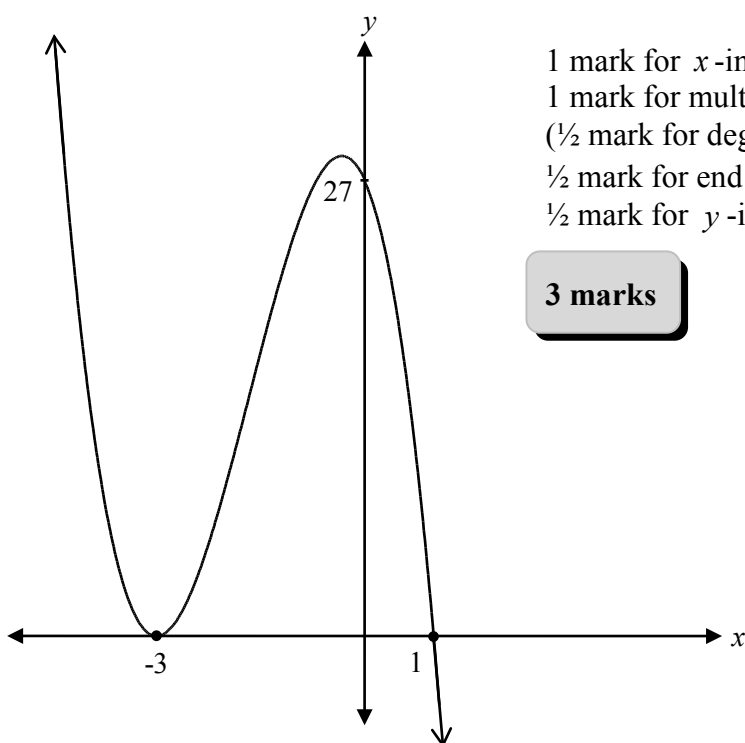
1 mark for $P(\theta)$ ($\frac{1}{2}$ mark for each coordinate)

$$P(\theta) = \left(\frac{-2}{\sqrt{53}}, \frac{7}{\sqrt{53}} \right)$$

2 marks

Sketch a graph of $P(x)$ that satisfies all of the following conditions:

- $P(x)$ is a polynomial function of degree 3.
- $P(x)$ has a zero at -3 with a multiplicity of 2.
- $P(x)$ has a zero at 1 .
- $P(x)$ has a leading coefficient of -3 .

Solution

1 mark for x -intercepts
1 mark for multiplicity
($\frac{1}{2}$ mark for degree 2 at $x = -3$, $\frac{1}{2}$ mark for degree 1 at $x = 1$)
 $\frac{1}{2}$ mark for end behaviour
 $\frac{1}{2}$ mark for y -intercept

3 marks

Question 46

P1

Determine how many 4-digit numbers greater than 4000 can be made using the digits 2, 3, 4, 5, and 6 if repetitions are not allowed.

Solution

$$\frac{3}{4, 5, 6} \cdot \frac{4}{\text{remaining options}} \cdot \frac{3}{\text{remaining options}} \cdot \frac{2}{\text{remaining options}} = 72$$

1 mark for restriction of first digit

1 mark for fundamental counting principle

2 marks

Question 1

P1

A group of 7 friends decide to go to a movie.

Determine how many ways the friends can sit in a row if two of the friends refuse to sit next to each other.

Solution

$$7! - 6! \cdot 2! = 3600 \text{ ways}$$

$\frac{1}{2}$ mark for $7!$

1 mark for product of $6! \cdot 2!$

($\frac{1}{2}$ mark for $6!$, $\frac{1}{2}$ mark for $2!$)

$\frac{1}{2}$ mark for subtraction

2 marks

Question 2

R10

Gabrielle listens to her radio at a sound level of 80 dB. She attended a music concert that had a sound level of 115 dB. Determine how many times more intense the music concert was than the radio.

You may use the formula:

$$\beta = 10 \log \left(\frac{I}{I_0} \right)$$

where β is the intensity level of sound, measured in dB

I is the intensity of sound

I_0 is the standard minimum intensity that a person can hear

Solution

Radio:

$$80 = 10 \log \left(\frac{I}{I_0} \right)$$

$$8 = \log \left(\frac{I}{I_0} \right)$$

$$10^8 = \frac{I}{I_0} \quad \frac{1}{2} \text{ mark for exponential form}$$

$$10^8 I_0 = I$$

Music concert:

$$115 = 10 \log \left(\frac{I}{I_0} \right)$$

$$11.5 = \log \left(\frac{I}{I_0} \right)$$

$$10^{11.5} = \frac{I}{I_0} \quad \frac{1}{2} \text{ mark for exponential form}$$

$$10^{11.5} I_0 = I$$

$$\begin{aligned} \frac{\text{intensity of music concert}}{\text{intensity of radio}} &= \frac{10^{11.5} I_0}{10^8 I_0} \\ &= 10^{3.5} \\ &= 3162.27766 \\ &= 3162.278 \end{aligned}$$

1 mark for comparison

2 marks

Question 3

R10

Solve, algebraically.

$$2(7)^x = 3^{2x-3}$$

Solution

$$\log\left(2(7)^x\right) = \log 3^{2x-3}$$

½ mark for applying logarithms

$$\log 2 + x \log 7 = (2x - 3) \log 3$$

1 mark for product law

1 mark for power law

$$\log 2 + x \log 7 = 2x \log 3 - 3 \log 3$$

$$\log 2 + 3 \log 3 = 2x \log 3 - x \log 7$$

½ mark for collecting terms with x

$$\log 2 + 3 \log 3 = x(2 \log 3 - \log 7)$$

$$\frac{\log 2 + 3 \log 3}{2 \log 3 - \log 7} = x$$

½ mark for isolating x

$$15.872\ 483 = x$$

$$15.872 = x$$

½ mark for evaluating quotient of logarithms

4 marks

Solve for θ , algebraically, over the interval $[0, 2\pi]$.

$$\csc^2 \theta + 2 \csc \theta - 8 = 0$$

Solution

$$(\csc \theta + 4)(\csc \theta - 2) = 0$$

$$\csc \theta = -4 \quad \csc \theta = 2$$

$$\sin \theta = -\frac{1}{4} \quad \sin \theta = \frac{1}{2}$$

$$\theta_r = 0.252 \ 680$$

$$\theta = 3.394 \quad \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\theta = 6.031$$

or

$$\theta = 3.394 \quad \theta = 0.524$$

$$\theta = 6.031 \quad \theta = 2.618$$

1 mark for solving for $\csc \theta$

1 mark for reciprocal

2 marks for solving for θ ($\frac{1}{2}$ mark for each value)

4 marks

Question 5

P1

You have forgotten the code to unlock your cell phone. You know the code is made up of four numbers from 0 to 9.

Determine the number of possible codes, if repetition is allowed.

Solution

$$\underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} = 10\,000$$

1 mark

In the binomial expansion of $\left(\frac{7}{x^3} - 3x^7\right)^n$, the 5th term contains x^7 .

Determine the value of n .

Solution

$$x^7 = \left(\frac{1}{x^3}\right)^{n-4} (x^7)^4$$

1 mark for $k = 4$
½ mark for substitution

$$x^7 = (x^{-3})^{n-4} (x^7)^4$$

$$x^7 = x^{-3n+12+28}$$

$$7 = -3n + 40$$

$$-33 = -3n$$

$$11 = n$$

½ mark for solving for n

2 marks

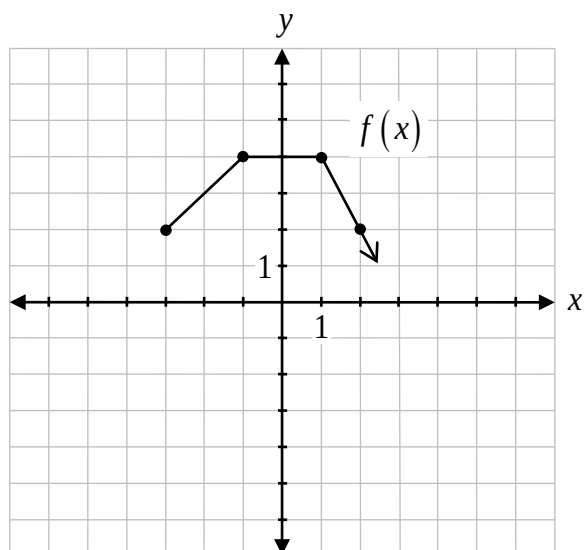
Given the domain of $f(x)$ is $\{-6, 1, 3, 4\}$ and the range of $f(x)$ is $\{-4, 7, 10, 15\}$, state the domain of $f^{-1}(x)$.

Solution $\{-4, 7, 10, 15\}$ **1 mark**

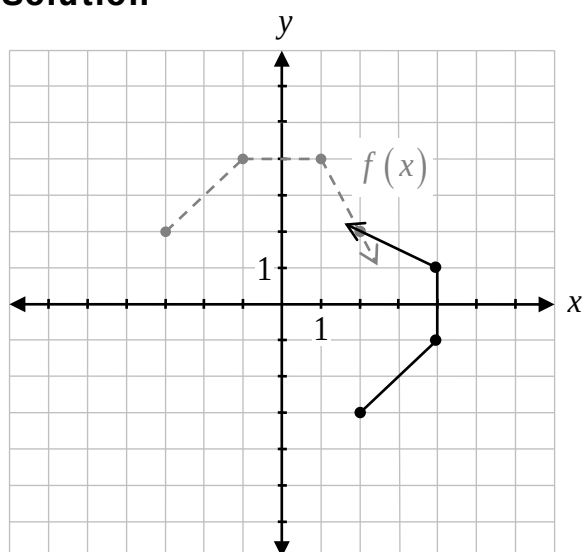
Question 8

R6

Given the graph of $y = f(x)$, sketch the graph of its inverse.



Solution



1 mark

Prove the following identity for all permissible values of θ .

$$\frac{1 + \cos \theta}{1 - \sin^2 \theta} = \sec \theta + \tan^2 \theta + 1$$

Solution

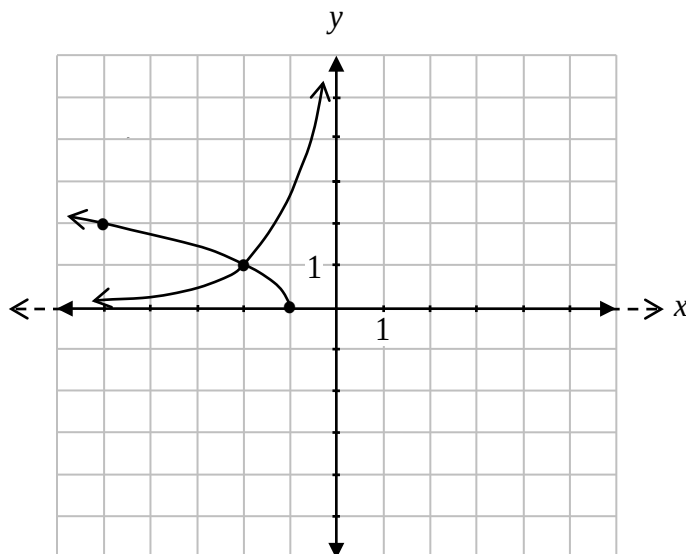
Method 1

Left-Hand Side	Right-Hand Side
$\frac{1 + \cos \theta}{1 - \sin^2 \theta}$	$\sec \theta + \tan^2 \theta + 1$
$\frac{1 + \cos \theta}{\cos^2 \theta}$	1 mark for algebraic strategies
$\frac{1}{\cos^2 \theta} + \frac{1}{\cos \theta}$	1 mark for logical process to prove the identity
$\sec^2 \theta + \sec \theta$	1 mark for correct substitution of appropriate identities
$\tan^2 \theta + 1 + \sec \theta$	3 marks

Solution**Method 2**

Left-Hand Side	Right-Hand Side	
$\frac{1 + \cos \theta}{\cos^2 \theta}$	$\frac{1}{\cos \theta} + \sec^2 \theta$	
	$\frac{1}{\cos \theta} + \frac{1}{\cos^2 \theta}$	1 mark for algebraic strategies
	$\frac{\cos \theta + 1}{\cos^2 \theta}$	1 mark for logical process to prove the identity
		1 mark for correct substitution of appropriate identities
		3 marks

Thomas used graphs to solve the equation $e^{x+2} = \sqrt{-(x+1)}$.



He incorrectly states the solution as $(-2, 1)$.

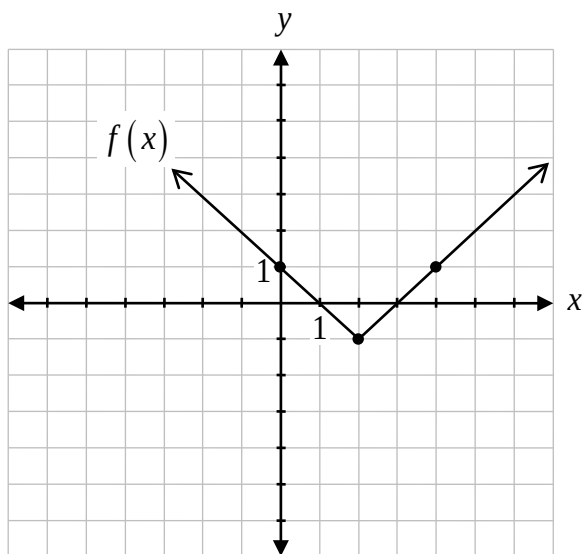
Describe how Thomas should have stated the solution.

Solution

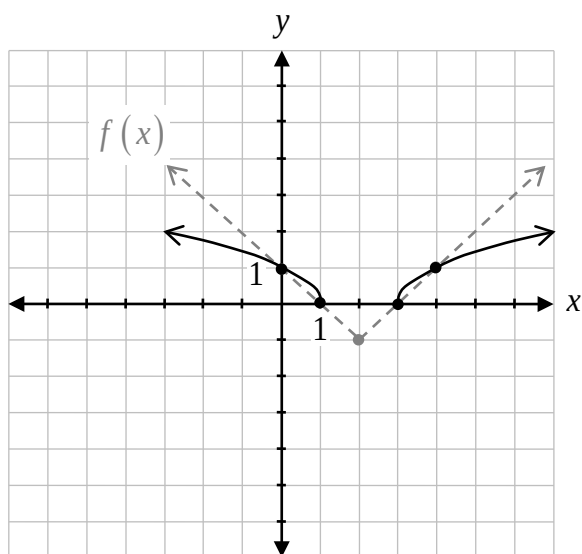
He stated his solution as a coordinate point; his solution should have only been the value of x .

1 mark

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



Solution



1 mark for restricting domain
 $\frac{1}{2}$ mark for shape between both pairs of invariant points
 $\frac{1}{2}$ mark for shape above both pairs of invariant points

2 marks

Question 12

R11

When a polynomial, $P(x)$, is divided by $(x - 2)$ the resulting equation is

$$\frac{P(x)}{x - 2} = x^2 - x + 1 + \frac{3}{x - 2}.$$

- a) Explain why $x - 2$ is not a factor of $P(x)$.
- b) Determine the equation for the polynomial function $P(x)$.

Solution

- a) There is a remainder when $P(x)$ is divided by $x - 2$.

1 mark

b) $P(x) = \underline{(x - 2)(x^2 - x + 1) + 3}$

1 mark

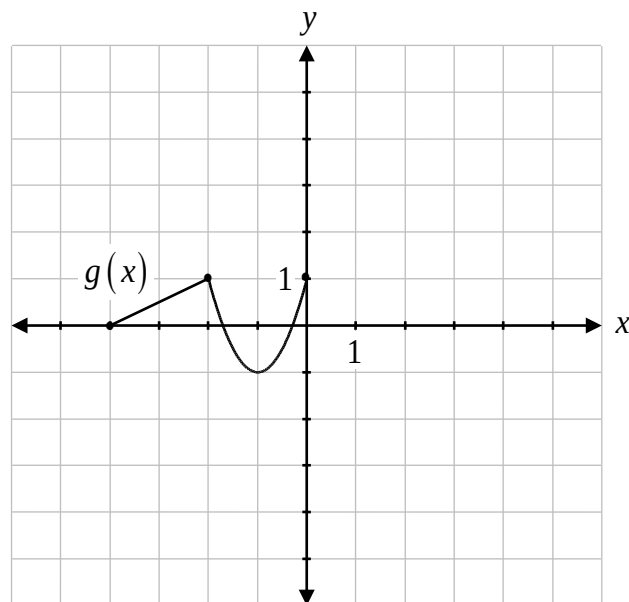
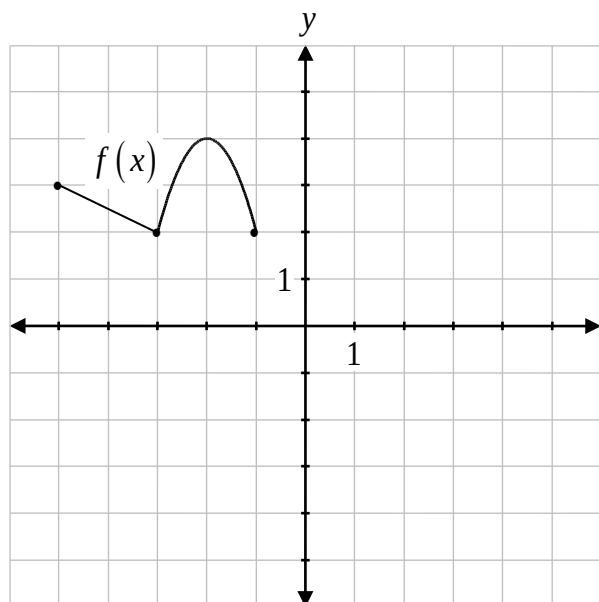
or

$$P(x) = \underline{x^3 - 3x^2 + 3x + 1}$$

Question 13

R2, R5

Determine the equation for $g(x)$ in terms of $f(x)$.



Solution

$$g(x) = -f(x-1) + 3$$

1 mark for vertical reflection
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Explain why the binomial expansion of $(2x + y)^9$ does not have a middle term.

Solution

The expansion contains $n + 1$ terms. Since n equals 9, there are 10 terms, which would not allow for a middle term.

1 mark

Using the laws of logarithms, completely expand the expression $\log\left(\frac{5\sqrt{a}}{b^3}\right)$.

Solution

$$\log 5 + \frac{1}{2} \log a - 3 \log b$$

1 mark for product law
1 mark for power law (½ mark for each)
1 mark for quotient law

3 marks

Evaluate the following expression.

$$\tan\left(\frac{2\pi}{3}\right)\csc\left(\frac{-2\pi}{3}\right) + \cos(3\pi)$$

Solution

$$\begin{array}{l} (-\sqrt{3})\left(-\frac{2}{\sqrt{3}}\right) + (-1) \\ \frac{2-1}{1} \end{array}$$

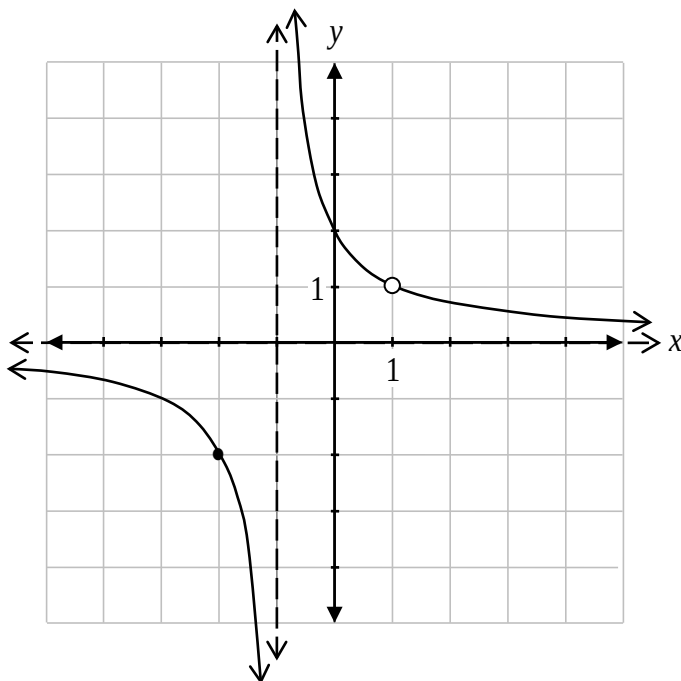
1 mark for $\tan\left(\frac{2\pi}{3}\right)$ (½ mark for quadrant, ½ mark for value)

1 mark for $\csc\left(-\frac{2\pi}{3}\right)$ (½ mark for quadrant, ½ mark for value)

1 mark for $\cos(3\pi)$

3 marks

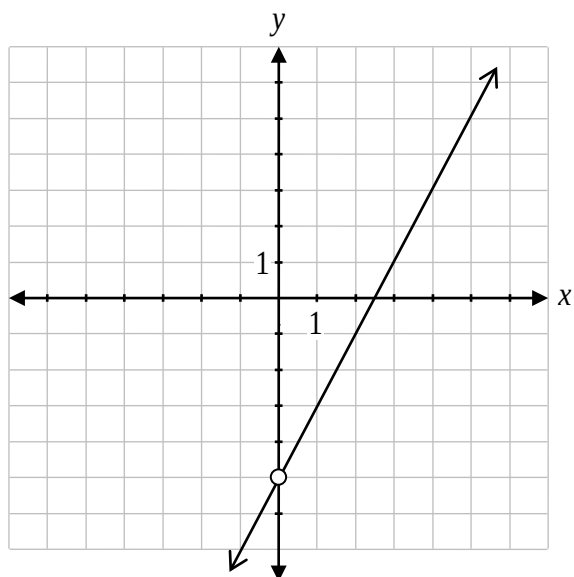
State the range of the graph below.

**Solution**

Range: $\{y \in \mathbb{R}, y \neq 0 \text{ and } y \neq 1\}$ 1 mark ($\frac{1}{2}$ mark for $y \neq 0$, $\frac{1}{2}$ mark for $y \neq 1$)

1 mark

Sketch the graph of the function $f(x) = \frac{2x^2 - 5x}{x}$.

Solution

1 mark for point of discontinuity (hole) at $(0, -5)$

($\frac{1}{2}$ mark for $x = 0$, $\frac{1}{2}$ mark for $y = -5$)

1 mark for shape of a linear function

2 marks

State a possible value of n if the polynomial function $P(x) = (x-1)^2(x+2)^n$ has a range of $[0, \infty)$.

Solution

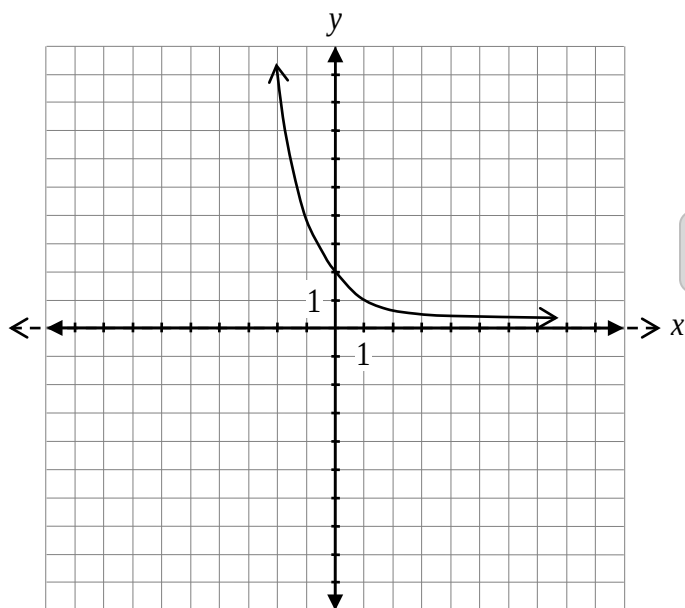
$$n = 2$$

1 mark

Note(s):

- Accept any even positive value of n , including zero.

Sketch the graph of $y = \left(\frac{1}{2}\right)^{x-1}$.

Solution

1 mark for decreasing exponential function
1 mark for horizontal translation

2 marks

Solve.

$$\log_x 27 = 3$$

Solution

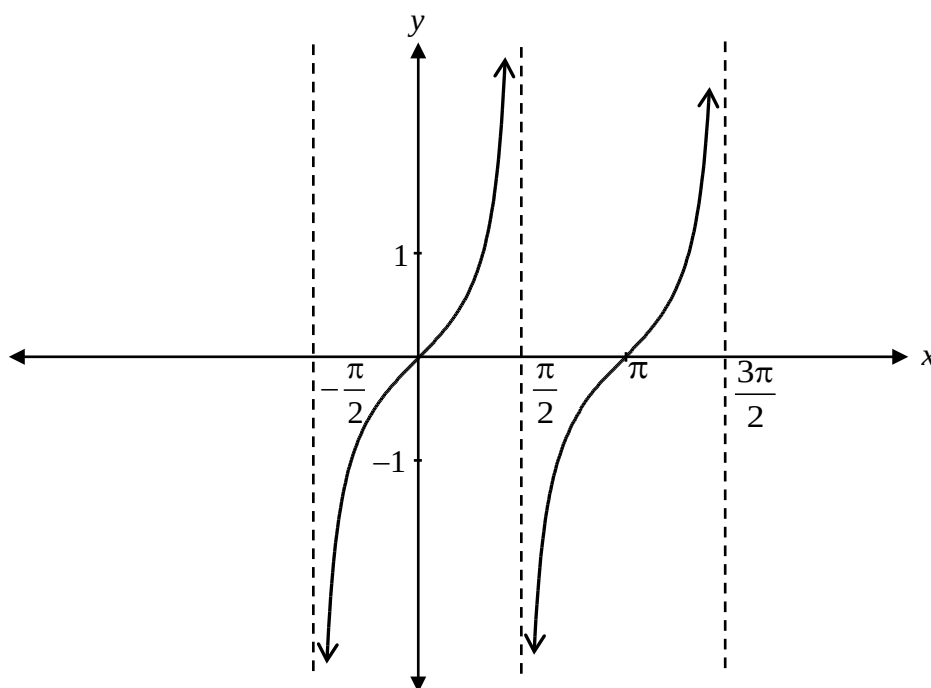
$$x^3 = 27$$

1 mark for exponential form

$$x = 3$$

1 mark

Sketch at least two periods of the graph $y = \tan x$.

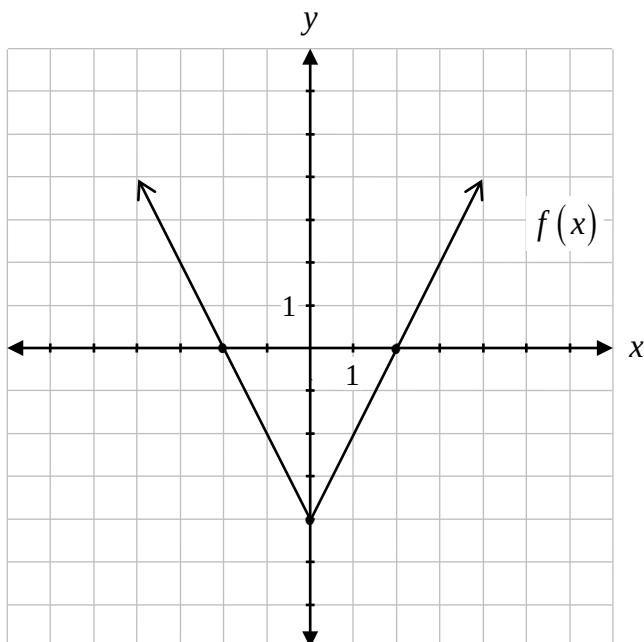
Solution

1 mark for increasing trigonometric function

1 mark for asymptotic behaviour approaching $x = \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$

2 marks

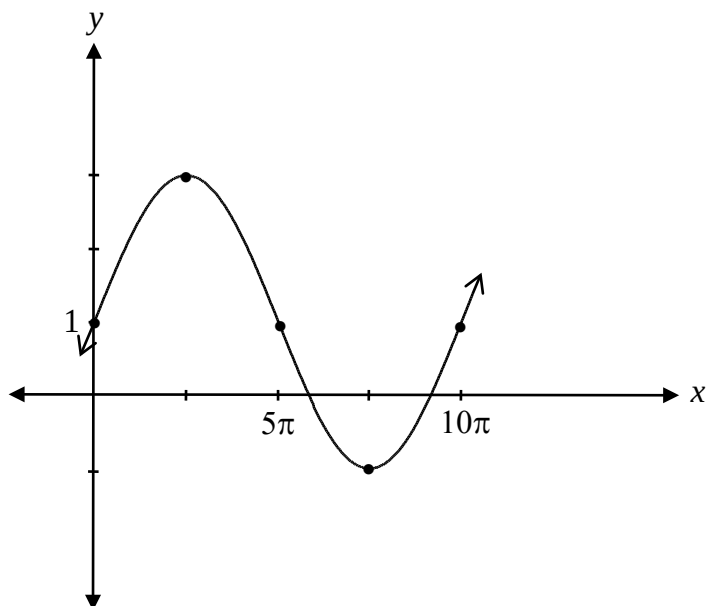
Given the graph of $f(x)$, state the domain of $\frac{1}{f(x)}$.

**Solution**

Domain: $\{x \in \mathbb{R}, x \neq \pm 2\}$ 1 mark (½ mark for $x \neq 2$, ½ mark for $x \neq -2$)

1 mark

Determine the values of A, B, and D of the sinusoidal function in the form $y = A \sin(Bx) + D$.

**Solution**

$$A = \underline{\quad 2 \quad}$$

1 mark for A

$$B = \underline{\quad \frac{1}{5} \quad}$$

1 mark for B

$$D = \underline{\quad 1 \quad}$$

1 mark for D

3 marks

Determine if the point $\left(-\frac{\sqrt{7}}{5}, \frac{2}{5}\right)$ is on the unit circle.

Justify your answer.

Solution

$$x^2 + y^2 = 1$$

$$\text{Left-hand side} = \left(-\frac{\sqrt{7}}{5}\right)^2 + \left(\frac{2}{5}\right)^2$$

$$= \frac{7}{25} + \frac{4}{25}$$

$$= \frac{11}{25}$$

$$\frac{11}{25} \neq 1$$

$\therefore$ not on the unit circle

1 mark for justification

1 mark

Solve, algebraically.

$$\frac{{}_n C_5}{{}_n C_4} = 6$$

Solution

$$\frac{\frac{n!}{(n-5)!5!}}{\frac{n!}{(n-4)!4!}} = 6$$

½ mark for substitution into equation

$$\frac{n!(n-4)!4!}{n!(n-5)!5!} = 6$$

$$\frac{\cancel{n!} (n-4) \cancel{(n-5)!} \cancel{4!}}{\cancel{n!} \cancel{(n-5)!} 5 \cdot \cancel{4!}} = 6$$

1 mark for factorial expansion

(½ mark for numerical factors; ½ mark for factors with variables)

1 mark for simplification of factorials

(½ mark for numerical factors; ½ mark for factors with variables)

$$\frac{n-4}{5} = 6$$

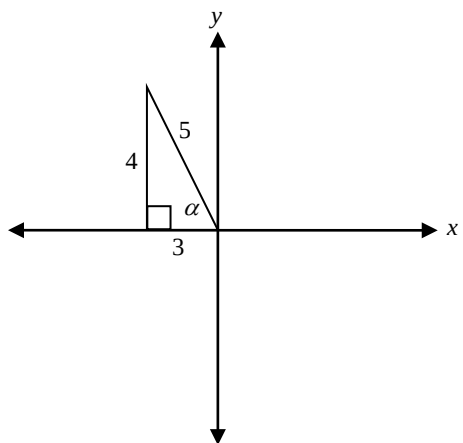
$$n-4 = 30$$

$$n = 34$$

½ mark for solving for n

3 marks

Given $\sin \alpha = \frac{4}{5}$, where α is in quadrant II, determine the exact value of $\sin 2\alpha$.

Solution

$$x^2 + y^2 = r^2$$

$$x^2 + 16 = 25$$

$$x^2 = 9$$

$$x = \pm 3$$

$$x = -3$$

$$\cos \alpha = -\frac{3}{5}$$

½ mark for value of x

½ mark for $\cos \alpha$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$= 2 \left(\frac{4}{5} \right) \left(-\frac{3}{5} \right)$$

$$= -\frac{24}{25}$$

1 mark for substitution into correct identity

2 marks

Note(s):

- Accept any of the following values for x : $x = \pm 3$; $x = -3$; or $x = 3$.

Given the functions $f(x) = x + 1$ and $g(x) = \sqrt{x}$,

a) determine the equation of $g(f(x))$.

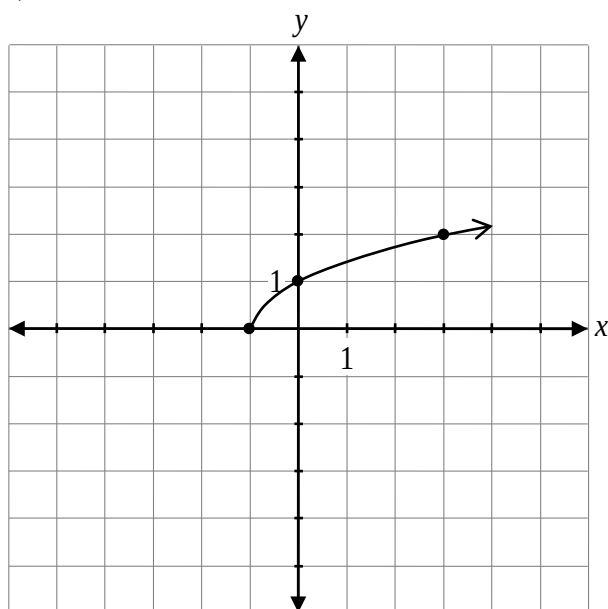
b) sketch the graph of $g(f(x))$.

Solution

a) $g(f(x)) = \sqrt{x+1}$ _____

1 mark

b)



1 mark for domain of $g(f(x))$

1 mark for shape consistent with $g(f(x))$

2 marks

Steve is asked to determine an equation with a larger period than the period of the graph of $y = \cos(2x)$.

Justify why Steve's answer of $y = \cos(6x)$ is incorrect.

Solution

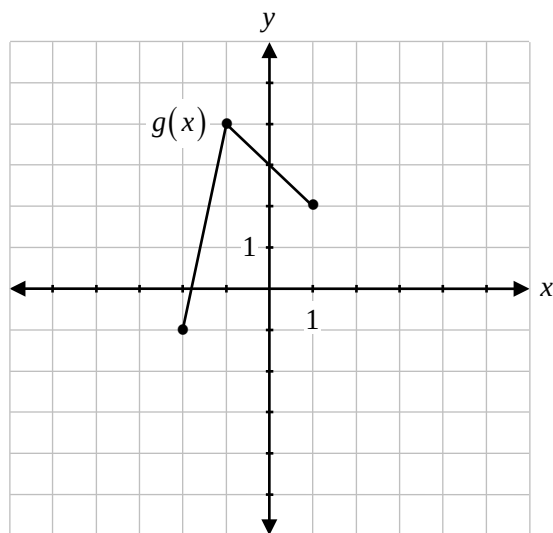
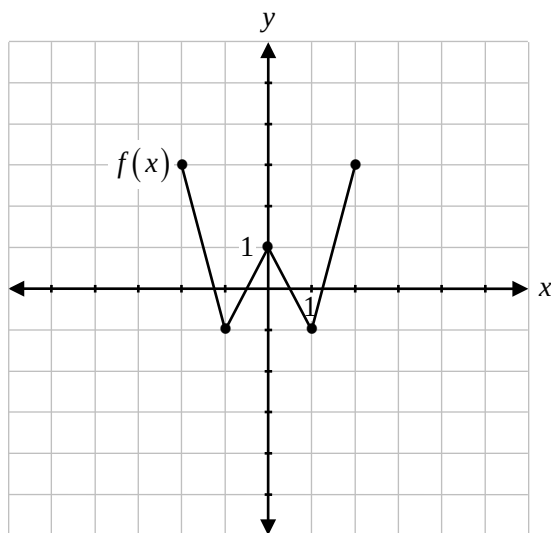
Steve's equation needs to have a value of $|b|$ less than 2.

1 mark

or

Steve's graph would have a period of $\frac{2\pi}{6} = \frac{\pi}{3}$, which is smaller than $\frac{2\pi}{2} = \pi$, the period of the given graph.

Given the graphs of $f(x)$ and $g(x)$,



a) determine the value of $(f \cdot g)(-1)$.

b) determine the value of $g(f(0))$.

Solution

$$\begin{aligned} \text{a) } (f \cdot g)(-1) &= (-1)(4) \\ &= -4 \end{aligned}$$

1 mark for value of $(f \cdot g)(-1)$

1 mark

$$\text{b) } f(0) = 1$$

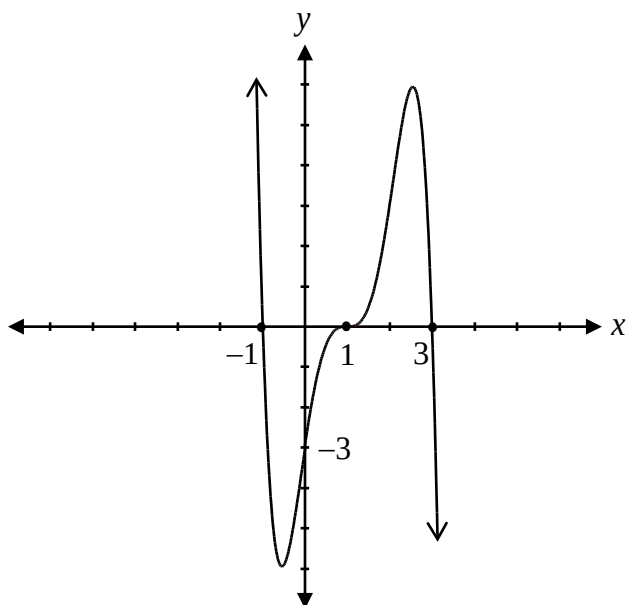
$\frac{1}{2}$ mark for $f(0)$

$$g(f(0)) = 2$$

$\frac{1}{2}$ mark for $g(f(0))$ consistent with $f(0)$ value

1 mark

Sketch the graph of $P(x) = -(x-1)^3(x-3)(x+1)$.

Solution

1 mark for x-intercepts

½ mark for y-intercept

1 mark for multiplicity (degree 3 at $x = 1$)

½ mark for end behaviour

3 marks

The point $(-\sqrt{3}, 1)$ is on the terminal arm of an angle θ , in standard position.

a) Determine $\tan \theta$.

b) Determine a possible value of θ , in radians.

Solution

a) $\tan \theta = -\frac{1}{\sqrt{3}}$

1 mark

b) $\theta = \frac{5\pi}{6}$

1 mark

Describe the transformation used to obtain the graph of $y = \log_5 x$ given the graph of $y = 5^x$.

Solution

The graph of $y = \log_5 x$ is obtained by reflecting the graph of $y = 5^x$ over the line $y = x$.

or

1 mark

The graph of $y = \log_5 x$ is the inverse of $y = 5^x$.

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$, where $\theta \in \mathbb{R}$.

Solution

$$\theta = \frac{4\pi}{3}, \frac{5\pi}{3}$$

1 mark for values of θ ($\frac{1}{2}$ mark for each value)

$$\theta = \left\{ \begin{array}{l} \frac{4\pi}{3} + 2k\pi, k \in \mathbb{Z} \\ \frac{5\pi}{3} + 2k\pi, k \in \mathbb{Z} \end{array} \right\}$$

1 mark for general solution

2 marks

or

$$\theta = 240^\circ, 300^\circ$$

$$\theta = \left\{ \begin{array}{l} 240^\circ + 360^\circ k, k \in \mathbb{Z} \\ 300^\circ + 360^\circ k, k \in \mathbb{Z} \end{array} \right\}$$

Given that the point (a, b) is on the graph of $f(x)$, describe how you would determine the corresponding point on the graph of $y = \sqrt{f(x)}$.

Solution

The value of a stays the same, square root the value of b .

1 mark

Evaluate.

$$\cos\left(\frac{\pi}{20}\right)\cos\left(\frac{\pi}{5}\right) - \sin\left(\frac{\pi}{20}\right)\sin\left(\frac{\pi}{5}\right)$$

Solution

$$\cos\left(\frac{\pi}{20} + \frac{\pi}{5}\right)$$

½ mark for substitution of an appropriate identity

$$\cos\left(\frac{\pi}{20} + \frac{4\pi}{20}\right)$$

$$\cos\left(\frac{5\pi}{20}\right)$$

$$\cos\left(\frac{\pi}{4}\right)$$

$$\frac{\sqrt{2}}{2}$$

½ mark for exact value

1 mark

Describe the transformations used to obtain the graph of the function $y = f(-x + 6) - 8$ from the graph of $y = f(x)$.

Solution

Reflect the graph of $y = f(x)$ over the y -axis and then translate 6 units right and 8 units down.

1 mark for horizontal reflection
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Note(s):

- Deduct 1 mark if correct transformations are given in the wrong order.

State the equations of all the asymptotes of the function, $y = \frac{1}{3x+1}$.

Solution

$$y = 0$$

1 mark for horizontal asymptote

$$x = -\frac{1}{3}$$

1 mark for vertical asymptote

2 marks

Determine the zeros of the polynomial function $P(x) = 2x^3 + 5x^2 - 4x - 3$.

Solution

$$P(1) = 2(1)^3 + 5(1)^2 - 4(1) - 3 \quad 1 \text{ mark for identifying one possible value of } x$$

$$P(1) = 0$$

$(x - 1)$ is a factor

$$\begin{array}{r|rrrr} 1 & 2 & 5 & -4 & -3 \\ & \downarrow & & & \\ \hline & 2 & 7 & 3 & 0 \end{array}$$

1 mark for synthetic division (or for any equivalent strategy)

$$P(x) = (x - 1)(2x^2 + 7x + 3) \quad \frac{1}{2} \text{ mark for consistent factors}$$

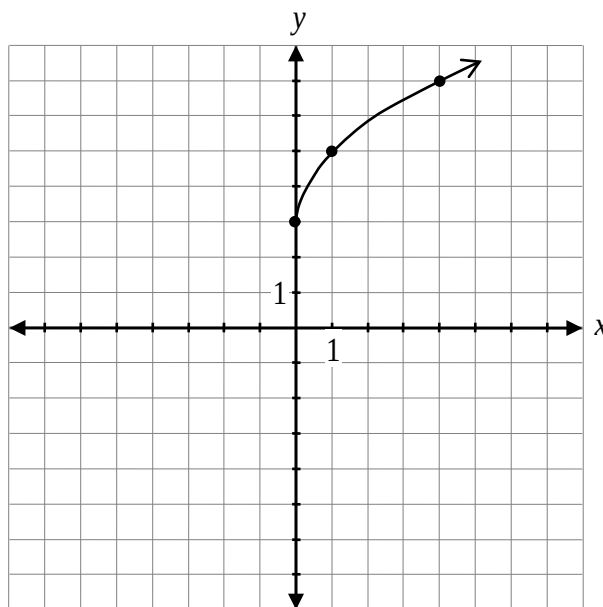
$$0 = (x - 1)(2x + 1)(x + 3)$$

$$x = 1 \quad x = -\frac{1}{2} \quad x = -3$$

$\frac{1}{2}$ mark for all zeros

3 marks

Determine the equation of the radical function represented by the graph.

**Solution**

$$y = 2\sqrt{x} + 3$$

1 mark for vertical stretch
1 mark for vertical translation

or

$$y = \sqrt{4x} + 3$$

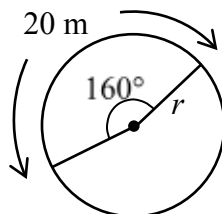
1 mark for horizontal stretch
1 mark for vertical translation

2 marks

Question 1

T1

Determine the length of the radius, r , given an arc length of 20 metres and a central angle of 160° .



Solution

$$\theta = (160) \left(\frac{\pi}{180} \right)$$

$$= \frac{8\pi}{9}$$

1 mark for conversion

$$r = \frac{s}{\theta}$$

$$r = \frac{20}{\left(\frac{8\pi}{9} \right)}$$

1 mark for substitution

$$r = \frac{180}{8\pi} \text{ m}$$

2 marks

or

$$r = 7.162 \text{ m}$$

Question 2

P1

There are eight cars parked in a row. Determine the number of possible arrangements of these eight cars if Mrs. Jones must always park in the third spot and Mr. Rodriguez must always park in the last spot.

Solution

$$\underline{6} \cdot \underline{5} \cdot \underline{1} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} \cdot \underline{1}$$

720

1 mark

Question 3

R10

Bill wins \$1 300 000 in a lottery and invests the entire amount at an annual interest rate of 2.5% compounded quarterly. He will withdraw \$10 000 at the end of every three months.

Determine, algebraically, the total number of withdrawals, including the partial amount, that Bill can make until there is no money left. Express your answer as a whole number.

Use the formula:

$$PV = \frac{R[1 - (1 + i)^{-n}]}{i}$$

where PV = the present value deposited

R = the amount of each withdrawal

n = the number of equal withdrawals

$i = \frac{\text{the annual interest rate (in decimal form)}}{\text{the number of compounding periods}}$

Solution

$$1300000 = \frac{10000 \left[1 - \left(1 + \frac{0.025}{4} \right)^{-n} \right]}{\frac{0.025}{4}}$$

½ mark for substitution

$$8125 = 10000 \left[1 - (1.00625)^{-n} \right]$$

$$0.8125 = 1 - (1.00625)^{-n}$$

$$-0.1875 = -(1.00625)^{-n}$$

½ mark for simplification

$$\log(0.1875) = -n \log 1.00625$$

½ mark for applying logarithms

$$-\frac{\log(0.1875)}{\log(1.00625)} = n$$

1 mark for power law

$$268.672348 = n$$

½ mark for solving for n

Bill can make 269 withdrawals.

3 marks

Determine and simplify the 12th term in the binomial expansion of $\left(x^3 - \frac{1}{2x^2}\right)^{12}$.

Solution

$$t_{12} = {}_{12}C_{11} \left(x^3\right)^1 \left(-\frac{1}{2x^2}\right)^{11} \quad \text{2 marks (1 mark for } {}_{12}C_{11}; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$t_{12} = (12) \left(x^3\right) \left(-\frac{1}{2048x^{22}}\right)$$

$$t_{12} = -\frac{12}{2048x^{19}}$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient; $\frac{1}{2}$ mark for exponent)

or

$$t_{12} = -\frac{3}{512x^{19}}$$

or

$$t_{12} = -0.006x^{-19}$$

3 marks

Solve, algebraically.

$$e^{2x-3} = 7^{x+1}$$

Solution

$$\ln e^{2x-3} = \ln 7^{x+1}$$

½ mark for applying logarithms

$$(2x - 3) \ln e = (x + 1) \ln 7$$

1 mark for power law

$$2x - 3 = x \ln 7 + \ln 7$$

$$2x - x \ln 7 = \ln 7 + 3$$

½ mark for collecting terms with x

$$x(2 - \ln 7) = \ln 7 + 3$$

$$x = \frac{\ln 7 + 3}{2 - \ln 7}$$

½ mark for isolating x

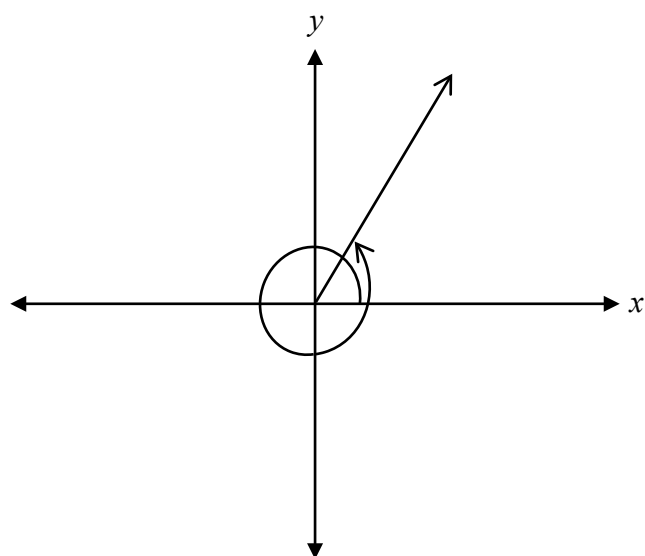
$$x = 91.438\ 783$$

½ mark for evaluating quotient of logarithms

$$x = 91.439$$

3 marks

Sketch the angle $\frac{7\pi}{3}$ in standard position.

Solution

$\frac{1}{2}$ mark for an appropriate angle in quadrant I
 $\frac{1}{2}$ mark for correct number of revolutions

1 mark

Note:

- If the directional arrow is not indicated, deduct an E1 error (final answer not stated).

Question 7

R11

Determine, algebraically, all of the zeros of the polynomial function $P(x) = x^4 - 5x^3 - 4x^2 + 20x$.

Solution

$$P(x) = x(x^3 - 5x^2 - 4x + 20)$$

$$P(2) = 2(2^3 - 5(2)^2 - 4(2) + 20)$$

1 mark for identifying one possible value of x

$$P(2) = 0$$

$\therefore (x - 2)$ is a factor

$$\begin{array}{r|rrrr} 2 & 1 & -5 & -4 & 20 \\ & \downarrow & 2 & -6 & -20 \\ \hline & 1 & -3 & -10 & 0 \end{array}$$

1 mark for synthetic division
(or any other equivalent strategy)

$$P(x) = x(x - 2)(x^2 - 3x - 10)$$

$$0 = x(x - 2)(x - 5)(x + 2)$$

$\frac{1}{2}$ mark for identifying all the factors

$$x = 0, x = 2, x = 5, x = -2$$

$\frac{1}{2}$ mark for consistent zeros

3 marks

Justify why four of the terms in the binomial expansion of $(-x + y)^6$ are positive.

Solution

$$(-x)^6(y)^0, (-x)^5(y)^1, (-x)^4(y)^2, \dots$$

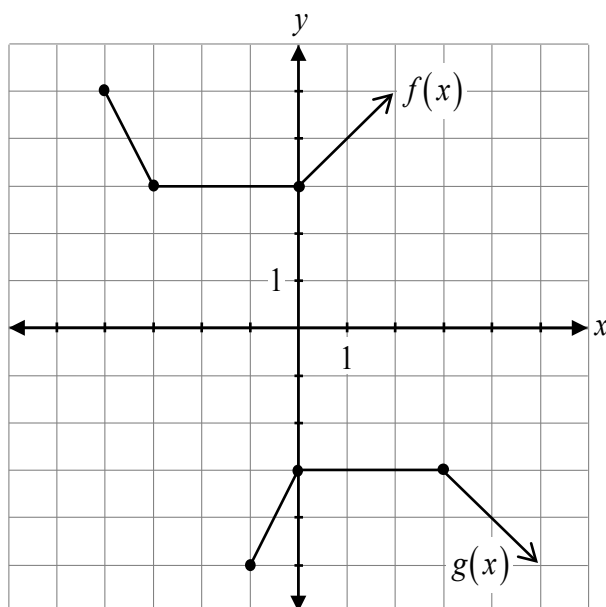
$$x^6, -x^5y, x^4y^2, \dots$$

There are 7 terms. The first term is positive and the signs alternate.

1 mark for justification

1 mark

Determine the equation of $g(x)$ in terms of $f(x)$.



Solution

$$g(x) = -f(x - 3)$$

1 mark for vertical reflection

1 mark for horizontal translation

2 marks

Question 10

T6

Prove the identity for all permissible values of x .

$$\frac{\sin x + \tan x}{\cot x + \csc x} = \frac{\sin^2 x}{\cos x}$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\sin x + \frac{\sin x}{\cos x}}{\frac{\cos x}{\sin x} + \frac{1}{\sin x}}$	$\frac{\sin^2 x}{\cos x}$
$\frac{\frac{\sin x \cos x + \sin x}{\cos x}}{\frac{\cos x + 1}{\sin x}}$	
$\frac{\sin x (\cancel{\cos x + 1})}{\cos x} \cdot \frac{\sin x}{(\cancel{\cos x + 1})}$	
$\frac{\sin^2 x}{\cos x}$	

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Solution**Method 2**

Left-Hand Side	Right-Hand Side
$\frac{\frac{\sin x + \tan x}{1}}{\frac{1}{\tan x} + \frac{1}{\sin x}}$	$\frac{\sin^2 x}{\cos x}$
$\frac{\frac{\sin x + \tan x}{\sin x + \tan x}}{\sin x \tan x}$	
$\frac{\cancel{\sin x + \tan x}}{\cancel{\sin x + \tan x}} \cdot \sin x \tan x$	
$\sin x \cdot \frac{\sin x}{\cos x}$	
$\frac{\sin^2 x}{\cos x}$	

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Given that the point $(-2, 1)$ is on the graph of $y = f(x)$, describe how the coordinates of the corresponding point on the graph of $y = f(4x)$ are different.

Solution

The x -value is divided by 4.

1 mark

Using the laws of logarithms, completely expand the expression:

$$\log \left(\frac{x^2 \sqrt{y}}{w-1} \right)$$

Solution

$$2 \log x + \frac{1}{2} \log y - \log (w-1)$$

1 mark for product law

1 mark for power law (½ mark for each)

1 mark for quotient law

3 marks

Given $\sec \theta = -\frac{5}{4}$ and $\tan \theta > 0$, state the quadrant in which θ terminates.

Justify your answer.

Solution

Since $\sec \theta$ is negative in quadrants II and III, and $\tan \theta$ is positive in quadrants I and III, θ terminates in quadrant III.

1 mark for justification

1 mark

State an equation of a rational function that has a vertical asymptote at $x = -8$ and a horizontal asymptote at $y = 9$.

Solution

$$y = \frac{9x}{x + 8}$$

1 mark for vertical asymptote
1 mark for horizontal asymptote

2 marks

Note:

- Other equations are possible.

Given the point $(5, 1)$, state the coordinates of the corresponding point after a reflection across the line $y = x$.

Solution

$(1, 5)$

1 mark

Simplify.

$$\frac{(n-13)!}{(n-12)!}$$

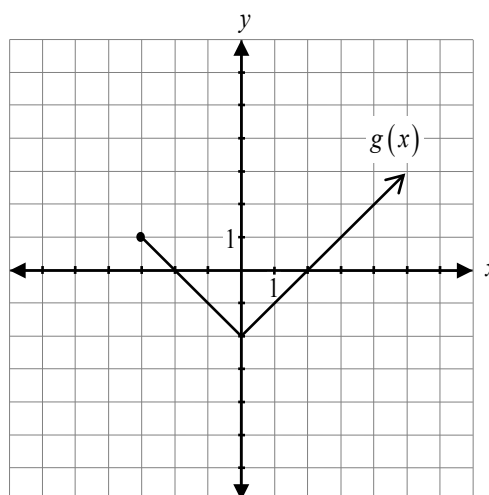
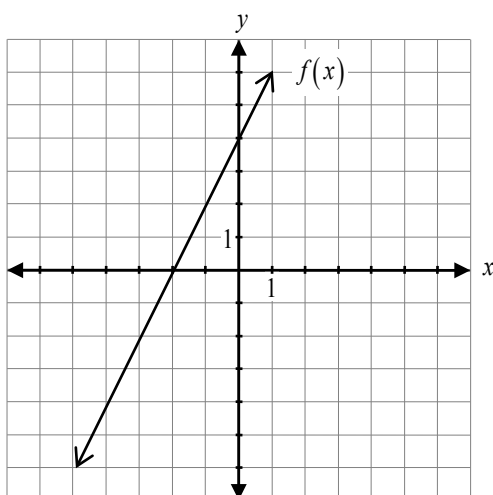
Solution

$$\frac{\cancel{(n-13)!}}{(n-12) \cancel{(n-13)!}} \\ \frac{1}{n-12}$$

1 mark for factorial expansion

1 mark

Given the graphs of $y = f(x)$ and $y = g(x)$,



Solution

a) determine the value of $f(g(2))$.

$$g(2) = 0$$

$\frac{1}{2}$ mark for the value of $g(2)$

$$f(0) = 4$$

$\frac{1}{2}$ mark for the value of $f(g(2))$ consistent with $g(2)$

1 mark

b) determine the value of $(g - f)(-3)$.

$$g(-3) - f(-3)$$

$$1 - (-2)$$

$$3$$

$\frac{1}{2}$ mark for the values of $g(-3)$ and $f(-3)$

$\frac{1}{2}$ mark for value of $(g - f)(-3)$ consistent with $g(-3)$ and $f(-3)$

1 mark

Given $\csc \theta = -\frac{8}{5}$, determine the exact value of $\cos 2\theta$.

Solution

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$= 1 - 2\left(-\frac{5}{8}\right)^2$$

$$= \frac{14}{64}$$

or

$$= \frac{7}{32}$$

1 mark for the value of $\sin \theta$

1 mark for substitution into correct identity

2 marks

Determine the period of the sinusoidal function, $f(x) = -6 \cos\left(\frac{\pi}{6}(x+1)\right) + 5$.

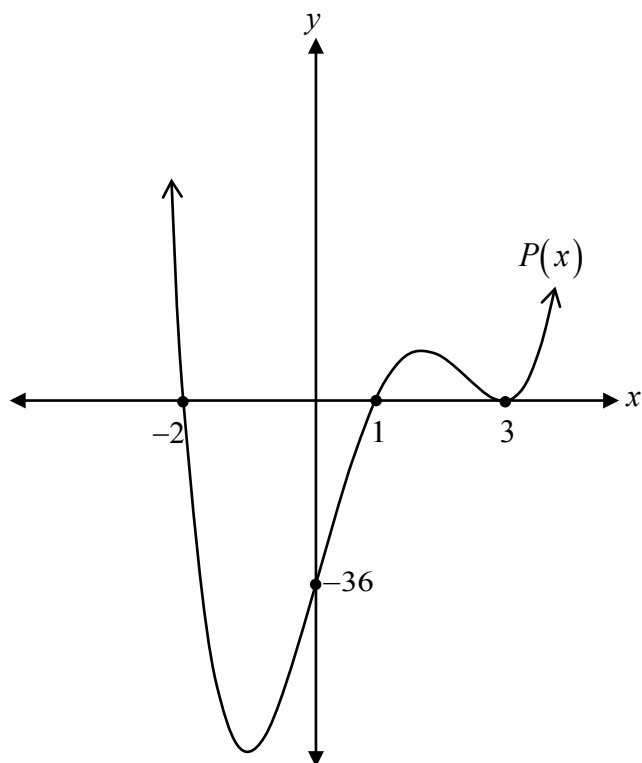
Solution

$$\text{Period} = \frac{2\pi}{\left(\frac{\pi}{6}\right)}$$

$$= 12$$

1 mark

Determine, algebraically, the equation of $P(x)$, given the graph of the polynomial function $P(x)$.



Solution

$$P(x) = a(x-3)^2(x-1)(x+2)$$

$\frac{1}{2}$ mark for factors of $P(x)$

$$-36 = a(-3)^2(-1)(2)$$

$\frac{1}{2}$ mark for multiplicity of 2 at $x = 3$

$\frac{1}{2}$ mark for substitution of $P(0) = -36$

$$-36 = -18a$$

$$a = 2$$

$\frac{1}{2}$ mark for correct value of a

$$P(x) = \underline{2(x-3)^2(x-1)(x+2)}$$

2 marks

Solve $2 \sin^2 \theta - 7 \sin \theta - 4 = 0$ where $\theta \in \mathbb{R}$.

Solution

$$(2 \sin \theta + 1)(\sin \theta - 4) = 0$$

$$\sin \theta = -\frac{1}{2}$$

$$\sin \theta = 4$$

1 mark for solving for $\sin \theta$ ($\frac{1}{2}$ mark for each branch)

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

No solution

2 marks for solving for θ (1 mark for each branch)

$$\theta = \frac{7\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

$$\theta = \frac{11\pi}{6} + 2k\pi, k \in \mathbb{Z}$$

1 mark for general solution

4 marks

Justify that the shapes of the graphs of $f(x) = (x+1)^2(x-1)$ and $g(x) = (x+1)^2(x-1)^3$ are different as they approach the x -intercept at $x=1$.

Solution

At $x=1$, both graphs pass through the x -axis, however, the graph of $g(x)$ flattens as it passes through the x -axis.

1 mark

Determine the exact value of $\cot \theta$ if $\cos \theta = -\frac{4}{7}$ and $\sin \theta$ is positive.

Solution

$$x^2 + y^2 = r^2$$

$$(-4)^2 + y^2 = (7)^2 \quad \frac{1}{2} \text{ mark for substitution}$$

$$y^2 = 49 - 16$$

$$y = \pm\sqrt{33} \quad \frac{1}{2} \text{ mark for solving for } y$$

$$\cot \theta = \frac{-4}{\sqrt{33}} \quad 1 \text{ mark for consistent value of } \cot \theta \text{ (}\frac{1}{2} \text{ mark for quadrant; } \frac{1}{2} \text{ mark for value)}$$

or

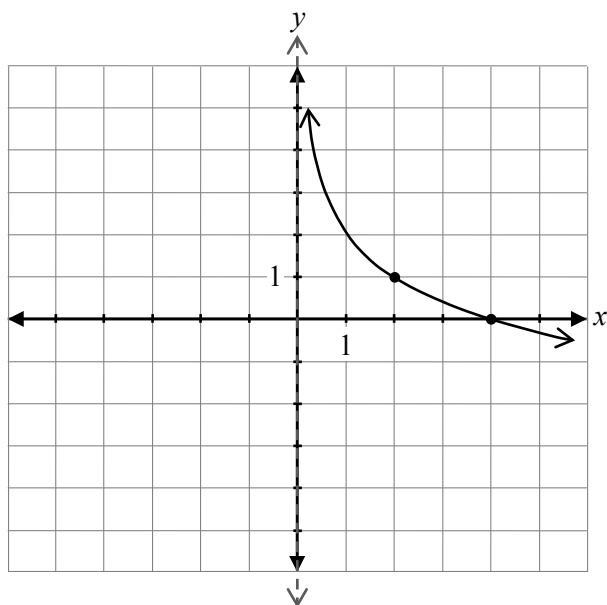
2 marks

$$\cot \theta = \frac{-4\sqrt{33}}{33}$$

Note:

- Accept any of the following values for y : $y = \pm\sqrt{33}$, $y = \sqrt{33}$, or $y = -\sqrt{33}$.

Sketch the graph of $f(x) = -\log_2(x) + 2$.

Solution

1 mark for asymptotic behaviour approaching $x = 0$

1 mark for vertical reflection

1 mark for vertical translation

3 marks

State the range of $f(x) = \sqrt{x+4}$.

Solution

Range: $[0, \infty)$ _____

1 mark

or

Range: $\{y \in \mathbb{R} \mid y \geq 0\}$ _____

Sophie correctly solved the logarithmic equation, $\log_7(x-1) = \log_7(2x-2)$.

$$x-1 = 2x-2$$

$$-1+2 = 2x-x$$

$$1 = x$$

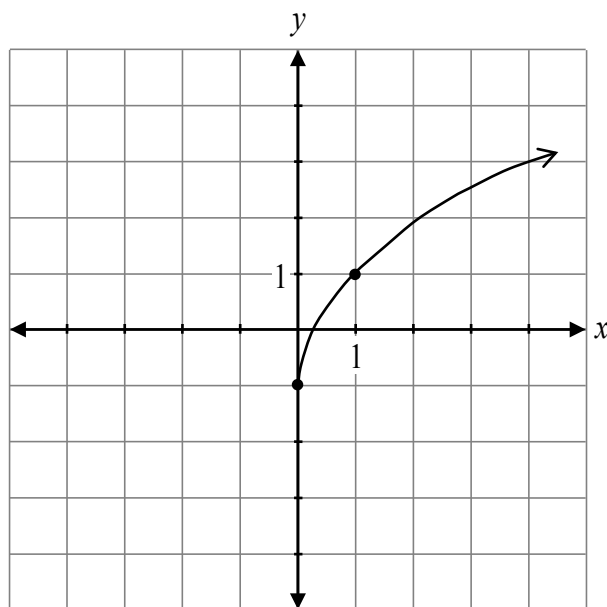
Explain why $x=1$ is an extraneous root.

Solution

The root, $x=1$, is an extraneous root because the argument of a logarithm cannot be zero.

1 mark

Sketch the graph of $f(x) = \sqrt{4x} - 1$.

Solution

1 mark for shape of a radical function
1 mark for horizontal compression
1 mark for vertical translation

3 marks

Solve, algebraically.

$${}_nC_2 = 2n + 7$$

Solution

$$\frac{n!}{(n-2)!2!} = 2n + 7 \quad \frac{1}{2} \text{ mark for substitution into equation}$$

$$\frac{n(n-1)\cancel{(n-2)!}}{\cancel{(n-2)!}} = 2!(2n + 7) \quad \begin{array}{l} \frac{1}{2} \text{ mark for factorial expansion} \\ \frac{1}{2} \text{ mark for simplification of factorial} \end{array}$$

$$n(n-1) = 2(2n + 7)$$

$$n^2 - n = 4n + 14$$

$$n^2 - 5n - 14 = 0 \quad \frac{1}{2} \text{ mark for simplification}$$

$$(n+2)(n-7) = 0$$

$$\cancel{n+2} \quad n = 7$$

$\frac{1}{2}$ mark for the permissible value of n
 $\frac{1}{2}$ mark for showing the rejection of the extraneous root

3 marks

Given $f(x) = x^2 - 1$ and $g(x) = x - 3$, explain why the domain of $h(x) = \frac{f(x)}{g(x)}$ has a restriction when $x = 3$.

Solution

When $x = 3$, the denominator is equal to zero and it is not possible to divide by zero.

1 mark

Evaluate.

$$\frac{\cot\left(\frac{11\pi}{6}\right)\sin\left(-\frac{4\pi}{3}\right)}{\cos\left(\frac{2\pi}{3}\right)}$$

Solution

$$\frac{(-\sqrt{3})\left(\frac{\sqrt{3}}{2}\right)}{-\frac{1}{2}}$$

1 mark for $\cot\left(\frac{11\pi}{6}\right)$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

1 mark for $\sin\left(-\frac{4\pi}{3}\right)$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

1 mark for $\cos\left(\frac{2\pi}{3}\right)$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

$$\left(-\frac{3}{2}\right)\left(-\frac{2}{1}\right)$$

3 marks

3

Solve, algebraically.

$$\log_2(\log_3 x) = 2$$

Solution

$$2^2 = \log_3 x$$

$\frac{1}{2}$ mark for exponential form

$$4 = \log_3 x$$

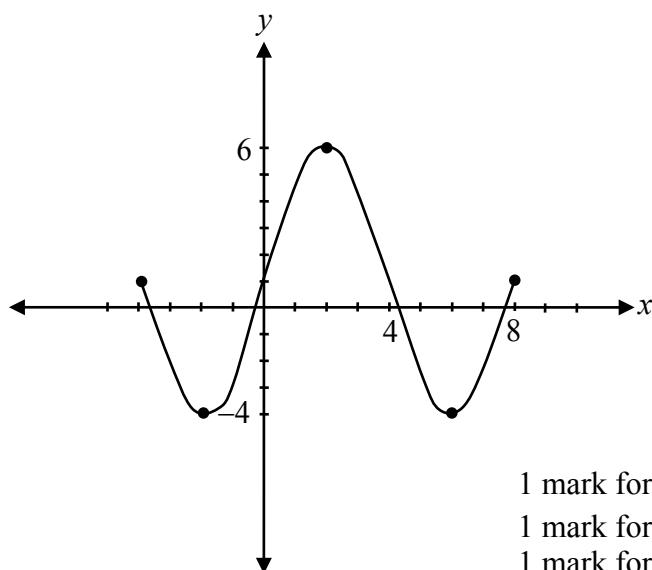
$$x = 3^4$$

$\frac{1}{2}$ mark for exponential form

$$x = 81$$

1 mark

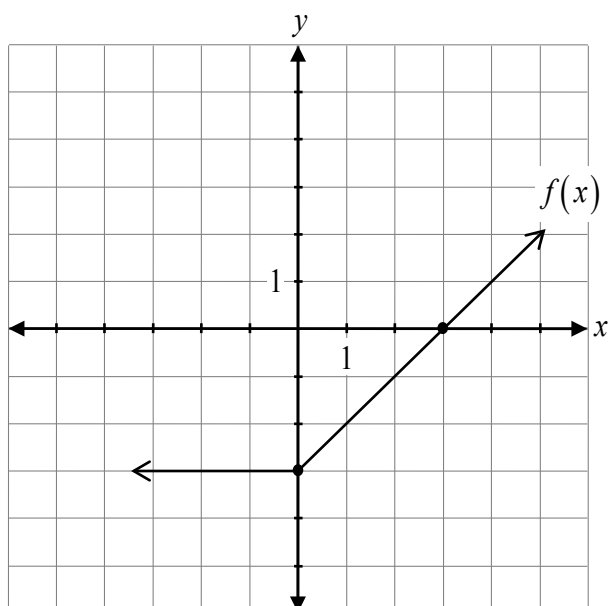
Sketch the graph of the function $y = 5 \sin\left(\frac{\pi}{4}x\right) + 1$ over the domain $[-4, 8]$.

Solution

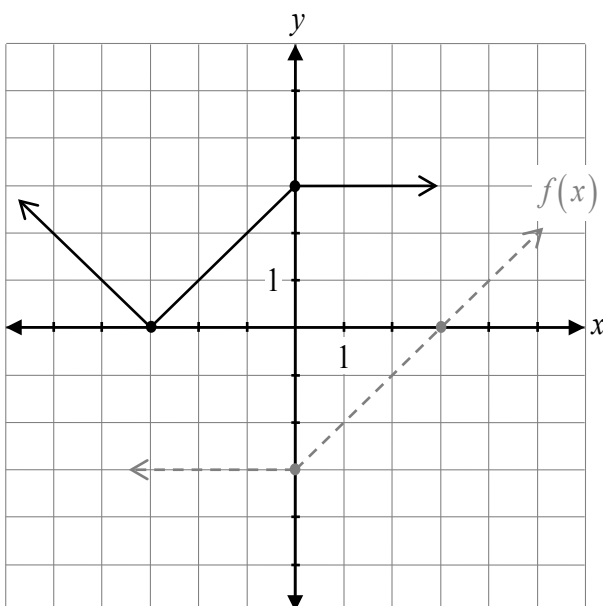
1 mark for shape of $y = \sin x$
1 mark for amplitude
1 mark for period
1 mark for vertical translation

4 marks

Given the graph of $y = f(x)$, sketch the graph of $y = |f(-x)|$.



Solution

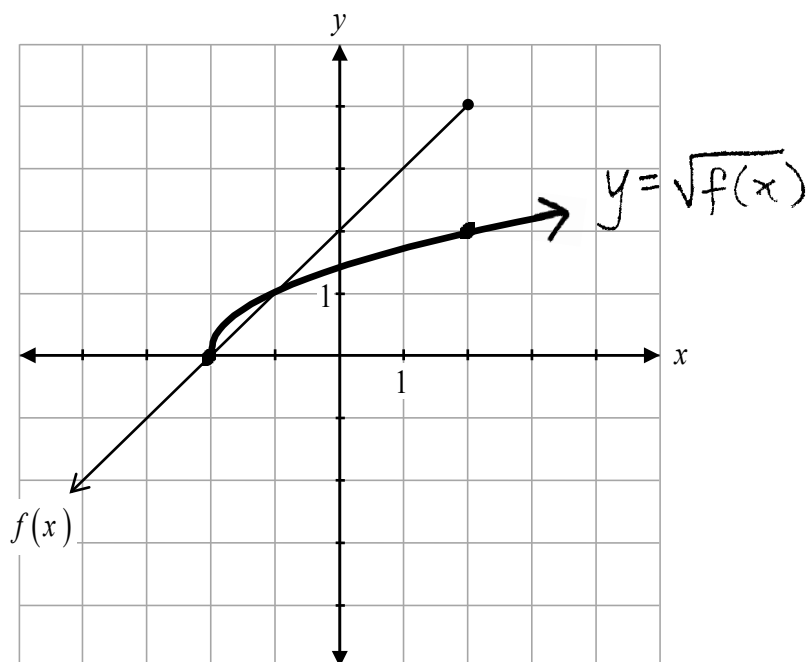


1 mark for horizontal reflection

1 mark for absolute value

2 marks

Savannah used the graph of $y = f(x)$ to sketch the graph of $y = \sqrt{f(x)}$. Her solution is given below. Describe her error.

**Solution**

Savannah did not restrict the domain at $x = 2$.

1 mark

Determine the exact value of $\sin\left(\frac{13\pi}{12}\right)$.

Solution

$$\sin\left(\frac{3\pi}{4} + \frac{\pi}{3}\right) = \sin\left(\frac{3\pi}{4}\right)\cos\left(\frac{\pi}{3}\right) + \cos\left(\frac{3\pi}{4}\right)\sin\left(\frac{\pi}{3}\right)$$

1 mark for substitution into correct identity

$$= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) + \left(-\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)$$

2 marks (½ mark for each exact value)

$$= \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4}$$

$$= \frac{\sqrt{2} - \sqrt{6}}{4}$$

3 marks

Note:

- Other combinations are possible.

State the equation of the horizontal asymptote of $f(x) = \frac{3x}{x-1}$.

Solution

$$y = 3$$

1 mark

Sketch the graph of $f(x) = \frac{5x-10}{x^2+x-6}$.

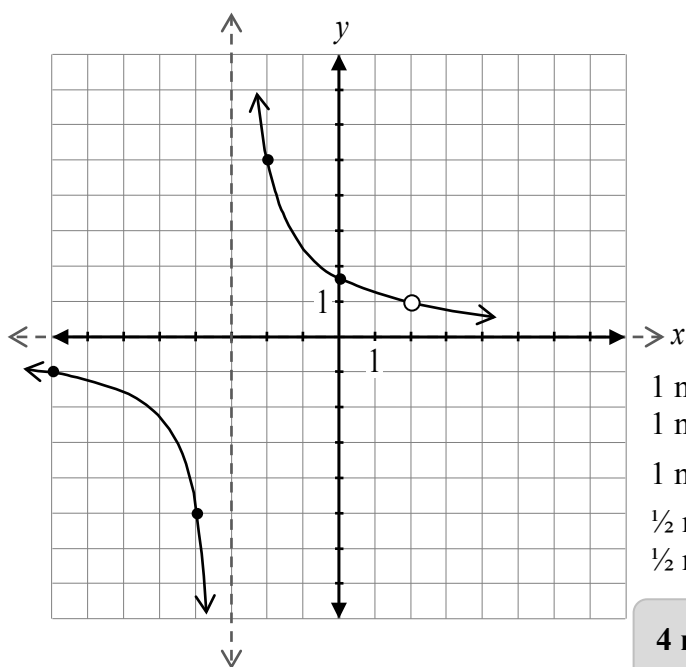
Solution

$$\begin{aligned} f(x) &= \frac{5x-10}{x^2+x-6} \\ &= \frac{5(x-2)}{(x-2)(x+3)} \\ &= \frac{5}{x+3}, x \neq 2 \end{aligned}$$

$\therefore$ there is a point of discontinuity (hole) at $(2,1)$

vertical asymptote at $x = -3$

horizontal asymptote at $y = 0$



1 mark for asymptotic behaviour approaching $x = -3$

1 mark for asymptotic behaviour approaching $y = 0$

1 mark for a point of discontinuity (hole) at $(2,1)$

$\frac{1}{2}$ mark for graph left of $x = -3$

$\frac{1}{2}$ mark for graph right of $x = -3$

4 marks

Determine, algebraically, the inverse of $f(x) = 3x + 4$.

Solution

$$\text{Let } f(x) = y$$

$$y = 3x + 4$$

$$x = 3y + 4$$

1 mark for switching x and y -values

$$x - 4 = 3y$$

$$\frac{x - 4}{3} = y$$

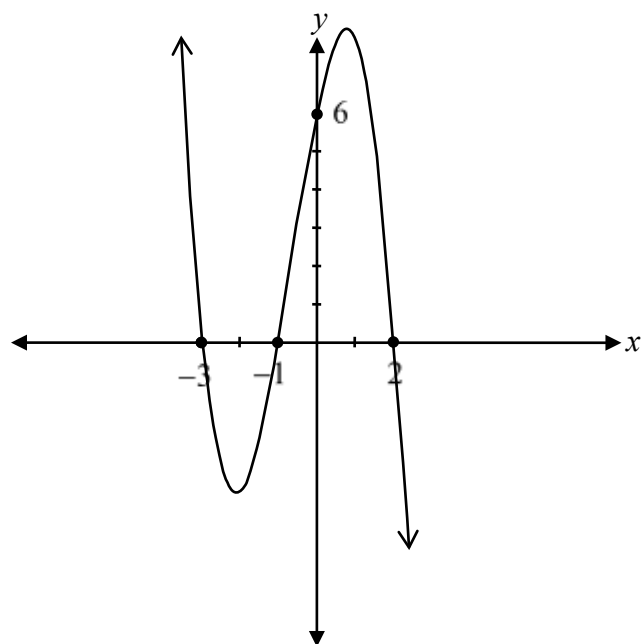
$\frac{1}{2}$ mark for solving for y

$$f^{-1}(x) = \frac{x - 4}{3}$$

$\frac{1}{2}$ mark for writing equation of $f^{-1}(x)$

2 marks

Sketch the graph of $P(x) = -(x+1)(x-2)(x+3)$.

Solution

1 mark for x -intercepts
 $\frac{1}{2}$ mark for y -intercept
 $\frac{1}{2}$ mark for end behaviour

2 marks

Determine and simplify the 8th term in the binomial expansion of $\left(x - \frac{2}{x^3}\right)^{10}$.

Solution

$$t_8 = {}_{10}C_7 (x)^3 \left(-\frac{2}{x^3}\right)^7 \quad 2 \text{ marks (1 mark for } {}_{10}C_7; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$= 120 \left(x^3\right) \left(-\frac{128}{x^{21}}\right)$$

$$= -\frac{15\,360}{x^{18}}$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient; $\frac{1}{2}$ mark for exponent)

3 marks

Question 2

R10

The temperature of hot chocolate is recorded as it cools down. The data shows that the temperature cools according to the equation:

$$T = 82(0.87)^t + 16$$

where T is the temperature of the hot chocolate, in degrees Celsius after t minutes and

t is the time, in minutes, after the hot chocolate is made.

In order to avoid burns, hot chocolate should not be served at a temperature higher than 71°C . Determine, algebraically, the amount of time it will take the hot chocolate to reach this temperature.

Solution

$$71 = 82(0.87)^t + 16$$

$$55 = 82(0.87)^t$$

$$\frac{55}{82} = 0.87^t$$

$$\log\left(\frac{55}{82}\right) = t \log(0.87)$$

$\frac{1}{2}$ mark for applying logarithms
1 mark for power law

$$t = \frac{\log\left(\frac{55}{82}\right)}{\log(0.87)}$$

$$t = 2.867\ 874$$

$$t = 2.868 \text{ minutes}$$

$\frac{1}{2}$ mark for evaluating quotient of logarithms

2 marks

Solve for θ , algebraically, over the interval $[0, 2\pi]$.

$$3\sin^2\theta + 6\sin\theta + 2 = 0$$

Solution

$$\sin\theta = \frac{-6 \pm \sqrt{6^2 - 4(3)(2)}}{2(3)}$$

½ mark for substitution into quadratic formula

$$\sin\theta = \frac{-6 \pm \sqrt{12}}{6}$$

½ mark for solving for $\sin\theta$

$$\sin\theta = -0.422\ 649\dots$$

$$\sin\theta = -1.577\ 350\dots$$

$$\theta_r = 0.436\ 367\dots$$

no solution

$$\theta = 3.578, 5.847$$

2 marks for solving for θ

(½ mark for each value; 1 mark for indicating no solution)

3 marks

Question 4

P3

A committee of 4 people is to be selected from 6 high school students and 5 middle years students.

Determine the number of possible committees if the committee must have at least 3 high school students.

Solution

Case 1: ${}_6C_3 \cdot {}_5C_1 = 100$

1 mark for Case 1

Case 2: ${}_6C_4 \cdot {}_5C_0 = 15$

1 mark for Case 2

$100 + 15 = 115$ possible committees

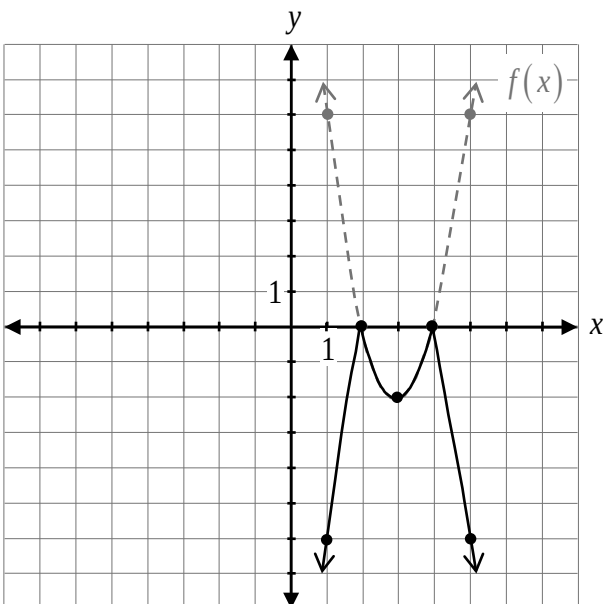
1 mark for addition of cases

3 marks

Note:

- ${}_5C_0$ does not need to be shown.

Given the graph of $y = f(x)$, sketch the graph of $y = -|f(x)|$.

Solution

1 mark for absolute value
1 mark for vertical reflection

2 marks

Describe the transformations used to obtain the graph of the function $y = 4f\left(\frac{x}{2}\right) - 3$ from the graph of $y = f(x)$.

Solution

The graph of $y = f(x)$ is vertically stretched by a factor of 4, horizontally stretched by a factor of 2, and translated down 3 units.

1 mark for vertical stretch

1 mark for horizontal stretch

1 mark for vertical translation

3 marks

Note:

- Deduct a maximum of 1 mark for concept error of incorrect order of vertical transformations.

If $\log 4 = m$ and $\log 3 = n$, express $\log 48$ in terms of m and n .

Solution

$$\log 48 = \log(4^2 \cdot 3)$$

$$= 2\log 4 + \log 3$$

$$= 2m + n$$

1 mark for power law

1 mark for product law

2 marks

State the domain and range of the radical function, $f(x) = -3\sqrt{x-8} + 1$.

Solution

Domain: $[8, \infty)$

1 mark for domain

Range: $(-\infty, 1]$

1 mark for range

2 marks

Explain why the graph of $y = \tan x$ does not have an amplitude.

Solution

The range of $y = \tan x$ is all real numbers.

1 mark

or

As the graph of $y = \tan x$ approaches the vertical asymptotes, the function value approaches positive and negative infinity.

Prove the following identity for all permissible values of θ .

$$\frac{\sec \theta - \sin^2 \theta \sec \theta}{\tan \theta \sin \theta} = \csc^2 \theta - 1$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\frac{1}{\cos \theta} - \frac{\sin^2 \theta}{\cos \theta}}{\left(\frac{\sin \theta}{\cos \theta}\right)(\sin \theta)}$ $\frac{1 - \sin^2 \theta}{\cancel{\cos \theta}}$ $\frac{\sin^2 \theta}{\cancel{\cos \theta}}$ $\frac{\cos^2 \theta}{\sin^2 \theta}$ $\cot^2 \theta$ $\csc^2 \theta - 1$	$\csc^2 \theta - 1$

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Method 2

Left-Hand Side	Right-Hand Side
$\frac{\sec \theta (1 - \sin^2 \theta)}{\left(\frac{\sin \theta}{\cos \theta}\right)(\sin \theta)}$ $\frac{\left(\frac{1}{\cos \theta}\right)(\cos^2 \theta)}{\frac{\sin^2 \theta}{\cos \theta}}$ $\frac{\cos^2 \theta}{\cancel{\cos \theta} \frac{\sin^2 \theta}{\cancel{\cos \theta}}}$ $\cot^2 \theta$ $\csc^2 \theta - 1$	$\csc^2 \theta - 1$

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 11

R11

Given that $x = 3$ is one of the zeros of $p(x) = x^3 - 7x - 6$, express $p(x)$ in completely factored form.

Solution

$$\begin{array}{r|rrrr} 3 & 1 & 0 & -7 & -6 \\ & & 3 & 9 & 6 \\ \hline & 1 & 3 & 2 & 0 \end{array}$$

1 mark for synthetic division (or equivalent strategy)

$$p(x) = (x - 3)(x^2 + 3x + 2)$$

$\frac{1}{2}$ mark for factor of $(x - 3)$

$$p(x) = \underline{(x - 3)(x + 2)(x + 1)}$$

$\frac{1}{2}$ mark for other consistent factors

2 marks

Solve, algebraically.

$${}_{n+1}P_2 = 6$$

Solution

$$\frac{(n+1)!}{(n+1-2)!} = 6$$

½ mark for substitution

$$\frac{(n+1)!}{(n-1)!} = 6$$

$$\frac{(n+1)(n)\cancel{(n-1)!}}{\cancel{(n-1)!}} = 6$$

1 mark for factorial expansion

½ mark for simplification of factorials

$$n^2 + n = 6$$

$$n^2 + n - 6 = 0$$

$$(n+3)(n-2) = 0$$

$$\cancel{n+3} \quad n = 2$$

½ mark for the permissible value of n

½ mark for showing the rejection of the extraneous root

3 marks

Express $\log(3x - 1) - \log x + \log 9$ as a single logarithm.

Solution

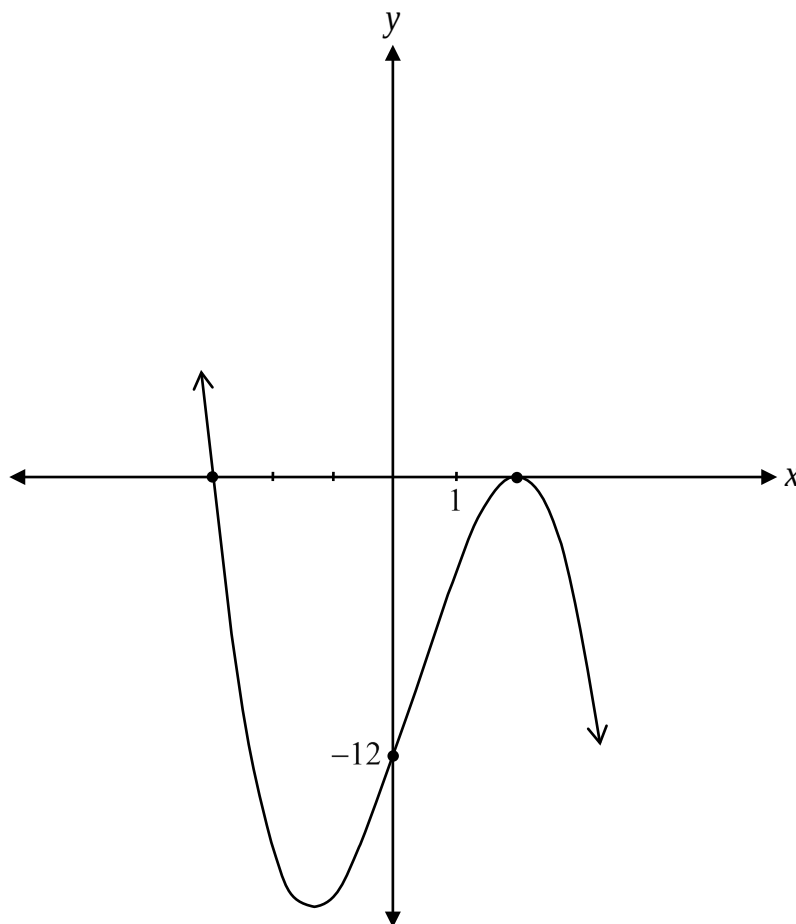
$$\log\left(\frac{9(3x - 1)}{x}\right)$$

1 mark for product law
1 mark for quotient law

2 marks

Sketch the graph of $P(x) = -(x-2)^2(x+3)$.

Solution



1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

1 mark for multiplicity of 2 at $x = 2$

$\frac{1}{2}$ mark for end behaviour

3 marks

Given $f(x) = \frac{1}{2}x - 3$, state the coordinates of an invariant (unchanged) point when sketching the graph of $y = \sqrt{f(x)}$.

Solution

$$\frac{1}{2}x - 3 = 0$$

$$\frac{1}{2}x = 3$$

$$x = 6$$

$$(6, 0)$$

1 mark**or**

$$\frac{1}{2}x - 3 = 1$$

$$\frac{1}{2}x = 4$$

$$x = 8$$

$$(8, 1)$$

Solve, algebraically.

$$125^{3x+4} = \left(\frac{1}{5}\right)^x$$

Solution

$$\left(5^3\right)^{3x+4} = \left(5^{-1}\right)^x$$

1 mark for changing to a common base (½ mark for each side)

$$5^{9x+12} = 5^{-x}$$

½ mark for exponent law

$$9x + 12 = -x$$

½ mark for equating exponents

$$10x = -12$$

$$x = -\frac{12}{10}$$

2 marks

or

$$x = -\frac{6}{5}$$

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ and $\cos \theta = -\frac{\sqrt{2}}{2}$, justify that the value of $\tan(2\theta)$ is undefined.

Solution**Method 1**

$$\begin{aligned}\tan(2\theta) &= \frac{2 \tan \theta}{1 - \tan^2 \theta} \\ &= \frac{2(1)}{1-1} \\ &= \frac{2}{0}\end{aligned}$$

1 mark for value of $\tan \theta$

$\therefore$ undefined

1 mark for justification

2 marks

Method 2

$$\tan \theta = 1$$

$$\theta = \frac{5\pi}{4}$$

1 mark for value of θ

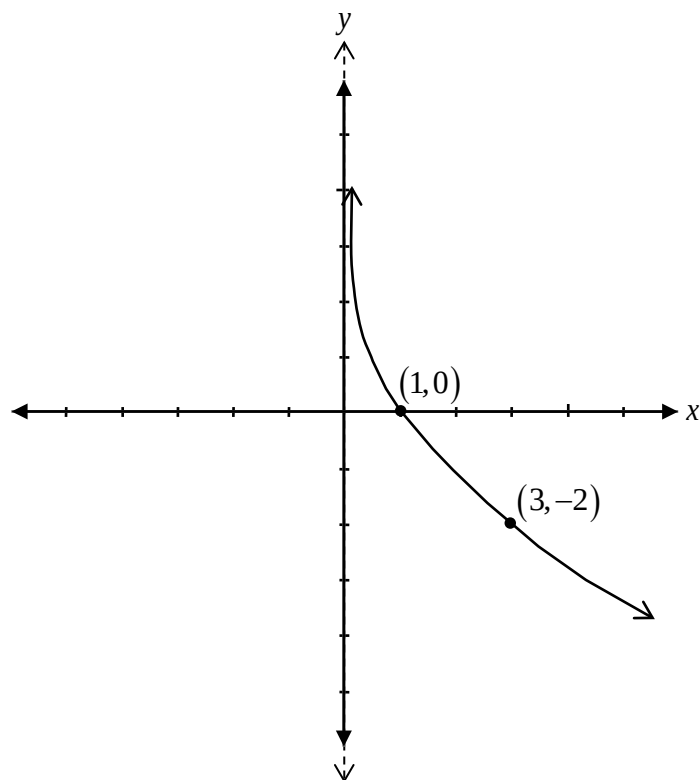
$$\begin{aligned}\tan(2\theta) &= \tan\left(2\left(\frac{5\pi}{4}\right)\right) \\ &= \tan \frac{5\pi}{2}\end{aligned}$$

$\therefore$ undefined

1 mark for justification

2 marks

Sketch the graph of $y = -2\log_3 x$.

Solution

1 mark for asymptotic behaviour
approaching $x = 0$

1 mark for vertical reflection

1 mark for vertical stretch

3 marks

Evaluate.

$$\cos\left(\pi \cdot \sin\left(-\frac{\pi}{6}\right)\right)$$

Solution

$$\cos\left(\pi\left(-\frac{1}{2}\right)\right)$$

1 mark for $\sin\left(-\frac{\pi}{6}\right)$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

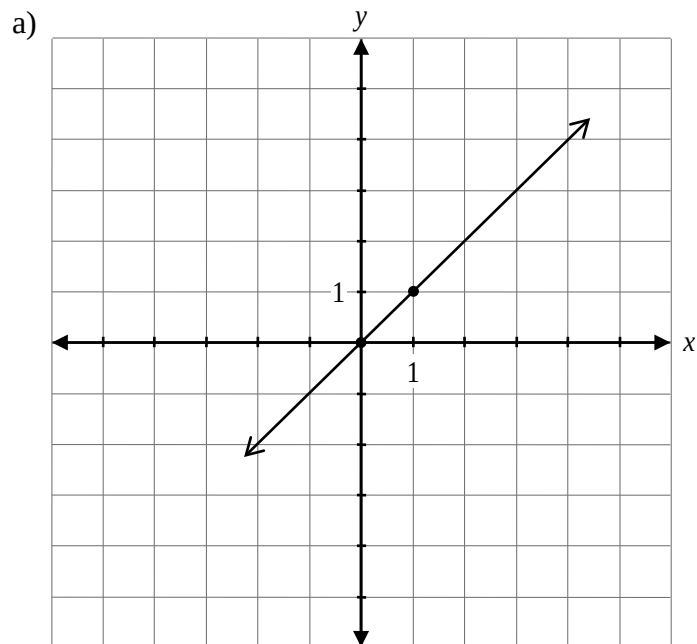
$$\cos\left(-\frac{\pi}{2}\right)$$

1 mark for consistent value

$$0$$

2 marks

- a) Given $f(x) = -x + 2$, sketch the graph of $h(x) = f(f(x))$.
- b) Explain why the domain of $h(x) = f(f(x))$ does not have any restrictions.

Solution**1 mark**

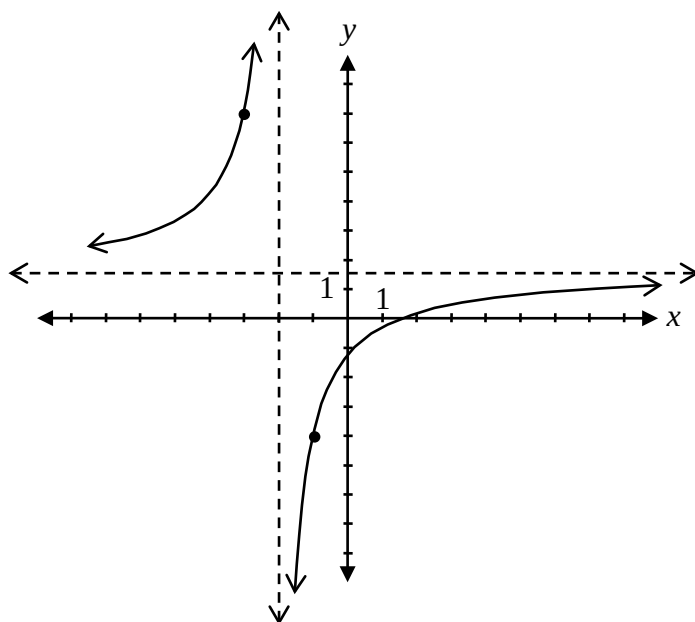
- b) Since the domain of $y = f(x)$ does not have any restrictions, then the domain of the composition does not have any restrictions.

1 mark

a) Sketch the graph of $f(x) = \frac{3x-5}{2x+4}$.

b) State the range of $f(x)$.

Solution



1 mark for asymptotic behaviour approaching $x = -2$

1 mark for asymptotic behaviour approaching $y = \frac{3}{2}$

$\frac{1}{2}$ mark for graph right of $x = -2$

$\frac{1}{2}$ mark for graph left of $x = -2$

3 marks

b) Range: $\left\{ y \in \mathbb{R} \mid y \neq \frac{3}{2} \right\}$

1 mark for range consistent with graph

1 mark

Determine, algebraically, the equation of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 4
- $p(x)$ has a zero at 3 with a multiplicity of 2
- $p(x)$ has zeroes at -1 and -2
- $p(x)$ passes through the point $(2, 24)$.

Solution

$$p(x) = a(x+1)(x+2)(x-3)^2$$

$\frac{1}{2}$ mark for factors of $p(x)$

$\frac{1}{2}$ mark for multiplicity of 2 at $x = 3$

$$24 = a(2+1)(2+2)(2-3)^2$$

$\frac{1}{2}$ mark for substitution of $p(2) = 24$

$$24 = 12a$$

$$a = 2$$

$\frac{1}{2}$ mark for consistent value of a

$$p(x) = \underline{2(x+1)(x+2)(x-3)^2}$$

2 marks

Verify, by substitution, that the equation $\frac{\cos \theta + \cot \theta}{\cot \theta} = 1 + \sin \theta$ is true for $\theta = \frac{2\pi}{3}$.

Solution

Left-Hand Side	Right-Hand Side
$\frac{\cos\left(\frac{2\pi}{3}\right) + \cot\left(\frac{2\pi}{3}\right)}{\cot\left(\frac{2\pi}{3}\right)}$	$1 + \sin\left(\frac{2\pi}{3}\right)$ ½ mark for substitution
$\frac{-\frac{1}{2} - \frac{1}{\sqrt{3}}}{-\frac{1}{\sqrt{3}}}$	$1 + \frac{\sqrt{3}}{2}$ 1½ marks for exact values (½ mark for each)
$\frac{-\frac{\sqrt{3}-2}{2\sqrt{3}}}{-\frac{1}{\sqrt{3}}}$	$\frac{2}{2} + \frac{\sqrt{3}}{2}$
$\frac{-\sqrt{3}-2}{-2}$	$\frac{2+\sqrt{3}}{2}$
$\frac{-(\sqrt{3}+2)}{-2}$	$\frac{\sqrt{3}+2}{2}$ 1 mark for simplification (½ mark for LHS; ½ mark for RHS)
$\frac{\sqrt{3}+2}{2}$	3 marks

Note:

- Deduct a maximum of 1 mark if student proves identity without substitution of $\theta = \frac{2\pi}{3}$.

Jyugo was asked to state the equation of the vertical asymptote(s) on the graph of $f(x) = \frac{3x-15}{x^2-5x}$.

His solution:

$$f(x) = \frac{3(x-5)}{x(x-5)}$$

$$x=0 \quad x=5$$

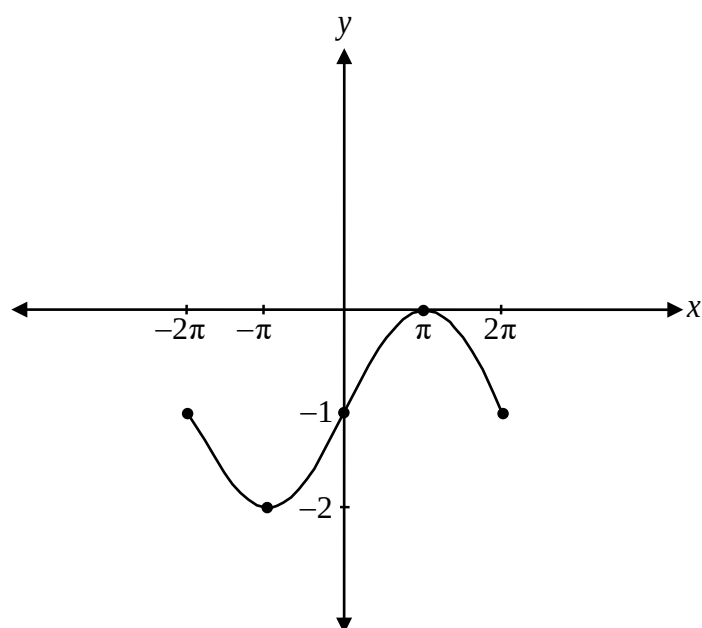
Explain why his solution is incorrect.

Solution

His solution at $x=5$ is the x -value of a point of discontinuity (hole), not a vertical asymptote.

1 mark

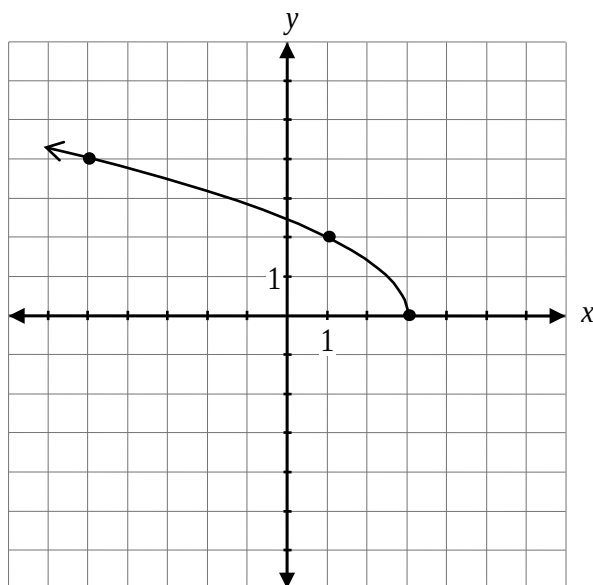
Sketch the graph of $y = \sin\left(\frac{1}{2}x\right) - 1$ over the interval $[-2\pi, 2\pi]$.

Solution

1 mark for shape of $y = \sin x$
1 mark for period
1 mark for vertical translation

3 marks

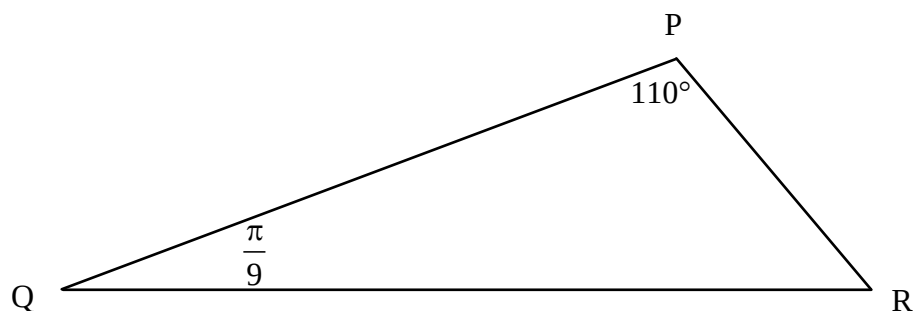
Sketch the graph of $y = \sqrt{-2x + 6}$.

Solution

- 1 mark for shape of a radical function
- 1 mark for horizontal reflection
- 1 mark for horizontal compression
- 1 mark for horizontal translation

4 marks

Determine the measure of angle R, in radians.

**Solution**

$$(110)\left(\frac{\pi}{180}\right) = \frac{11\pi}{18} \quad \text{1 mark for conversion}$$

$$\angle R = \pi - \left(\frac{\pi}{9} + \frac{11\pi}{18}\right)$$

$$\angle R = \frac{18\pi}{18} - \left(\frac{2\pi}{18} + \frac{11\pi}{18}\right)$$

$$\angle R = \frac{5\pi}{18} \quad \text{1 mark for angle R}$$

2 marks

Given $f(x) = \sqrt{x-4}$, state the equation of the resulting function, $g(x)$, after a reflection over the x -axis.

Solution

$$g(x) = -\sqrt{x-4}$$

1 mark for vertical reflection

1 mark

a) Determine the coterminal angle of $\frac{29\pi}{12}$ over the interval $[0, 2\pi]$.

b) Determine the exact value of $\sin\left(\frac{29\pi}{12}\right)$.

Solution

a) $\frac{5\pi}{12}$

1 mark

b)

$$\sin\left(\frac{5\pi}{12}\right) = \sin\left(\frac{\pi}{6} + \frac{\pi}{4}\right)$$

1 mark for substitution into correct identity

$$= \sin\frac{\pi}{6}\cos\frac{\pi}{4} + \cos\frac{\pi}{6}\sin\frac{\pi}{4}$$

$$= \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

2 marks ($\frac{1}{2}$ mark for each exact value)

$$= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4}$$

$$= \frac{\sqrt{2} + \sqrt{6}}{4}$$

3 marks

or

$$= \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

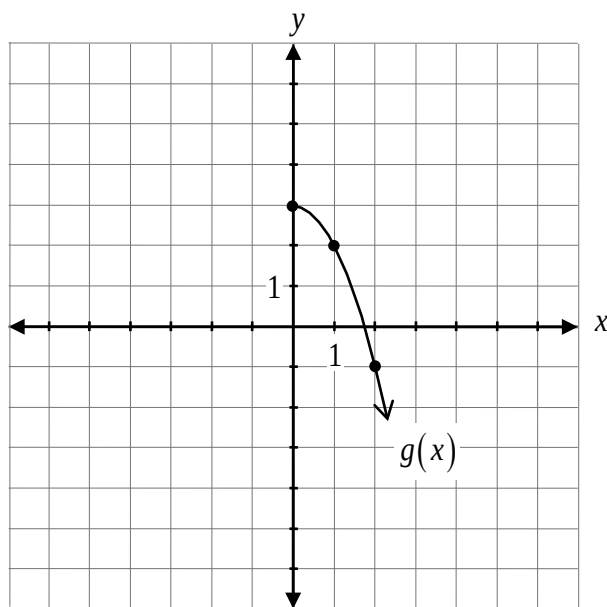
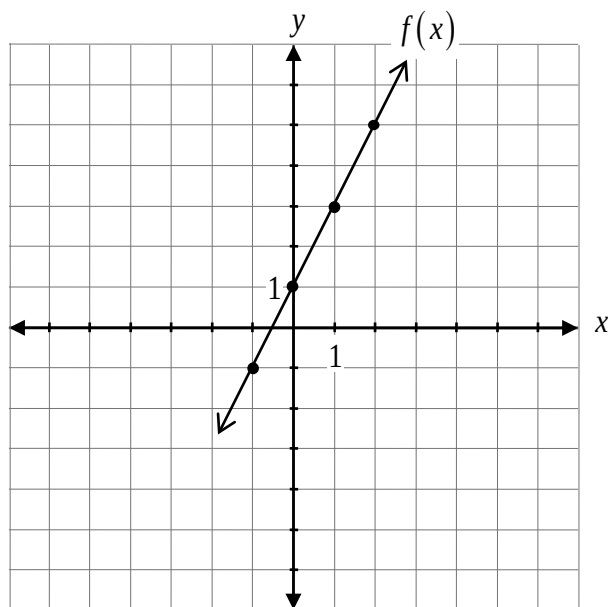
Note:

- Other combinations are possible.

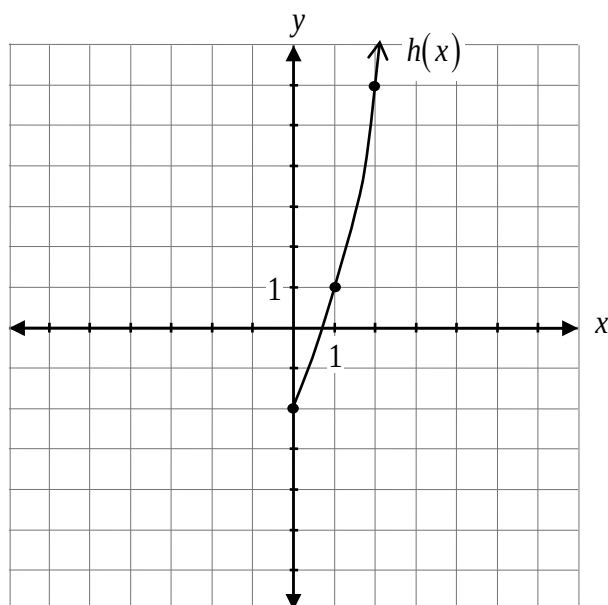
Question 38

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) - g(x)$.



Solution



1 mark for operation of subtraction
1 mark for restricted domain

2 marks

Determine an equation of a sinusoidal function that has the following characteristics:

- an amplitude of 3
- a period of 6
- a minimum value of -5

Solution

$$y = A \sin(Bx) + D$$

$$y = 3 \sin\left(\frac{\pi}{3}x\right) - 2$$

or

$$y = A \cos(Bx) + D$$

$$y = 3 \cos\left(\frac{\pi}{3}x\right) - 2$$

1 mark for value of A

1 mark for value of B

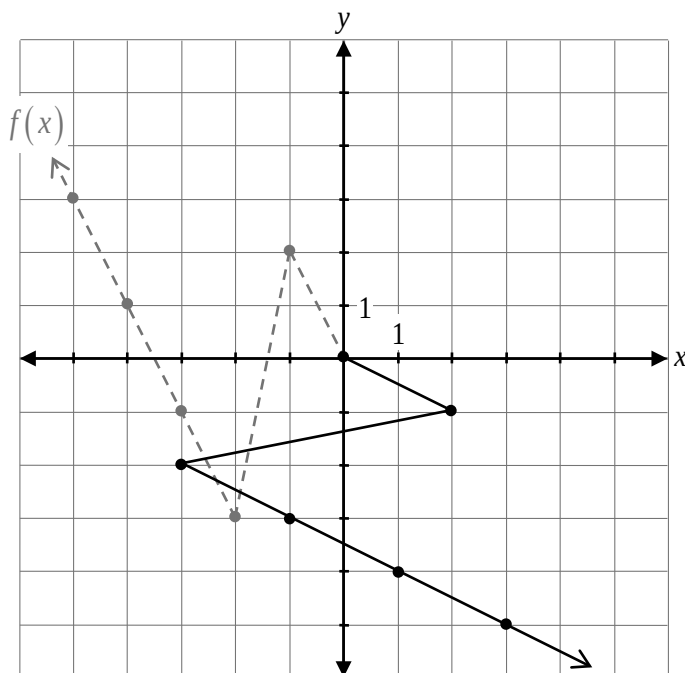
1 mark for value of D

3 marks

Note:

- Other equations are possible.

Given the graph of $y = f(x)$, sketch the graph after a reflection over the line $y = x$.

Solution

1 mark for reflection over the line $y = x$

1 mark

a) Determine the remainder when $3x^3 + 5x^2 - 13x - 3$ is divided by $(x + 3)$.

b) Is $(x + 3)$ a factor of $3x^3 + 5x^2 - 13x - 3$? Explain your reasoning.

Solution

a) $3(-3)^3 + 5(-3)^2 - 13(-3) - 3$

1 mark for remainder theorem

$$3(-27) + 5(9) - 13(-3) - 3$$

$$0$$

or

-3	3	5	-13	-3
		-9	12	3
	3	-4	-1	0

1 mark for synthetic division
(or any equivalent strategy)

The remainder is 0.

1 mark

b) Yes, because the remainder is 0.

1 mark for explanation

1 mark

State the range, the y-intercept, and the equation of the asymptote of the exponential function,
 $f(x) = 3^{x-1} + 2$.

SolutionRange: $(2, \infty)$

1 mark for range

y-intercept: $\frac{7}{3}$

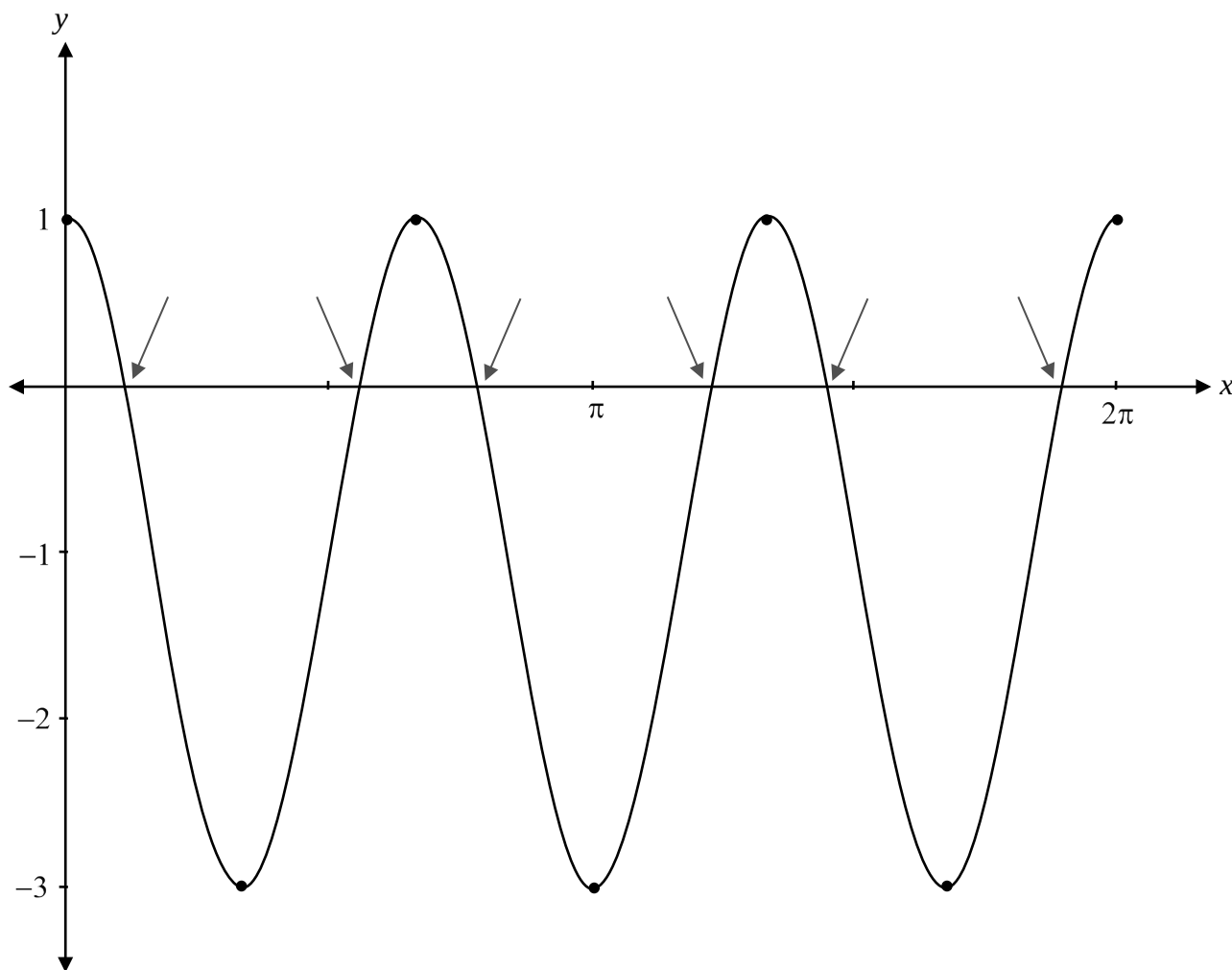
1 mark for y-intercept

Equation of the asymptote: $y = 2$

1 mark for equation of the asymptote

3 marks

The graph of $y = 2 \cos(3x) - 1$ below can be used to solve the equation $0 = 2 \cos(3x) - 1$ over the interval $[0, 2\pi]$. Indicate on the graph where to find at least one solution to the equation $0 = 2 \cos(3x) - 1$.

**Solution**

See arrows on graph above. 1 mark for indicating a solution on graph

1 mark

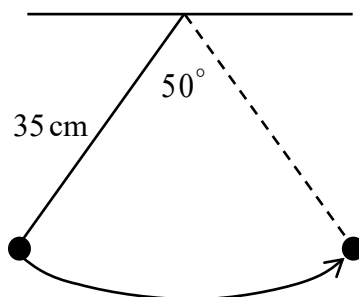
Note:

- Students do not need to show all six arrows.

Question 1

T1

A pendulum that is 35 cm long swings through an angle of 50° . Determine the length of the arc through which the pendulum swings.

**Solution**

$$\theta = 50 \left(\frac{\pi}{180} \right)$$

1 mark for conversion

$$= \frac{5\pi}{18}$$

or

$$= 0.872\ 664\dots$$

$$s = \theta r$$

$$= \left(\frac{5\pi}{18} \right) (35)$$

1 mark for substitution

$$= \frac{175\pi}{18} \text{ cm}$$

2 marks**or**

$$= 30.543 \text{ cm}$$

Solve algebraically, where $0 \leq \theta \leq 2\pi$.

$$2 \cos^2 \theta = \sin^2 \theta - 2 \cos \theta$$

Solution

$$2 \cos^2 \theta = 1 - \cos^2 \theta - 2 \cos \theta$$

1 mark for substitution of appropriate identity

$$3 \cos^2 \theta + 2 \cos \theta - 1 = 0$$

$$(\cos \theta + 1)(3 \cos \theta - 1) = 0$$

$$\cos \theta = -1 \quad \cos \theta = \frac{1}{3}$$

1 mark for solving for $\cos \theta$

$$\theta = \pi \quad \theta_r = 1.230\,959\dots$$

$$\theta = 1.231, 5.052$$

2 marks for solving for θ (1 mark for each branch)

4 marks

Determine the number of arrangements of the letters in the word ATTENTION which begin with the letter A.

Solution

$$\frac{8!}{3!2!} = 3360$$

1 mark for 8!

1 mark for division by 3!2! ($\frac{1}{2}$ mark for 3!; $\frac{1}{2}$ mark for 2!)

2 marks

Solve for x , algebraically.

$$e^{2x+1} = 5^x$$

Solution

$$\ln(e^{2x+1}) = \ln(5^x)$$

½ mark for applying logarithms

$$(2x+1)\ln e = x\ln 5$$

1 mark for power law (½ mark for each)

$$2x+1 = x\ln 5$$

$$2x - x\ln 5 = -1$$

½ mark for collecting terms with x

$$x(2 - \ln 5) = -1$$

$$x = \frac{-1}{2 - \ln 5}$$

½ mark for isolating x

$$x = -2.560\ 412\dots$$

$$x = -2.560$$

½ mark for evaluating quotient of logarithms

3 marks

There are 10 teachers and 17 students who would like to attend a field trip.

Determine the number of ways that 3 teachers and 9 students can be selected given that Mr. Jones and Mrs. Carol, two of the teachers, must be selected to attend the field trip.

Solution

$${}_2C_2 \cdot {}_8C_1 \cdot {}_{17}C_9$$

$$194\,480$$

1 mark for ${}_8C_1$

$\frac{1}{2}$ mark for ${}_{17}C_9$

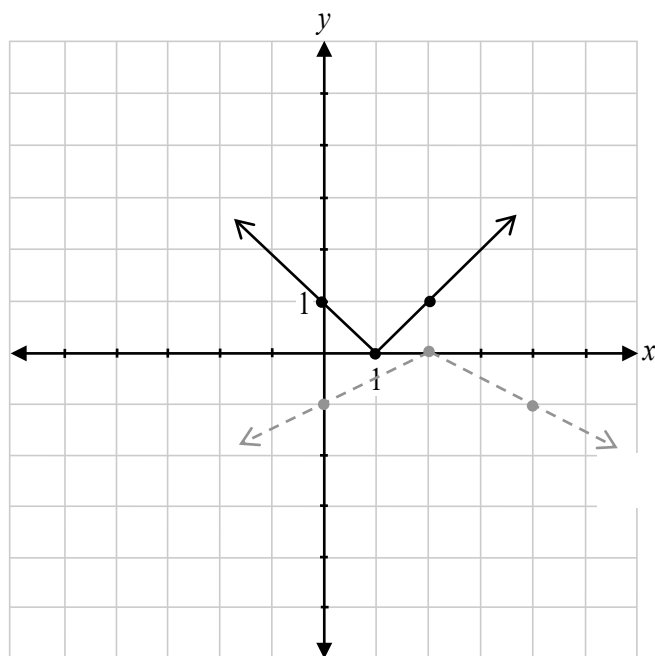
$\frac{1}{2}$ mark for product of combinations

2 marks

Note:

- ${}_2C_2$ does not need to be shown.

Given the graph of $f(x)$, sketch the graph of $y = -f(2x)$.

Solution

1 mark for vertical reflection
1 mark for horizontal compression

2 marks

There are 5 roads between Anneville and Berrybourg, and 2 roads between Berrybourg and Carriton.

Determine how many ways Blake can travel from Anneville to Carriton and back to Anneville, given the following conditions:

- he must travel through Berrybourg in both directions
- he cannot use the same road twice

Solution

$$\frac{5}{A \rightarrow B} \cdot \frac{2}{B \rightarrow C} \cdot \frac{1}{C \rightarrow B} \cdot \frac{4}{B \rightarrow A}$$

40 ways

$\frac{1}{2}$ mark for correct number of routes between towns
 $\frac{1}{2}$ mark for product

1 mark

Determine which term contains x^0 in the binomial expansion of $\left(x^2 + \frac{1}{x}\right)^6$.

Solution**Method 1**

$$x^0 = \left(x^2\right)^{6-k} \left(\frac{1}{x}\right)^k$$

½ mark for substitution

$$x^0 = x^{12-2k} x^{-k}$$

$$x^0 = x^{12-3k}$$

$$3k = 12$$

$$k = 4$$

½ mark for solving for k

∴ the fifth term

1 mark for the fifth term (or a term consistent with the value of k)

2 marks

Method 2

$$\left(x^2\right)^6, \left(x^2\right)^5 \left(\frac{1}{x}\right), \left(x^2\right)^4 \left(\frac{1}{x}\right)^2$$

1 mark for determining the pattern

$$x^{12}, x^9, x^6 \dots$$

∴ the fifth term

1 mark for the fifth term (or a term consistent with the pattern)

2 marks

Given $f(x) = x^3 + 1$, determine the equation of $f^{-1}(x)$.

Solution

Let $y = f(x)$

$$y = x^3 + 1$$

To determine the inverse of $f(x)$, switch x and y .

$$x = y^3 + 1$$

1 mark for switching x and y

$$x - 1 = y^3$$

$$\sqrt[3]{x-1} = y$$

$\frac{1}{2}$ mark for isolating y

$$f^{-1}(x) = \sqrt[3]{x-1}$$

$\frac{1}{2}$ mark for writing equation in terms of $f^{-1}(x)$

2 marks

Guillermo was asked to determine the number of ways to select a president, a vice president, and a treasurer from a group of 11 people.

His solution: ${}_{11}C_3$.

Explain why he should have used a permutation instead of a combination.

Solution

He should have used a permutation because the people are being selected for specific positions.

1 mark

Prove the following identity for all permissible values of x .

$$\frac{\csc^2 x \sec x}{\tan x + \cot x} = \csc x$$

Solution

Left-Hand Side	Right-Hand Side
$\frac{\left(\frac{1}{\sin^2 x}\right)\left(\frac{1}{\cos x}\right)}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$ $\frac{1}{\frac{\sin^2 x \cos x}{\sin^2 x + \cos^2 x}}$ $\frac{1}{\sin^2 x \cancel{\cos x}} \cdot \frac{\cancel{\sin x} \cancel{\cos x}}{1}$ $\frac{1}{\sin x}$ $\csc x$	$\csc x$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Determine the value of x , algebraically.

$$5\log_a 2 - \frac{1}{4}\log_a 16 = \log_a x$$

Solution

$$\log_a 2^5 - \log_a 16^{\frac{1}{4}} = \log_a x$$

1 mark for power law ($\frac{1}{2}$ mark for each)

$$\log_a \left(\frac{32}{2} \right) = \log_a x$$

1 mark for quotient law

$$\log_a 16 = \log_a x$$

$$x = 16$$

1 mark for equating arguments

3 marks

Tamara must determine the factors of $x^4 - 13x + 2x^3 - 14x^2 + 24$.

Explain why the coefficients Tamara used to set up her synthetic division are not written correctly.

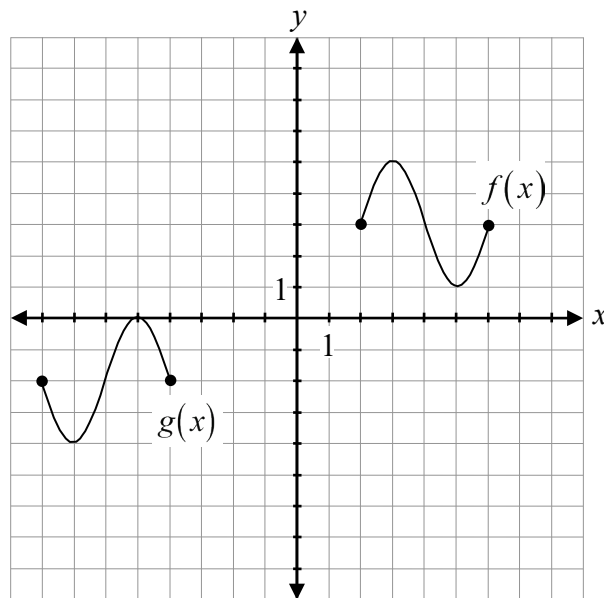
$$\begin{array}{r|rrrrr} & 1 & -13 & 2 & -14 & 24 \end{array}$$

Solution

Tamara did not arrange the coefficients of the terms in order of descending degree.

1 mark

Determine the equation of the graph of $g(x)$ in terms of $f(x)$.



Solution

$$g(x) = \underline{f(-(x+2)) - 5}$$

1 mark for horizontal reflection
1 mark for horizontal translation
1 mark for vertical translation

or

$$g(x) = \underline{-f(x+10) + 1}$$

1 mark for vertical reflection
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Expand, using the laws of logarithms.

$$\log_2 \left[\frac{(x-1)(x-2)}{x} \right]$$

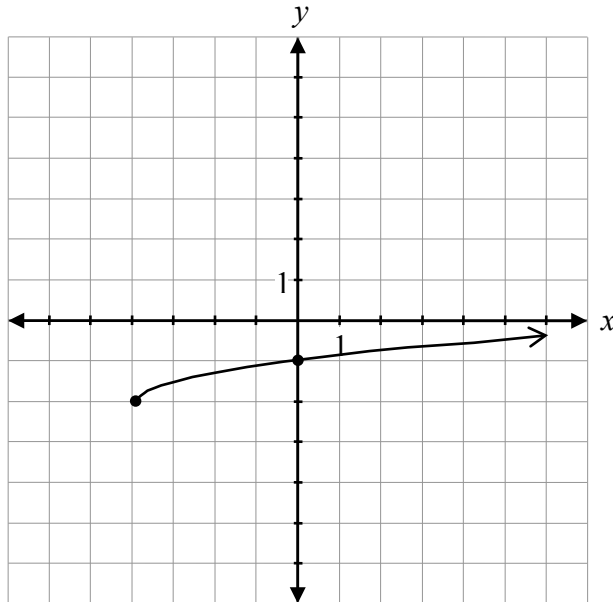
Solution

$$\log_2 (x-1) + \log_2 (x-2) - \log_2 x$$

1 mark for product law
1 mark for quotient law

2 marks

Determine the equation of the radical function represented by the graph.

**Solution**

$$y = \frac{1}{2}\sqrt{x+4} - 2$$

1 mark for vertical compression
1 mark for horizontal translation
1 mark for vertical translation

3 marks

or

$$y = \sqrt{\frac{1}{4}(x+4)} - 2$$

1 mark for horizontal stretch
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Determine the exact value of x .

$$\sec\left(\frac{2\pi}{3}\right)\left(\sin\left(-\frac{5\pi}{3}\right)\right)(x) = 3$$

Solution

$$(-2)\left(\frac{\sqrt{3}}{2}\right)(x) = 3$$

$$-\sqrt{3}(x) = 3$$

$$x = -\frac{3}{\sqrt{3}}$$

or

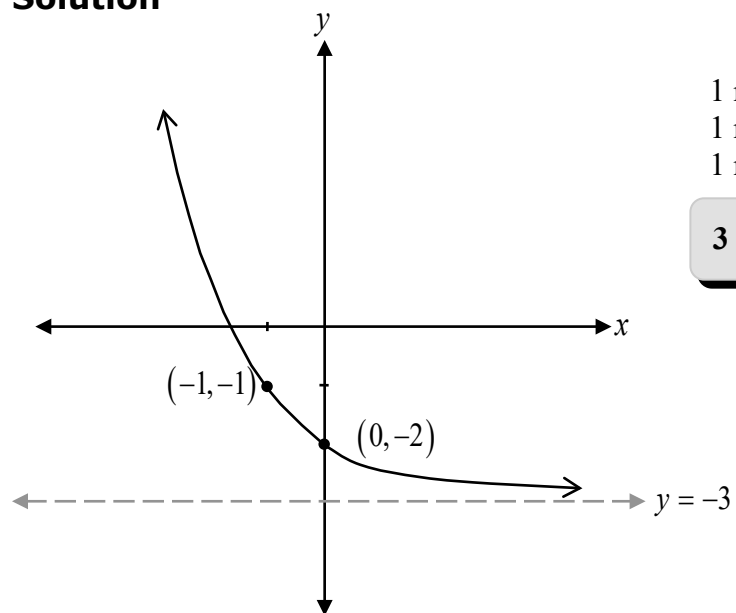
$$x = -\sqrt{3}$$

1 mark for $\sec\left(\frac{2\pi}{3}\right)$ (½ mark for value; ½ mark for quadrant)

1 mark for $\sin\left(-\frac{5\pi}{3}\right)$ (½ mark for value; ½ mark for quadrant)

2 marks

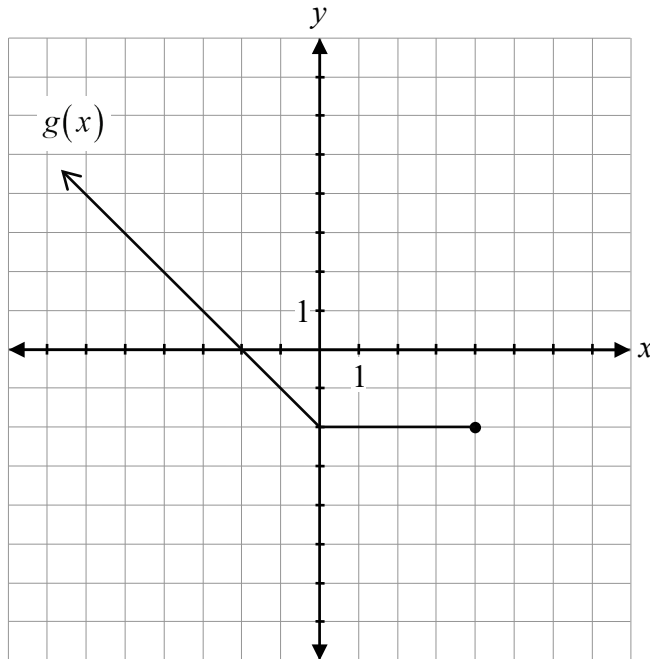
Sketch the graph of $y = 2^{-x} - 3$.

Solution

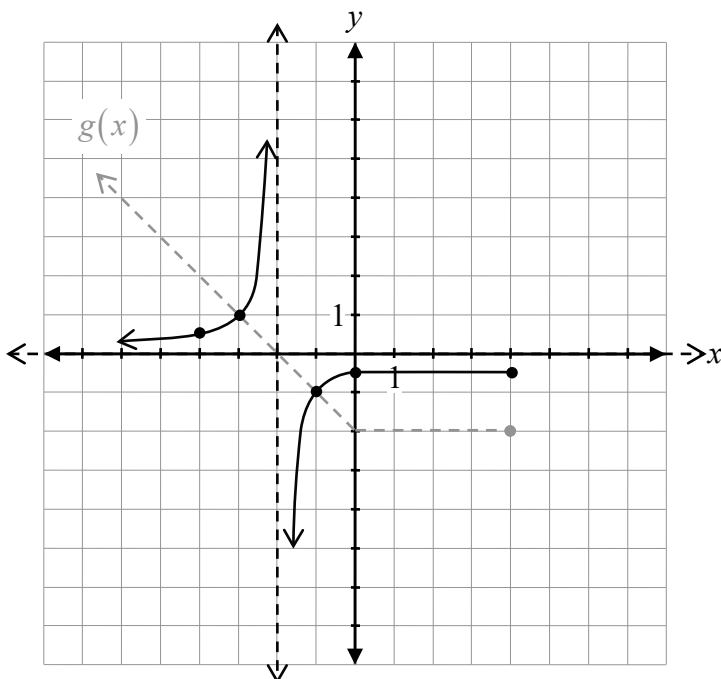
1 mark for shape of an exponential function
1 mark for horizontal reflection
1 mark for vertical translation

3 marks

Given the graph of $y = g(x)$, sketch the graph of $y = \frac{1}{g(x)}$.



Solution



1 mark for asymptotic behaviour approaching $x = -2$ ($\frac{1}{2}$ mark for each side)

$\frac{1}{2}$ mark for asymptotic behaviour approaching $y = 0$

$\frac{1}{2}$ mark for graph left of $x = -2$

$\frac{1}{2}$ mark for graph right of $x = -2$

$\frac{1}{2}$ mark for restricting domain

3 marks

Determine the exact value of $\tan\left(\frac{\pi}{12}\right)$.

Solution

$$\tan\left(\frac{\pi}{12}\right) = \tan\left(\frac{4\pi}{12} - \frac{3\pi}{12}\right)$$

$$\tan\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \frac{\tan\frac{\pi}{3} - \tan\frac{\pi}{4}}{1 + \tan\frac{\pi}{3}\tan\frac{\pi}{4}}$$

1 mark for substitution into correct identity

$$= \frac{\sqrt{3} - 1}{1 + \sqrt{3}}$$

1 mark for exact values (½ mark for each value)

or

$$= 2 - \sqrt{3}$$

or

$$= \frac{3 - \sqrt{3}}{3 + \sqrt{3}}$$

2 marks

Note:

- Other combinations are possible.

Explain why the graph of $g(x) = \frac{3}{x^2 + 4}$ does not have a vertical asymptote.

Solution

The graph of a rational function has a vertical asymptote when the denominator is equal to zero. The denominator of $g(x)$ will never be equal to zero.

1 mark

Solve, algebraically.

$$\log_3 x + \log_3 (x+8) = 2$$

Solution

Method 1

$$\log_3 [x(x+8)] = 2$$

1 mark for product law

$$x^2 + 8x = 3^2$$

1 mark for exponential form

$$x^2 + 8x - 9 = 0$$

$$(x+9)(x-1) = 0$$

$$\cancel{x+9} \quad x = 1$$

½ mark for the permissible value of x

½ mark for showing the rejection of the extraneous root

3 marks

Method 2

$$\log_3 x + \log_3 (x+8) = \log_3 3^2$$

1 mark for logarithmic form

$$\log_3 [x(x+8)] = \log_3 9$$

1 mark for product law

$$x^2 + 8x = 9$$

$$x^2 + 8x - 9 = 0$$

$$(x+9)(x-1) = 0$$

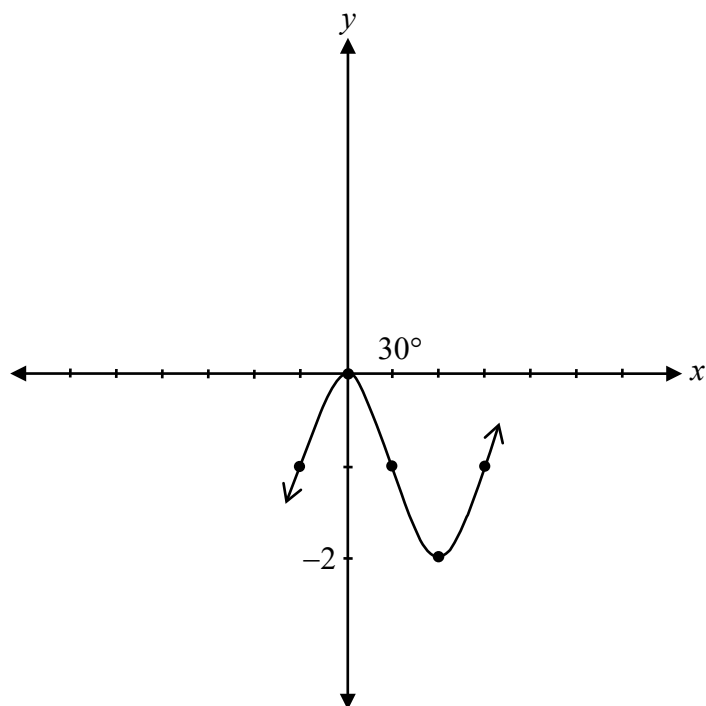
$$\cancel{x+9} \quad x = 1$$

½ mark for the permissible value of x

½ mark for showing the rejection of the extraneous root

3 marks

Sketch at least one period of the graph of the function $y = \sin(3(x + 30^\circ)) - 1$.

Solution

1 mark for shape of a sinusoidal function with correct amplitude

1 mark for period

1 mark for horizontal translation

1 mark for vertical translation

4 marks

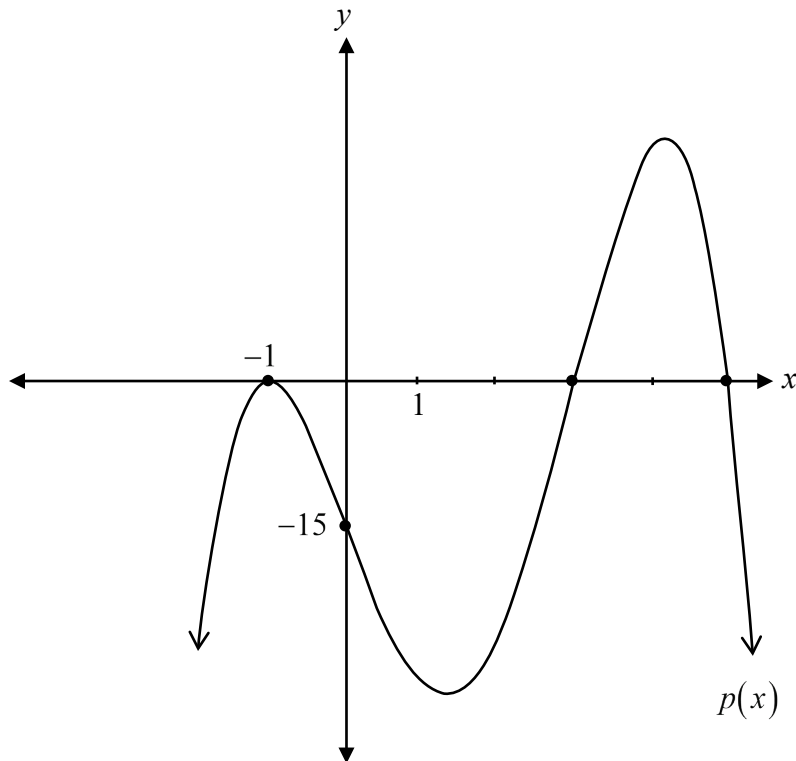
Explain why the domain of the function, $f(x) = \log(x-3)$, is $x > 3$.

Solution

The argument of a logarithm cannot be negative or zero.

1 mark

Sketch the graph of $p(x) = -(x-3)(x+1)^2(x-5)$.

Solution

1 mark for x -intercepts

1 mark for multiplicity of 2 at $x = -1$

$\frac{1}{2}$ mark for end behaviour

$\frac{1}{2}$ mark for y -intercept

3 marks

Given that $\sin \theta = -\frac{2}{3}$ and $\tan \theta > 0$, determine the exact value of $\sin 2\theta$.

Solution

$$x^2 = r^2 - y^2$$

$$x^2 = 9 - 4$$

$$x^2 = 5$$

$$x = \pm\sqrt{5}$$

$\frac{1}{2}$ mark for value of x

$$x = -\sqrt{5}$$

$$\cos \theta = -\frac{\sqrt{5}}{3}$$

$\frac{1}{2}$ mark for value of $\cos \theta$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \left(-\frac{2}{3} \right) \left(-\frac{\sqrt{5}}{3} \right)$$

1 mark for substitution into correct identity

$$= \frac{4\sqrt{5}}{9}$$

2 marks

Note:

- Accept any of the following values for x : $x = \pm\sqrt{5}$, $x = -\sqrt{5}$, or $x = \sqrt{5}$.

Justify whether $\frac{5\pi}{8}$ and $-\frac{11\pi}{4}$ are coterminal angles.

Solution**Method 1**

$$\frac{5\pi}{8} - \left(-\frac{11\pi}{4}\right)$$

$$\frac{5\pi}{8} + \frac{11\pi}{4}$$

$$\frac{5\pi}{8} + \frac{22\pi}{8}$$

$$\frac{27\pi}{8}$$

Since $\frac{27\pi}{8}$ is not a multiple of 2π they are not coterminal angles.

1 mark**Method 2**

Coterminal angles of $\frac{5\pi}{8}$:

$$\frac{5\pi}{8} - \frac{16\pi}{8} = -\frac{11\pi}{8}$$

$$-\frac{11\pi}{8} - \frac{16\pi}{8} = -\frac{27\pi}{8}$$

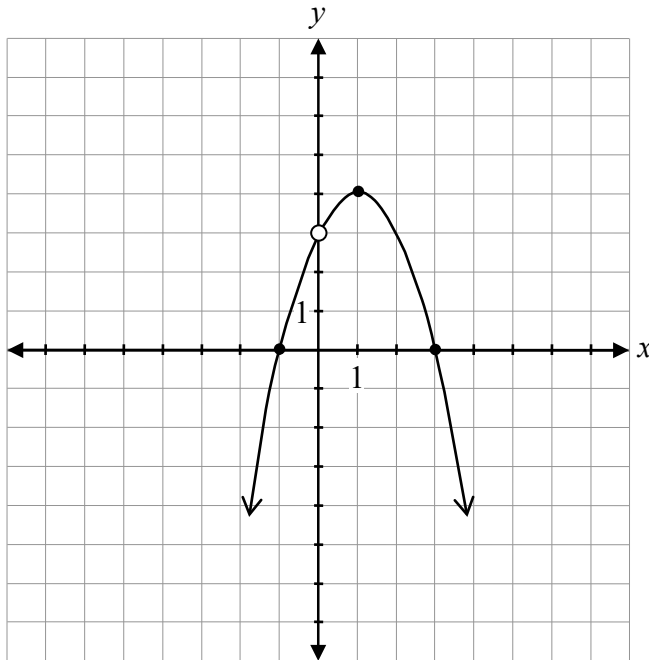
Coterminal angles of $-\frac{11\pi}{4}$:

$$-\frac{11\pi}{4} \left(\frac{2}{2}\right) = -\frac{22\pi}{8}$$

$\therefore$ Angles are not coterminal since $-\frac{27\pi}{8} \neq -\frac{22\pi}{8}$.

1 mark

Sketch the graph of $f(x) = \frac{-2x(x+1)(x-3)}{2x}$.

Solution

1 mark for point of discontinuity (hole) at $x = 0$
½ mark for shape of a parabola with correct
x-intercepts
½ mark for end behaviour

2 marks

Note:

- Deduct ½ mark for procedural error (incorrect y-value for point of discontinuity (hole)).

Given $\frac{\sin \theta + \cos \theta \csc \theta}{\sin \theta}$, determine the non-permissible values of θ , where $\theta \in \mathbb{R}$.

Solution**Method 1**

$$\sin \theta = 0$$

$\frac{1}{2}$ mark for $\sin \theta = 0$

$$\theta = 0, \pi, 2\pi$$

$\frac{1}{2}$ mark for any non-permissible value of θ

$$\theta = \pi k, k \in \mathbb{Z}$$

1 mark for all non-permissible values of θ

2 marks

Method 2

$$\sin \theta = 0$$

$\frac{1}{2}$ mark for $\sin \theta = 0$

$$\theta = 0, \pi, 2\pi$$

$\frac{1}{2}$ mark for any non-permissible value of θ

$$\theta = 2\pi k, k \in \mathbb{Z}$$

$$\theta = \pi + 2\pi k, k \in \mathbb{Z}$$

1 mark for all non-permissible values of θ

2 marks

Write an equation of a rational function that has a horizontal asymptote at $y = 0$ and a vertical asymptote at $x = 6$.

Solution

$$f(x) = \frac{1}{x-6}$$

1 mark for horizontal asymptote

1 mark for vertical asymptote

2 marks

Note:

- Other answers are possible.

Given the functions $f(x) = \sqrt{x-1}$ and $g(x) = x^2$,

a) state the equation of $g(f(x))$.

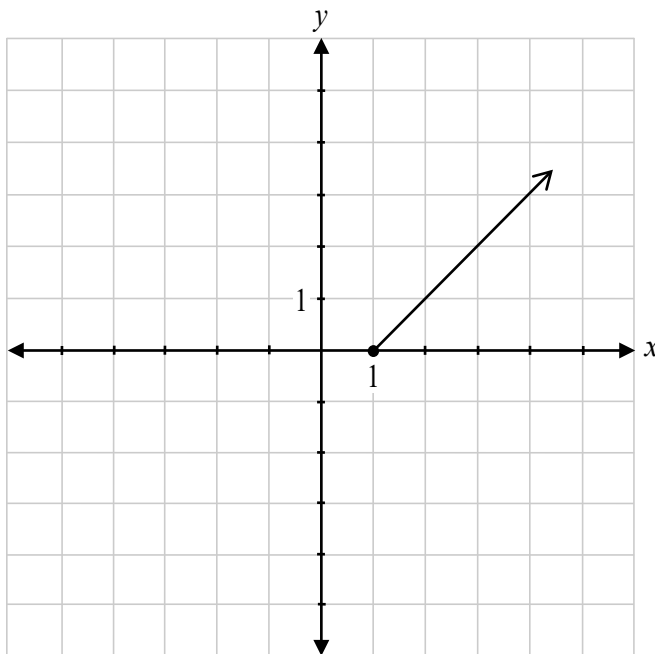
b) sketch the graph of $g(f(x))$.

Solution

a) $g(f(x)) = x - 1, x \geq 1$

1 mark

b)



1 mark for shape of graph consistent with a)
1 mark for restricted domain

2 marks**Notes:**

- Deduct a maximum of 1 mark for the concept error of not restricting the domain.
- Deduct ½ mark for procedural error (not stating domain) if graph shows correct domain.

Suzanne was asked to determine the value of $\tan \theta$, given that $\sec \theta = -\frac{8}{3}$ and θ terminates in quadrant II.

Her solution:

$$\begin{aligned}(-3)^2 + y^2 &= (8)^2 \\ y^2 &= 55 \\ y &= \sqrt{55} \\ \tan \theta &= \frac{\sqrt{55}}{3}\end{aligned}$$

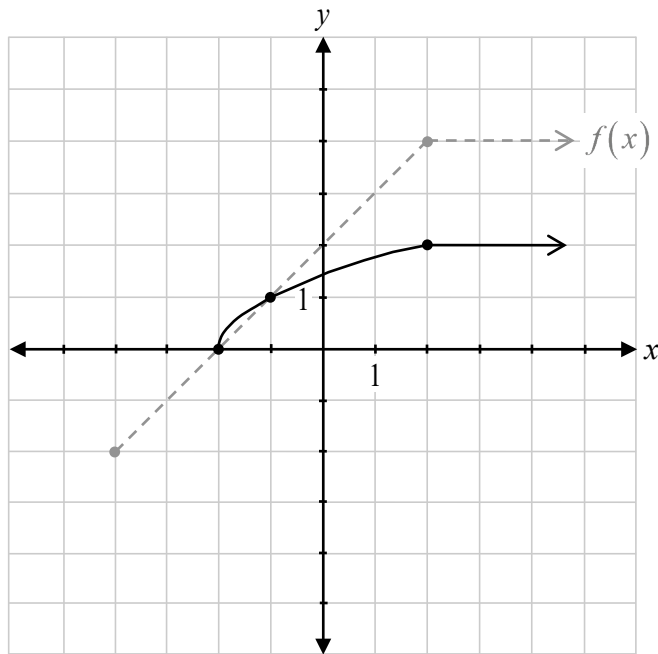
Describe her error.

Solution

Suzanne did not consider that the value of $\tan \theta$ is negative in quadrant II.

1 mark

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.

Solution

1 mark for restricting domain
 $\frac{1}{2}$ mark for shape between invariant points
 $\frac{1}{2}$ mark for shape to the right of invariant points

2 marks

The point $P(\theta) = (0, -1)$ lies on the unit circle. State the angle θ , over the interval $[2\pi, 4\pi]$.

Solution

$$\theta = \frac{7\pi}{2}$$

1 mark

Describe how the transformations of $f(x)$ on the graphs of $g(x) = f(3x - 6)$ and $h(x) = f(3(x - 6))$ are different.

Solution

The graph of $g(x)$ is a horizontal translation of $f(x)$ two units to the right and the graph of $h(x)$ is a horizontal translation of $f(x)$ six units to the right.

1 mark

a) Solve.

$$\sqrt{2x+5}-3=0$$

b) Describe how the solution in a) relates to the graph of $y = \sqrt{2x+5} - 3$.

Solution

a) $(\sqrt{2x+5})^2 = 3^2$

$$2x+5=9$$

$$2x=4$$

$$x=2$$

1 mark

b) The solution is the x -intercept of the graph.

1 mark

Determine all of the zeros of the function $p(x) = x^3 - 2x^2 - 9x + 18$.

Solution

$$\begin{aligned} p(2) &= 2^3 - 2(2)^2 - 9(2) + 18 \\ &= 0 \end{aligned}$$

1 mark for identifying one possible zero of $p(x)$

$\therefore (x-2)$ is a factor of $p(x)$

$$\begin{array}{r|rrrr} 2 & 1 & -2 & -9 & 18 \\ & & 2 & 0 & -18 \\ \hline & 1 & 0 & -9 & 0 \end{array}$$

1 mark for synthetic division (or equivalent strategy)

$$(x-2)(x^2-9) = 0$$

$\frac{1}{2}$ mark for consistent factors

$$(x-2)(x-3)(x+3) = 0$$

$$x = 2, x = 3, x = -3$$

$\frac{1}{2}$ mark for consistent zeros

3 marks

Given that the point $\left(\frac{\sqrt{23}}{6}, y\right)$ is on the unit circle, determine the exact value(s) of y .

Solution

$$\left(\frac{\sqrt{23}}{6}\right)^2 + y^2 = 1 \quad \frac{1}{2} \text{ mark for substitution}$$

$$\frac{23}{36} + y^2 = 1$$

$$y^2 = \frac{36}{36} - \frac{23}{36}$$

$$y = \pm \frac{\sqrt{13}}{6} \quad \frac{1}{2} \text{ mark for exact values of } y$$

1 mark

State one zero of the function $y = \tan x$.

Solution

Any solution where $x = \pi k$, $k \in \mathbb{Z}$, is acceptable.

1 mark

Question 1

P1

Youyi wants to buy a new cell phone from a company that offers 33 different types of cell phones. Each cell phone is available in 5 different colours. Determine the total number of available options.

Solution

$$\underline{33} \bullet \underline{5} = 165 \text{ options}$$

1 mark

Question 2

T1

Determine the central angle subtended by an arc length of 30 cm on a circle with a diameter of 20.6 cm. Express your answer in degrees.

Solution

$$s = \theta r$$

1 mark for substitution

$$30 = \theta(10.3)$$

$$\theta = 2.912\ 621\dots$$

$$\theta = (2.912\ 621)\left(\frac{180^\circ}{\pi}\right)$$

1 mark for conversion

$$\theta = 166.880\ 911\dots^\circ$$

$$\theta = 166.881^\circ$$

2 marks

Question 3

P3

A choir has 14 sopranos. Karin and Ellen are two of them. Determine the number of possible ensembles of 3 sopranos that could be formed if either Karin or Ellen, or both, must be included.

Solution

Method 1

$$\text{Case 1: } {}_1C_1 \cdot {}_{12}C_2 = 66$$

1 mark for Case 1 and Case 2

$$\text{Case 2: } {}_1C_1 \cdot {}_{12}C_2 = 66$$

$$\text{Case 3: } {}_2C_2 \cdot {}_{12}C_1 = 12$$

1 mark for Case 3

$$66 + 66 + 12 = 144$$

1 mark for addition of cases

3 marks

Method 2

$${}_{14}C_3 - {}_{12}C_3 = 144$$

1 mark for ${}_{14}C_3$

1 mark for ${}_{12}C_3$

1 mark for subtraction of cases

3 marks

Question 4

R10

Loïc invests \$5000 at an annual interest rate of 6.75% compounded semi-annually. The value of his investment grows according to the following formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

where A is the value of the investment after t years

P is the amount of the initial deposit

r is the annual interest rate (as a decimal)

n is the number of compounding periods per year

t is the time in years

Determine, algebraically, the length of time it will take Loïc's investment to grow to \$9500.

Solution

$$9500 = 5000 \left(1 + \frac{0.0675}{2} \right)^{2t} \quad \frac{1}{2} \text{ mark for substitution}$$

$$1.9 = (1.03375)^{2t}$$

$$\log 1.9 = \log(1.03375)$$

1 mark for applying logarithms

$$\log 1.9 = 2t \log 1.03375$$

1 mark for power law (or appropriate logarithm laws)

$$t = \frac{\log 1.9}{2 \log 1.03375}$$

$$t = 9.668\ 522\dots$$

$\frac{1}{2}$ mark for evaluating a quotient of logarithms

$$t = 9.669 \text{ years}$$

$$\therefore t = 10 \text{ years}$$

3 marks

Note:

Accept $t = 9.669$ years as a final answer (or consistent value of t).

Question 5

T5

Solve, algebraically, over the interval $[0, 2\pi]$.

$$\csc^2 \theta - \csc \theta - 6 = 0$$

Solution

Method 1

$$(\csc \theta - 3)(\csc \theta + 2) = 0$$

$$\csc \theta = 3$$

$$\csc \theta = -2$$

1 mark for solving for $\csc \theta$

$$\sin \theta = \frac{1}{3}$$

$$\sin \theta = -\frac{1}{2}$$

1 mark for reciprocal

$$\theta_r = 0.339\ 836\dots$$

$$\theta = 0.340; 2.802$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

2 marks for solving for θ ($\frac{1}{2}$ mark for each value)

4 marks

Method 2

$$\frac{1}{\sin^2 \theta} - \frac{1}{\sin \theta} - 6 = 0$$

1 mark for reciprocal

$$1 - \sin \theta - 6 \sin^2 \theta = 0$$

$$6 \sin^2 \theta + \sin \theta - 1 = 0$$

$$(3 \sin \theta - 1)(2 \sin \theta + 1) = 0$$

$$\sin \theta = \frac{1}{3}$$

$$\sin \theta = -\frac{1}{2}$$

1 mark for solving for $\sin \theta$

$$\theta_r = 0.339\ 836\dots$$

$$\theta = 0.340; 2.802$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

2 marks for solving for θ ($\frac{1}{2}$ mark for each value)

4 marks

Question 6

P1

Determine how many 4-digit numbers can be made using the digits 0, 1, 2, 3, 4, and 5, if the third digit must be 3, and repetition is not allowed.

Solution

$$\frac{4}{\text{not 0 or 3}} \cdot \frac{4}{\quad} \cdot \frac{1}{3} \cdot \frac{3}{\quad} = 48$$

½ mark for restriction of first digit

½ mark for restriction of third digit

1 mark for fundamental counting principle

2 marks

Question 7

R3, R5

Describe the transformations required to obtain the graph of the function $y = -4f\left(\frac{1}{3}x\right)$ from the graph of $y = f(x)$.

Solution

- Reflection over the x -axis
- Vertical stretch by a factor of 4
- Horizontal stretch by a factor of 3

1 mark for vertical reflection

1 mark for vertical stretch

1 mark for horizontal stretch

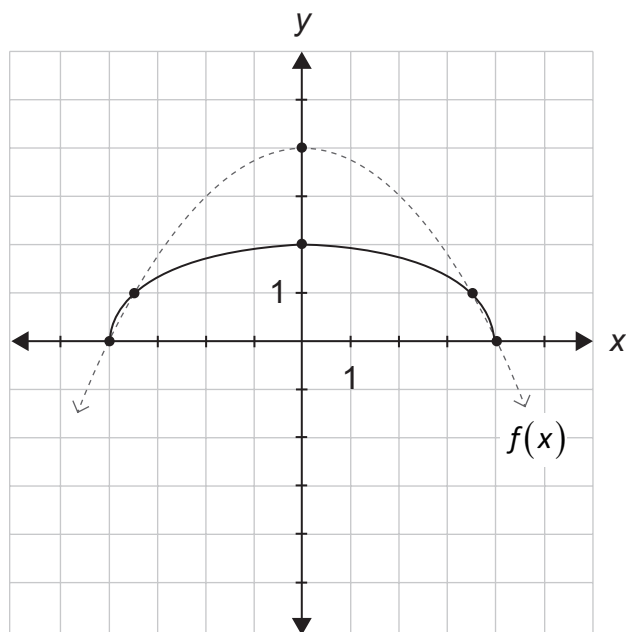
3 marks

Question 8

R13

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.

Solution



1 mark for restricting domain

$\frac{1}{2}$ mark for shape between invariant points,
 $\{0 \leq y \leq 1\}$

$\frac{1}{2}$ mark for shape above invariant points,
 $\{1 \leq y \leq 2\}$

2 marks

Question 9

R11

Describe how to determine if $(x - a)$ is a factor of the polynomial function, $p(x)$.

Solution

$(x - a)$ is a factor of $p(x)$, if and only if $p(a) = 0$.

1 mark

Question 10

R8

Express $\frac{1}{2}\log x - 2\log y + \log z$ as a single logarithm.

Solution

$$\log\left(\frac{\sqrt{x} \cdot z}{y^2}\right)$$

1 mark for product law

1 mark for quotient law

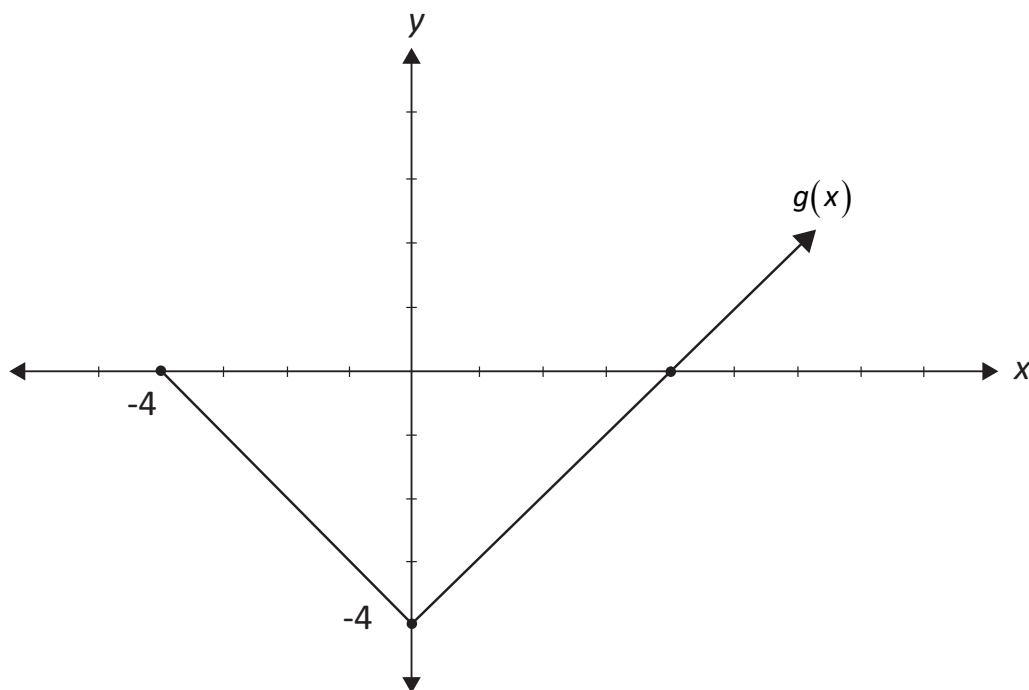
1 mark for power law ($\frac{1}{2}$ mark for each)

3 marks

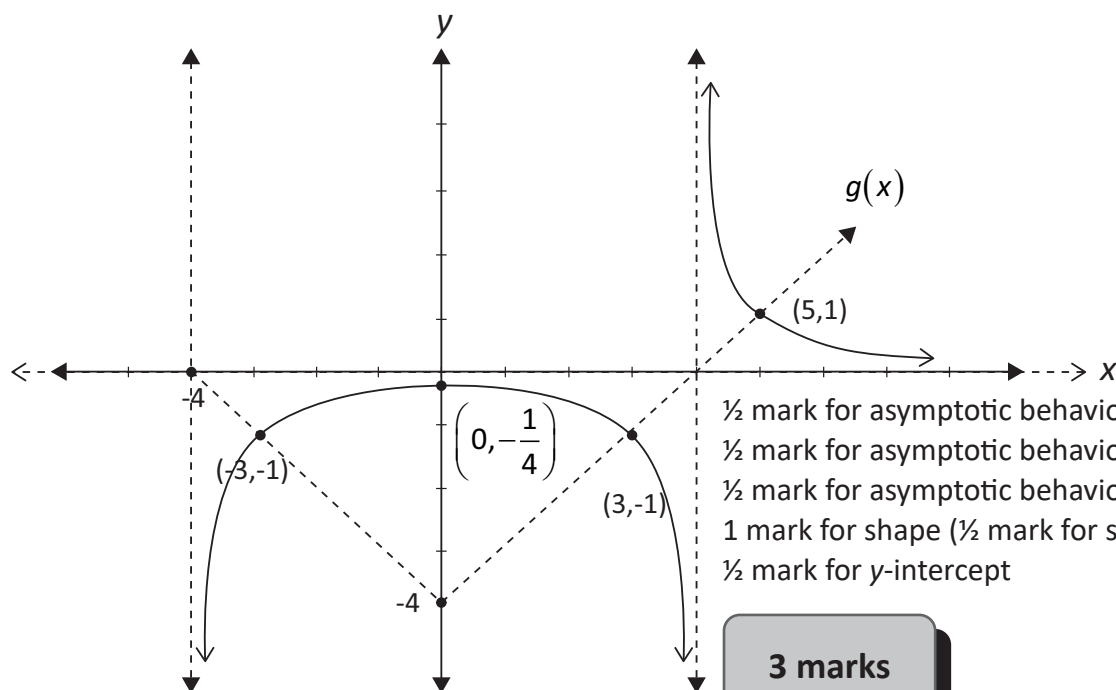
Question 11

R1

Given the graph of $g(x)$, sketch the graph of $f(x) = \frac{1}{g(x)}$.



Solution



$\frac{1}{2}$ mark for asymptotic behaviour approaching $x = -4$
 $\frac{1}{2}$ mark for asymptotic behaviour approaching $x = 4$
 $\frac{1}{2}$ mark for asymptotic behaviour approaching $y = 0$
 1 mark for shape ($\frac{1}{2}$ mark for shape in each section)
 $\frac{1}{2}$ mark for y-intercept

3 marks

Question 12

T6

Given the identity $\frac{1 - \tan^2 \theta}{\sec^2 \theta} = \cos 2\theta$,

- state a non-permissible value of θ .
- prove the identity for all permissible values of θ .

Solution

a) Any value where $\theta = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$ or $\theta = 90^\circ + 180^\circ k, k \in \mathbb{Z}$, is acceptable.

1 mark

b)

Method 1

Left-Hand Side	Right-Hand Side
$ \begin{array}{r} 1 - \frac{\sin^2 \theta}{\cos^2 \theta} \\ \hline \frac{1}{\cos^2 \theta} \\ \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta} \\ \hline \frac{1}{\cos^2 \theta} \\ \frac{\cos 2\theta}{\cancel{\cos^2 \theta}} \\ \hline \frac{1}{\cancel{\cos^2 \theta}} \\ \cos 2\theta \end{array} $	$\cos 2\theta$

1 mark for correct substitution of identities

1 mark for algebraic strategies

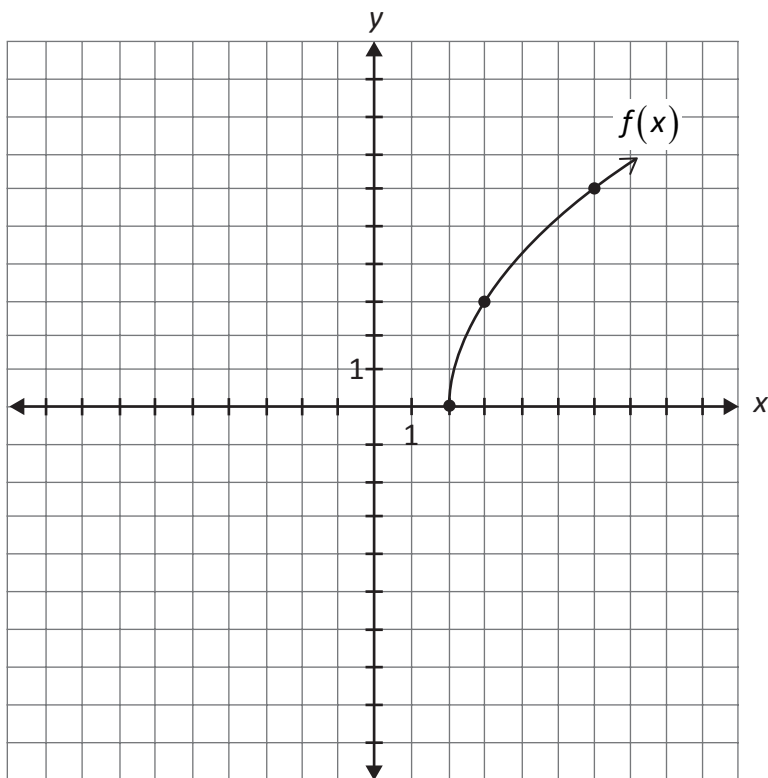
1 mark for logical process to prove the identity

3 marks

Question 13

R13

Determine an equation of the radical function represented by the graph of $f(x)$.



Solution

$$f(x) = \underline{3\sqrt{x-2}}$$

1 mark for vertical stretch

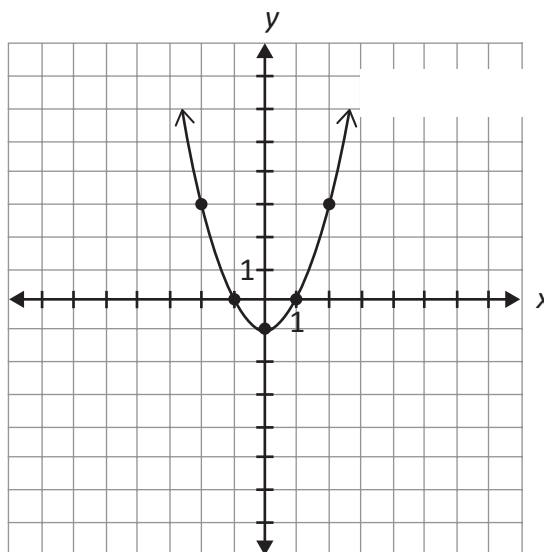
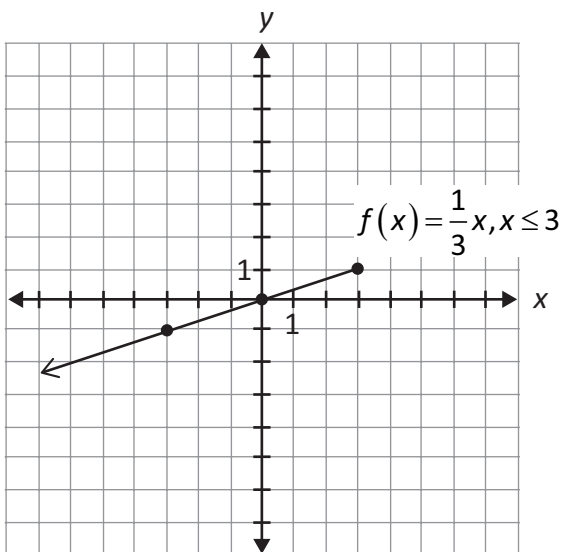
1 mark for horizontal translation

2 marks

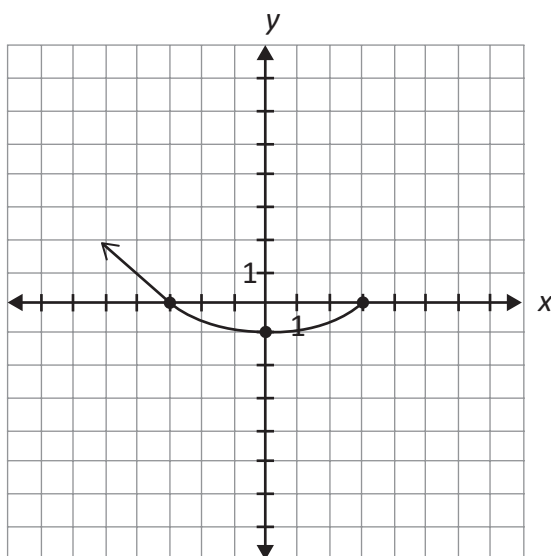
Question 14

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $g(f(x))$.



Solution



1 mark for $g(f(x))$

1 mark for restricted domain

2 marks

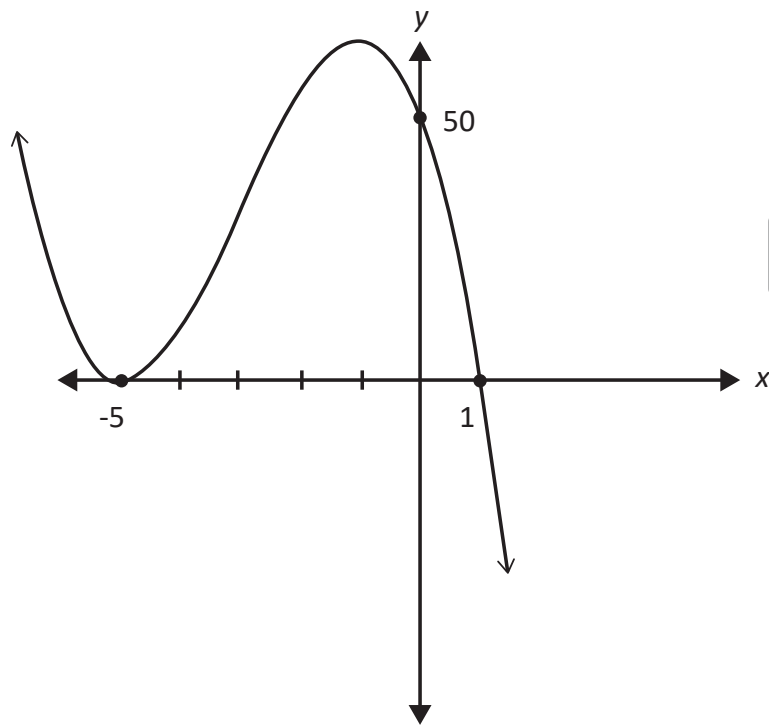
Question 15

R12

Sketch a graph of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 3
- $p(x)$ has a zero at -5 with a multiplicity of 2
- $p(x)$ has a zero at 1
- $p(x)$ has a leading coefficient of -2

Solution



$\frac{1}{2}$ mark for x-intercepts
1 mark for multiplicity of 2 at $x = -5$
 $\frac{1}{2}$ mark for y-intercept
 $\frac{1}{2}$ mark for end behaviour
 $\frac{1}{2}$ mark for shape of cubic function

3 marks

Question 16

R14

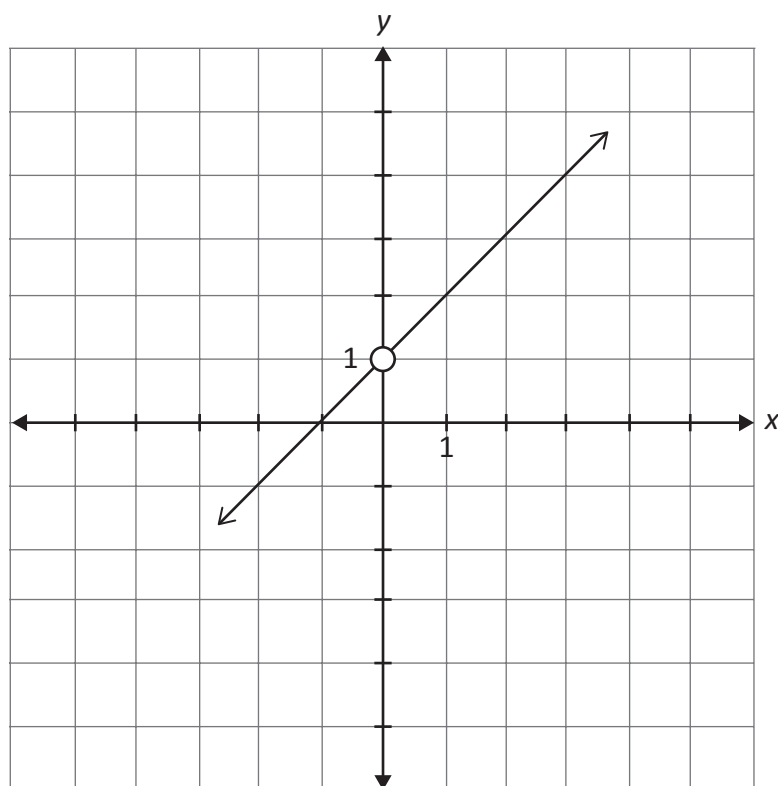
Sketch the graph of $f(x) = \frac{x^2 + x}{x}$.

Solution

$$f(x) = \frac{\cancel{x}(x+1)}{\cancel{x}}$$

$$f(x) = x + 1, x \neq 0$$

$\therefore$ there is a point of discontinuity (hole) at $(0,1)$.



1 mark for point of discontinuity (hole) at $x = 0$

1 mark for shape of $y = x + 1$

2 marks

Question 17

R2

Identify the function that has a different range than $f(x) = x^2$.

a. $y = 4f(x)$

b. $y = f(x) + 4$

c. $y = f(x + 4)$

d. $y = f(-x)$

Question 18

T3

Identify the exact value of $\sin\left(-\frac{7\pi}{3}\right)$.

a. $-\frac{\sqrt{3}}{2}$

b. $-\frac{1}{2}$

c. $\frac{1}{2}$

d. $\frac{\sqrt{3}}{2}$

Question 19

R7

Identify the value of a in the equation, $\log_a 3 = \frac{1}{4}$.

a. $\frac{1}{64}$

b. $3^{\frac{1}{4}}$

c. 12

d. 81

Question 20**T3**

Identify a value of x where the graphs of $y = \cos x$ and $y = \sin x$ intersect.

a. $-\frac{3\pi}{4}$

b. $-\frac{\pi}{4}$

c. 0

d. $\frac{3\pi}{4}$

Question 21**P4**

Identify the number of negative terms in the binomial expansion of $(2x - y)^6$.

a. 0

b. 3

c. 4

d. 6

Question 22**T5**

Identify the general solution of the equation, $\sin x = 1$.

a. $x = k\pi, k \in \mathbb{Z}$

b. $x = 2k\pi, k \in \mathbb{Z}$

c. $x = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

d. $x = \frac{\pi}{2} + 2k\pi, k \in \mathbb{Z}$

Question 23**R12**

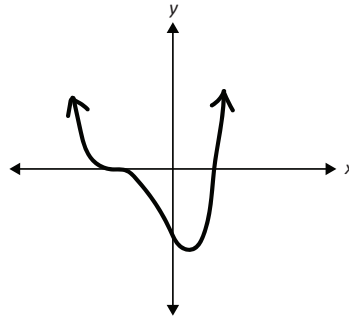
Identify a possible degree of the polynomial function represented below.

a. 2

b. 3

c. 4

d. 5

**Question 24****T6**

Identify the expression that is equivalent to $\cos x \cot x$.

a. $\frac{1 - \sin x}{\sin x}$ b. $\frac{\sin^2 x - 1}{\sin x}$ c. $\sin x$ d. $\frac{1 - \sin^2 x}{\sin x}$ **Question 25****R1**

Given that the point $(-7, -5)$ is on the graph of $y = g(x)$, identify the corresponding point on the graph of $y = |g(x)|$.

a. $(7, 5)$ b. $(-7, 5)$ c. $(7, -5)$ d. $(-5, -7)$

Question 26

R11

Given that $x = 1$ is one of the zeros of $p(x) = x^3 + x^2 - 10x + 8$, express $p(x)$ in completely factored form.

Solution

$$p(x) = x^3 + x^2 - 10x + 8$$

1		1	1	-10	8
		↓	1	2	-8
		1	2	-8	0

1 mark for synthetic division (or any equivalent strategy)

$$p(x) = (x - 1)(x^2 + 2x - 8)$$

$$p(x) = (x - 1)(x + 4)(x - 2)$$

1 mark for consistent factors

2 marks

Question 27

T3

Determine the exact value of $\frac{\csc^2 \theta - 1}{1 - \cos^2 \theta}$ if $\theta = \frac{5\pi}{6}$.

Solution

Method 1

$$\frac{\csc^2\left(\frac{5\pi}{6}\right) - 1}{1 - \cos^2\left(\frac{5\pi}{6}\right)}$$

$$\frac{(2)^2 - 1}{1 - \left(-\frac{\sqrt{3}}{2}\right)^2}$$

$$\frac{4 - 1}{1 - \frac{3}{4}}$$

$$\frac{3}{\frac{1}{4}}$$

$$12$$

1 mark for $\csc\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

1 mark for $\cos\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

2 marks

Method 2

$$\frac{\cot^2 \theta}{\sin^2 \theta}$$

$$\frac{\cot^2\left(\frac{5\pi}{6}\right)}{\sin^2\left(\frac{5\pi}{6}\right)}$$

$$\frac{(-\sqrt{3})^2}{\left(\frac{1}{2}\right)^2}$$

$$\frac{3}{\frac{1}{4}}$$

$$12$$

1 mark for $\cot\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

1 mark for $\sin\left(\frac{5\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

2 marks

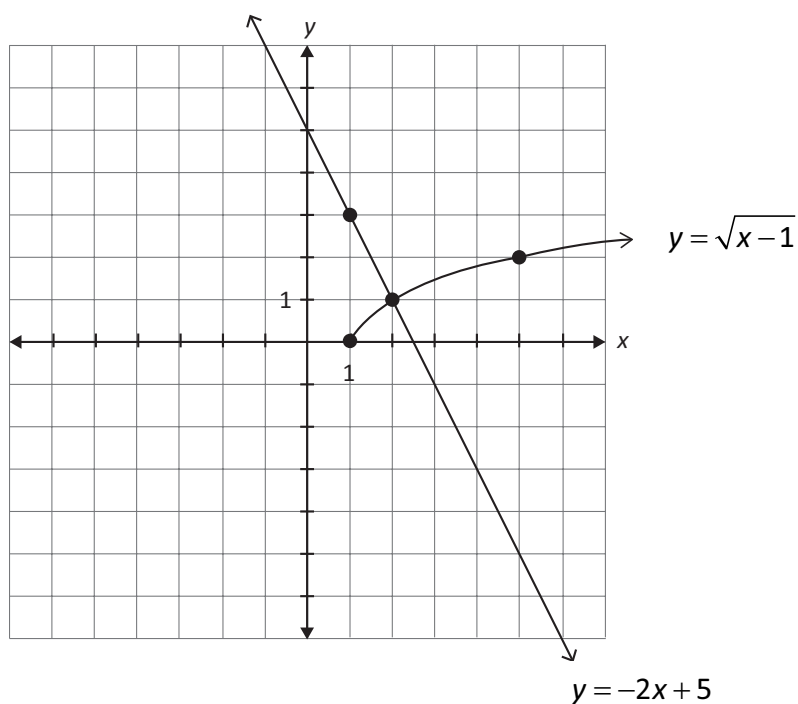
Question 28

R13

Solve, graphically.

$$-2x + 5 = \sqrt{x - 1}$$

Solution



$$\therefore x = 2$$

1 mark for shape of a radical function

1 mark for horizontal translation

1 mark for x-value consistent with intersection point of the two graphs

3 marks

Question 29

R8

If $\log_a 2 = p$ and $\log_a 5 = q$, express $\log_a 40$ in terms of p and q .

Solution

$$\begin{aligned}\log_a 40 &= \log_a (2^3 \cdot 5) \\ &= 3\log_a 2 + \log_a 5 \\ &= 3p + q\end{aligned}$$

1 mark for power law

1 mark for product law

2 marks

Question 30

R6

Describe how to sketch the graph of $y=f^{-1}(x)$, given the graph of a linear function, $y=f(x)$.

Solution

The graph of $y=f^{-1}(x)$ is obtained by reflecting the graph of $y=f(x)$ over the line $y=x$.

or

To sketch the inverse, $y=f^{-1}(x)$, transform all points on the graph of $y=f(x)$ from (x,y) to (y,x) .

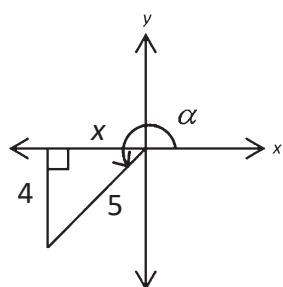
1 mark

Question 31

T6

Given that $\sin \alpha = -\frac{4}{5}$ where α is in quadrant III, and $\cos \beta = \frac{12}{13}$ where β is in quadrant IV, determine the exact value of $\cos(\alpha - \beta)$.

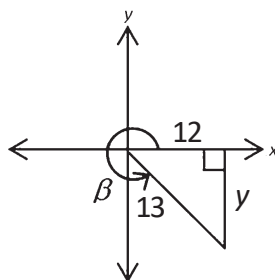
Solution



$$x^2 + 4^2 = 5^2$$

$$x^2 = 9$$

$$x = \pm 3$$



$$12^2 + y^2 = 13^2$$

$$y^2 = 25$$

$$y = \pm 5$$

½ mark for value of x

½ mark for value of y

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$= \left(-\frac{3}{5}\right)\left(\frac{12}{13}\right) + \left(-\frac{4}{5}\right)\left(-\frac{5}{13}\right)$$

$$= \frac{-36 + 20}{65}$$

$$= -\frac{16}{65}$$

½ mark for consistent value of $\cos \alpha$

½ mark for consistent value of $\sin \beta$

1 mark for substitution into correct identity

3 marks

Note:

Accept any of the following values for x : $x = \pm 3$; $x = -3$; or $x = 3$.

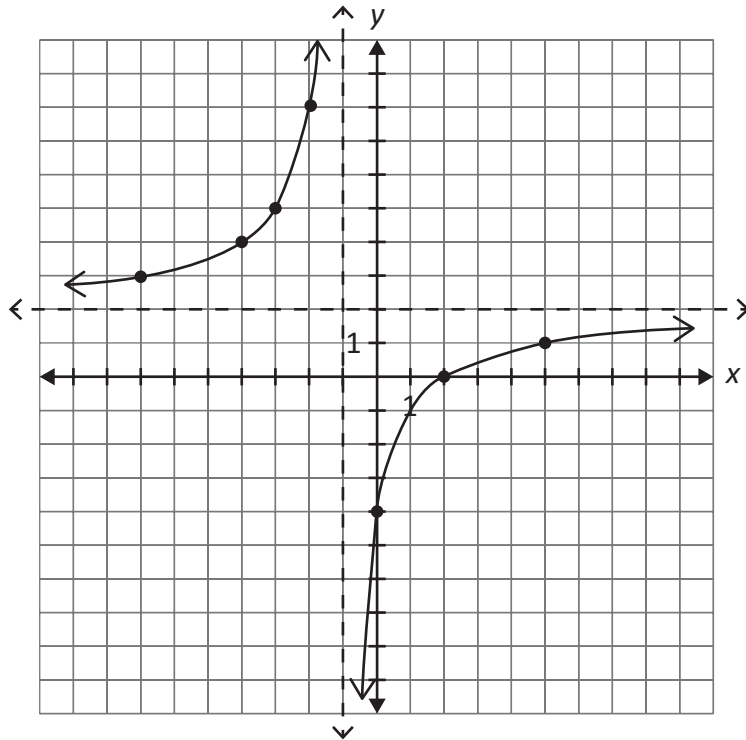
Accept any of the following values for y : $y = \pm 5$; $y = -5$; or $y = 5$.

Question 32

R14

Sketch the graph of $f(x) = \frac{2x-4}{x+1}$.

Solution



1 mark for asymptotic behaviour approaching $x = -1$

1 mark for asymptotic behaviour approaching $y = 2$

1 mark for shape of rational function

($\frac{1}{2}$ mark for graph left of $x = -1$, $\frac{1}{2}$ mark for graph right of $x = -1$)

3 marks

Question 33

T3, T4

Due to the tidal variations, the depth of the water in a bay is represented by the function,

$$d(t) = -2\cos\left[\frac{\pi}{6}(t+4)\right] + 16$$

where d is the depth of the water, in metres and
 t is the time, in hours.

- a) Determine the depth of the water when $t = 2$.
- b) State the minimum depth of the water.
- c) Determine the length of time between two consecutive low tides (minimum depths).

Solution

$$\begin{aligned}\text{a) } d(2) &= -2\cos\left[\frac{\pi}{6}(2+4)\right] + 16 \\ &= -2\cos\pi + 16 \\ &= -2(-1) + 16 \\ &= 18 \text{ metres}\end{aligned}$$

1 mark for exact value

1 mark

b) 14 metres

1 mark

$$\begin{aligned}\text{c) period} &= \frac{2\pi}{\frac{\pi}{6}} \\ &= 12 \text{ hours}\end{aligned}$$

1 mark for period

1 mark

Question 34

T3

Evaluate.

$$\sin^2\left(\frac{5\pi}{4}\right)\sec\left(\frac{13\pi}{3}\right)-\cot\left(\frac{7\pi}{4}\right)$$

Solution

$$\left(-\frac{\sqrt{2}}{2}\right)^2\left(\frac{2}{1}\right)-(-1)$$

$$\frac{1+1}{2}$$

1 mark for $\sin\left(\frac{5\pi}{4}\right)$ (½ mark for value; ½ mark for quadrant)

1 mark for $\sec\left(\frac{13\pi}{3}\right)$ (½ mark for value; ½ mark for quadrant)

1 mark for $\cot\left(\frac{7\pi}{4}\right)$ (½ mark for value; ½ mark for quadrant)

3 marks

Question 35

R14

a) State the non-permissible value(s) of the function, $f(x) = \frac{x-3}{x^2-9}$.

b) Describe what each non-permissible value represents on the graph of $f(x) = \frac{x-3}{x^2-9}$.

Solution

a) $x = 3, -3$

1 mark for non-permissible values ($\frac{1}{2}$ mark for each)

1 mark

b) At $x = 3$, there is a point of discontinuity (hole) on the graph of $f(x)$. At $x = -3$, there is a vertical asymptote on the graph of $f(x)$.

$\frac{1}{2}$ mark for point of discontinuity (hole) at $x = 3$

$\frac{1}{2}$ mark for vertical asymptote at $x = -3$

1 mark

Question 36

R7

Determine which of the given values is greater. Justify your answer.

$$\log_3 38 \text{ or } \log_4 50$$

Solution

$$3^3 = 27$$

$$3^4 = 81 \quad \log_3 38 > 3$$

$$4^2 = 16$$

$$4^3 = 64 \quad \log_4 50 < 3$$

$\therefore \log_3 38$ is greater

1 mark for justification

1 mark

Question 37

T2

Determine if the point $\left(\frac{3}{4}, \frac{\sqrt{5}}{4}\right)$ is on the unit circle. Justify your answer.

Solution

$$x^2 + y^2 = 1$$

$$\text{LHS} = \left(\frac{3}{4}\right)^2 + \left(\frac{\sqrt{5}}{4}\right)^2$$

$$= \frac{9}{16} + \frac{5}{16}$$

$$= \frac{14}{16}$$

$$\frac{14}{16} \neq 1$$

$\therefore$ The point is not on the unit circle.

1 mark for justification

1 mark

Question 38

P4

Determine which term contains x^5 in the binomial expansion of $\left(2x + \frac{3}{x}\right)^{11}$.

Solution

Method 1

$$x^{11}, (x^{10})\left(\frac{1}{x}\right), (x^9)\left(\frac{1}{x^2}\right), \dots$$

1 mark for determining a pattern

$$x^{11}, x^9, x^7, \dots$$

$\therefore$ The fourth term contains x^5 .

1 mark for fourth term
(or a term consistent with the pattern)

2 marks

Method 2

$$(x)^{11-k} (x^{-1})^k = x^5$$

$\frac{1}{2}$ mark for substitution

$$x^{11-2k} = x^5$$

$$11 - 2k = 5$$

$$6 = 2k$$

$$k = 3$$

$\frac{1}{2}$ mark for solving for k

$\therefore$ The fourth term contains x^5 .

1 mark for fourth term
(or a term consistent with the value of k)

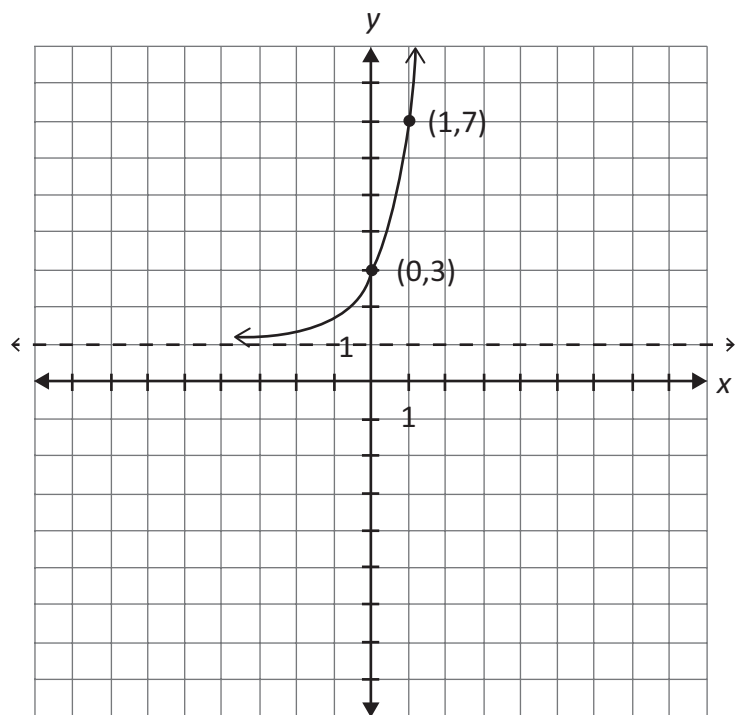
2 marks

Question 39

R9

Sketch the graph of $y = 2(3^x) + 1$.

Solution



1 mark for shape of an increasing exponential function

1 mark for vertical stretch

1 mark for vertical translation

3 marks

Question 40

R8, R10

Solve, algebraically.

$$\log(2x + 1) = \log(x - 2) + 1$$

Solution

Method 1

$$\log(2x + 1) - \log(x - 2) = 1$$

$$\log\left(\frac{2x + 1}{x - 2}\right) = 1$$

$$10^1 = \frac{2x + 1}{x - 2}$$

$$10x - 20 = 2x + 1$$

$$8x = 21$$

$$x = \frac{21}{8}$$

1 mark for quotient law

1 mark for exponential form

2 marks

Method 2

$$\log(2x + 1) = \log(x - 2) + \log 10$$

$$\log(2x + 1) = \log(10(x - 2))$$

$$2x + 1 = 10(x - 2)$$

$$2x + 1 = 10x - 20$$

$$21 = 8x$$

$$x = \frac{21}{8}$$

½ mark for logarithmic form

1 mark for product law

½ mark for equating arguments

2 marks

Question 41

R1

Given the functions, $f(x) = x + 2$ and $g(x) = x - 2$,

a) state the equation of $h(x) = f(x) \cdot g(x)$.

b) sketch the graph of $h(x) = f(x) \cdot g(x)$.

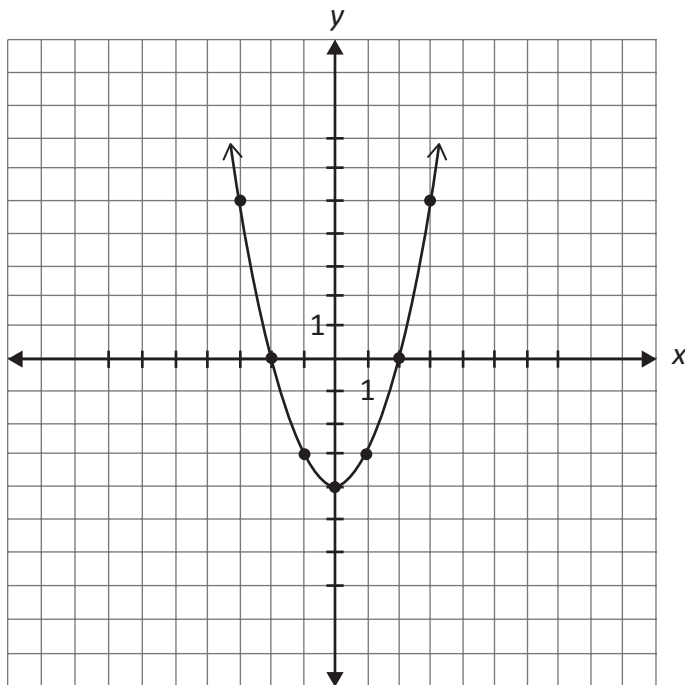
Solution

a) $h(x) = (x + 2)(x - 2)$

or

$$h(x) = x^2 - 4$$

b)



1 mark

1 mark for graph consistent with a)

1 mark

Question 42

R11

Given $p(x) = -2x^3 + 3x + 1$, determine the remainder when $p(x)$ is divided by $(x - 2)$.

Solution

Method 1

$$p(2) = -2(2)^3 + 3(2) + 1$$

$$p(2) = -2(8) + 6 + 1$$

$$p(2) = -9$$

1 mark for remainder theorem

1 mark

Method 2

2		-2	0	3	1
		↓	-4	-8	-10
		-2	-4	-5	-9

1 mark for synthetic division (or any equivalent strategy)

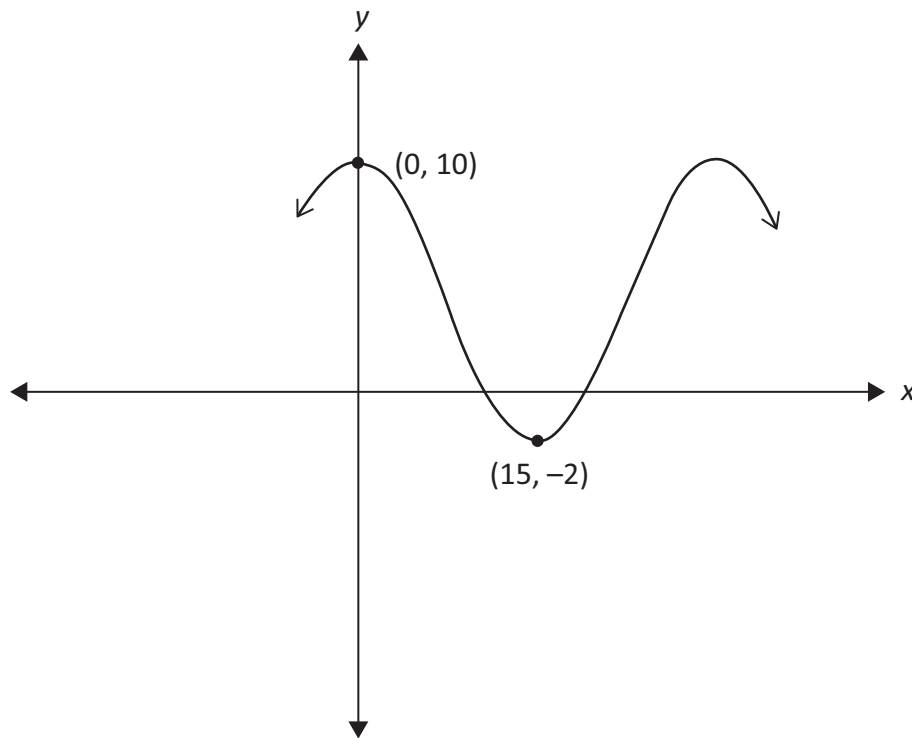
1 mark

∴ The remainder is -9

Question 43

T4

Determine the equation of the sinusoidal function in the form $y = A\cos(Bx) + D$.



Solution

$$y = 6\cos\left(\frac{\pi}{15}x\right) + 4$$

1 mark for the value of A

1 mark for the value of B ($\frac{1}{2}$ mark for period; $\frac{1}{2}$ mark for consistent value of B)

1 mark for the value of D

3 marks

Question 44

P2

Determine the total number of arrangements of the letters in the word TORONTO.

State your answer as a whole number.

Solution

$$\frac{7!}{2!3!}$$
$$\frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot \cancel{3!}}{2! \cancel{3!}}$$

$$\frac{7 \cdot 6 \cdot 5 \cdot 4}{2}$$

$$7 \cdot 6 \cdot 5 \cdot 2$$

$$420$$

½ mark for 7!

½ mark for division by 2!3!

½ mark for factorial expansion

½ mark for consistent number of arrangements

2 marks

Question 1

T1

Liam went for a ride on a Ferris wheel with a radius of 5 metres. Determine the distance he travelled if the Ferris wheel stopped after rotating 1260° .

Solution

$$\theta = 1260^\circ \cdot \frac{\pi}{180^\circ} \quad 1 \text{ mark for conversion}$$

$$\theta = 7\pi$$

or

$$\theta = 21.991148\dots$$

$$s = \theta r$$

$$s = 7\pi(5) \quad 1 \text{ mark for substitution}$$

$$s = 35\pi \text{ metres}$$

or

$$s = 109.956 \text{ metres}$$

2 marks

Determine and simplify the term that contains x^{12} in the binomial expansion of $(2x^4 - 3)^7$.

Solution

Method 1

$$(x^4)^7, (x^4)^6, (x^4)^5, \dots$$

$$x^{28}, x^{24}, x^{20}, \dots$$

1 mark for determining a pattern

$$t_5 = {}_7C_4 (2x^4)^3 (-3)^4$$

$$= 22\,680x^{12}$$

2 marks (1 mark for ${}_7C_4$; ½ mark for each consistent factor)

1 mark for simplification (½ mark for coefficient, ½ mark for exponent)

4 marks

Method 2

$$x^{12} = (x^4)^{7-k} (x^0)^k$$

½ mark for substitution

$$12 = 28 - 4k$$

$$-16 = -4k$$

$$4 = k$$

½ mark for solving for k

$$t_5 = {}_7C_4 (2x^4)^3 (-3)^4$$

$$= 22\,680x^{12}$$

2 marks (1 mark for ${}_7C_4$; ½ mark for each consistent factor)

1 mark for simplification (½ mark for coefficient, ½ mark for exponent)

4 marks

Question 3

R10

The normal healing of wounds can be modeled by an exponential function. The area of the wound decreases according to the formula,

$$A = A_0 e^{-0.35t}$$

where: A is the area of the wound in cm^2 , after t days,
 A_0 is the initial area of the wound in cm^2 , and
 t is the time, in days, after healing begins.

If a wound has an initial area of 25 cm^2 , determine:

- a) the area of the wound after 3 days.
- b) the amount of time it will take for the area of the wound to decrease to 4 cm^2 .

Solution

a) $A = 25e^{-0.35(3)}$

$$A = 8.748 \text{ cm}^2$$

1 mark

b) $4 = 25e^{-0.35t}$

$$0.16 = e^{-0.35t}$$

$$\ln 0.16 = \ln e^{-0.35t}$$

1 mark for applying logarithms

$$\ln 0.16 = -0.35t \ln e$$

½ mark for power law

$$\frac{\ln 0.16}{-0.35} = t$$

$$t = 5.235947...$$

$$t = 5.236 \text{ days}$$

½ mark for the value of t

2 marks

Question 4

P3

During a hockey practice, 4 players are wearing a green jersey and 5 players are wearing a red jersey. Determine the number of ways three players can be selected if at least two of the players must be wearing a red jersey.

Solution

Case 1: ${}_5C_2 \cdot {}_4C_1 = 40$

1 mark for Case 1

Case 2: ${}_5C_3 \cdot {}_4C_0 = 10$

1 mark for Case 2

$40 + 10 = 50$ ways

1 mark for addition of cases

3 marks

Note:

${}_4C_0$ does not need to be shown.

Question 5

T5

Solve, algebraically, over the interval $[0, 2\pi]$.

$$2\sin^2 x + 5\cos x + 1 = 0$$

Solution

$$2(1 - \cos^2 x) + 5\cos x + 1 = 0$$

1 mark for substitution of an appropriate identity

$$2 - 2\cos^2 x + 5\cos x + 1 = 0$$

$$-2\cos^2 x + 5\cos x + 3 = 0$$

$$2\cos^2 x - 5\cos x - 3 = 0$$

$$(2\cos x + 1)(\cos x - 3) = 0$$

$$\cos x = -\frac{1}{2} \quad \cos x = 3$$

1 mark for solving for $\cos x$

$$x = \frac{2\pi}{3}, \frac{4\pi}{3} \quad \text{no solution}$$

2 marks for solving for x ($\frac{1}{2}$ mark for each value; 1 mark for indicating no solution)

4 marks

Question 6

R3

The function, $y = f(x)$, has a domain of $[-8, 4]$. The domain of the graph of $y = f(bx)$ is $[-4, 2]$.

State the value of b .

Solution

$$b = 2$$

1 mark

Question 7**R14**

Describe the difference between the graph of $f(x) = \frac{(x+3)(x-4)}{(x+3)}$ and the graph of $g(x) = x - 4$.

Solution

The graph of $f(x)$ has a point of discontinuity at $x = -3$, where as the graph of $g(x)$ does not.

1 mark

Question 8

T2

Determine if the point $\left(-\frac{3}{5}, \frac{2}{\sqrt{3}}\right)$ is on the unit circle. Justify your answer.

Solution

Method 1

$$x^2 + y^2 = 1$$

$$\begin{aligned}\text{Left-Hand Side} &= \left(-\frac{3}{5}\right)^2 + \left(\frac{2}{\sqrt{3}}\right)^2 \\ &= \frac{9}{25} + \frac{4}{3} \\ &= \frac{27}{75} + \frac{100}{75}\end{aligned}$$

$$\frac{127}{75} \neq 1$$

Therefore the point $\left(-\frac{3}{5}, \frac{2}{\sqrt{3}}\right)$ is not on the unit circle.

1 mark for justification

1 mark

Method 2

Since $\frac{2}{\sqrt{3}} > 1$, the point is not on the unit circle.

1 mark for justification

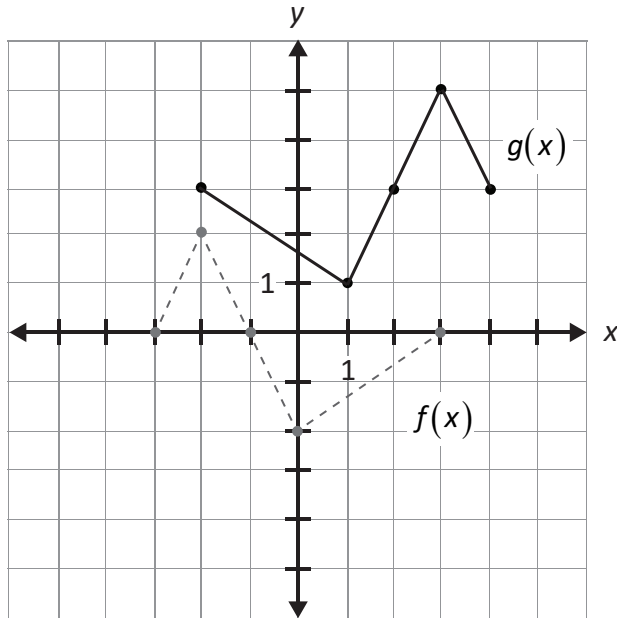
1 mark

Question 9

R2, R5

Given the graph of $f(x)$, sketch the graph of $g(x) = f(-(x-1)) + 3$.

Solution



1 mark for horizontal reflection
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Question 10

R11

Bonnie was asked to determine the factors of $p(x) = -3x^4 + 2x^2 - 4x + 5$.

Describe the error she made in her set up of synthetic division.

$$\begin{array}{r|rrrrr} 1 & & -3 & 2 & -4 & 5 \end{array}$$

Solution

Bonnie did not write the coefficient of 0 for the missing term, x^3 .

1 mark

Question 11

R1

Given the functions, $f(x) = \frac{1}{x-1}$ and $g(x) = x - 3$,

a) state the equation of $f(g(x))$.

b) state the domain of $f(g(x))$.

Solution

a) $f(g(x)) = \frac{1}{x-4}$

1 mark

b) Domain: $\{x \mid x \neq 4, x \in \mathbb{R}\}$

or

Domain: $(-\infty, 4) \cup (4, \infty)$

Question 12

T6

Prove the identity for all permissible values of x .

$$\tan 2x = \frac{2}{\cot x - \tan x}$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\tan 2x$	$\frac{2}{\frac{\cos x}{\sin x} - \frac{\sin x}{\cos x}}$ $\frac{2}{\frac{\cos^2 x - \sin^2 x}{\sin x \cos x}}$ $\frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x}$ $\frac{\sin 2x}{\cos 2x}$ $\tan 2x$

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 12

T6

Method 2

Left-Hand Side	Right-Hand Side
$\frac{2 \tan x}{1 - \tan^2 x}$	$\frac{2}{\frac{\cos x}{\sin x} - \frac{\sin x}{\cos x}}$
$\frac{\frac{2 \sin x}{\cos x}}{1 - \frac{\sin^2 x}{\cos^2 x}}$	$\frac{2}{\frac{\cos^2 x - \sin^2 x}{\sin x \cos x}}$
$\frac{\frac{2 \sin x}{\cos x}}{\frac{\cos^2 x - \sin^2 x}{\cos^2 x}}$	$\frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x}$
$\frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x}$	

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 12

T6

Method 3

Left-Hand Side	Right-Hand Side
$\frac{2 \tan x}{1 - \tan^2 x}$	$\frac{2}{\frac{1}{\tan x} - \tan x}$
	$\frac{2}{\frac{1 - \tan^2 x}{\tan x}}$
	$\frac{2 \tan x}{1 - \tan^2 x}$

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 13

R1

Given $f(x) = -x^2 + 2$ and $g(x) = x^2 - 1$,

a) determine the equation $h(x) = g(x) - f(x)$.

b) state the range of $h(x)$.

Solution

a) $h(x) = g(x) - f(x)$

1 mark for subtraction of $g(x) - f(x)$

$$h(x) = (x^2 - 1) - (-x^2 + 2)$$

$$h(x) = x^2 - 1 + x^2 - 2$$

$$h(x) = \underline{2x^2 - 3}$$

1 mark

b) Range: $\underline{\{y | y \geq -3, y \in \mathbb{R}\}}$

1 mark for range consistent with $h(x)$

or

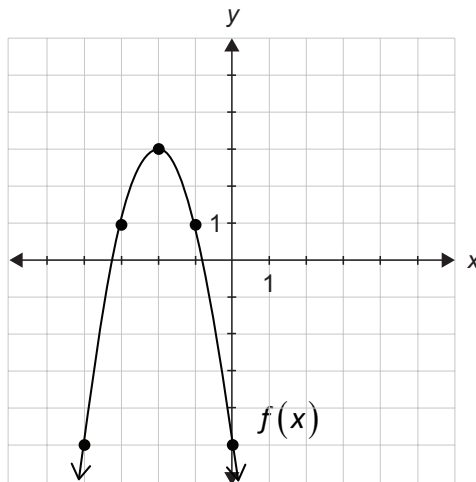
1 mark

Range: $\underline{[-3, \infty)}$

Question 14

R5, R6

Given the graph of $f(x)$,

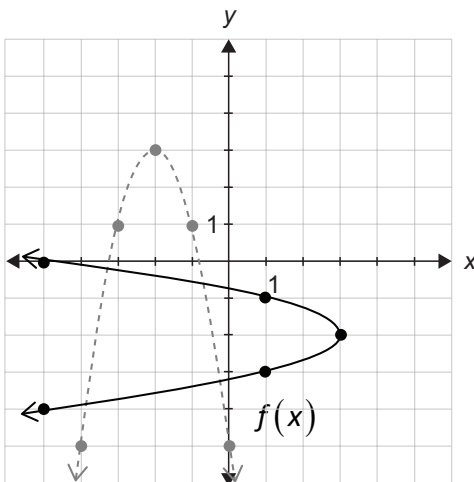


a) sketch the graph after a reflection over the line $y = x$.

b) state a restriction on the domain of $f(x)$ in order for its inverse to be a function.

Solution

a)



1 mark

b) $\{x | x \geq -2, x \in \mathbb{R}\}$

or

$\{x | x \leq -2, x \in \mathbb{R}\}$

1 mark

Note:

Other answers are possible for b).

Question 15

R4, R5

Describe a transformation of a function that would not affect its domain.

Solution

vertical stretch by a factor of a

or

vertical reflection

or

vertical translation k units up/down

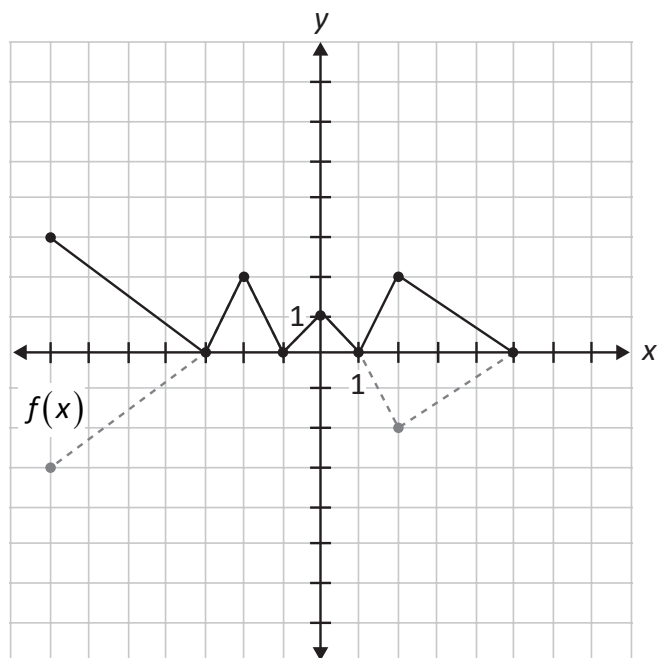
1 mark

Question 16

R1

Given the graph of $f(x)$, sketch the graph of the function $y = |f(x)|$.

Solution



1 mark

Question 17

R11

Determine, algebraically, all of the zeros of the polynomial function, $p(x) = 3x^3 + x^2 - 20x + 12$.

Solution

$$p(x) = 3x^3 + x^2 - 20x + 12$$

$$p(2) = 3(2)^3 + (2)^2 - 20(2) + 12$$

$$p(2) = 0$$

1 mark for identifying one zero of $p(x)$

$\therefore (x - 2)$ is a factor

2	3	1	-20	12
		6	14	-12
	3	7	-6	0

1 mark for synthetic division (or equivalent strategy)

$$p(x) = (x - 2)(3x^2 + 7x - 6)$$

$$0 = (x - 2)(3x - 2)(x + 3)$$

$\frac{1}{2}$ mark for consistent factors

$$x = 2, \quad x = \frac{2}{3}, \quad x = -3$$

$\frac{1}{2}$ mark for consistent zeros

3 marks

Question 18**T2**

Given $\theta = 240^\circ$, identify the coordinates of the point, $P(\theta)$, on the unit circle.

a. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

b. $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

c. $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

d. $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

Question 19**R1**

Given the functions, $f(x) = -3x + 5$ and $g(x) = x^2 + x - 1$, identify the value of $g(f(2))$.

a. -10

b. -5

c. -3

d. -1

Question 20**R3**

Given $f(x) = \sqrt{x} + 5$, identify the equation of the transformed graph that has the same y -intercept.

a. $y = f(x-3)$

b. $y = -3f(x)$

c. $y = f(-3x)$

d. $y = f(x) - 3$

Question 21**R12**

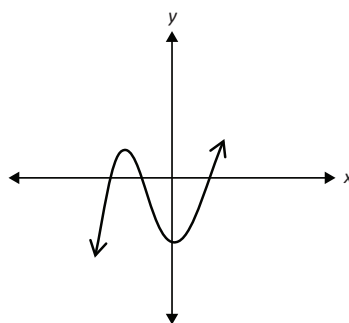
Given the graph of $p(x) = ax^3 + bx^2 + cx + d$, identify the statement that is true.

a. $a > 0, d < 0$

b. $a > 0, d > 0$

c. $a < 0, d < 0$

d. $a < 0, d > 0$

**Question 22****T5**

Identify the general solution of the equation, $\csc \theta = -1$.

a. $\theta = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$

b. $\theta = \frac{3\pi}{2} + 2k\pi, k \in \mathbb{Z}$

c. $\theta = \frac{3\pi}{2} + k\pi, k \in \mathbb{Z}$

d. $\theta = \pi + 2k\pi, k \in \mathbb{Z}$

Question 23**R13**

Identify the equation of a radical function with a domain of $[-6, \infty)$ and a range of $(-\infty, 3]$.

a. $y = -\sqrt{x+6} - 3$

b. $y = -\sqrt{x+6} + 3$

c. $y = \sqrt{-(x+3)} - 6$

d. $y = \sqrt{x-6} + 3$

Question 24**P3**

Identify the value of n in the equation, ${}_nC_3 = {}_nC_7$.

a. 3

b. 4

c. 7

d. 10

Question 25**R11**

Identify the remainder when $p(x) = x^5 - 1$ is divided by $(x+1)$.

a. -2

b. -1

c. 0

d. 4

Question 26

R13

Match each function with its corresponding graph.

Solution

Write the corresponding letter in this column.

$$y = \sqrt{x} - 3$$

 B

$$y = \sqrt{-x} + 3$$

 D

$$y = -\sqrt{x+3}$$

 C

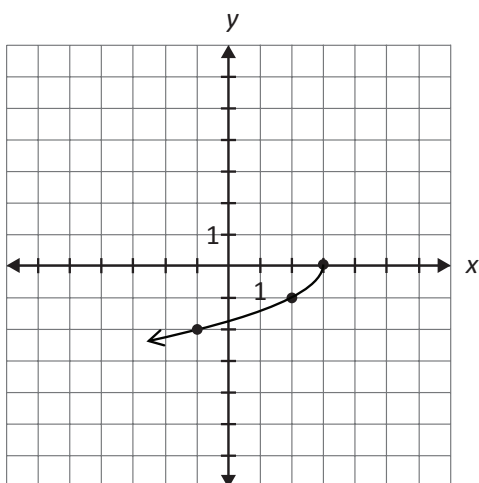
$$y = -\sqrt{-(x-3)}$$

 A

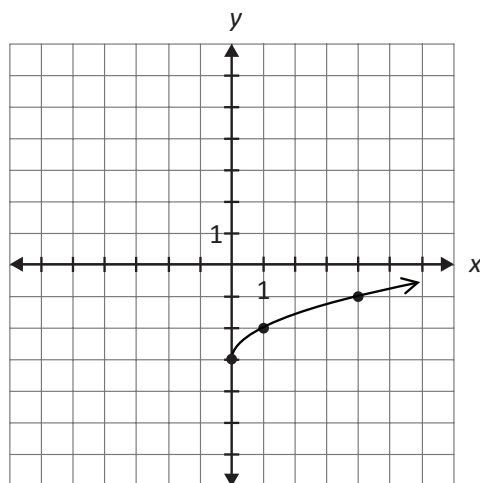
½ mark for each correct answer

2 marks

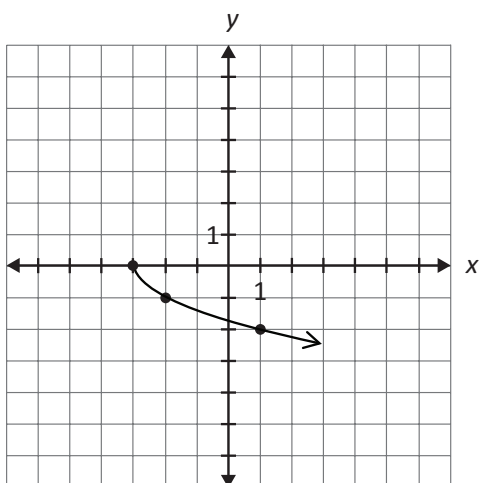
A)



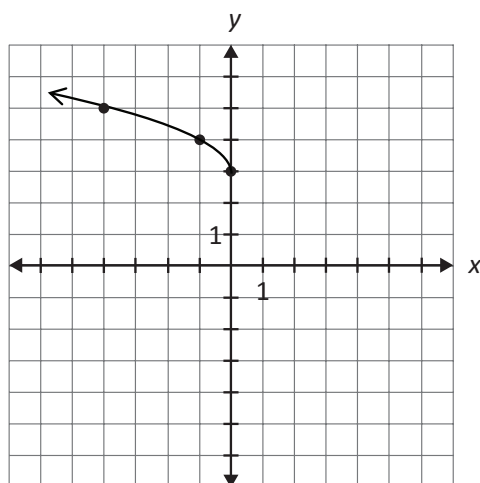
B)



C)



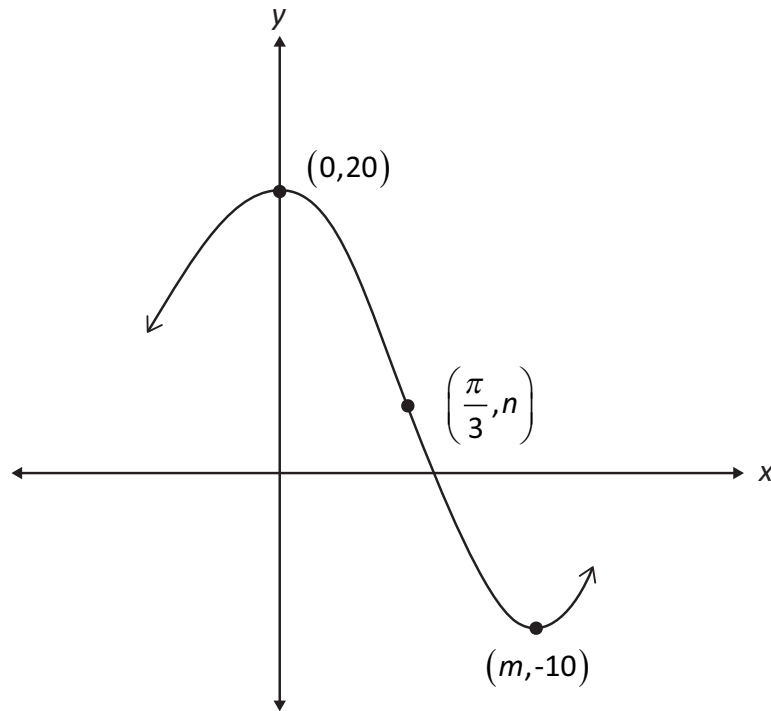
D)



Question 27

T4

The amplitude, A , of the sinusoidal function can be determined using the equation $A = 20 - n$. State the values of m and n .



Solution

$$m = \frac{2\pi}{3}$$

1 mark for value of m

$$n = 5$$

1 mark for value of n

2 marks

Question 28

P2

Solve, algebraically.

$${}_{n-1}P_2 = 20$$

Solution

$$\frac{(n-1)!}{(n-1-2)!} = 20$$

½ mark for substitution

$$\frac{(n-1)!}{(n-3)!} = 20$$

$$\frac{(n-1)(n-2)\cancel{(n-3)!}}{\cancel{(n-3)!}} = 20$$

1 mark for factorial expansion

½ mark for simplification of factorials

$$n^2 - 3n + 2 = 20$$

$$n^2 - 3n - 18 = 0$$

$$(n-6)(n+3) = 0$$

$$n = 6 \quad \cancel{n = -3}$$

½ mark for the permissible value of n

½ mark for showing the rejection of the extraneous root

3 marks

Question 29

T6

Determine the exact value of $\sin\left(\frac{23\pi}{12}\right)$.

Solution

$$\sin\left(\frac{23\pi}{12}\right) = \sin\left(\frac{21\pi}{12} + \frac{2\pi}{12}\right)$$

$$\sin\left(\frac{7\pi}{4} + \frac{\pi}{6}\right) = \sin\left(\frac{7\pi}{4}\right)\cos\left(\frac{\pi}{6}\right) + \cos\left(\frac{7\pi}{4}\right)\sin\left(\frac{\pi}{6}\right)$$

$$= \left(-\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$$

$$= \frac{-\sqrt{6} + \sqrt{2}}{4}$$

1 mark for substitution into correct identity

2 marks for exact values (½ mark for each)

3 marks

Note:

Other combinations are possible.

Question 30

R10

Cameron was asked to solve the equation, $\log_3(x-4)=2$.

Cameron's solution:

$$x - 4 = 2^3$$

$$x - 4 = 8$$

$$x = 12$$

Describe Cameron's error.

Solution

Cameron mixed up the exponent and the base when changing to exponential form.

1 mark

Question 31

T6

Given the identity $\sin^2 \theta = \frac{\tan \theta \sin \theta}{\sec \theta}$, state the non-permissible value of θ over the interval $[\pi, 2\pi]$.

Solution

$$\cos \theta = 0$$

$$\theta = \frac{3\pi}{2}$$

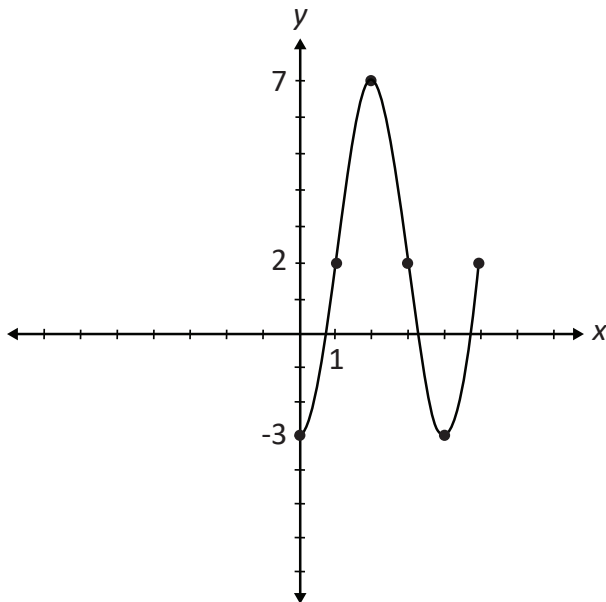
1 mark

Question 32

T4

Sketch the graph of $y = -5\cos\left(\frac{\pi}{2}x\right) + 2$ over the domain $[0, 5]$.

Solution



1 mark for shape of a sinusoidal function with correct amplitude

1 mark for vertical reflection

1 mark for vertical translation

1 mark for period

4 marks

Question 33

R12

Describe the behaviour of the graph of the polynomial function, $p(x) = (x-3)^2(x-1)$, as it approaches the x-intercept at $x=3$.

Solution

As it approaches $x=3$, the graph touches but does not cross the x-axis.

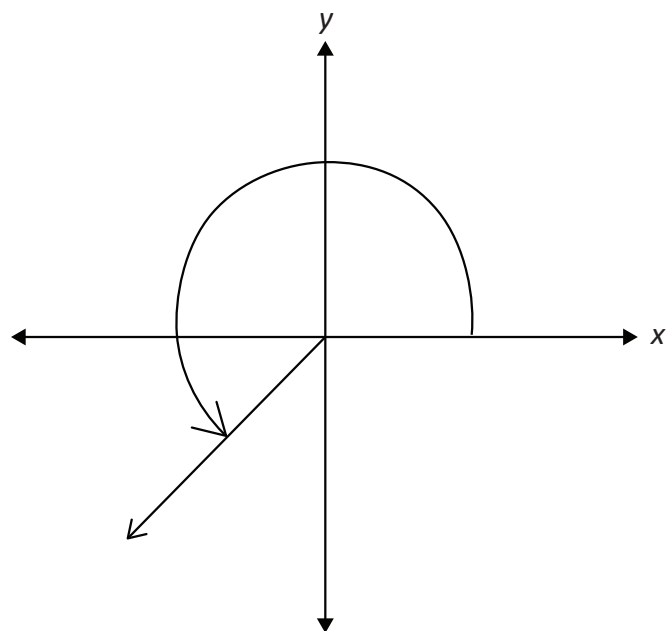
1 mark

Question 34

T1

Sketch the angle of 4 radians in standard position.

Solution



$\frac{1}{2}$ mark for an appropriate angle
in quadrant III

$\frac{1}{2}$ mark for the correct direction of an
angle in standard position

1 mark

Question 35

R10

Solve, algebraically.

$$\log_5 12 + 2\log_5 x = \log_5 48$$

Solution

Method 1

$$\log_5 12 + 2\log_5 x = \log_5 48$$

$$\log_5 12 + \log_5 x^2 = \log_5 48$$

1 mark for power law

$$\log_5 12x^2 = \log_5 48$$

1 mark for product law

$$12x^2 = 48$$

1 mark for equating arguments

$$x^2 = 4$$

$$x = 2 \quad \cancel{x = -2}$$

½ mark for permissible value of x

½ mark for showing the rejection of the extraneous root

4 marks

Method 2

$$\log_5 12 + 2\log_5 x = \log_5 48$$

$$\log_5 12 + \log_5 x^2 - \log_5 48 = 0$$

1 mark for power law

$$\log_5 \left(\frac{12x^2}{48} \right) = 0$$

½ mark for product law

½ mark for quotient law

$$\frac{12x^2}{48} = 5^0$$

1 mark for exponential form

$$x^2 = 4$$

$$x = 2 \quad \cancel{x = -2}$$

½ mark for permissible value of x

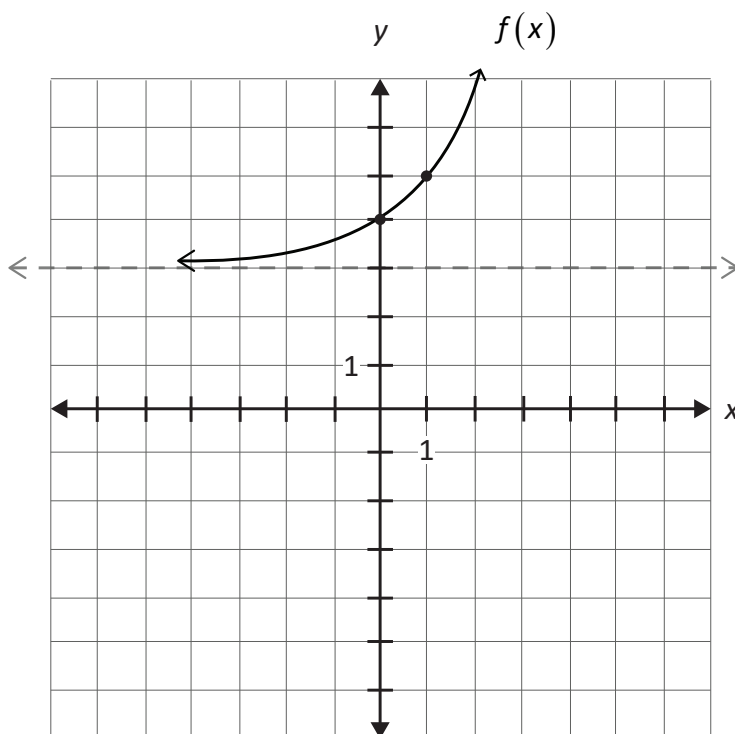
½ mark for showing the rejection of the extraneous root

4 marks

Question 36

R9

State the equation of the exponential function represented by the graph of $f(x)$.



Solution

$$f(x) = 2^x + 3$$

1 mark for exponential function with a base of 2
1 mark for vertical translation

2 marks

Question 37

T3

Evaluate.

$$\cos\left(\frac{11\pi}{3}\right) + \csc\left(-\frac{\pi}{3}\right) \cot\left(\frac{11\pi}{6}\right)$$

Solution

$$\left(\frac{1}{2}\right) + \left(-\frac{2}{\sqrt{3}}\right) \left(-\frac{\sqrt{3}}{1}\right)$$

$$\frac{1}{2} + 2$$

$$\frac{5}{2}$$

1 mark for $\cos\left(\frac{11\pi}{3}\right)$ ($\frac{1}{2}$ mark for value; $\frac{1}{2}$ mark for quadrant)

1 mark for $\csc\left(-\frac{\pi}{3}\right)$ ($\frac{1}{2}$ mark for value; $\frac{1}{2}$ mark for quadrant)

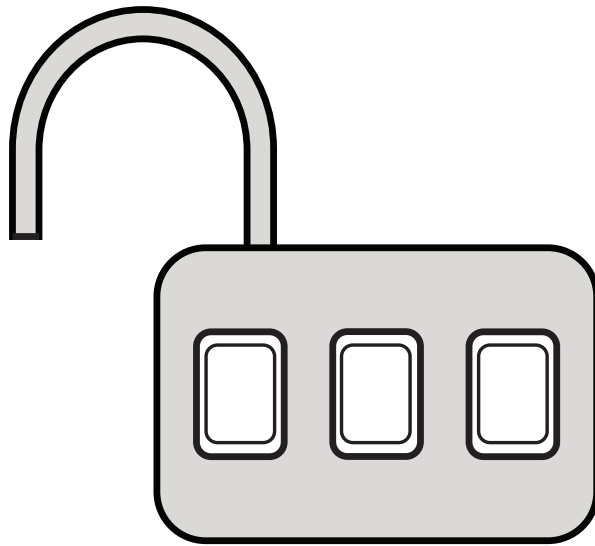
1 mark for $\cot\left(\frac{11\pi}{6}\right)$ ($\frac{1}{2}$ mark for value; $\frac{1}{2}$ mark for quadrant)

3 marks

Question 38

P1

Min Li has a 3-digit code for his lock. Determine the number of possible codes for his lock, if repetition is allowed.



Solution

$$\underline{10} \cdot \underline{10} \cdot \underline{10} = 1000$$

1 mark

Question 39

R10

Solve, algebraically.

$$\left(\frac{1}{6}\right)^{3x+2} = 6^{2x}$$

Solution

$$(6^{-1})^{3x+2} = 6^{2x}$$

1 mark for changing to a common base

$$6^{-3x-2} = 6^{2x}$$

$$-3x - 2 = 2x$$

1 mark for equating exponents

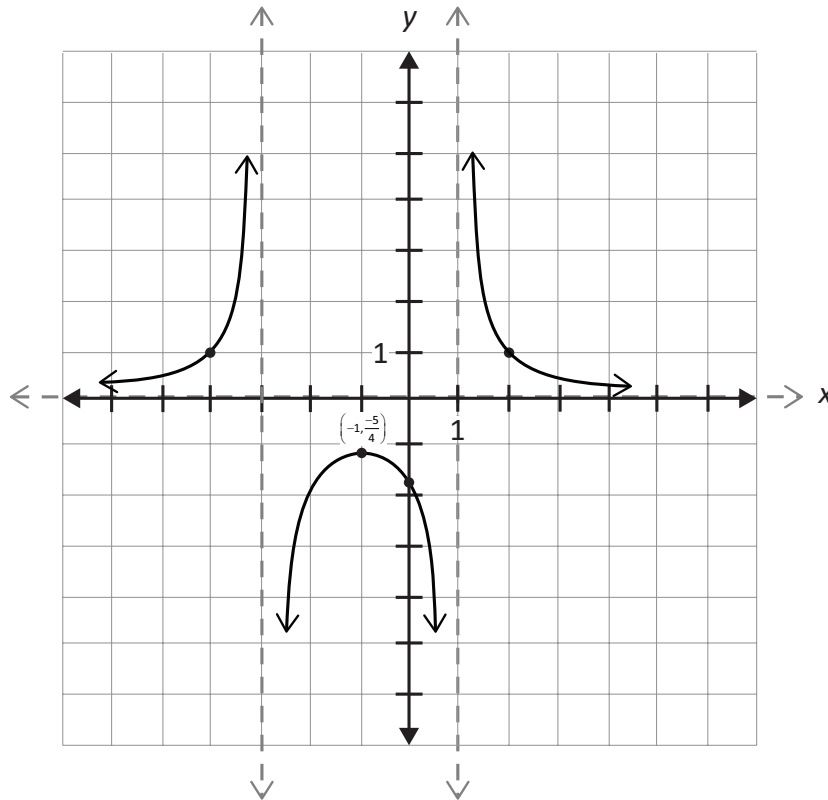
$$-5x = 2$$

$$x = -\frac{2}{5}$$

2 marks

Question 40**R14**

Sketch the graph of $f(x) = \frac{5}{(x-1)(x+3)}$ and state the y-intercept.

Solution

y-intercept: $\left(0, -\frac{5}{4}\right)$

1 mark for vertical asymptotic behaviour

(½ mark for behaviour approaching $x = -3$; ½ mark for behaviour approaching $x = 1$)

1 mark for horizontal asymptotic behaviour approaching $y = 0$

1½ marks for shape (½ mark for shape of each branch)

½ mark for y-intercept

4 marks

Question 41

R7, R10

Given $\log_2 x = 5$, determine the value of $\log_4 (2x)$.

Solution

$$x = 2^5$$

1 mark for solving for x

$$x = 32$$

$$\log_4 (2(32))$$

$$\log_4 (64)$$

$$3$$

1 mark for evaluating the logarithm

2 marks

Question 42

T6

Henry was asked to prove the identity, $\csc^2 \theta - 2\cos^2 \theta + 1 = \cot^2 \theta + 2\sin^2 \theta$, for all permissible values of θ .

His work:

Left-Hand side	Right-Hand side
$\csc^2\left(\frac{\pi}{4}\right) - 2\cos^2\left(\frac{\pi}{4}\right) + 1$	$\cot^2\left(\frac{\pi}{4}\right) + 2\sin^2\left(\frac{\pi}{4}\right)$
$(\sqrt{2})^2 - 2\left(\frac{\sqrt{2}}{2}\right)^2 + 1$	$(1)^2 + 2\left(\frac{\sqrt{2}}{2}\right)^2$
$2 - 1 + 1$	$1 + 1$
2	2
LHS	RHS ✓

Explain why his proof is insufficient.

Solution

Henry proved the identity for $\theta = \frac{\pi}{4}$, but not for all permissible values of θ .

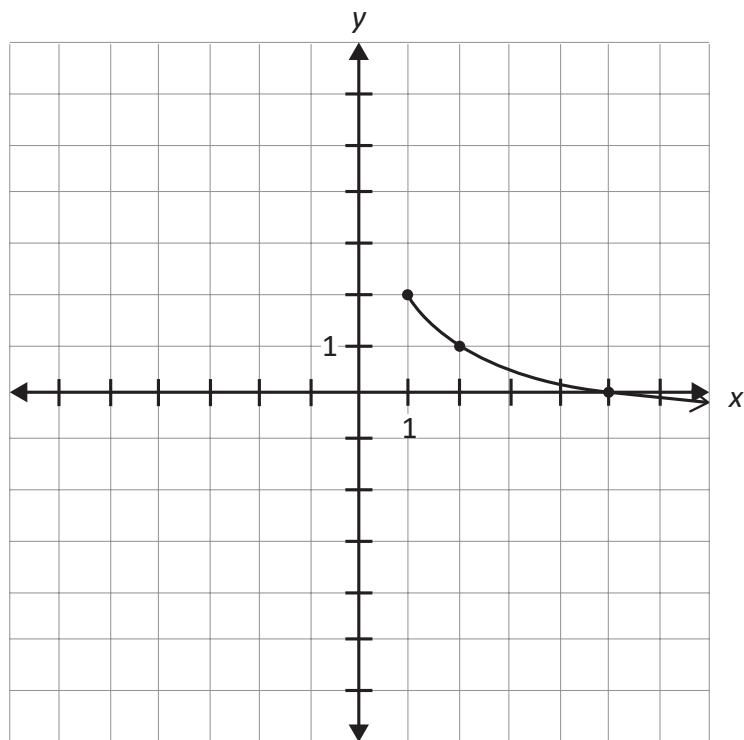
1 mark

Question 43

R13

Sketch the graph of $y - 2 = -\sqrt{x - 1}$.

Solution



1 mark for shape of a radical function
1 mark for vertical reflection
1 mark for horizontal translation
1 mark for vertical translation

4 marks

Question 44

R7, R8

Evaluate.

$$\log_3 54 - \log_3 6$$

Solution

$$\log_3 \left(\frac{54}{6} \right)$$

1 mark for quotient law

$$\log_3 9$$

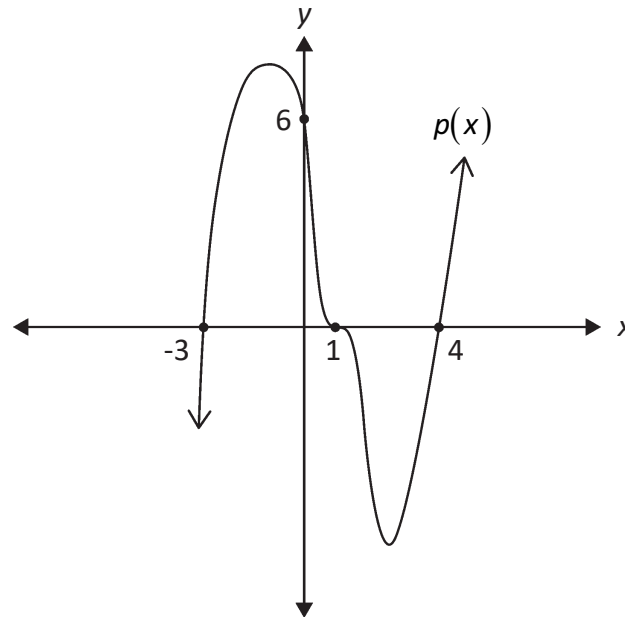
$$2$$

1 mark for evaluating the logarithm

2 marks

Question 45**R12**

Determine, algebraically, the equation of the graph of the polynomial function, $p(x)$.

**Solution**

$$p(x) = a(x-1)^3(x+3)(x-4)$$

½ mark for factors of $p(x)$

½ mark for multiplicity of 3 at $x = 1$

$$p(0) = a(0-1)^3(0+3)(0-4)$$

$$6 = a(-1)^3(3)(-4)$$

$$6 = 12a$$

$$\frac{1}{2} = a$$

1 mark for consistent value of a

$$p(x) = \frac{1}{2}(x-1)^3(x+3)(x-4)$$

2 marks

Question 1

T1

A central angle of a circle subtends an arc length of 5π cm.

Given the circle has a radius of 9 cm, find the measure of the central angle in degrees.

Solution

$$s = \theta r$$

$$5\pi = \theta(9)$$

$\frac{1}{2}$ mark for substitution into correct formula

$$\theta = \frac{5\pi}{9}$$

$\frac{1}{2}$ mark for solving for θ

$$\theta \text{ (in degrees)} = \frac{5\pi}{9} \cdot \frac{180^\circ}{\pi}$$

$$= 100^\circ$$

1 mark for conversion to degrees

2 marks

Solve the equation $\csc^2 \theta + 3 \csc \theta - 4 = 0$ over the interval $[0, 2\pi]$.
Express your answers as exact values or correct to 3 decimal places.

Solution**Method 1**

$$\csc^2 \theta + 3 \csc \theta - 4 = 0$$

$$(\csc \theta - 1)(\csc \theta + 4) = 0$$

$$\csc \theta = 1$$

$$\csc \theta = -4$$

1 mark for solving for $\csc \theta$

$$\sin \theta = 1$$

$$\sin \theta = -\frac{1}{4}$$

1 mark for reciprocal of $\csc \theta$

$$\theta = \frac{\pi}{2}$$

$$\theta_r = 0.252\ 680$$

or

$$\theta = 1.570\ 796$$

$$\theta = 3.394\ 273, 6.030\ 505$$

$$\theta = \frac{\pi}{2}, 3.394, 6.031$$

2 marks (1 mark for consistent solutions of each trigonometric equation)

or

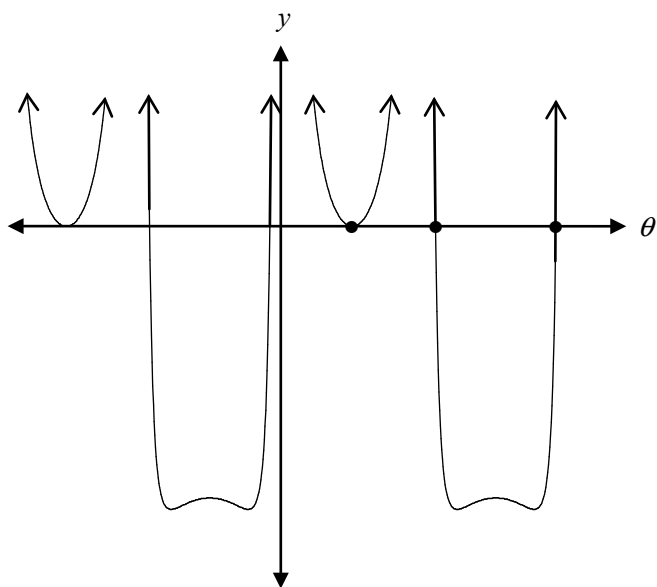
$$\theta = 1.571, 3.394, 6.031$$

4 marks

Method 2–Graphing Calculator

$$y = \left(\frac{1}{\sin \theta} \right)^2 + \frac{3}{\sin \theta} - 4$$

1 mark for equation



1 mark for justification

Find all zeros from $[0, 2\pi]$.

1 mark for restricted domain

$$\theta = 1.571, 3.394, 6.031$$

1 mark for solutions

4 marks

Jess invests \$12 000 at a rate of 4.75% compounded monthly.

How long will it take for Jess to triple her investment?

Express your answer in years, correct to 3 decimal places.

Solution

Method 1

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$36\,000 = 12\,000\left(1 + \frac{0.0475}{12}\right)^{12t} \quad \frac{1}{2} \text{ mark for substitution}$$

$$3 = \left(1 + \frac{0.0475}{12}\right)^{12t}$$

$$\ln 3 = \ln\left(1 + \frac{0.0475}{12}\right)^{12t} \quad \frac{1}{2} \text{ mark for applying logarithms}$$

$$\ln 3 = 12t \ln\left(1 + \frac{0.0475}{12}\right) \quad 1 \text{ mark for power rule}$$

$$t = \frac{\ln 3}{12 \ln\left(1 + \frac{0.0475}{12}\right)} \quad \frac{1}{2} \text{ mark for isolating } t$$

$$t = 23.174\,425$$

$$t = 23.174 \text{ years}$$

$\frac{1}{2}$ mark for evaluating quotient of logarithms

3 marks

Note(s):

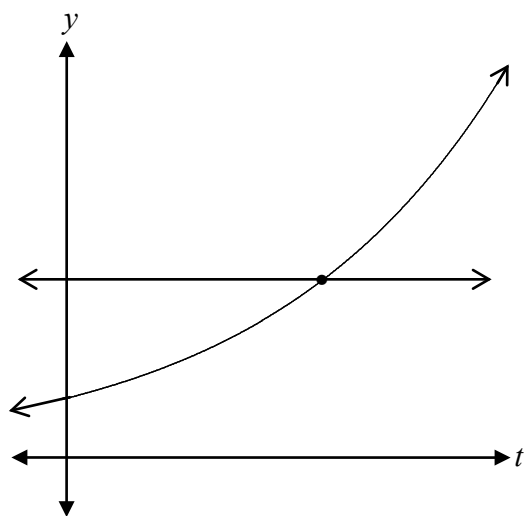
- award a maximum of 2 marks for the formula $A = Pe^{rt}$ used correctly

Method 2–Graphing Calculator

$$y = 3$$

$$y = \left(1 + \frac{0.0475}{12}\right)^{12t}$$

1 mark for equations

**or**

1 mark for justification

Find the value of t at the point of intersection of these two functions.

$$t = 23.174 \text{ years}$$

1 mark for solution

3 marks

The 4th term in the binomial expansion of $\left(qx^2 - \frac{3}{x}\right)^{10}$ is $414\,720x^{11}$.

Determine the value of q algebraically.

Solution

$$t_4 = {}_{10}C_3 \left(qx^2\right)^7 \left(-\frac{3}{x}\right)^3 \quad 2 \text{ marks (1 mark for } {}_{10}C_3, \frac{1}{2} \text{ mark for each consistent factor)}$$

$$414\,720x^{11} = 120 \left(q^7 x^{14}\right) \left(-\frac{27}{x^3}\right) \quad \frac{1}{2} \text{ mark for comparing coefficients}$$

$$414\,720 = -3240q^7$$

$$q^7 = -128$$

$$q = -2$$

$\frac{1}{2}$ mark for solving for q

3 marks

Question 5

P1

Bella has 2 pairs of shoes, 3 pairs of pants, and 10 shirts.

Carey has 4 pairs of shoes, 4 pairs of pants, and 4 shirts.

An outfit is made up of one pair of shoes, one pair of pants, and one shirt.

Who can make more outfits? Justify your answer.

Solution

Bella: $2 \times 3 \times 10 = 60$ outfits

Carey: $4 \times 4 \times 4 = 64$ outfits

$\therefore$ Carey can make more outfits.

1 mark for justification

1 mark

Question 6

P4

In the binomial expansion of $(x - y)^{10}$, how many terms will be positive?

Justify your answer.

Solution

Six terms will be positive.

1 mark for six terms

The term will be positive when “ $-y$ ” has an even exponent.

1 mark for justification

2 marks

Solve the following equation algebraically where $180^\circ \leq \theta \leq 360^\circ$.

$$2\sin^2 \theta + 5\cos \theta + 1 = 0$$

Solution

$$2(1 - \cos^2 \theta) + 5\cos \theta + 1 = 0$$

1 mark for identity

$$2 - 2\cos^2 \theta + 5\cos \theta + 1 = 0$$

$$2\cos^2 \theta - 5\cos \theta - 3 = 0$$

$$(2\cos \theta + 1)(\cos \theta - 3) = 0$$

$$\cos \theta = -\frac{1}{2}$$

$$\cos \theta = 3$$

1 mark for solving for $\cos \theta$

$$\theta_r = 60^\circ$$

$\therefore$ no solution

1 mark for indicating no solution

$$\theta = 240^\circ$$

1 mark for solving for θ

4 marks

Note(s):

- award a maximum of 3 marks if not solved algebraically

Solve the following equation algebraically:

$$\log_3(x-4) + \log_3(x-2) = 1$$

Solution

Method 1

$$\log_3(x-4) + \log_3(x-2) = 1$$

$$\log_3(x-4)(x-2) = 1$$

1 mark for product rule

$$3^1 = (x-4)(x-2)$$

1 mark for exponential form

$$3 = x^2 - 6x + 8$$

$$0 = x^2 - 6x + 5$$

$$0 = (x-5)(x-1)$$

$$x = 5 \quad \cancel{x=1}$$

½ mark for solving for x within a quadratic equation
½ mark for rejecting extraneous root

3 marks

Method 2

$$\log_3(x-4) + \log_3(x-2) = 1$$

$$\log_3(x-4)(x-2) = 1$$

1 mark for product rule

$$\log_3(x^2 - 6x + 8) = \log_3 3$$

½ mark for logarithmic form

$$x^2 - 6x + 8 = 3$$

½ mark for equating arguments

$$x^2 - 6x + 5 = 0$$

$$(x-1)(x-5) = 0$$

$$\cancel{x=1} \quad x = 5$$

½ mark for solving for x within a quadratic equation
½ mark for rejecting extraneous roots

3 marks

Given that $f(x) = \{(1, 3), (2, 5), (3, 4), (4, 2)\}$, find $f(f(3))$.

Solution

$$\begin{aligned} f(f(3)) &= f(4) \\ &= 2 \end{aligned}$$

½ mark for $f(3) = 4$

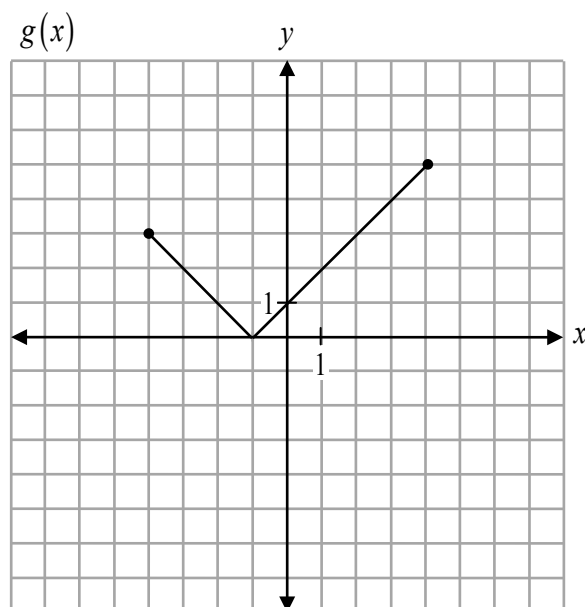
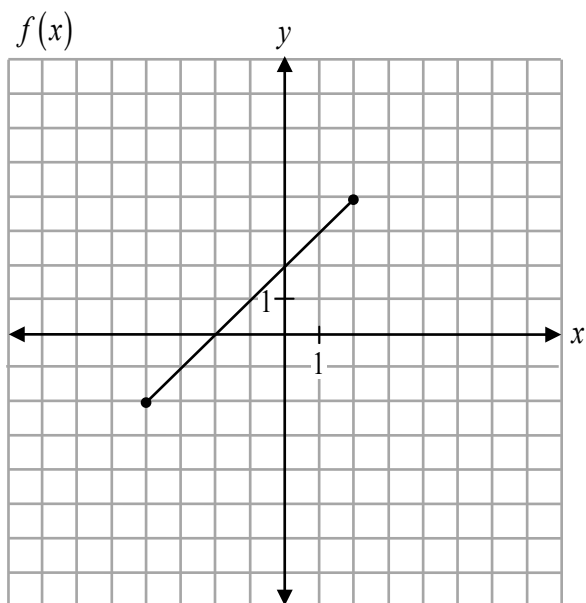
½ mark for $f(4) = 2$

1 mark

Question 10

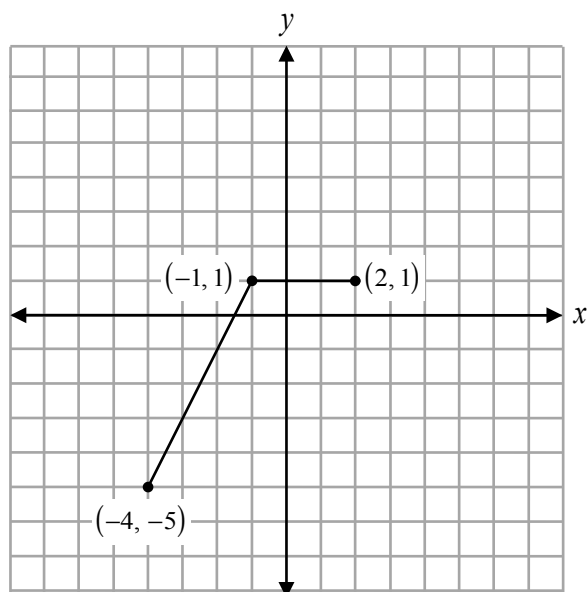
R1

Given the graphs of $f(x)$ and $g(x)$ below,



sketch the graph of $y = f(x) - g(x)$.

Solution



x	$f(x)$	$g(x)$	$f(x) - g(x)$
-4	-2	3	-5
-2	0	1	-1
-1	1	0	1
0	2	1	1
2	4	3	1

1 mark for subtraction of $f(x) - g(x)$

1 mark for restricted domain

2 marks

Question 11

R2, R3

Given the graph of $y = f(x)$, describe the transformations to obtain the graph of the function $y = f(2x - 6)$.

Solution

Method 1

Factor out the 2.

$$y = f(2(x - 3))$$

Horizontally compress by a factor of 2.
Then shift 3 units to the right.

1 mark for starting with a horizontal compression by a factor of 2
1 mark for ending with a horizontal shift of 3 units to the right

Method 2

$$y = f(2x - 6)$$

Shift 6 units to the right.
Then horizontally compress by a factor of 2.

1 mark for starting with a horizontal shift of 6 units to the right
1 mark for ending with a horizontal compression by a factor of 2

2 marks

2 marks

Question 12

R5

Given $f(x) = \{(-3, 4), (2, 7), (8, 6)\}$, state the domain of the resulting function after $f(x)$ is reflected through the line $y = x$.

Solution

Domain: $\{4, 6, 7\}$

1 mark for correct domain

1 mark

Note(s):

- award $\frac{1}{2}$ mark for stating the inverse of the function: $f^{-1}(x) = \{(4, -3), (7, 2), (6, 8)\}$

Question 13

R8

Determine the value of y in the following equation:

$$\log_x 27 - \log_x 3 = 2 \log_x y$$

Solution

$$\log_x 27 - \log_x 3 = 2 \log_x y$$

$$\log_x \frac{27}{3} = 2 \log_x y$$

1 mark for quotient rule

$$\log_x 9 = \log_x y^2$$

1 mark for power rule

$$9 = y^2$$

$$y = \pm 3$$

$$y = 3 \quad \cancel{y = -3}$$

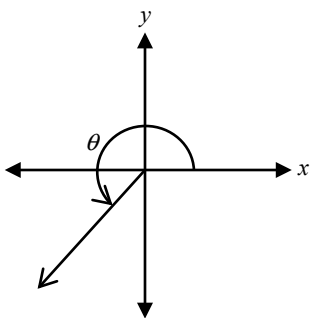
 $\frac{1}{2}$ mark for positive value of y $\frac{1}{2}$ mark for negative value of y and rejecting extraneous root**3 marks**

Question 14

T1

Angle θ , measuring $\frac{5\pi}{4}$, is drawn in standard position as shown below.

Determine the measures of all angles in the interval $[-4\pi, 2\pi]$ that are coterminal with θ .

**Solution**

$$\theta = -\frac{3\pi}{4} \quad \frac{1}{2} \text{ mark}$$

$$\theta = -\frac{11\pi}{4} \quad \frac{1}{2} \text{ mark}$$

1 mark

Prove the identity below for all permissible values of x :

$$\frac{\sin^2 x}{\sec x + 1} = \cos x - \cos^2 x$$

Solution

Method 1

$$\text{LHS} = \frac{1 - \cos^2 x}{\frac{1}{\cos x} + 1}$$

$$= \frac{1 - \cos^2 x}{\frac{1 + \cos x}{\cos x}}$$

$$= (1 - \cos^2 x) \left(\frac{\cos x}{1 + \cos x} \right)$$

$$= (1 - \cos x)(1 + \cos x) \left(\frac{\cos x}{1 + \cos x} \right)$$

$$= (1 - \cos x)(\cos x)$$

$$= \cos x - \cos^2 x$$

$$= \text{RHS}$$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Method 2

$$\begin{aligned}\text{LHS} &= \frac{\sin^2 x}{\sec x + 1} \cdot \frac{(\sec x - 1)}{(\sec x - 1)} \\&= \frac{\sin^2 x (\sec x - 1)}{\sec^2 x - 1} \\&= \frac{\sin^2 x (\sec x - 1)}{\tan^2 x} \\&= \frac{\sin^2 x (\sec x - 1)}{\frac{\sin^2 x}{\cos^2 x}} \\&= \cos^2 x (\sec x - 1) \\&= \cos^2 x \left(\frac{1}{\cos x} - 1 \right) \\&= \cos x - \cos^2 x \\&= \text{RHS}\end{aligned}$$

1 mark for correct substitution of identities
1 mark for algebraic strategies
1 mark for logical process to prove the identity

3 marks

Solve algebraically:

$${}_nC_2 = 4n + 5$$

Solution

$${}_nC_2 = 4n + 5$$

$$\frac{n!}{(n-2)!2!} = 4n + 5$$

½ mark for factorial notation

$$\frac{n(n-1)\cancel{(n-2)!}}{\cancel{(n-2)!}2!} = 4n + 5$$

½ mark for factorial expansion

½ mark for simplification of factorial

$$n(n-1) = 2!(4n+5)$$

$$n^2 - n = 8n + 10$$

$$n^2 - 9n - 10 = 0$$

½ mark for simplification

$$(n-10)(n+1) = 0$$

$$n = 10 \quad \cancel{n = -1}$$

½ mark for both values of n

½ mark for rejecting extraneous root

3 marks

Question 20

T4

Which of the following is true about the periods of the three functions below?

$$f(\theta) = 2 \sin 3\left(\theta - \frac{\pi}{2}\right)$$

$$g(\theta) = \sin 3\theta + 6$$

$$k(\theta) = 3 \sin \theta + 6$$

a) The graphs of $f(\theta)$ and $g(\theta)$ have the same period.

b) The graphs of $g(\theta)$ and $k(\theta)$ have the same period.

c) All of the graphs have the same period.

d) None of the graphs have the same period.

Question 21

T5

Which of the following represents the general solution to the equation $\tan \theta = -1$?

a) $\theta = \frac{\pi}{4} + 2k\pi, k \in \mathbb{I}$

b) $\theta = \frac{\pi}{4} + k\pi, k \in \mathbb{I}$

c) $\theta = \frac{3\pi}{4} + 2k\pi, k \in \mathbb{I}$

d) $\theta = \frac{3\pi}{4} + k\pi, k \in \mathbb{I}$

Question 22

R4

If $(3, -2)$ is a point on the graph of $y = f(x)$, what point must be on the graph of $y = 2f(x+1)$?

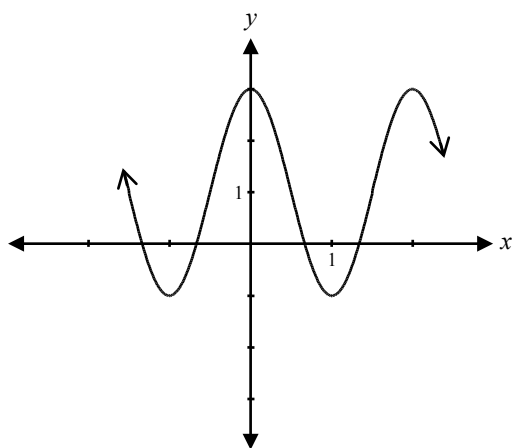
a) $(4, -1)$

b) $(4, -4)$

c) $(2, 1)$

d) $(2, -4)$

Given the graph of $y = 2\cos \pi x + 1$ below, determine another equation that will produce the same graph.



Solution

Some sample equations are:

$$y = 2\cos \pi(x - 2) + 1$$

1 mark for correct equation

$$y = -2\cos \pi(x - 1) + 1$$

1 mark

$$y = -2\cos \pi(x + 1) + 1$$

$$y = 2\sin \pi\left(x + \frac{1}{2}\right) + 1$$

$$y = 2\sin \pi\left(x - \frac{3}{2}\right) + 1$$

Other answers are possible.

Question 26

R1

Given $f(x) = 3$ and $g(x) = x + 2$, determine the domain and range of $h(x) = \frac{f(x)}{g(x)}$.

Solution

Domain: $\{x \mid x \in \mathbb{R}, x \neq -2\}$

1 mark for domain

Range: $\{y \mid y \in \mathbb{R}, y \neq 0\}$

1 mark for range

2 marks

Question 27

T3

Explain how to find the exact value of $\sec\left(\frac{19\pi}{6}\right)$.

Solution

Find the exact value of $\cos\left(\frac{19\pi}{6}\right)$.

1 mark for $\cos\left(\frac{19\pi}{6}\right)$

Then take the reciprocal of the value of $\cos\left(\frac{19\pi}{6}\right)$.

1 mark for reciprocal

2 marks

Given $f(x) = 4 - x$, verify that $f^{-1}(x) = f(x)$.

Solution

Method 1

$$y = 4 - x$$

To find $f^{-1}(x)$, switch x and y values.

$$x = 4 - y$$

$$-y = x - 4$$

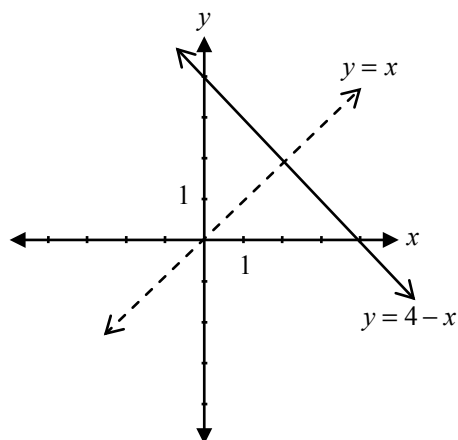
$$y = 4 - x$$

$$f^{-1}(x) = 4 - x$$

1 mark for verifying $f^{-1}(x) = f(x)$

1 mark

Method 2



When $y = 4 - x$ is reflected over the line $y = x$ it produces the same graph.

1 mark

Method 3

Assume $f^{-1}(x) = 4 - x$.

$$f(f^{-1}(x)) = 4 - (4 - x)$$

$$= x$$

$\therefore f(x)$ and $f^{-1}(x)$ are inverses of one another.

1 mark

Sketch the graph of:

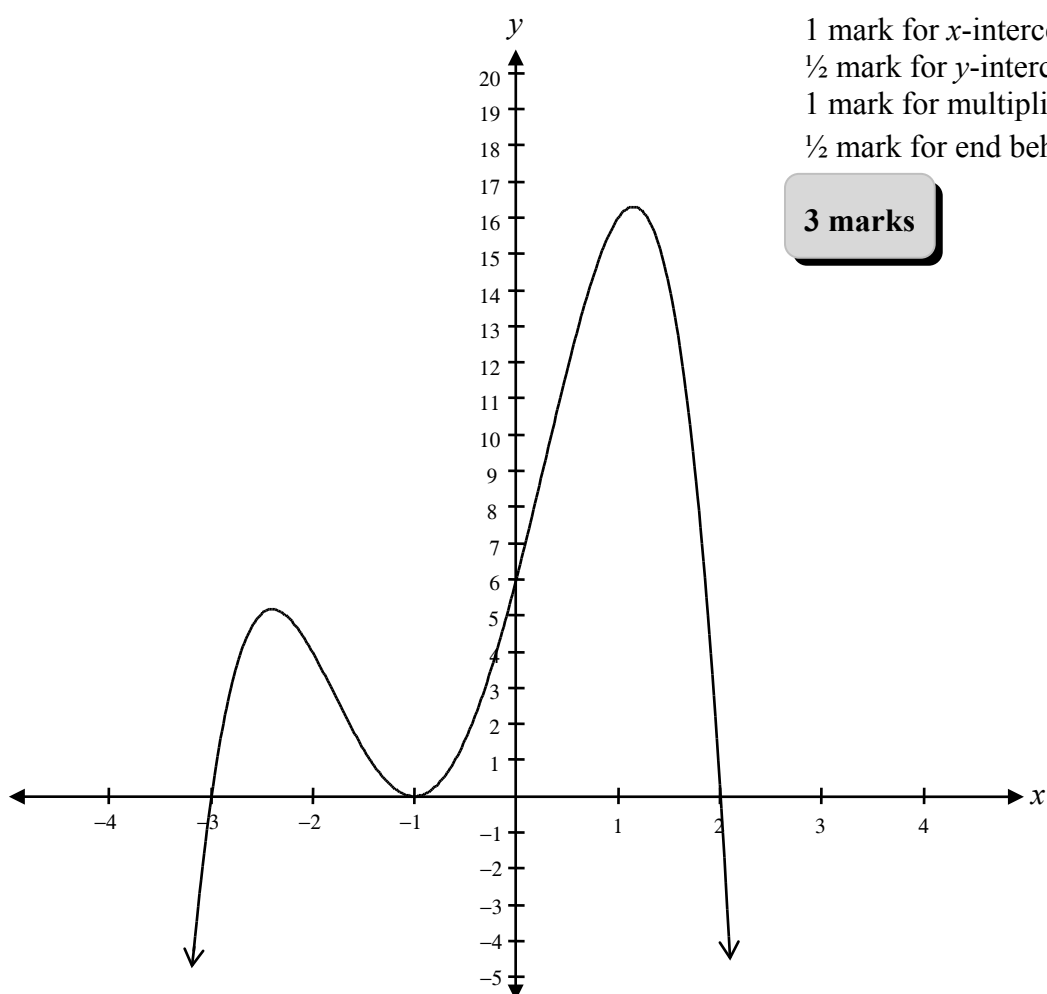
$$f(x) = (2 - x)(x + 3)(x + 1)^2$$

Label the x -intercepts and y -intercept.

Solution

x -intercepts: -3 , -1 , and 2

y -intercept: 6



1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

1 mark for multiplicity of 2 at $x = -1$ only

$\frac{1}{2}$ mark for end behaviour

3 marks

Which expression has a larger value?

$$\log_2 36 \text{ or } \log_3 80$$

Justify your answer.

Solution

Method 1

$$\log_2 36 \quad 2^? = 36 \begin{cases} 2^5 = 32 \\ 2^6 = 64 \end{cases} \approx 5.1$$

$$\log_3 80 \quad 3^? = 80 \begin{cases} 3^3 = 27 \\ 3^4 = 81 \end{cases} \approx 3.9$$

$\therefore \log_2 36$ is the larger value

1 mark for justification

1 mark

Method 2

$$\log_2 32 = 5 \therefore \log_2 36 \text{ is a little more than } 5$$

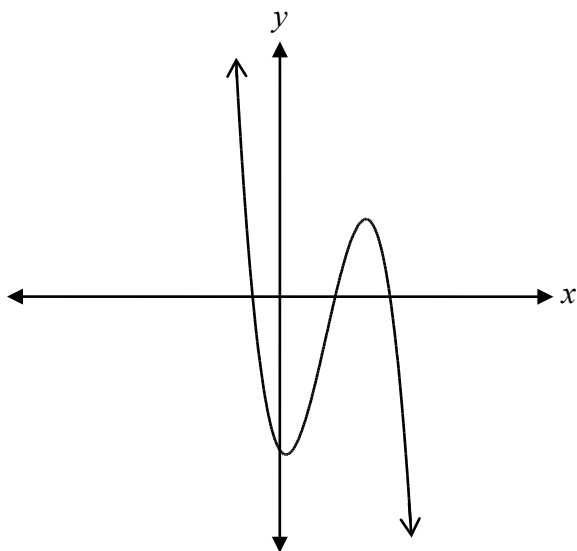
$$\log_3 81 = 4 \therefore \log_3 80 \text{ is a little less than } 4$$

$\therefore \log_2 36$ is the larger value

1 mark for justification

1 mark

The graph below represents the equation $y = ax^3 + 6x^2 + 5x - 10$.



What must be true about the value of a ? Explain your reasoning.

Solution

a is any negative number.

½ mark

Explanation with reference to end behaviour.

½ mark for explanation

1 mark

or

a cannot be zero.

½ mark

The graph is of a cubic function,
not a quadratic function.

½ mark for explanation

1 mark

The terminal arm of an angle θ , in standard position, intersects the unit circle in Quadrant IV at a point

$P\left(\frac{\sqrt{5}}{4}, y\right)$. Determine the value of $\sin \theta$.

Solution

Method 1

The point $P(\theta)$ on the unit circle has coordinates $(\cos \theta, \sin \theta)$.

$$\cos^2 \theta + \sin^2 \theta = 1$$

$\frac{1}{2}$ mark for showing $y = \sin \theta$

$$\left(\frac{\sqrt{5}}{4}\right)^2 + \sin^2 \theta = 1$$

$\frac{1}{2}$ mark for substitution

$$\sin^2 \theta = 1 - \frac{5}{16}$$

$$\sqrt{\sin^2 \theta} = \sqrt{\frac{11}{16}}$$

$$\sin \theta = \pm \frac{\sqrt{11}}{4}$$

$\frac{1}{2}$ mark for solving for $\sin \theta$

$$\sin \theta = -\frac{\sqrt{11}}{4}$$

$\frac{1}{2}$ mark for a negative $\sin \theta$ value in Quadrant IV

2 marks

Method 2

$$(\sqrt{5})^2 + y^2 = 4^2$$

$\frac{1}{2}$ mark for substitution

$$5 + y^2 = 16$$

$$y^2 = 11$$

$$y = \pm \sqrt{11}$$

$\frac{1}{2}$ mark for solving for y

$$\sin \theta = -\frac{\sqrt{11}}{4}$$

$\frac{1}{2}$ mark for using the value of y to find the value of $\sin \theta$

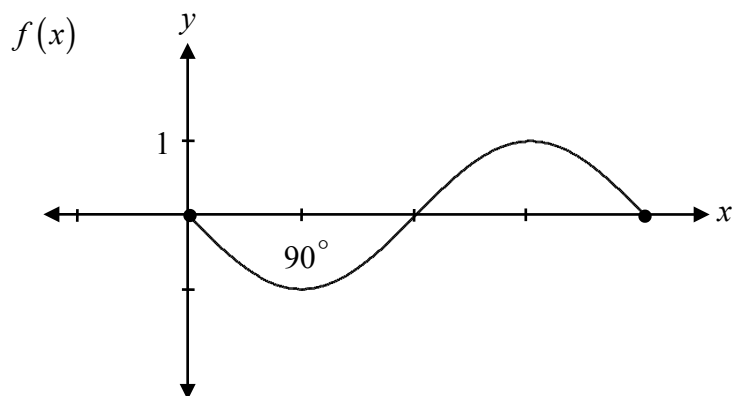
$\frac{1}{2}$ mark for a negative $\sin \theta$ value in Quadrant IV

2 marks

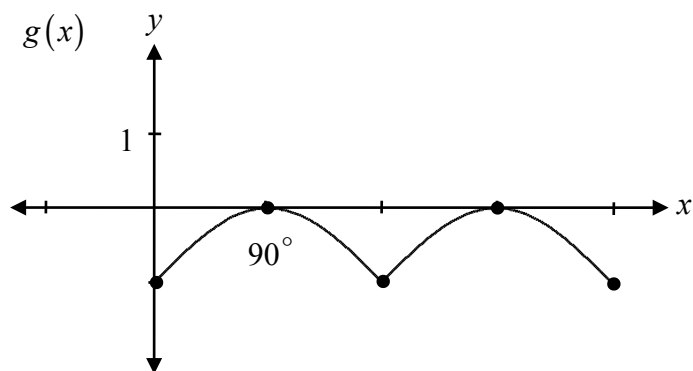
Question 33

R1, R2

Given the sinusoidal function $f(x)$ below, sketch the graph of $g(x) = |f(x)| - 1$.



Solution



1 mark for absolute value
1 mark for vertical shift

2 marks

Question 34

R14

The graph of a rational function, $f(x)$, has a point of discontinuity when $x = 2$ and an asymptote when $x = 4$. Write a possible equation for $f(x)$.

Solution

A possible equation is:

$$f(x) = \frac{x-2}{(x-2)(x-4)}$$

1 mark for $\frac{x-2}{x-2}$ (point of discontinuity when $x = 2$)

1 mark for $x-4$ in denominator (asymptote when $x = 4$)

2 marks

Question 35

R11

Given that $(x-1)$ is one of the factors, express $x^3 - 57x + 56$ as a product of factors.

Solution

$$\begin{array}{r|rrrr} 1 & 1 & 0 & -57 & 56 \\ & \downarrow & 1 & 1 & -56 \\ \hline & 1 & 1 & -56 & 0 \end{array}$$

$\frac{1}{2}$ mark for $x = 1$

1 mark for synthetic division (or for any other equivalent strategy)

$$(x-1)(x^2 + x - 56)$$

$\frac{1}{2}$ mark for consistent factors

or

$$(x-1)(x+8)(x-7)$$

2 marks

Give an example using values for A and B , in degrees or radians, to verify that $\cos(A + B) = \cos A + \cos B$ is **not** an identity.

Solution

Method 1

Let $A = 45^\circ$ and $B = 90^\circ$.

LHS	RHS
$\cos(45^\circ + 90^\circ)$	$\cos 45^\circ + \cos 90^\circ$
$\cos(135^\circ)$	$\cos 45^\circ + \cos 90^\circ$
$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2} + 0$
$-\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$

1 mark for simplification of $\cos(A + B)$

1 mark for simplification of $\cos A + \cos B$

2 marks

$\text{LHS} \neq \text{RHS} \therefore \cos(A + B) = \cos A + \cos B$ is not an identity.

Method 2

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

Let $A = 60^\circ$ and $B = 30^\circ$.

$$\cos(60^\circ + 30^\circ) = \cos 60^\circ \cos 30^\circ - \sin 60^\circ \sin 30^\circ$$

$$= \left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right)$$

$$= \frac{\sqrt{3} - \sqrt{3}}{4}$$

$$= 0$$

1 mark for simplification of $\cos(A + B)$

$$\cos A + \cos B = \cos 60^\circ + \cos 30^\circ$$

$$= \frac{1}{2} + \frac{\sqrt{3}}{2}$$

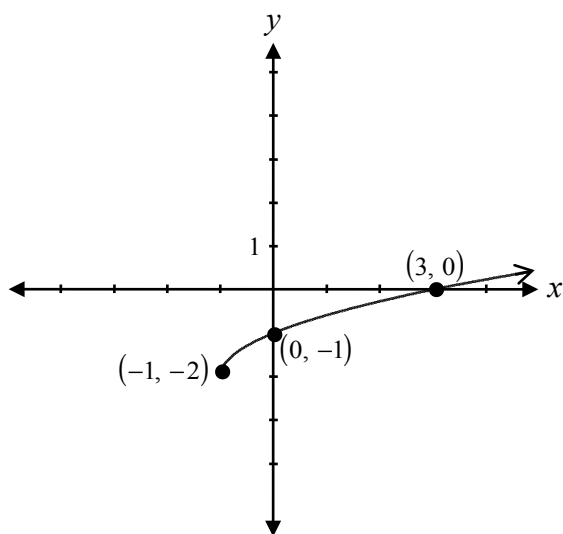
$$= \frac{1 + \sqrt{3}}{2}$$

1 mark for simplification of $\cos A + \cos B$

2 marks

These two solutions are not equal $\therefore \cos(A + B) = \cos A + \cos B$ is not an identity.

Sketch the graph of $y = \sqrt{x+1} - 2$ and verify that the value of the x -intercept is the same as the solution to the equation $\sqrt{x+1} - 2 = 0$.

Solution

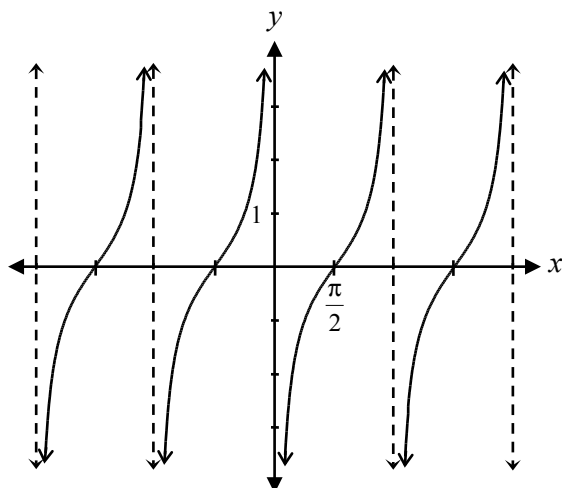
1 mark for general shape
 $\frac{1}{2}$ mark for horizontal shift
 $\frac{1}{2}$ mark for vertical shift

$$\begin{array}{ll} \sqrt{x+1} = 2 & \sqrt{x+1} - 2 = 0 \\ (\sqrt{x+1})^2 = (2)^2 & \text{or} \quad \sqrt{3+1} - 2 = 0 \\ x+1 = 4 & \sqrt{4} - 2 = 0 \\ x = 3 & 0 = 0 \end{array}$$

1 mark for verification

3 marks

Mohamed is asked to sketch the graph of $y = \tan x$.
His graph is shown below.



Explain why his graph is incorrect.

Solution

The graph of $y = \tan x$ should have zeros at $k\pi$, $k \in \mathbb{I}$.

or

1 mark for explanation

1 mark

The graph of $y = \tan x$ should have asymptotes at $(2k + 1)\frac{\pi}{2}$ or $\frac{\pi}{2} + k\pi$, $k \in \mathbb{I}$.

or

Mohamed sketched the incorrect graph. He sketched the graph of $y = \tan\left(x - \frac{\pi}{2}\right)$.

On the interval $0 \leq \theta < 2\pi$, identify the non-permissible values of θ for the trigonometric identity:

$$\tan \theta = \frac{1}{\cot \theta}$$

Solution

$$\frac{\sin \theta}{\cos \theta} = \frac{1}{\frac{\cos \theta}{\sin \theta}}$$

$\therefore$ the above identity is non-permissible when $\cos \theta = 0$ or $\sin \theta = 0$.

1 mark for identifying non-permissible values
($\frac{1}{2}$ mark for $\cos \theta = 0$, $\frac{1}{2}$ mark for $\sin \theta = 0$)

$$\cos \theta \neq 0$$

$$\theta \neq \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\sin \theta \neq 0$$

$$\theta \neq 0, \pi$$

1 mark for solving for θ ($\frac{1}{2}$ mark for each solution set)

2 marks

$$\theta \neq 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$$

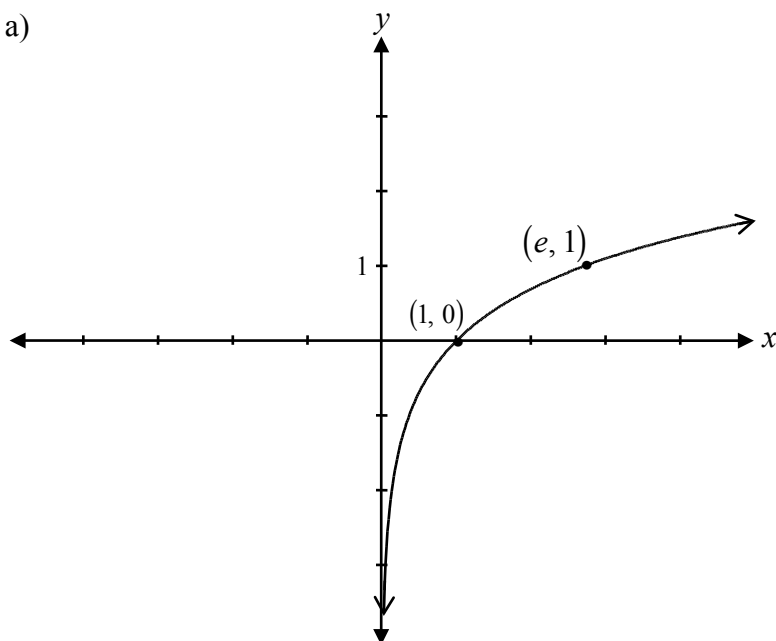
Question 40

a) R9 b) R2, R5

- a) Sketch the graph of $y = \ln(x)$.
 b) Sketch the graph of $y = -\ln(x - 2)$.

Solutions

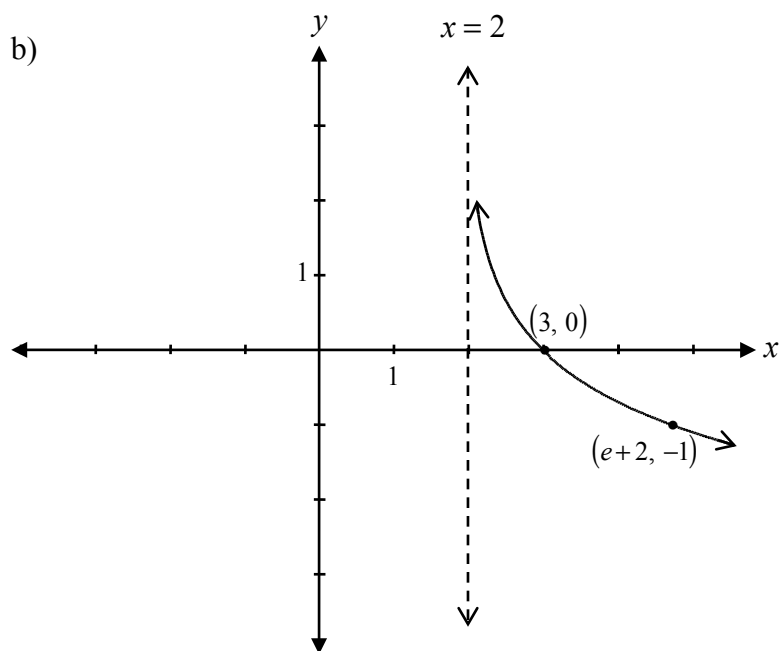
a)



$\frac{1}{2}$ mark for increasing logarithmic function
 $\frac{1}{2}$ mark for x -intercept at $(1, 0)$
 $\frac{1}{2}$ mark for consistent point on logarithmic function
 $\frac{1}{2}$ mark for vertical asymptotic behaviour

2 marks

b)



1 mark for reflection in x -axis
 1 mark for horizontal shift

2 marks

Given $f(x) = \sqrt{x-2}$ and $g(x) = 3x$, write the equation for $h(x) = f(g(x))$.

What are the restrictions on the domain of $h(x)$?

Explain your reasoning.

Solution

$$h(x) = \sqrt{3x-2}$$

1 mark for $h(x) = f(g(x))$

$$3x - 2 \geq 0$$

$$3x \geq 2$$

$$x \geq \frac{2}{3}$$

$\frac{1}{2}$ mark for identifying restriction

Since we cannot find a square root of a negative number, there is a restriction

$\frac{1}{2}$ mark for explanation

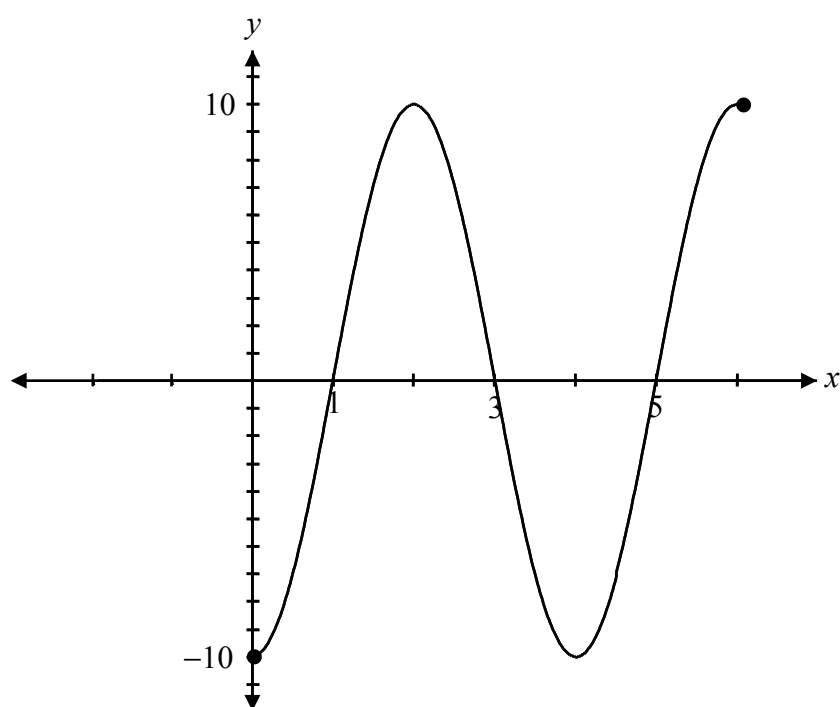
on the domain, $x \geq \frac{2}{3}$.

2 marks

Sketch the graph of $y = 10 \cos\left[\frac{\pi}{2}(x - 2)\right]$ over the interval $[0, 6]$.

Solution

$$\text{period} = \frac{2\pi}{\frac{\pi}{2}} = 4$$



1 mark for amplitude
1 mark for period
1 mark for horizontal shift

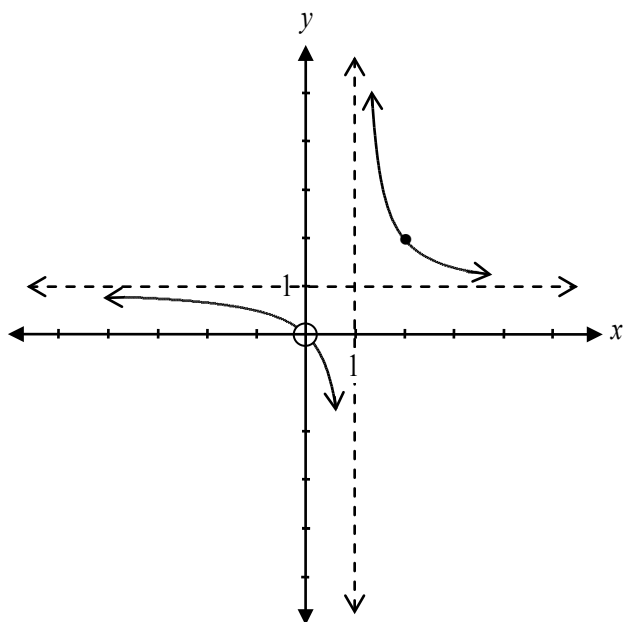
3 marks

Note(s):

- deduct ½ mark if the interval $[0, 6]$ is not completely sketched

Sketch the graph of the function $f(x) = \frac{x^2}{x^2 - x}$.

Solution



$$f(x) = \frac{x^2}{x(x-1)}$$

$$= \frac{x}{x-1} \text{ with a point of discontinuity where } x = 0$$

$$\text{point of discontinuity: } f(0) = \frac{0}{0-1} = 0$$

$\therefore$ there is a point of discontinuity at $(0, 0)$.

divide:

$$\begin{array}{r} 1 \\ x-1 \overline{) x+0} \\ \underline{x-1} \\ 1 \end{array}$$

or

$$\begin{aligned} f(x) &= \frac{x}{x-1} \\ &= \frac{x-1+1}{x-1} \\ &= 1 + \frac{1}{x-1} \end{aligned}$$

$$\therefore f(x) = \frac{1}{x-1} + 1$$

$\therefore$ horizontal asymptote at $y = 1$

$\therefore$ vertical asymptote at $x = 1$

1 mark for vertical asymptote at $x = 1$

1 mark for horizontal asymptote at $y = 1$

1 mark for point of discontinuity at $(0, 0)$ or a point of discontinuity consistent with graph

$\frac{1}{2}$ mark for graph left of vertical asymptote

$\frac{1}{2}$ mark for graph right of vertical asymptote

4 marks

Is $(x - 3)$ a factor of $x^4 - x^3 - 3x^2 + x - 1$?

Justify your answer.

Solution

Method 1

$$x = 3$$

$$\therefore (3)^4 - (3)^3 - 3(3)^2 + (3) - 1 = 81 - 27 - 27 + 3 - 1 = 29$$

The remainder does not equal zero,
therefore $(x - 3)$ is not a factor.

$\frac{1}{2}$ mark for $x = 3$

1 mark for remainder theorem

$\frac{1}{2}$ mark for explanation

2 marks

Method 2

<u>3</u>	1	-1	-3	1	-1
	↓	3	6	9	30
	1	2	3	10	29

The remainder does not equal zero,
therefore $(x - 3)$ is not a factor.

$\frac{1}{2}$ mark for $x = 3$

1 mark for synthetic division

$\frac{1}{2}$ mark for explanation

2 marks

Given $f(x) = x - 1$ and $g(x) = x^2$, write the equation of $y = f(g(x))$ and sketch the graph.

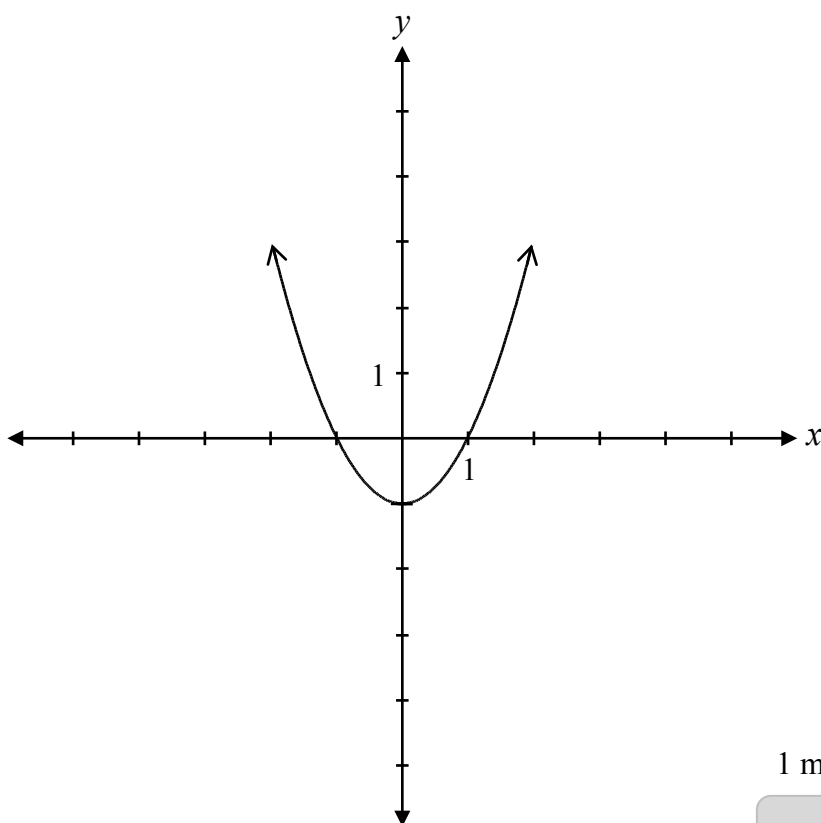
Solution

$$f(g(x)) = x^2 - 1$$

1 mark for composition

or

$$y = x^2 - 1$$



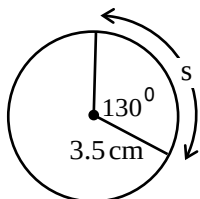
1 mark for consistent graph

2 marks

Question 1

T1

Use the information in the diagram to determine the value of the arc length “s”.



Solution

$$130^\circ \times \frac{\pi}{180^\circ} = \frac{13\pi}{18}$$

1 mark for conversion

$$s = \theta r$$

$$s = \frac{13\pi}{18}(3.5)$$

1 mark for substitution

$$s = 7.941 \text{ 248}$$

$$s = 7.941 \text{ cm}$$

2 marks

Question 2

T5

Solve the following equation over the interval $[0, 2\pi)$.

$$\tan^2 \theta + 2.8 \tan \theta + 1.96 = 0$$

Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ for $ax^2 + bx + c = 0$.

Solution

$$\tan \theta = \frac{-2.8 \pm \sqrt{(2.8)^2 - 4(1)(1.96)}}{2(1)}$$

½ mark for substitution

$$\tan \theta = \frac{-2.8 \pm 0}{2}$$

$$\tan \theta = -1.4$$

½ mark for solving for $\tan \theta$

$$\theta_r = \tan^{-1}(1.4)$$

$$= 0.950\ 546$$

$$\theta = 2.191 \qquad \theta = 5.333$$

1 mark (½ mark for each value of θ)

2 marks

Question 3

R10

Determine how many monthly investments of \$50 would have to be deposited into a savings account that pays 3% annual interest, compounded monthly, for the account's future value to be \$50,000.

Use the formula:

$$FV = \frac{R \left[(1+i)^n - 1 \right]}{i}$$

where FV = the future value

R = the investment amount

$$i = \frac{\text{the annual interest rate}}{\text{the number of compounding periods per year}}$$

n = the number of investments

Express your answer as a whole number.

Solution

$$50\,000 = \frac{50 \left[\left(1 + \frac{0.03}{12} \right)^n - 1 \right]}{\frac{0.03}{12}}$$

½ mark for substitution

$$50\,000 = \frac{50 \left[(1 + 0.0025)^n - 1 \right]}{0.0025}$$

$$50\,000 = 20\,000 (1.0025^n - 1)$$

$$2.5 = 1.0025^n - 1$$

$$3.5 = 1.0025^n$$

½ mark for simplification

$$\log 3.5 = \log 1.0025^n$$

½ mark for applying logarithms

$$\log 3.5 = n \log 1.0025$$

1 mark for power rule

$$n = \frac{\log 3.5}{\log 1.0025}$$

$$n = 501.73$$

½ mark for solving for n

∴ 502 monthly investments are needed.

3 marks

Question 4

P1

There are 5 men and 4 women to be seated in a row.

How many arrangements are possible if two men must sit at the beginning of the row and two men must sit at the end of the row?

Solution

$$\frac{5}{m} \times \frac{4}{m} \times \frac{5!}{1} \times \frac{3}{m} \times \frac{2}{m} = 14\,400$$

1 mark for correctly restricting men at beginning and end of the row

or

1 mark for 5!

$$\frac{5}{m} \times \frac{4}{m} \times \frac{5}{1} \times \frac{4}{1} \times \frac{3}{1} \times \frac{2}{1} \times \frac{1}{1} \times \frac{3}{m} \times \frac{2}{m} = 14\,400$$

2 marks

- a) In the binomial expansion of $\left(\frac{3}{x^2} - 4x^5\right)^8$, determine the 3rd term.
- b) In the binomial expansion of $\left(\frac{3}{x^2} - 4x^5\right)^n$, the 6th term contains x^{25} .
Solve for n .

Solution

a)
$$t_3 = {}_8C_2 \left(\frac{3}{x^2}\right)^{8-2} (-4x^5)^2$$

2 marks (1 mark for ${}_8C_2$, $\frac{1}{2}$ mark for each consistent factor)

$$t_3 = 28 \left(\frac{3^6}{x^{12}}\right) \left(\frac{16x^{10}}{1}\right)$$

$$= \frac{28}{1} \times \frac{729}{x^{12}} \times \frac{16x^{10}}{1}$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark for exponents)

$$= \frac{326\,592}{x^2}$$

3 marks

b)
$$x^{25} = (x^{-2})^{n-5} (x^5)^5$$

1 mark for substitution

$$x^{25} = x^{-2n+10+25}$$

$$25 = -2n + 35$$

1 mark for equating exponents

$$-10 = -2n$$

$$5 = n$$

2 marks

Given the following two functions, $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 1$, evaluate $g(f(3))$.

Solution**Method 1**

$$\begin{aligned} f(3) &= \sqrt{3-1} \\ &= \sqrt{2} \end{aligned}$$

½ mark for $f(3)$

$$\begin{aligned} g(\sqrt{2}) &= (\sqrt{2})^2 + 1 \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

½ mark for consistent value of $g(f(3))$

1 mark

Method 2

$$\begin{aligned} g(f(x)) &= (\sqrt{x-1})^2 + 1 \\ &= x - 1 + 1 \\ &= x \end{aligned}$$

½ mark for $g(f(x))$

$$g(f(3)) = 3$$

½ mark for evaluating $g(f(3))$

1 mark

If θ terminates in quadrant II and $\csc \theta = \frac{3}{2}$, determine the exact value of $\tan \theta$.

Solution

$$\csc \theta = \frac{r}{y}$$

$$x^2 + y^2 = r^2$$

$$x^2 + 2^2 = 3^2 \quad \frac{1}{2} \text{ mark for identifying } y = 2, r = 3$$

$$x^2 = 5$$

$$x = \pm\sqrt{5} \quad \frac{1}{2} \text{ mark for solving for } x$$

$$\tan \theta = -\frac{2}{\sqrt{5}}$$

1 mark for $\tan \theta$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

2 marks

Note(s):

§ accept any of the following values for x : $x = \pm\sqrt{5}$, $x = \sqrt{5}$, or $x = -\sqrt{5}$

- a) Determine the remainder when $x^4 - 3x^2 + 1$ is divided by $x + 2$.
 b) Is $x + 2$ a factor of $x^4 - 3x^2 + 1$? Explain your reasoning.

Solution

a) $(-2)^4 - 3(-2)^2 + 1$

$$16 - 12 + 1$$

$$5$$

1 mark for remainder theorem

or

-2	1	0	-3	0	1
		-2	4	-2	4
	1	-2	1	-2	5

or

1 mark for synthetic division

1 mark

The remainder is 5.

- b) For $x + 2$ to be a factor of $x^4 - 3x^2 + 1$, the remainder must be 0. Since the remainder is 5, $x + 2$ is not a factor.

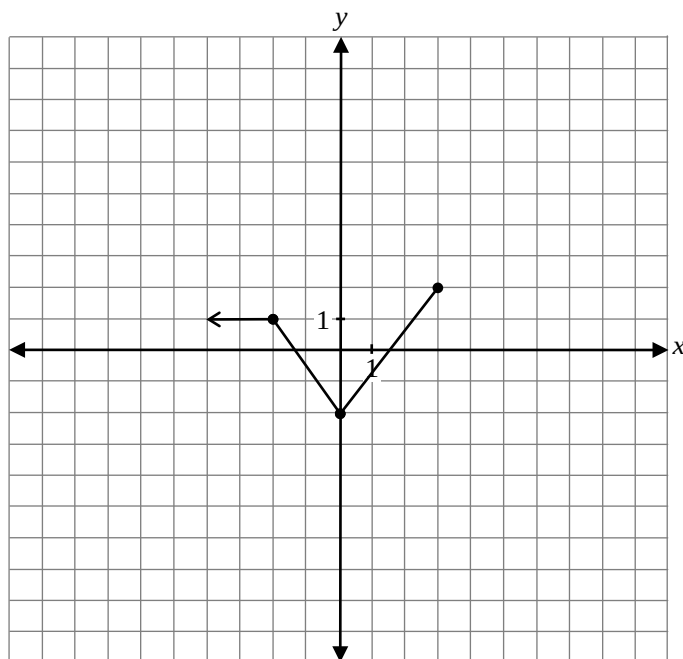
1 mark for explanation

1 mark

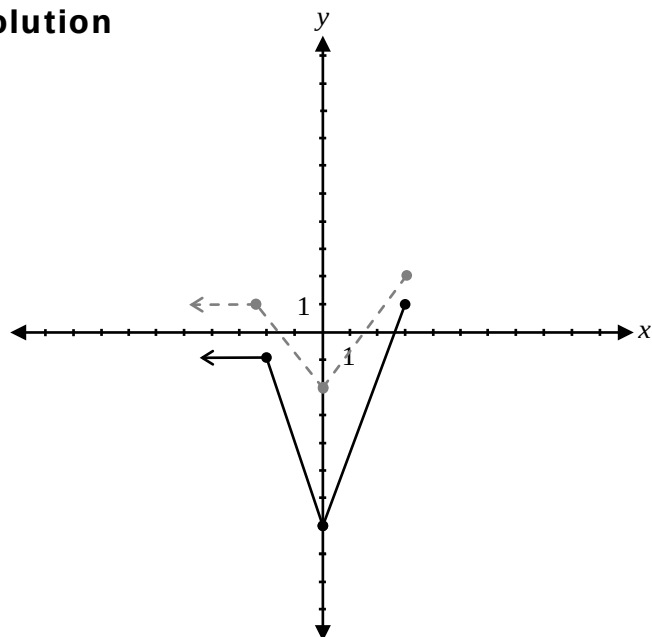
Question 9

R2, R3

Given the graph of $y = f(x)$ below, sketch the graph of $y = 2f(x) - 3$.



Solution



1 mark for vertical stretch
1 mark for vertical shift

2 marks

Determine one possible restriction for the domain of $f(x) = (x - 1)^2$ so that the inverse of $f(x)$ is a function.

Solution

$$x \geq 1$$

or

$$x < 1$$

1 mark

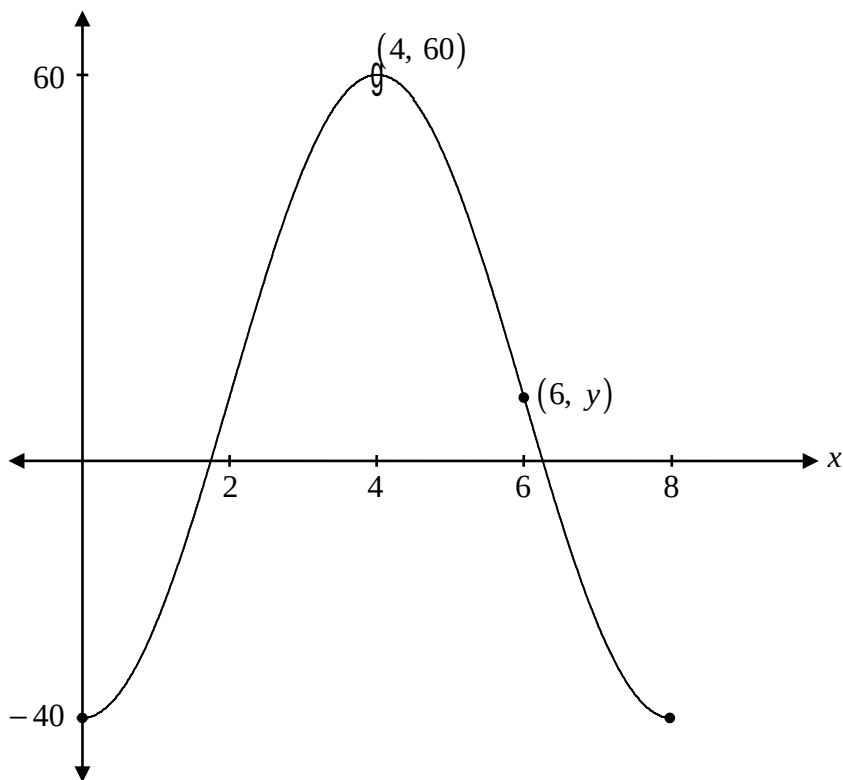
Note(s):

§ Many solutions are possible. Any solutions which restrict $f(x)$ to a one-to-one function are correct.

Question 11

T4

Using the graph of the sinusoidal function below, find the value of y in the point $(6, y)$.



Solution

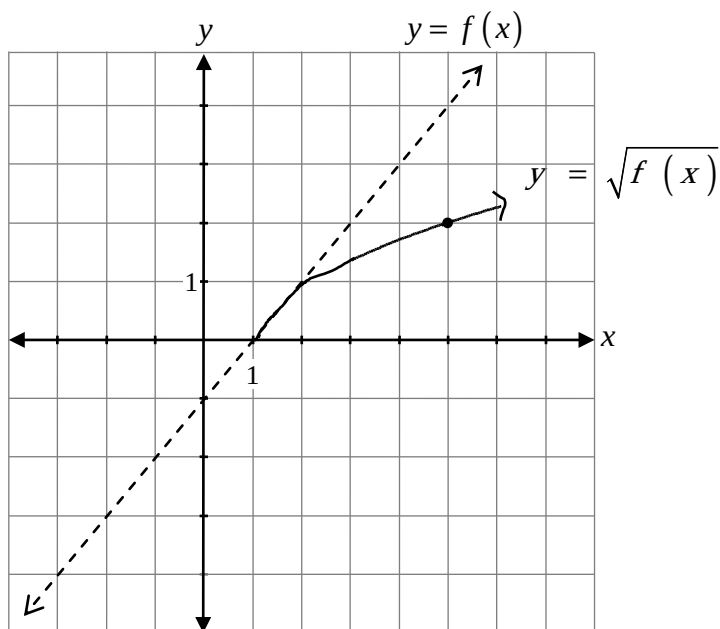
$$y = 10$$

1 mark

Billy was given the graph of $y = f(x)$.

He was asked to sketch the graph of $y = \sqrt{f(x)}$.

His answer is given on the graph below.



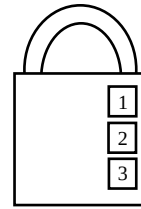
Explain the error Billy made when sketching the graph of $y = \sqrt{f(x)}$.

Solution

Billy's graph should be above the line $y = f(x)$ over the interval where x is between 1 and 2.

1 mark

Explain why a locker combination should really be called a locker permutation.

**Solution**

The order of the locker combination matters; therefore, it is a permutation.

1 mark

Question 14

R5

The graph of $f(x) = x^2 + 4$ is reflected over the x -axis.
Write the equation of the new function.

$$y = \underline{\hspace{2cm}}$$

Solution

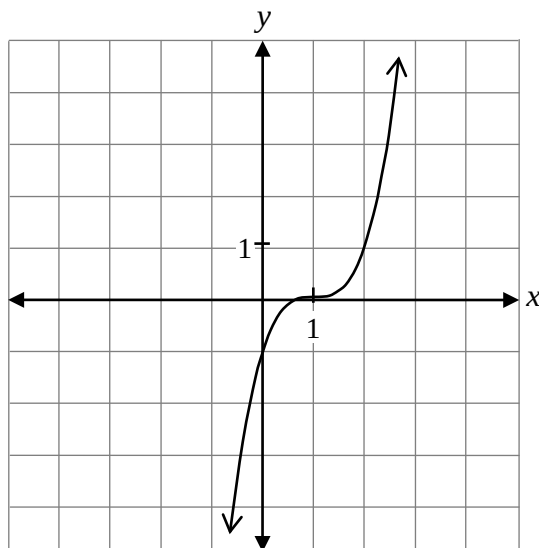
$$y = -x^2 - 4$$

1 mark

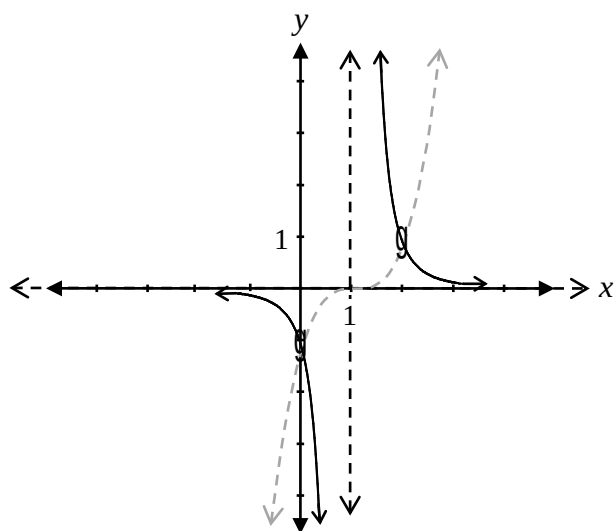
Question 15

R1

Given the graph of $y = f(x)$ below, sketch the graph of $y = \frac{1}{f(x)}$.



Solution



1 mark for vertical asymptote at $x = 1$
 $\frac{1}{2}$ mark for graph left of vertical asymptote
 $\frac{1}{2}$ mark for graph right of vertical asymptote

2 marks

Divide $(x^3 - 5x - 4)$ by $(x + 1)$.

Solution

Method 1

$$\begin{array}{r|rrrrr} -1 & 1 & 0 & -5 & -4 & \\ & & -1 & 1 & 4 & \\ \hline & 1 & -1 & -4 & 0 & \end{array}$$

$$x^2 - x - 4$$

1 mark for set-up of synthetic division using addition

1 mark for quotient

2 marks

Method 2

$$\begin{array}{r} x^2 - x - 4 \\ x+1 \overline{) x^3 + 0x^2 - 5x - 4} \\ \underline{-(x^3 + x^2)} \\ -x^2 - 5x \\ \underline{-(-x^2 - x)} \\ -4x - 4 \\ \underline{-(-4x - 4)} \\ 0 \end{array}$$

1 mark for set-up of long division

1 mark for quotient

2 marks

Method 3

$$\begin{array}{r|rrrrr} 1 & 1 & 0 & -5 & -4 & \\ & & 1 & -1 & -4 & \\ \hline & 1 & -1 & -4 & 0 & \end{array}$$

$$x^2 - x - 4$$

1 mark for set-up of synthetic division using subtraction

1 mark for quotient

2 marks

Question 17

P4

You are given the following row of Pascal's Triangle.

1 7 21 35 35 21 7 1

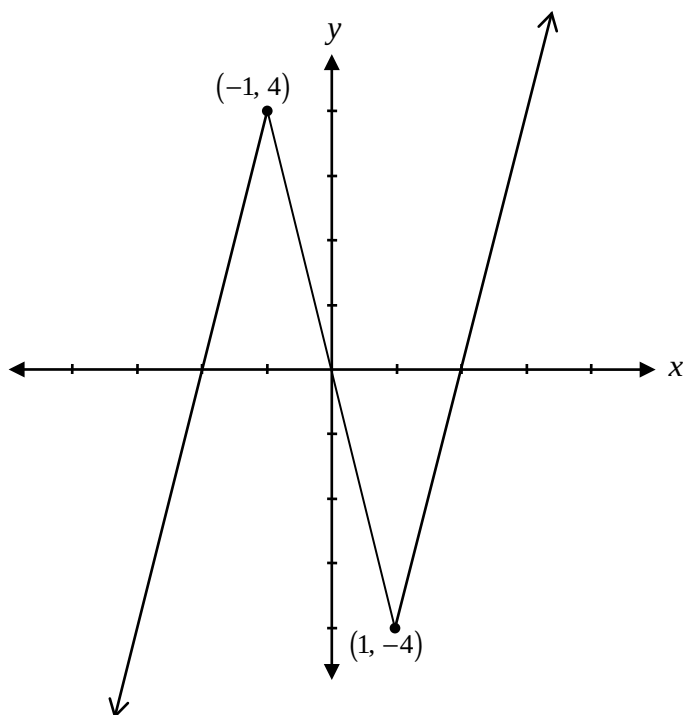
Determine the values of the next row.

Solution

1 8 28 56 70 56 28 8 1

1 mark

Given the graph of $y = f(x)$ below, state the domain and range of $y = \sqrt{f(x)}$.



Solution

Domain: $\{x \in \mathbb{R} \mid -2 \leq x \leq 0 \text{ or } x \geq 2\}$

or

$[-2, 0] \cup [2, \infty)$

1 mark for domain ($\frac{1}{2}$ mark for $-2 \leq x \leq 0$, $\frac{1}{2}$ mark for $x \geq 2$)

Range: $\{y \in \mathbb{R} \mid 0 \leq y\}$

or

$[0, \infty)$

1 mark for range

2 marks

Prove the identity below for all permissible values of θ :

$$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos 2\theta$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$	$\cos 2\theta$
$\frac{1 - \tan^2 \theta}{\sec^2 \theta}$	
$\frac{1}{\sec^2 \theta} - \frac{\tan^2 \theta}{\sec^2 \theta}$	
$\cos^2 \theta - \frac{\sin^2 \theta}{\cos^2 \theta}$	
$\cos^2 \theta - \sin^2 \theta$	
$\cos 2\theta$	

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Method 2

Left-Hand Side	Right-Hand Side
$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$	$\cos 2\theta$
$1 - \frac{\sin^2 \theta}{\cos^2 \theta}$	
$1 + \frac{\sin^2 \theta}{\cos^2 \theta}$	
$\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta}$	
$\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta}$	
$\cos^2 \theta - \sin^2 \theta$	
$\cos 2\theta$	

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Method 3

Left-Hand Side	Right-Hand Side
$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$ $\frac{1 - (\sec^2 \theta - 1)}{\sec^2 \theta}$ $\frac{2 - \sec^2 \theta}{\sec^2 \theta}$ $\frac{2 - \frac{1}{\cos^2 \theta}}{\frac{1}{\cos^2 \theta}}$ $\frac{2 \cos^2 \theta - 1}{\frac{1}{\cos^2 \theta}}$ $2 \cos^2 \theta - 1$ $\cos 2\theta$	$\cos 2\theta$

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 25

T5

The general solution to the equation $\cos \theta = -\frac{1}{2}$ is:

a) $\left. \begin{array}{l} \theta = \frac{\pi}{3} + 2\pi k \\ \theta = \frac{5\pi}{3} + 2\pi k \end{array} \right\} \text{ where } k \in \mathbb{I}$

b) $\left. \begin{array}{l} \theta = \frac{\pi}{3} + \pi k \\ \theta = \frac{5\pi}{3} + \pi k \end{array} \right\} \text{ where } k \in \mathbb{I}$

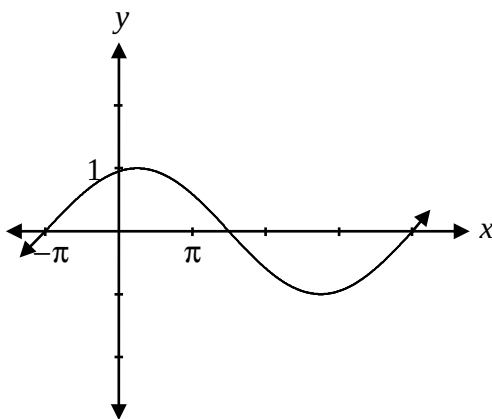
c) $\left. \begin{array}{l} \theta = \frac{2\pi}{3} + 2\pi k \\ \theta = \frac{4\pi}{3} + 2\pi k \end{array} \right\} \text{ where } k \in \mathbb{I}$

d) $\left. \begin{array}{l} \theta = \frac{2\pi}{3} + \pi k \\ \theta = \frac{4\pi}{3} + \pi k \end{array} \right\} \text{ where } k \in \mathbb{I}$

Question 26

T4

If the equation $y = \sin(b(x + \pi))$ is represented by the following graph, what is the value of b ?



a) $\frac{2}{5}$

b) $\frac{5}{2}$

c) $\frac{2\pi}{5}$

d) 5π

Determine one positive and one negative coterminal angle with the angle $\frac{5\pi}{6}$.

Solution

$$\frac{17\pi}{6} \text{ and } -\frac{7\pi}{6}$$

$\frac{1}{2}$ mark for one positive coterminal angle
 $\frac{1}{2}$ mark for one negative coterminal angle

1 mark

Note(s):

§ Other answers are possible.

§ Answers in degrees are acceptable.

Evaluate:

$$\left(\sin \frac{11\pi}{3}\right)\left(\sec \frac{11\pi}{6}\right)$$

Solution

$$\left(-\frac{\sqrt{3}}{2}\right)\left(\frac{2}{\sqrt{3}}\right) = -1$$

1 mark for $\sin \frac{11\pi}{3}$ (½ mark for quadrant, ½ mark for value)

1 mark for $\sec \frac{11\pi}{6}$ (½ mark for quadrant, ½ mark for value)

2 marks

Given the equation $2\sin^2 \theta - 3\sin \theta + 1 = 0$, verify that $\theta = \frac{\pi}{2}$ is a solution.

Solution

$$\text{Left-hand side} = 2\left(\sin \frac{\pi}{2}\right)^2 - 3\left(\sin \frac{\pi}{2}\right) + 1$$

$$= 2(1)^2 - 3(1) + 1$$

$$= 0$$

$$= \text{Right-hand side}$$

1 mark for verification

1 mark

Using the laws of logarithms, expand:

$$\log_a \left(\frac{xy}{z} \right)$$

Solution

$$\log_a x + \log_a y - \log_a z$$

1 mark for product rule

1 mark for quotient rule

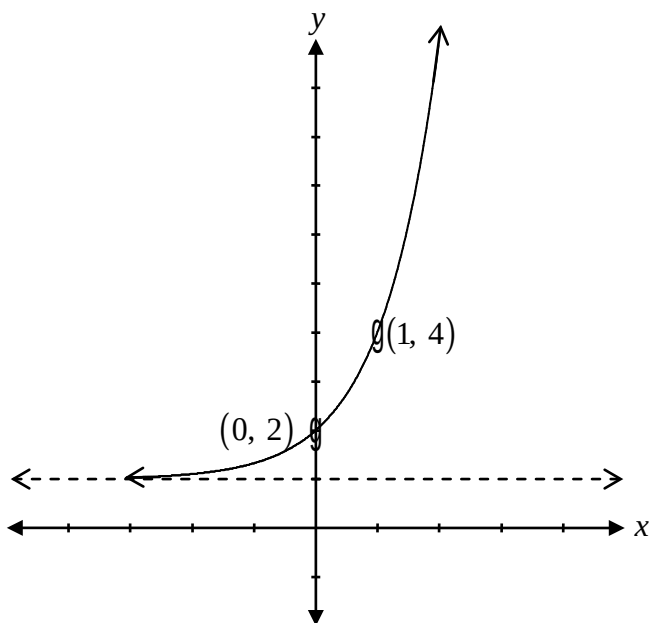
2 marks

a) Sketch the graph of $f(x) = 3^x + 1$.

b) Sketch the graph of $f^{-1}(x)$.

Solution

a)



½ mark for increasing exponential function

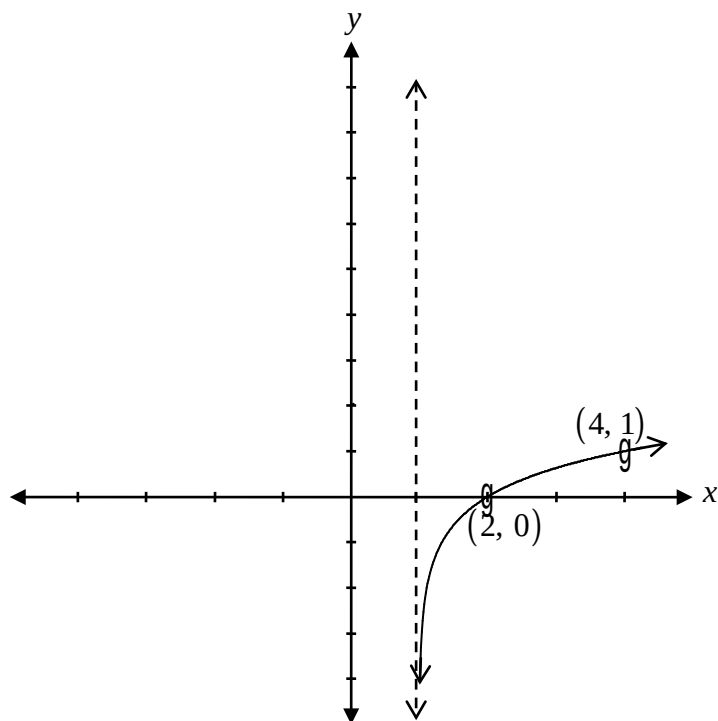
½ mark for y-intercept at (0, 2)

½ mark for asymptote at $y = 1$

½ mark for consistent point on exponential function

2 marks

b)



1 mark for consistent graph of the inverse

1 mark

Determine the x-intercept and y-intercept of $y = \log_2(x + 4) - 1$.

Solution

Substitute x with 0.

$$y = \log_2 4 - 1$$

$$y = 2 - 1$$

$$y = 1$$

$\therefore$ y-intercept is 1

½ mark for evaluating logarithm

½ mark for consistent value of y

Substitute y with 0.

$$0 = \log_2(x + 4) - 1$$

$$1 = \log_2(x + 4)$$

$$2 = x + 4$$

$$-2 = x$$

$\therefore$ x-intercept is -2

½ mark for exponential form

½ mark for consistent value of x

2 marks

Note(s):

§ award ½ mark if student substitutes x with 0 to find the y-intercept and y with 0 to find the x-intercept

Explain the error that was made when solving the following equation:

$$\sin 2\theta = \cos \theta, \text{ where } \theta \in \mathbb{R}$$

$$\begin{aligned} \sin 2\theta &= \cos \theta \\ 2\sin \theta \cos \theta &= \cos \theta \\ \frac{2\sin \theta \cos \theta}{\cos \theta} &= \frac{\cos \theta}{\cos \theta} \\ 2\sin \theta &= 1 \\ \sin \theta &= \frac{1}{2} \\ \theta &= \frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi, k \in \mathbb{I} \end{aligned}$$

Solution

The student divided by $\cos \theta$ instead of factoring out $\cos \theta$.

or

1 mark

There are two more solutions that come from the equation $\cos \theta = 0$.

or

The student cannot divide both sides by $\cos \theta$ since $\cos \theta$ could equal 0.

Question 37

R1

Given $f(x) = x^2 - 2x - 3$ and $g(x) = x + 1$:

- Write the equation of $y = f(g(x))$.
- Sketch the graph of $y = f(g(x))$.

Solution

a) $f(g(x)) = (x+1)^2 - 2(x+1) - 3$

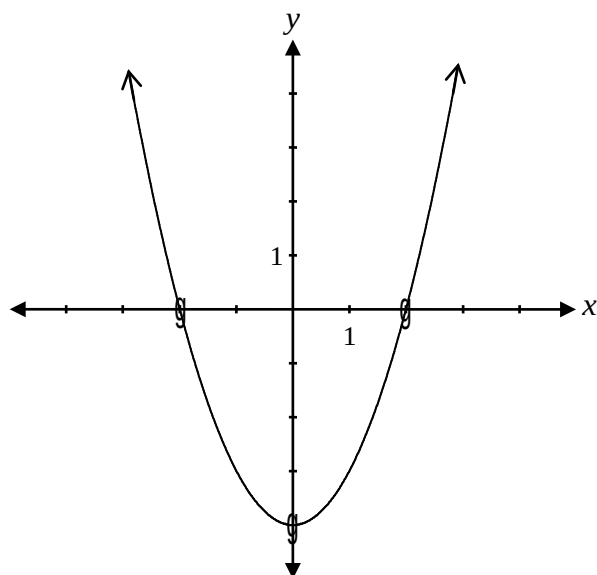
$$= x^2 + 2x + 1 - 2x - 2 - 3$$

$$= x^2 - 4$$

1 mark for composition

1 mark

b)



1 mark for graph (½ mark for x-intercepts, ½ mark for y-intercept)

1 mark

Is the point $\left(\frac{3}{4}, -\frac{\sqrt{3}}{4}\right)$ on the unit circle?

Justify your answer.

Solution

$$x^2 + y^2 = 1$$

$$\begin{aligned}\text{Left-hand side} &= \left(\frac{3}{4}\right)^2 + \left(-\frac{\sqrt{3}}{4}\right)^2 \\ &= \frac{9}{16} + \frac{3}{16} \\ &= \frac{12}{16}\end{aligned}$$

½ mark for substitution

$$\frac{12}{16} \neq 1 \therefore \text{not on unit circle}$$

½ mark for justification

1 mark

Explain why the equation $\sec \theta = \frac{1}{4}$ has no solution.

Solution

The value of $\sec \theta$ cannot be between -1 and 1 .

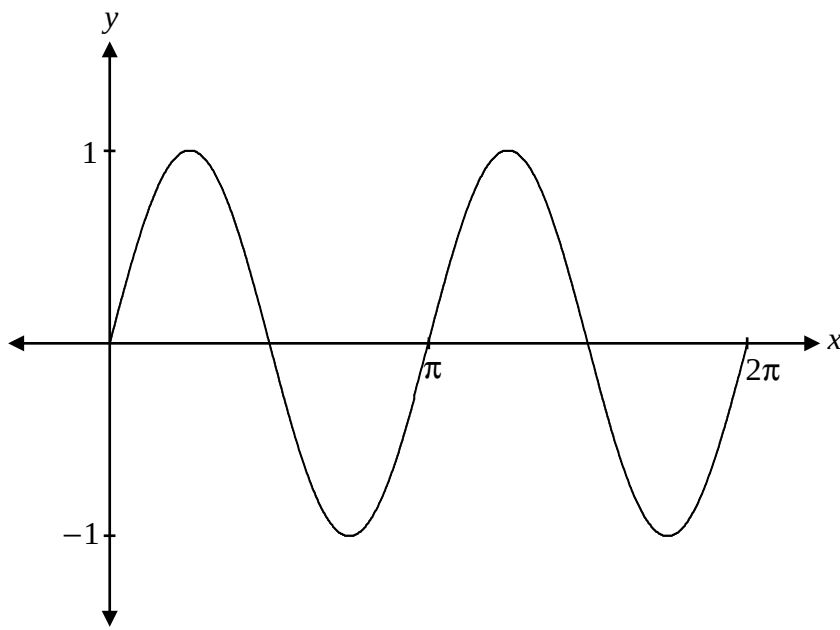
1 mark

or

$\cos \theta$ cannot be greater than 1 .

The graph of $y = \sin 2x$ is sketched below.

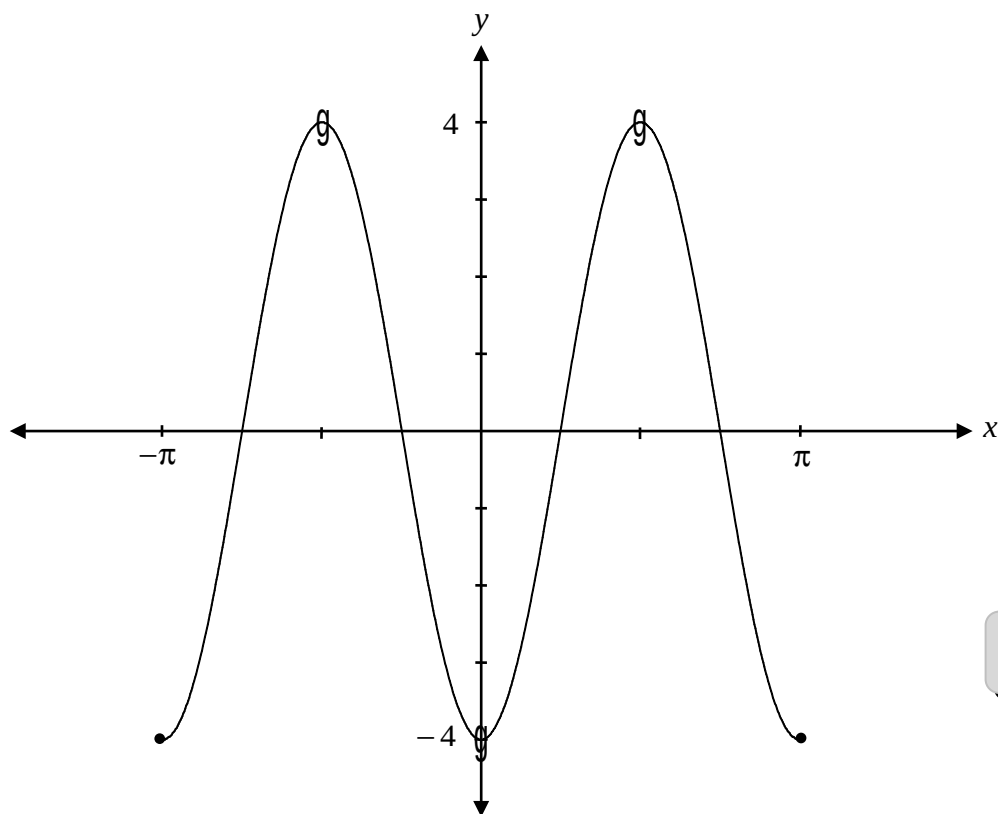
Explain how to use this graph to solve the equation $\sin 2x = \frac{1}{2}$ over the interval $[0, 2\pi]$.

**Solution**

Draw the line $y = \frac{1}{2}$. The solution will be the x -values where the two graphs intersect.

1 mark

Sketch the graph of $y = -4\cos(2x)$ over the interval $[-\pi, \pi]$.

Solution

1 mark for vertical reflection
1 mark for range
1 mark for period

3 marks

Write the equation for $f(x)$ that satisfies all of the following conditions:

- $f(x)$ is a polynomial function of degree 4
- $f(x)$ has a zero at 2 with a multiplicity of 3
- $f(x)$ has a zero at -5
- $f(x)$ has a y-intercept of 80

Solution

$$f(x) = a(x-2)^3(x+5)$$

$$80 = a(0-2)^3(0+5)$$

$$80 = a(-8)(5)$$

$$80 = -40a$$

$$a = -2$$

$\frac{1}{2}$ mark for the factors $(x-2)$ and $(x+5)$

$\frac{1}{2}$ mark for multiplicity of 3

$\frac{1}{2}$ mark for substitution/negative value of a

$\frac{1}{2}$ mark for value of 2 for a

2 marks

$$\therefore f(x) = -2(x-2)^3(x+5)$$

Find the exact value of $\sin\left(\frac{19\pi}{12}\right)$.

Solution

$$\sin\left(\frac{10\pi}{12} + \frac{9\pi}{12}\right) = \sin\left(\frac{5\pi}{6} + \frac{3\pi}{4}\right)$$

1 mark for combination

$$\sin\frac{19\pi}{12} = \sin\frac{5\pi}{6}\cos\frac{3\pi}{4} + \cos\frac{5\pi}{6}\sin\frac{3\pi}{4}$$

$$= \left(\frac{1}{2}\right)\left(-\frac{\sqrt{2}}{2}\right) + \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

2 marks ($\frac{1}{2}$ mark for each exact value)

$$= -\frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4}$$

$$= \frac{-\sqrt{2} - \sqrt{6}}{4}$$

3 marks

Note(s):

§ Other combinations are possible.

Solve the following equation:

$$2\log_2(x-1) - \log_2(x-5) = \log_2(x+1)$$

Solution

Method 1

$$\log_2 \frac{(x-1)^2}{x-5} = \log_2(x+1)$$

2 marks for logarithmic rules (1 mark for power rule, 1 mark for quotient rule)

$$\frac{(x-1)^2}{x-5} = x+1$$

1 mark for equating arguments

$$x^2 - 2x + 1 = x^2 - 4x - 5$$

$$2x = -6$$

$$\cancel{x = -3}$$

½ mark for solving for x

∴ no solution

½ mark for no solution

4 marks

Method 2

$$2\log_2(x-1) = \log_2(x+1) + \log_2(x-5)$$

$$\log_2(x-1)^2 = \log_2(x+1)(x-5)$$

2 marks for logarithmic rules (1 mark for power rule, 1 mark for product rule)

$$(x-1)^2 = (x+1)(x-5)$$

1 mark for equating arguments

$$x^2 - 2x + 1 = x^2 - 4x - 5$$

$$2x = -6$$

$$\cancel{x = -3}$$

½ mark for solving for x

∴ no solution

½ mark for no solution

4 marks

Method 3

$$\log_2 (x-1)^2 - \log_2 (x-5) - \log_2 (x+1) = 0$$

$$\log_2 \frac{(x-1)^2}{(x-5)(x+1)} = 0$$

$$2^0 = \frac{(x-1)^2}{(x-5)(x+1)}$$

$$x^2 - 4x - 5 = x^2 - 2x + 1$$

$$-6 = 2x$$

~~$$-3 = x$$~~

$\therefore$ no solution

2 marks for logarithmic rules (1 mark for power rule, 1 mark for quotient rule)

1 mark for exponential form

$\frac{1}{2}$ mark for solving for x

$\frac{1}{2}$ mark for no solution

4 marks

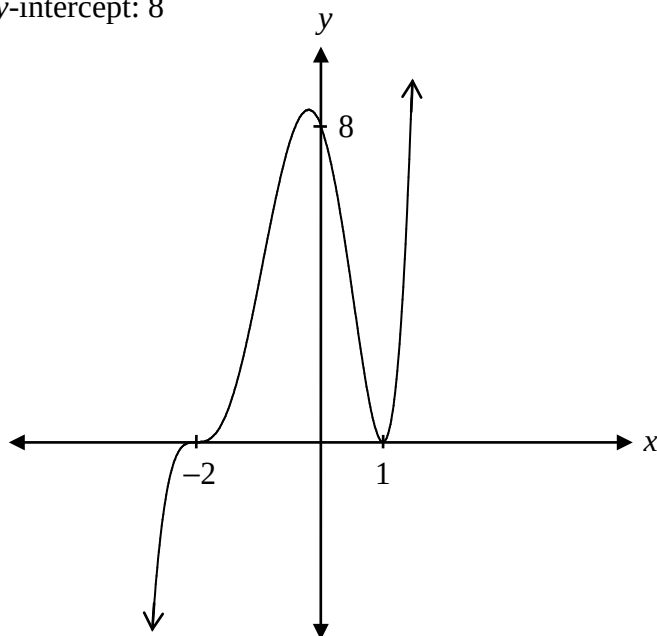
Sketch the graph of $f(x) = (x-1)^2(x+2)^3$.

Label the x -intercepts and the y -intercept.

Solution

x -intercepts: $-2, 1$

y -intercept: 8



1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

1 mark for multiplicity ($\frac{1}{2}$ mark for degree of 2, $\frac{1}{2}$ mark for degree of 3)

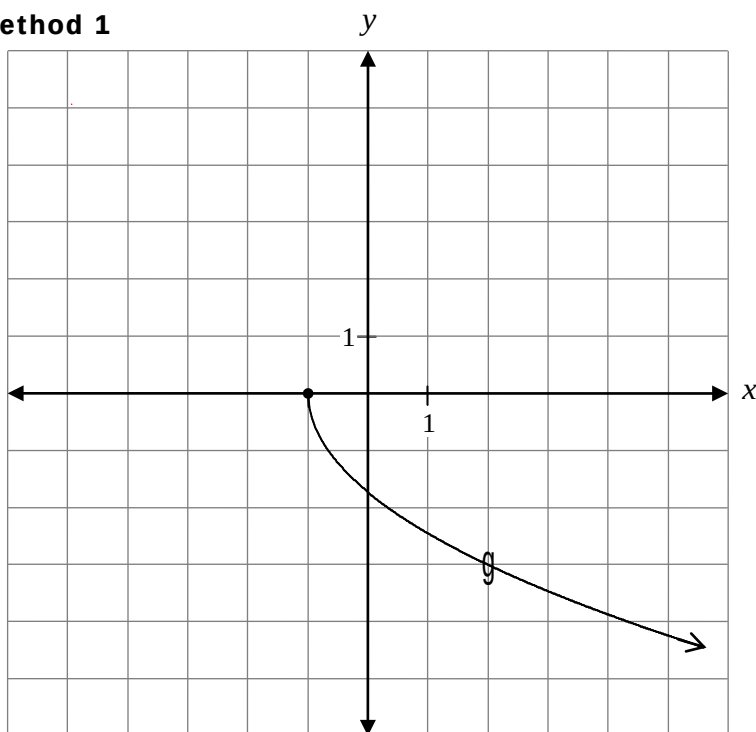
$\frac{1}{2}$ mark for end behaviour

3 marks

Sketch the graph of $y = -\sqrt{3(x+1)}$.

Solution

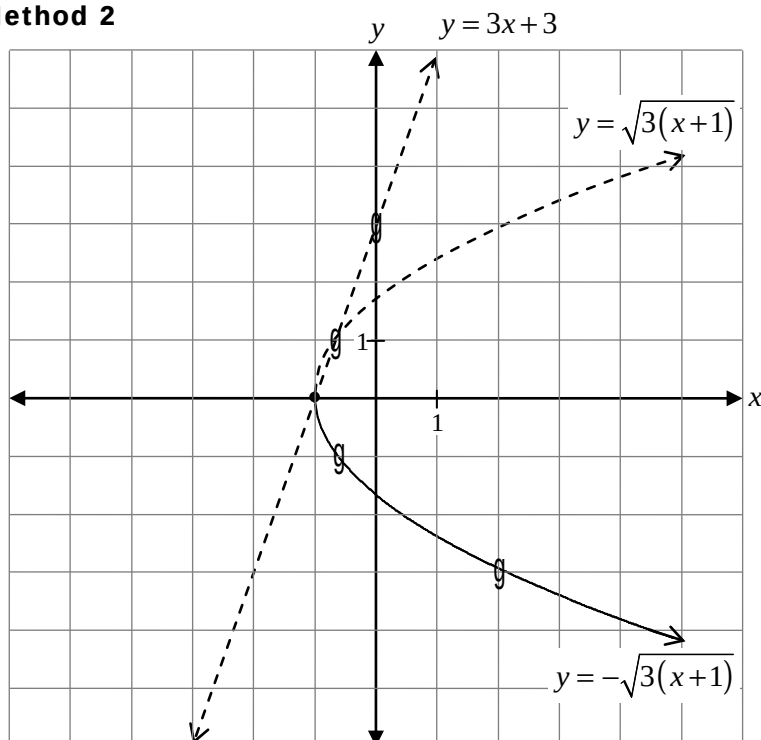
Method 1



- 1 mark for horizontal shift
- 1 mark for vertical reflection
- 1 mark for shape (graph of a radical function)
- 1 mark for horizontal compression

4 marks

Method 2



- 1 mark for invariant point where $y = 0$ and $y = 1$ ($\frac{1}{2}$ mark for each point)
- 1 mark for domain of $y = -\sqrt{3(x+1)}: [-1, \infty)$
- 1 mark for reflection in the x-axis
- $\frac{1}{2}$ mark for shape between invariant points
- $\frac{1}{2}$ mark for shape to the right of the invariant points

4 marks

Solve:

$${}_{n-1}P_2 = 42$$

Solution

$$\frac{(n-1)!}{(n-1-2)!} = 42$$

½ mark for substitution

$$\frac{(n-1)(n-2)(n-3)!}{(n-3)!} = 42$$

1 mark for factorial expansion

$$(n-1)(n-2) = 42$$

½ mark for simplification of factorials

$$n^2 - 3n + 2 = 42$$

$$n^2 - 3n - 40 = 0$$

$$(n-8)(n+5) = 0$$

$$n = 8 \quad \cancel{n = -5}$$

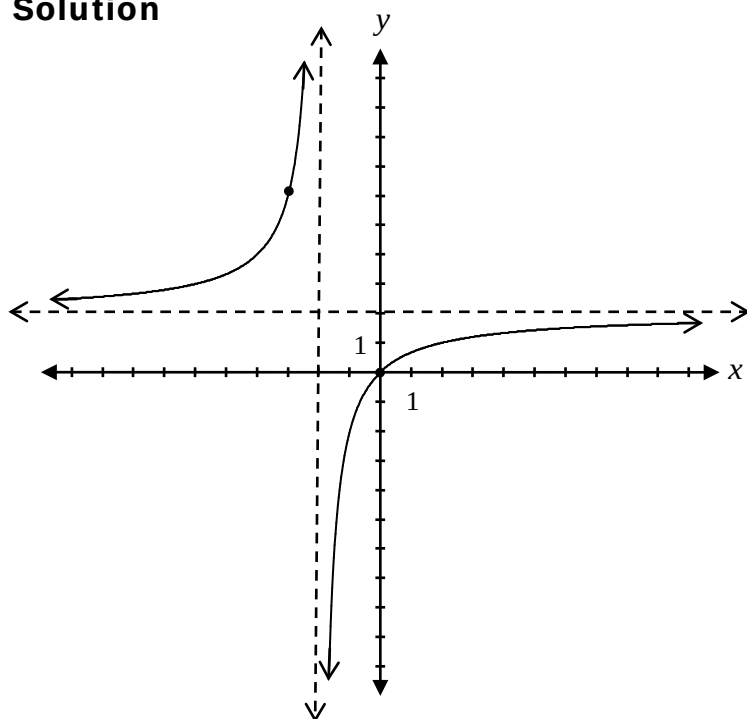
½ mark for solving for n

½ mark for rejecting extraneous root

3 marks

Sketch the graph of $y = \frac{2x}{x+2}$.

Solution



1 mark for horizontal asymptote at $y = 2$

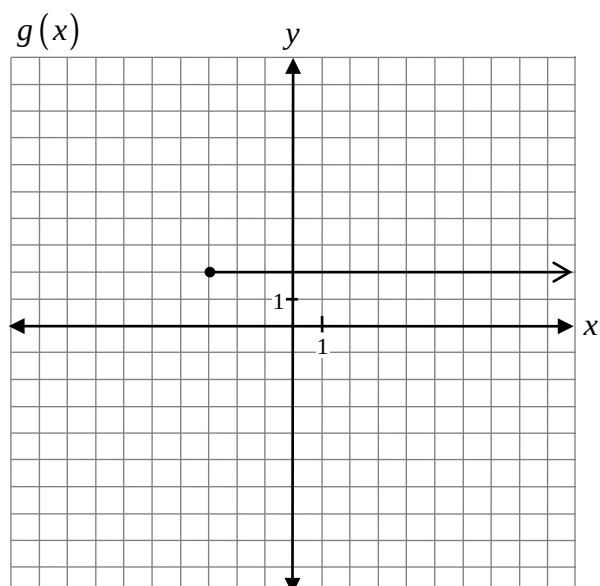
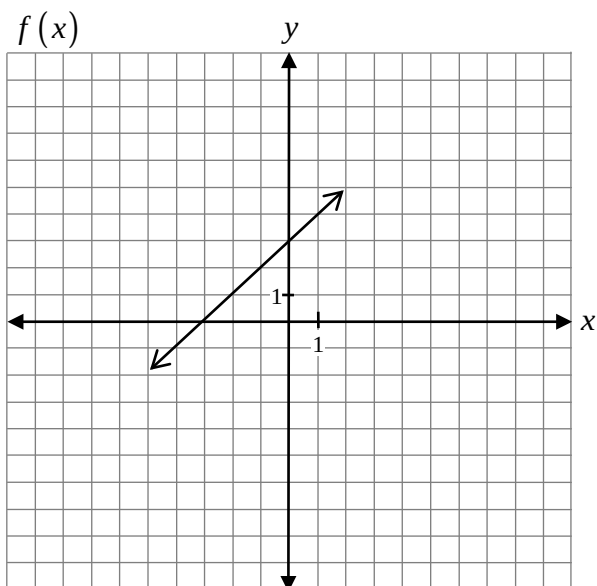
1 mark for vertical asymptote at $x = -2$

$\frac{1}{2}$ mark for graph left of vertical asymptote

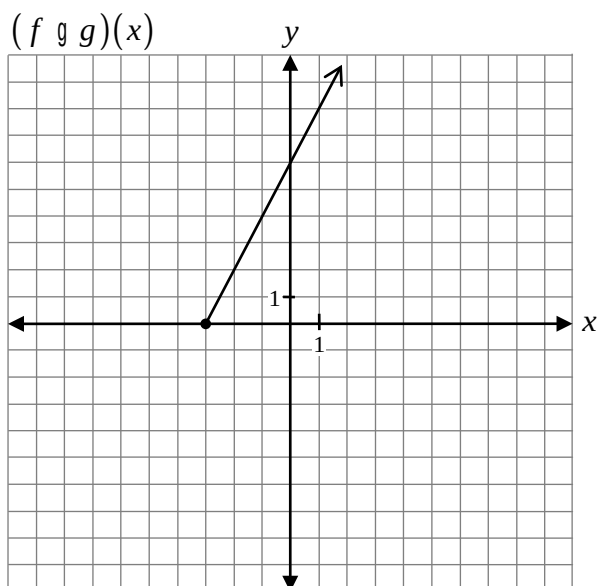
$\frac{1}{2}$ mark for graph right of vertical asymptote

3 marks

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $(f \circ g)(x)$.



Solution



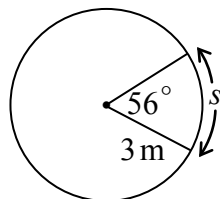
1 mark for operation of multiplication
1 mark for restricted domain

2 marks

Question 1

T1

Use the information in the diagram to determine the value of the arc length “ s ”, given the central angle of 56° .



Solution

$$\theta = 56^\circ \times \frac{\pi}{180^\circ}$$

1 mark for conversion

$$\theta = \frac{56\pi}{180} \text{ or } \frac{14\pi}{45}$$

$$s = \theta r$$

$$s = \left(\frac{14\pi}{45} \right) (3\text{ m})$$

1 mark for substitution

$$s = \frac{14\pi}{15} \text{ m or } 2.932 \text{ m}$$

2 marks

Solve $\tan^2 \theta - 5 \tan \theta + 4 = 0$ where $\theta \in \mathbb{R}$.

Solution

Method 1

$$(\tan \theta - 1)(\tan \theta - 4) = 0$$

$$\tan \theta = 1 \qquad \tan \theta = 4$$

$$\theta_r = 1.3258$$

$$\theta = \frac{\pi}{4}, \frac{5\pi}{4} \qquad \theta = 1.326, 4.467$$

$$\theta = 0.785, 3.927$$

1 mark for solving for $\tan \theta$ (½ mark for each branch)

2 marks (½ mark for each value of θ)

$$\left. \begin{aligned} \theta &= \frac{\pi}{4} + 2k\pi \\ &= 1.326 + 2k\pi \\ &= \frac{5\pi}{4} + 2k\pi \\ &= 4.467 + 2k\pi \end{aligned} \right\} \text{ where } k \in \mathbb{I}$$

1 mark for general solution

4 marks

Method 2

$$(\tan \theta - 1)(\tan \theta - 4) = 0$$

$$\tan \theta = 1 \qquad \tan \theta = 4$$

$$\theta_r = 1.3258$$

$$\theta = \frac{\pi}{4} \qquad \theta = 1.326$$

$$\theta = 0.785$$

1 mark for solving for $\tan \theta$ (½ mark for each branch)

2 marks (1 mark for each value of θ)

$$\left. \begin{aligned} \theta &= \frac{\pi}{4} + k\pi \\ &= 1.326 + k\pi \end{aligned} \right\} \text{ where } k \in \mathbb{I}$$

1 mark for general solution

4 marks

Solve:

$$2^{5x} = 3(5)^{x-3}$$

Solution

$$\log 2^{5x} = \log [3(5)^{x-3}]$$

$$5x \log 2 = \log 3 + (x-3) \log 5$$

$$5x \log 2 = \log 3 + x \log 5 - 3 \log 5$$

$$5x \log 2 - x \log 5 = \log 3 - 3 \log 5$$

$$x(5 \log 2 - \log 5) = \log 3 - 3 \log 5$$

$$x = \frac{\log 3 - 3 \log 5}{5 \log 2 - \log 5}$$

$$x = -2.009$$

½ mark for applying logarithms

1 mark for product rule

1 mark for power rule

½ mark for collecting like terms

½ mark for isolating x

½ mark for evaluating a quotient of logarithms

4 marks

Question 4

P3

David and Sarah are in a class of 10 boys and 8 girls.

A committee of 3 boys and 2 girls is to be selected from the students in this class.

Determine the number of possible committees if David and Sarah cannot be on the same committee.

Solution

Method 1

All: ${}_{10}C_3 \times {}_8C_2 = 3360$

1 mark for all possible committees

Both: ${}_9C_2 \times {}_7C_1 = 252$

1 mark for both on the committee

$$3360 - 252 = 3108$$

1 mark for subtraction of cases

3 marks

Method 2

Case 1: David, not Sarah ${}_9C_2 \times {}_7C_2 = 756$

$\frac{1}{2}$ mark for Case 1

Case 2: Sarah, not David ${}_9C_3 \times {}_7C_1 = 588$

$\frac{1}{2}$ mark for Case 2

Case 3: Neither David nor Sarah ${}_9C_3 \times {}_7C_2 = 1764$

1 mark for Case 3

$$756 + 588 + 1764 = 3108$$

1 mark for addition of cases

3 marks

In the binomial expansion of $\left(\frac{3}{x^2} - x^5\right)^{10}$, simplify the 7th term.

Solution

$$t_7 = {}_{10}C_6 \left(\frac{3}{x^2}\right)^4 (-x^5)^6$$

$$t_7 = 210 \left(\frac{81}{x^8}\right) (x^{30})$$

$$t_7 = 17\,010x^{22}$$

2 marks (1 mark for ${}_{10}C_6$, $\frac{1}{2}$ mark for each consistent factor)

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark for exponent)

3 marks

Question 6

R10

A lake affected by acid rain has a pH of 4.4.

A person suffering from heartburn has a stomach acid pH of 1.2.

The pH of a solution is defined as $\text{pH} = -\log[H^+]$ where $[H^+]$ is the hydrogen ion concentration.

How many times greater is the hydrogen ion concentration of the stomach than that of the lake?

Express your answer as a whole number.

Solution

Method 1

$$\text{pH} = -\log[H^+]$$

$$-\text{pH} = \log[H^+]$$

$$[H^+] = 10^{-\text{pH}}$$

1 mark for exponential form

$$\frac{[H^+]_{\text{stomach}}}{[H^+]_{\text{lake}}} = \frac{10^{-1.2}}{10^{-4.4}}$$

1 mark for comparison

$$\begin{aligned} &= 10^{3.2} \\ &= 1584.9 \\ &= 1585 \end{aligned}$$

2 marks

Method 2

Lake

$$4.4 = -\log[H^+]$$

$$-4.4 = \log[H^+]$$

$$10^{-4.4} = [H^+]$$

½ mark for exponential form

Stomach

$$1.2 = -\log[H^+]$$

$$-1.2 = \log[H^+]$$

$$10^{-1.2} = [H^+]$$

½ mark for exponential form

$$\frac{[H^+]_{\text{stomach}}}{[H^+]_{\text{lake}}} = \frac{10^{-1.2}}{10^{-4.4}}$$

1 mark for comparison

$$\begin{aligned} &= 10^{3.2} \\ &= 1585 \end{aligned}$$

2 marks

Solve the following equation algebraically over the interval $[0, 2\pi]$.

$$\cos 2\theta - 3 \sin \theta - 2 = 0$$

Solution

$$(1 - 2 \sin^2 \theta) - 3 \sin \theta - 2 = 0$$

1 mark for correct substitution of an appropriate identity

$$-2 \sin^2 \theta - 3 \sin \theta - 1 = 0$$

$$2 \sin^2 \theta + 3 \sin \theta + 1 = 0$$

$$(2 \sin \theta + 1)(\sin \theta + 1) = 0$$

$$\sin \theta = -\frac{1}{2} \qquad \sin \theta = -1$$

1 mark for solving for $\sin \theta$ ($\frac{1}{2}$ mark for each branch)

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6} \qquad \theta = \frac{3\pi}{2}$$

2 marks for solutions (1 mark for each branch; $\frac{1}{2}$ mark for each value in the left branch)

4 marks

Question 8

R12

Explain how the value of n affects the behaviour of the graph of the polynomial function

$p(x) = (x + 3)(x - 1)^n$, as $p(x)$ approaches the x -intercept at $x = 1$.

Solution

If n is even, the graph will turn at the x -axis at $x = 1$.

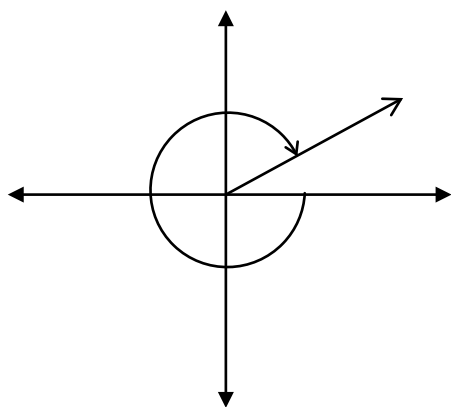
If n is odd, the graph will cross the x -axis at $x = 1$.

1 mark

or

As n increases, $p(x)$ will become flatter around the intercept at $x = 1$.

Sketch the angle -320° in standard position.

Solution

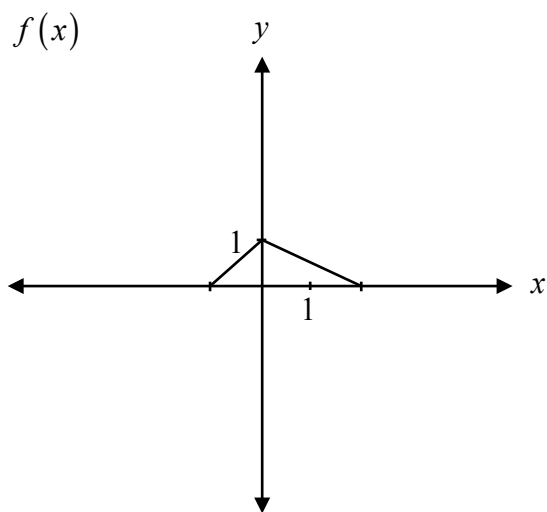
1 mark for angle drawn in quadrant I

1 mark

Note(s):

- If directional arrow is not indicated, it is a communication error, E1 (final answer not stated).

Given the graph of $y = f(x)$, explain how to graph $y = f(-x)$.

**Solution**

Multiply the x -coordinates by -1 .

1 mark

or

Reflect the graph over the y -axis.

Explain how the graph of $y = \frac{3(x-1)}{(x-1)}$ is different than the graph of $y = 3$.

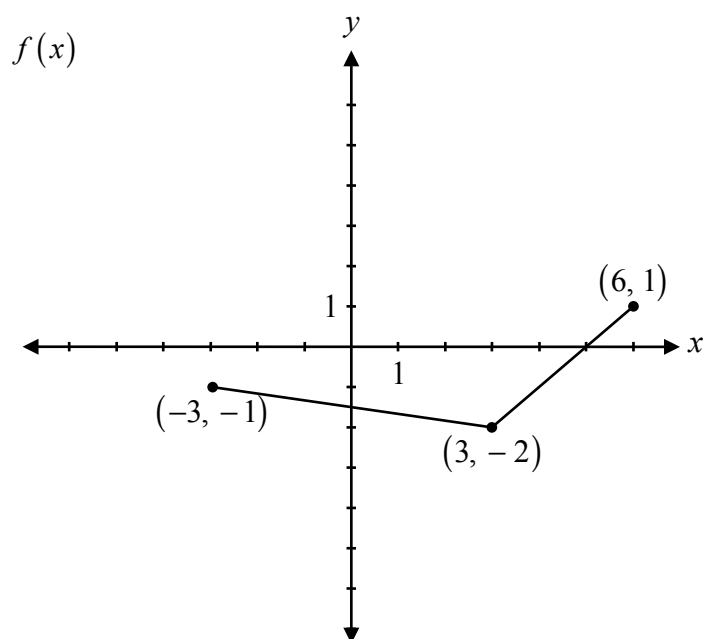
Solution

There is a point of discontinuity (hole) when $x = 1$ on the graph of $y = \frac{3(x-1)}{(x-1)}$. **1 mark**

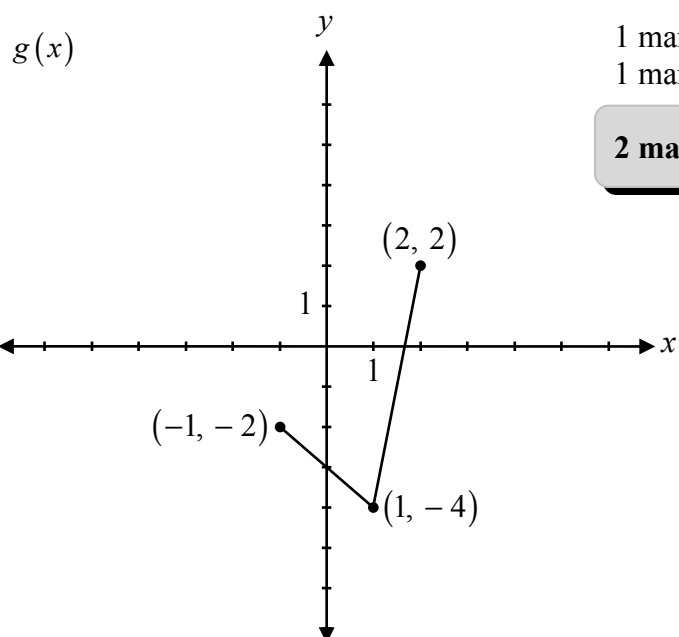
Question 12

R3

Given the graph of $f(x)$, sketch the graph of $g(x) = 2f(3x)$.



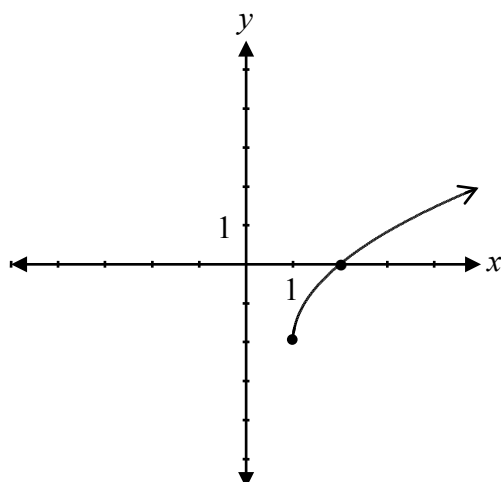
Solution



1 mark for horizontal stretch/compression
1 mark for vertical stretch

2 marks

Determine the equation of the radical function represented by the graph.

**Solution**

$$y = 2\sqrt{x-1} - 2$$

1 mark for vertical stretch
1 mark for horizontal translation
1 mark for vertical translation

3 marks

or

$$y = \sqrt{4(x-1)} - 2$$

1 mark for horizontal compression
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Solve:

$$2\log_4 x - \log_4(x+3) = 1$$

Solution

$$2\log_4 x - \log_4(x+3) = 1$$

$$\log_4 \left(\frac{x^2}{x+3} \right) = 1$$

1 mark for power rule

1 mark for quotient rule

$$4^1 = \frac{x^2}{x+3}$$

1 mark for exponential form

$$4x + 12 = x^2$$

$$x^2 - 4x - 12 = 0$$

$$(x-6)(x+2) = 0$$

$$x = 6 \quad \cancel{x = -2}$$

$\frac{1}{2}$ mark for solving for x

$\frac{1}{2}$ mark for rejecting extraneous root

4 marks

The following transformations are applied to $f(x)$, resulting in a new function, $g(x)$.

- reflection over the y -axis
- horizontal translation of 3 units to the right
- vertical translation of 4 units down

Write the equation of $g(x)$ in terms of $f(x)$.

Solution

$$g(x) = f(-(x-3)) - 4$$

or

$$g(x) = f(-x+3) - 4$$

1 mark for horizontal shift

1 mark for vertical shift

1 mark for reflection

3 marks

Question 25

T4

The height of a bicycle pedal as the bicycle is moving at a constant speed can be represented by the following function:

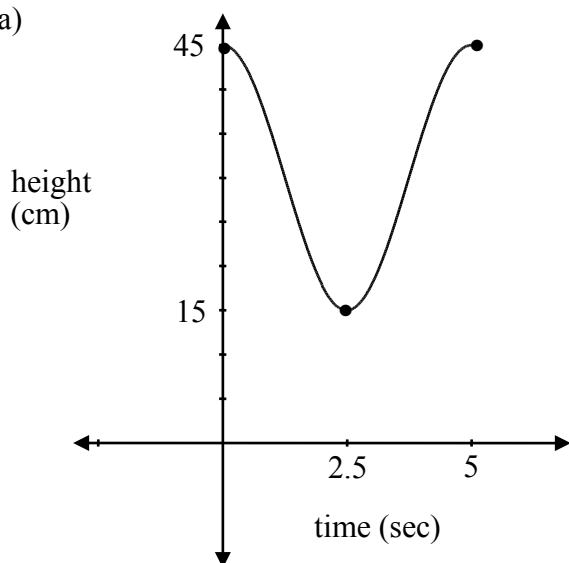
$$h(t) = 15 \cos \frac{2\pi}{5}t + 30$$

where h is the height of the pedal above the ground, in cm, and t is the time, in seconds.

- Sketch a graph of at least one period of this function, where $t \geq 0$.
- Determine the height of the bicycle pedal at 7.5 seconds.

Solution

a)



Amplitude = 15 1 mark for amplitude

Period = $\frac{2\pi}{\frac{2\pi}{5}}$ 1 mark for period
 $= 5$

Vertical Shift = 30 1 mark for vertical shift

3 marks

b) From the graph:

$$h(t) = 15 \text{ cm}$$

1 mark

or

From the equation:

$$\begin{aligned} h(t) &= 15 \cos \frac{2\pi}{5}(7.5) + 30 \\ &= 15 \cos 3\pi + 30 \\ &= 15(-1) + 30 \\ &= 15 \text{ cm} \end{aligned}$$

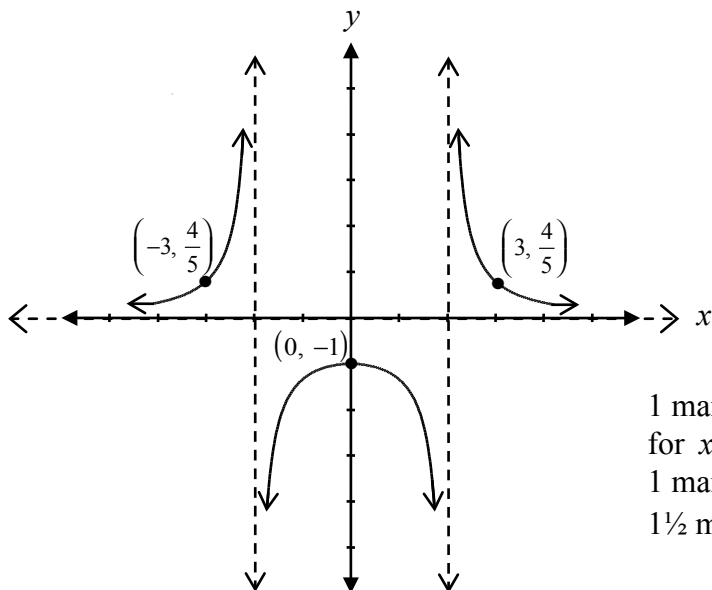
Question 26

R14

Sketch the graph of the function $f(x)$ and determine the y -intercept.

$$f(x) = \frac{4}{(x-2)(x+2)}$$

Solution



1 mark for vertical asymptotes ($\frac{1}{2}$ mark for $x = 2$, $\frac{1}{2}$ mark for $x = -2$)

1 mark for horizontal asymptote at $y = 0$

$1\frac{1}{2}$ marks for shape ($\frac{1}{2}$ mark for shape in each section)

y -intercept: -1

$\frac{1}{2}$ mark for y -intercept

4 marks

Question 27

R10

Kim solved the following logarithmic equation:

$$\log_2 \left(-\frac{x}{3} \right) = \log_2 (x - 4)$$

$$-\frac{x}{3} = x - 4$$

$$-x = 3x - 12$$

$$-4x = -12$$

$$\cancel{x = 3}$$

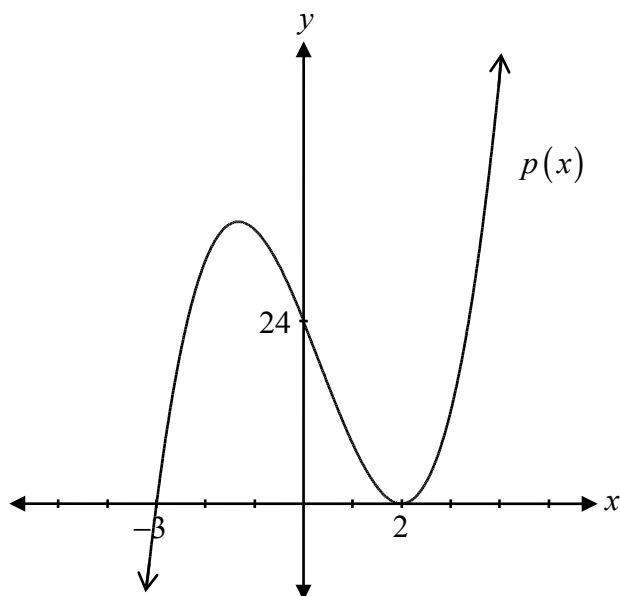
Explain why $x = 3$ is an extraneous solution.

Solution

$x = 3$ is an extraneous solution because the argument in a logarithmic equation cannot be negative.

1 mark

Determine the equation of the polynomial function represented by the graph.

**Solution**

$$p(x) = a(x - 2)^2(x + 3)$$

1 mark for x -intercepts ($\frac{1}{2}$ mark for each)

1 mark for multiplicity at $x = 2$

$$24 = a(0 - 2)^2(0 + 3)$$

$$24 = a(4)(3)$$

$$24 = 12a$$

$$a = 2$$

$$p(x) = 2(x - 2)^2(x + 3)$$

1 mark for leading coefficient

3 marks

Determine the coterminal angles with $\frac{2\pi}{3}$ over the interval $[-2\pi, 4\pi]$.

Solution

$$-\frac{4\pi}{3}, \frac{8\pi}{3}$$

1 mark (½ mark for each coterminal angle)

1 mark

- a) Solve the following equation:

$$0 = \sqrt{4x - 8} - 2$$

- b) Explain how your answer in a) is related to the graph of $y = \sqrt{4x - 8} - 2$.

Solution

- a) $4 = 4x - 8$
 $12 = 4x$
 $x = 3, x \geq 2$

1 mark

- b) The answer in a) is the x -intercept of the graph.

1 mark

Note(s):

- $x \geq 2$ does not need to be shown.

Determine the exact value of $\sin \frac{13\pi}{12}$.

Solution

$$\begin{aligned}\sin \frac{13\pi}{12} &= \sin \left(\frac{3\pi}{4} + \frac{\pi}{3} \right) \\ &= \left(\sin \frac{3\pi}{4} \right) \left(\cos \frac{\pi}{3} \right) + \left(\cos \frac{3\pi}{4} \right) \left(\sin \frac{\pi}{3} \right) \\ &= \left(\frac{\sqrt{2}}{2} \right) \left(\frac{1}{2} \right) + \left(-\frac{\sqrt{2}}{2} \right) \left(\frac{\sqrt{3}}{2} \right) \\ &= \frac{\sqrt{2} - \sqrt{6}}{4}\end{aligned}$$

1 mark for combination

2 marks for exact values (½ mark for each)

3 marks

Note(s):

- Other combinations are possible.

Given the functions $f(x) = x + 2$ and $g(x) = \frac{1}{x-5}$:

- a) Determine the equation of the composite function $f(g(x))$ and its domain.
b) Determine the x -intercept and y -intercept of $f(g(x))$.

Solution

a) $f(g(x)) = \frac{1}{x-5} + 2$

or

$$f(g(x)) = \frac{2x-9}{x-5}$$

domain: $\{x \in \mathbb{R}, x \neq 5\}$

1 mark for composition

1 mark for domain

2 marks

b) x -intercept: $\frac{9}{2}$

$\frac{1}{2}$ mark for x -intercept

y -intercept: $\frac{9}{5}$

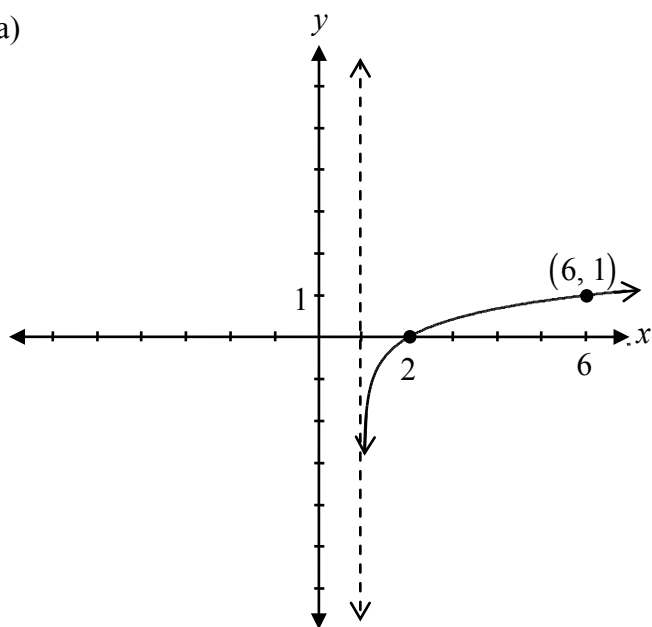
$\frac{1}{2}$ mark for y -intercept

1 mark

- a) Sketch the graph of $f(x) = \log_5(x-1)$.
- b) Sketch the graph of $f^{-1}(x)$.

Solution

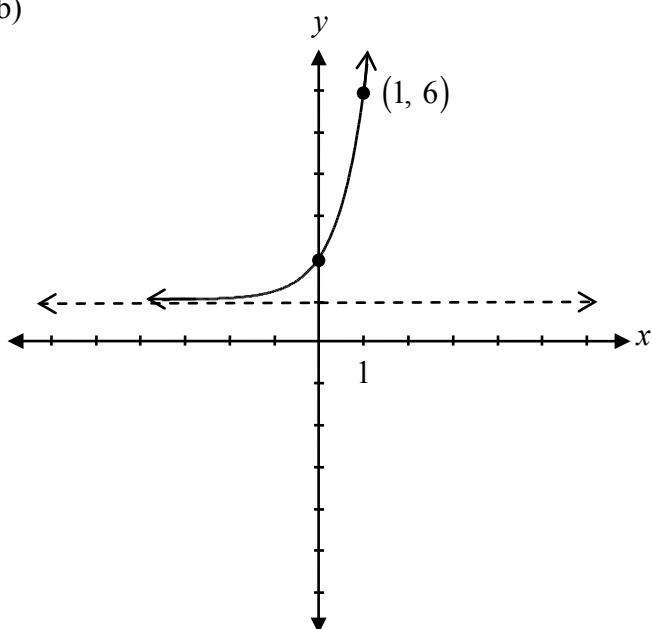
a)



$\frac{1}{2}$ mark for vertical asymptote at $x = 1$
 $\frac{1}{2}$ mark for x -intercept at $x = 2$
 $\frac{1}{2}$ mark for increasing logarithmic function
 $\frac{1}{2}$ mark for consistent point on the logarithmic graph

2 marks

b)



1 mark for the graph of the inverse function consistent with a)

1 mark

Explain what the graph of a rational function looks like near a vertical asymptote.

Solution

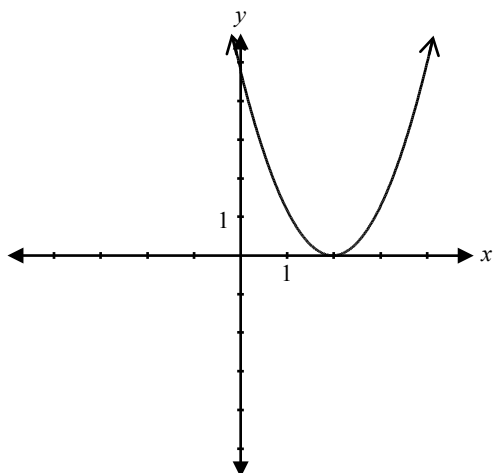
The graph approaches infinity (positive or negative) as it approaches the asymptote.

1 mark

or

The graph approaches the asymptote, but does not touch it.

Given the graph of $f(x) = (x - 2)^2$,



determine one possible restriction for the domain of $f(x)$ so that its inverse is a function.

Solution

Domain: $[2, \infty)$

1 mark**or**

Domain: $(-\infty, 2)$

Note(s):

- Other solutions are possible.

Over the interval $[0, 2\pi]$, determine the non-permissible values of θ in the expression $\csc \theta(\cos \theta + 1)$.

Solution

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\sin \theta \neq 0$$

$$\theta = 0, \pi, 2\pi$$

1 mark for correct substitution of an appropriate identity

½ mark for $\sin \theta \neq 0$

½ mark for consistent non-permissible values

2 marks

Explain why ${}_3C_8$ is undefined.

Solution

In the formula ${}_nC_r$, the number of objects, n , must be larger than or equal to the number of objects selected, r .

1 mark

or

You can't select 8 objects from a total of 3.

Solve:

$${}_nP_3 = 48(n-1)$$

Solution

$$\frac{n!}{(n-3)!} = 48(n-1)$$

½ mark for substitution in correct formula

$$\frac{(n)(n-1)(n-2)\cancel{(n-3)!}}{\cancel{(n-3)!}} = 48(n-1)$$

1 mark for expansion of factorial

½ mark for simplification of factorials

$$n(n-2) = 48$$

$$n^2 - 2n - 48 = 0$$

$$(n-8)(n+6) = 0$$

$$n = 8 \quad \cancel{n = -6}$$

½ mark for solving for both values of n

½ mark for rejecting extraneous solution

3 marks

Question 39

R11, R12

Christine dives off a diving board.

Her dive is modelled by the function $h(t) = t^3 - 3t^2 - t + 3$, where h is her height in metres, relative to the water surface and t is the time in seconds after diving off the diving board.

- Given that $(t + 1)$ is a factor for the function $h(t)$, determine the other factors.
- Sketch the graph of the function $h(t)$ for the time interval $t = 0$ to $t = 3$.
- Determine how many seconds Christine is underwater.

Solution

$$\begin{array}{r|rrrr} -1 & 1 & -3 & -1 & 3 \\ & \downarrow & -1 & 4 & -3 \\ \hline & 1 & -4 & 3 & 0 \end{array}$$

$$h(t) = (t + 1)(t^2 - 4t + 3)$$

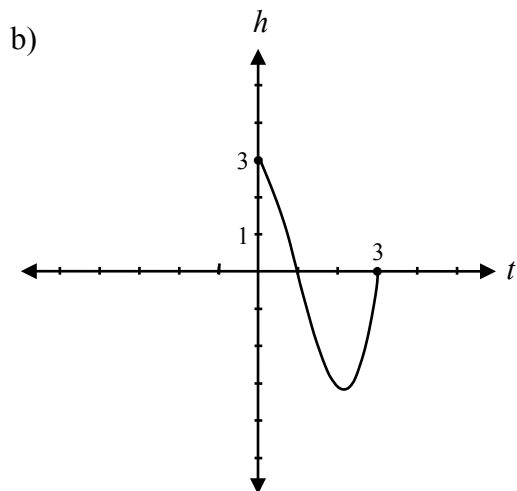
$$h(t) = (t + 1)(t - 1)(t - 3)$$

½ mark for $t = -1$

1 mark for synthetic division (or equivalent strategy)

½ mark for determining other factors

2 marks



½ mark for t -intercepts

½ mark for h -intercept

1 mark

- Christine is underwater for 2 seconds.

1 mark

Prove the identity for all permissible values of x .

$$\sec x + \tan x = \frac{\cos x}{1 - \sin x}$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\sec x + \tan x$	
$\frac{1}{\cos x} + \frac{\sin x}{\cos x}$	1 mark for correct substitution of appropriate identities
$\frac{(1 + \sin x)(1 - \sin x)}{\cos x(1 - \sin x)}$	1 mark for logical process to prove the identity
$\frac{1 - \sin^2 x}{(1 - \sin x)\cos x}$	
$\frac{\cos^2 x}{(1 - \sin x)\cos x}$	1 mark for algebraic strategies
$\frac{\cos x}{1 - \sin x}$	3 marks

Method 2

$$\sec x + \tan x = \frac{\cos x}{1 - \sin x}$$

Left-Hand Side	Right-Hand Side
	$\frac{\cos x}{1 - \sin x}$
	$\frac{\cos x(1 + \sin x)}{(1 - \sin x)(1 + \sin x)}$
	$\frac{\cos x(1 + \sin x)}{1 - \sin^2 x}$
	$\frac{\cos x(1 + \sin x)}{\cos^2 x}$
	$\frac{1 + \sin x}{\cos x}$
	$\frac{1}{\cos x} + \frac{\sin x}{\cos x}$
	$\sec x + \tan x$

1 mark for logical process to prove the identity

1 mark for correct substitution of appropriate identities

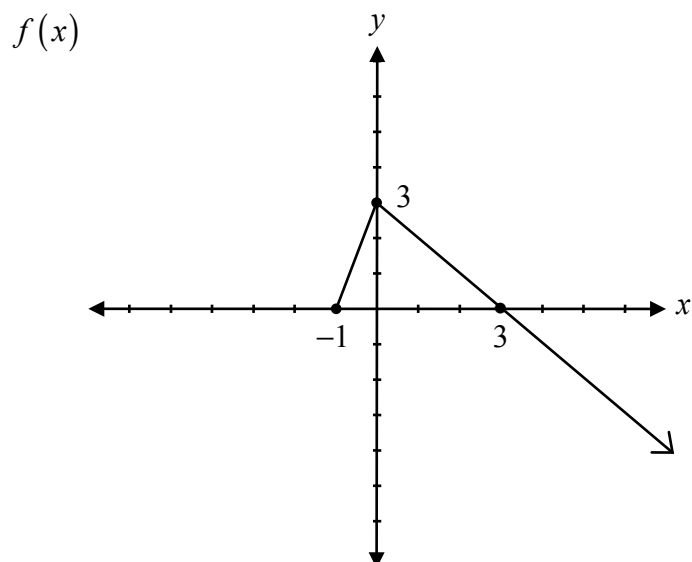
1 mark for algebraic strategies

3 marks

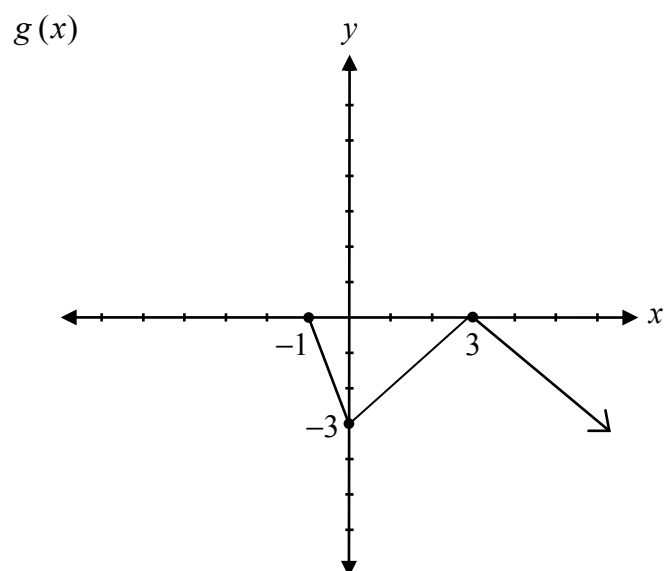
Question 41

R1, R5

Given the graph of $f(x)$, sketch the graph of the function $g(x) = -|f(x)|$.



Solution



1 mark for absolute value
1 mark for vertical reflection

2 marks

Given that $\cot \theta = -\frac{2}{5}$, where θ is in Quadrant IV, determine the exact value of $\sin \theta$.

Solution

$$P(2, -5)$$

$$r^2 = (2)^2 + (-5)^2$$

$$r^2 = 4 + 25$$

$$r = \sqrt{29}$$

$$\sin \theta = \frac{-5}{\sqrt{29}}$$

$\frac{1}{2}$ mark for substitution of $x = 2$, $y = -5$

$\frac{1}{2}$ mark for solving for r

1 mark for $\sin \theta$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

2 marks

A wheel has a diameter of 20 cm and rotates through a central angle of 252° .
Determine how far the wheel rolled.

Solution

$$\theta = (252^\circ) \left(\frac{\pi}{180^\circ} \right)$$

1 mark for conversion

$$= \frac{7\pi}{5}$$

$$s = \theta r$$

$$= \left(\frac{7\pi}{5} \right) \left(\frac{20}{2} \right)$$

1 mark for substitution

$$= 14\pi \text{ cm}$$

2 marks**or**

$$= 43.982 \text{ cm}$$

Solve the following equation over the interval $[0, 2\pi]$:

$$3\sin^2 \theta - 10\sin \theta - 8 = 0$$

Solution

$$3\sin^2 \theta - 10\sin \theta - 8 = 0$$

$$(3\sin \theta + 2)(\sin \theta - 4) = 0$$

$$\sin \theta = -\frac{2}{3} \quad \sin \theta = 4$$

1 mark for solving for $\sin \theta$ (½ mark for each branch)

$$\theta_r = 0.729 \ 728 \quad \text{No solution}$$

2 marks for solving for θ (1 mark for indicating no solution, ½ mark for each value)

$$\theta = 3.871$$

$$\theta = 5.553$$

3 marks

Determine and simplify the fourth term in the expansion of $(2x^4 - 3y)^8$.

Solution

$$t_4 = {}_8C_3 (2x^4)^5 (-3y)^3$$

2 marks (1 mark for ${}_8C_3$, $\frac{1}{2}$ mark for each consistent factor)

$$= 56(32x^{20})(-27y^3)$$

$$= -48\,384x^{20}y^3$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark for exponents)

3 marks

Question 4

R10

Sheeva's bank is lending her \$50 000 at an annual interest rate of 6%, compounded monthly, to purchase a car.

Given that the last payment will be a partial payment, determine how many full monthly payments of \$800 Sheeva will have to make.

The formula below may be used.

$$PV = \frac{R \left[1 - (1 + i)^{-n} \right]}{i}$$

where PV = the present value of the amount borrowed

R = the amount of each periodic payment

$i = \frac{\text{annual interest rate (as a decimal)}}{\text{the number of compounding periods per year}}$

n = the number of equal periodic payments

Express your answer as a whole number.

Solution

$$50\,000 = \frac{800 \left[1 - \left(1 + \frac{0.06}{12} \right)^{-n} \right]}{\frac{0.06}{12}}$$

½ mark for substitution

$$250 = 800 \left[1 - (1 + 0.005)^{-n} \right]$$

$$0.3125 = 1 - 1.005^{-n}$$

$$-0.6875 = -1.005^{-n}$$

$$0.6875 = 1.005^{-n}$$

½ mark for simplification

$$\log 0.6875 = -n \log 1.005$$

½ mark for applying logarithms

$$\frac{\log 0.6875}{-\log 1.005} = n$$

1 mark for power law

$$75.125\,880\,88 = n$$

½ mark for solving for n

∴ 75 full monthly payments are needed

3 marks

Question 5

P1

An employee asked 10 people in an ice cream shop to wait in line.

Determine the number of different arrangements possible if two of the people, Jamie and John, refused to stand next to each other in the line.

Solution

$$10! - 9!2!$$

$\frac{1}{2}$ mark for $10!$

1 mark for product of $9!2!$ ($\frac{1}{2}$ mark for $9!$, $\frac{1}{2}$ mark for $2!$)

$\frac{1}{2}$ mark for subtraction

$$2\,903\,040$$

2 marks

The point $(-2, 4)$ is on the graph of $f(x)$.

State the coordinates of the corresponding point when $f(x)$ is reflected over the y -axis.

Solution

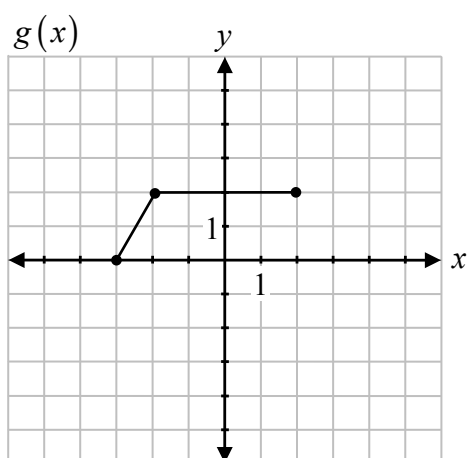
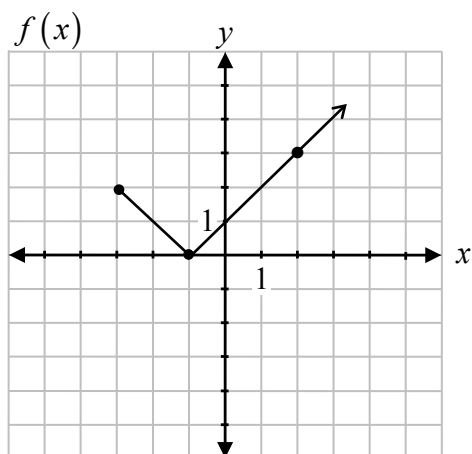
$(2, 4)$

1 mark

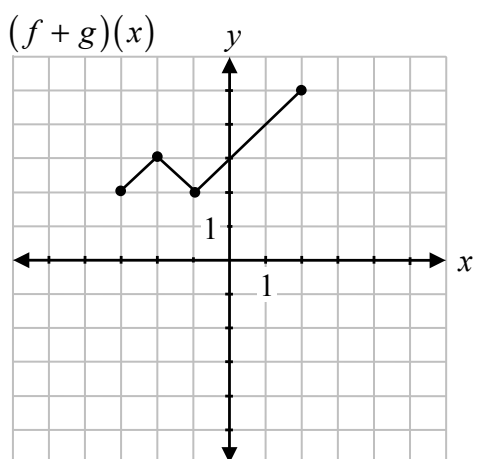
Question 7

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $(f + g)(x)$.



Solution



1 mark for operation of addition
1 mark for restricted domain

2 marks

Using the laws of logarithms, fully expand the expression:

$$\log_2 \left(\frac{w^3 x}{y-1} \right)$$

Solution

$$3\log_2 w + \log_2 x - \log_2 (y-1)$$

1 mark for power law
1 mark for product law
1 mark for quotient law

3 marks

Solve the following equation algebraically for θ , where $0 \leq \theta \leq 2\pi$:

$$2 \cos 2\theta = 1$$

Solution

Method 1

$$2(2 \cos^2 \theta - 1) = 1$$

1 mark for substitution of an appropriate identity

$$4 \cos^2 \theta - 2 = 1$$

$$\cos^2 \theta = \frac{3}{4}$$

$$\cos \theta = \pm \frac{\sqrt{3}}{2}$$

$$\cos \theta = \frac{\sqrt{3}}{2}$$

$$\cos \theta = -\frac{\sqrt{3}}{2}$$

1 mark for solving for $\cos \theta$ (½ mark for each value)

$$\theta = \frac{\pi}{6}, \frac{11\pi}{6}$$

$$\theta = \frac{5\pi}{6}, \frac{7\pi}{6}$$

2 marks (½ mark for each value of θ)

4 marks

Method 2

$$2(1 - 2 \sin^2 \theta) = 1$$

1 mark for substitution of an appropriate identity

$$2 - 4 \sin^2 \theta = 1$$

$$-4 \sin^2 \theta = -1$$

$$\sin^2 \theta = \frac{1}{4}$$

$$\sin \theta = \pm \frac{1}{2}$$

$$\sin \theta = \frac{1}{2}$$

$$\sin \theta = -\frac{1}{2}$$

1 mark for solving for $\sin \theta$ (½ mark for each value)

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$$

2 marks (½ mark for each value of θ)

4 marks

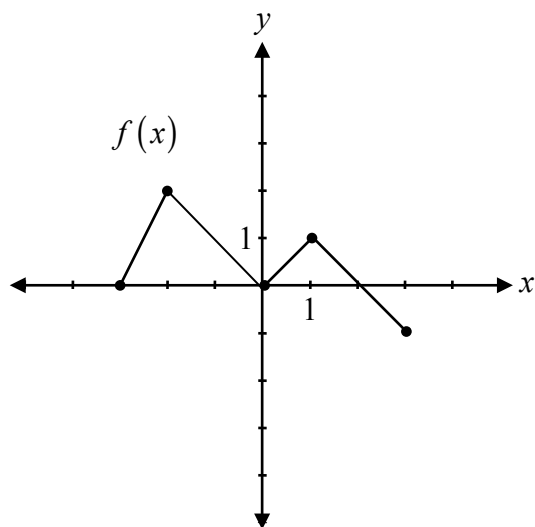
Note(s):

- Deduct a maximum of 1 mark if student omits second branch when taking the square root.

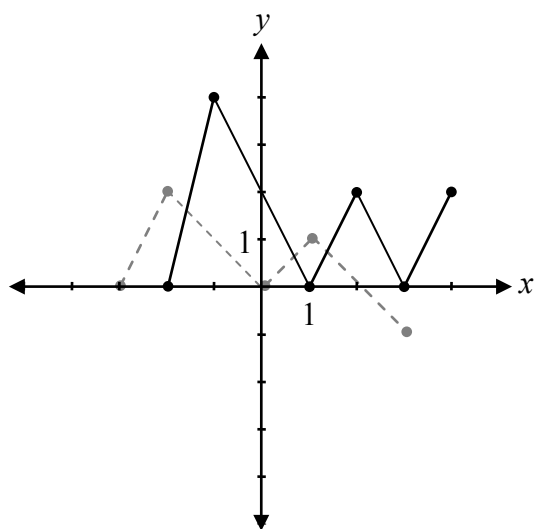
Question 10

R1, R2, R3

Given the graph of $y = f(x)$, sketch the graph of $y = 2|f(x-1)|$.



Solution



1 mark for vertical stretch
1 mark for horizontal translation
1 mark for absolute value

3 marks

Question 11

T6

Prove the identity for all permissible values of θ :

$$\cos \theta + \tan \theta \sin \theta = \frac{\tan \theta \sin \theta}{1 - \cos^2 \theta}$$

Solution

Left-Hand Side	Right-Hand Side	
$\cos \theta + \tan \theta \sin \theta$	$\frac{\tan \theta \sin \theta}{1 - \cos^2 \theta}$	
$\cos \theta + \frac{\sin \theta}{\cos \theta} \sin \theta$	$\frac{\frac{\sin \theta}{\cos \theta} \cdot \sin \theta}{\sin^2 \theta}$	1 mark for correct substitution of identities
$\cos \theta + \frac{\sin^2 \theta}{\cos \theta}$	$\frac{\sin^2 \theta}{\cos \theta} \cdot \frac{1}{\sin^2 \theta}$	1 mark for algebraic strategies
$\frac{\cos^2 \theta + \sin^2 \theta}{\cos \theta}$	$\frac{1}{\cos \theta}$	1 mark for logical process to prove the identity
$\frac{1}{\cos \theta}$		3 marks

Question 12

P1

Raoul has 8 shirts, 5 pairs of pants, and 3 hats. He adds the options together and determines that he has 16 different outfits to wear.

Raoul made an error in calculating the number of different outfits. Describe how to determine the correct number of outfits.

Solution

Raoul should have multiplied the number of clothing items to determine the total number of outfits.

1 mark

Given $f(x) = 2x - 1$ and $g(x) = x^2 + 1$:

a) Determine $f(x) \cdot g(x)$.

b) Determine $g(g(x))$.

Solution

$$\begin{aligned}\text{a) } f(x) \cdot g(x) &= (2x - 1)(x^2 + 1) \\ &= 2x^3 + 2x - x^2 - 1 \\ &= 2x^3 - x^2 + 2x - 1\end{aligned}$$

1 mark for product

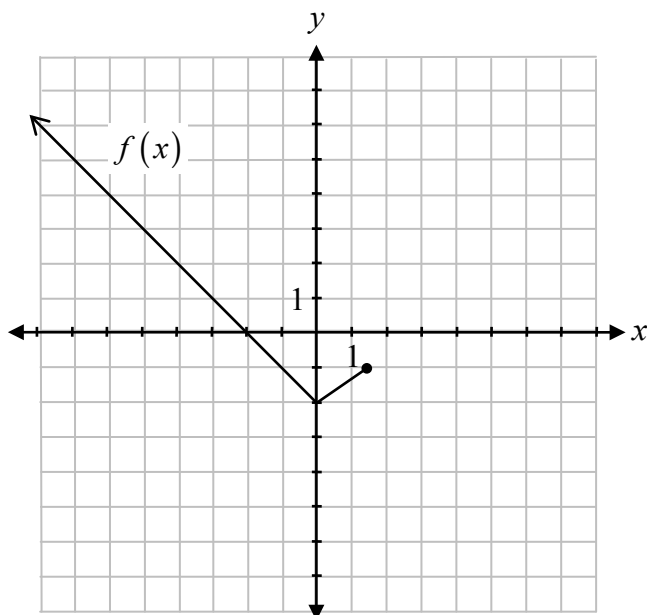
1 mark

$$\begin{aligned}\text{b) } g(g(x)) &= (x^2 + 1)^2 + 1 \\ &= x^4 + 2x^2 + 1 + 1 \\ &= x^4 + 2x^2 + 2\end{aligned}$$

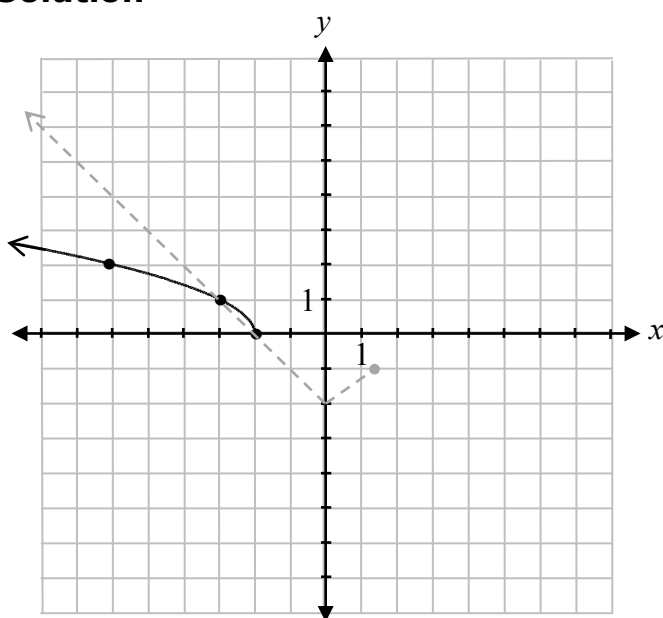
1 mark for composition

1 mark

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



Solution



1 mark for restricting the domain
 $\frac{1}{2}$ mark for shape between invariant points
 $\frac{1}{2}$ mark for shape to the left of invariant points

2 marks

Question 19

T5

Identify the equation that has a general solution of $\left. \begin{array}{l} \theta = \frac{\pi}{6} + 2\pi k \\ \theta = \frac{5\pi}{6} + 2\pi k \end{array} \right\}$ where $k \in \mathbb{Z}$.

a) $\sin \theta = \frac{1}{2}$

b) $\cos \theta = \frac{1}{2}$

c) $\sin \theta = \frac{\sqrt{3}}{2}$

d) $\cos \theta = \frac{\sqrt{3}}{2}$

Question 20

R13

Identify the function that has a domain of $x \leq -2$ and a range of $y \geq 3$.

a) $y = \sqrt{x+2} + 3$

b) $y = \sqrt{-(x+2)} + 3$

c) $y = -\sqrt{x-2} - 3$

d) $y = -\sqrt{-(x-2)} - 3$

Solve the following equation:

$$\log_3(x+3) + \log_3(x-5) = 2$$

Solution

$$\log_3[(x+3)(x-5)] = 2$$

1 mark for product law

$$(x+3)(x-5) = 3^2$$

1 mark for exponential form

$$x^2 - 2x - 15 = 9$$

$$x^2 - 2x - 24 = 0$$

$$(x-6)(x+4) = 0$$

$$x = 6 \quad \cancel{x = -4}$$

$\frac{1}{2}$ mark for solving for x

$\frac{1}{2}$ mark for rejecting extraneous root

3 marks

State a coterminal angle for $\theta = \frac{9\pi}{4}$.

Solution

$$\frac{9\pi}{4} + \frac{8\pi}{4} = \frac{17\pi}{4}$$

or

1 mark

$$405^\circ - 360^\circ = 45^\circ$$

Note(s):

- Other answers are possible.

Sketch the graph of the function $f(x) = \frac{2x+2}{x^2-1}$.

Solution

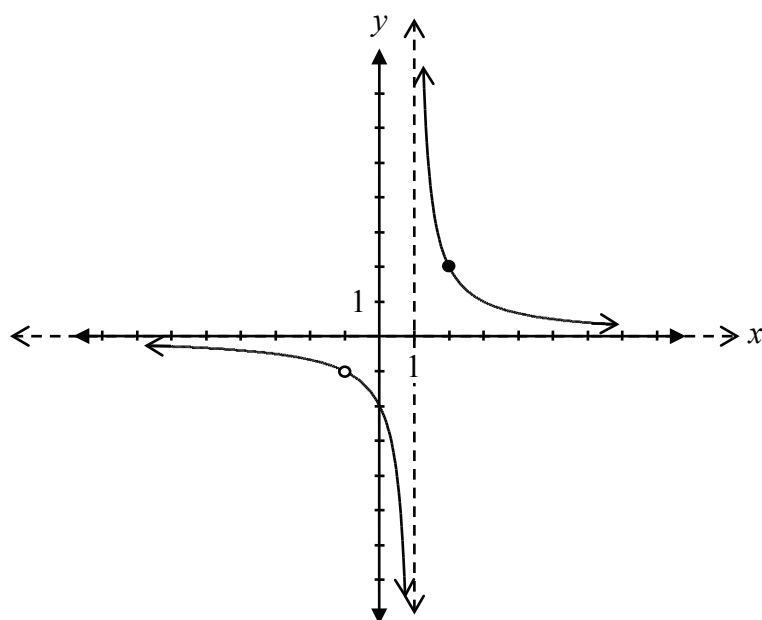
$$f(x) = \frac{2(\cancel{x+1})}{(x-1)(\cancel{x+1})}$$

$$= \frac{2}{x-1}$$

$\therefore$ there is a point of discontinuity (hole) at $(-1, -1)$

vertical asymptote at $x = 1$

horizontal asymptote at $y = 0$



1 mark for asymptotic behaviour at $x = 1$
 1 mark for asymptotic behaviour at $y = 0$
 1 mark for point of discontinuity (hole) at $(-1, -1)$ ($\frac{1}{2}$ mark for $x = -1$, $\frac{1}{2}$ mark for $y = -1$)
 $\frac{1}{2}$ mark for graph left of $x = 1$
 $\frac{1}{2}$ mark for graph right of $x = 1$

4 marks

Justify why the binomial expansion of $(x + x^3)^7$ does not have a term containing x^{10} .

Solution

Method 1

$$(x)^7, (x)^6(x^3)^1, (x)^5(x^3)^2, \dots$$

$$x^7, x^9, x^{11}, \dots$$

The exponents increase by 2.

Therefore x^{10} is not in the pattern.

1 mark for determining the pattern

1 mark for justification

2 marks

Method 2

$$x^{7-r}(x^3)^r = x^{10}$$

$$x^{7+2r} = x^{10}$$

$$7 + 2r = 10$$

$$2r = 3$$

$$r = \frac{3}{2}$$

½ mark for substitution

½ mark for solving for r

The binomial expansion does not contain x^{10} because the value of r must be a whole number.

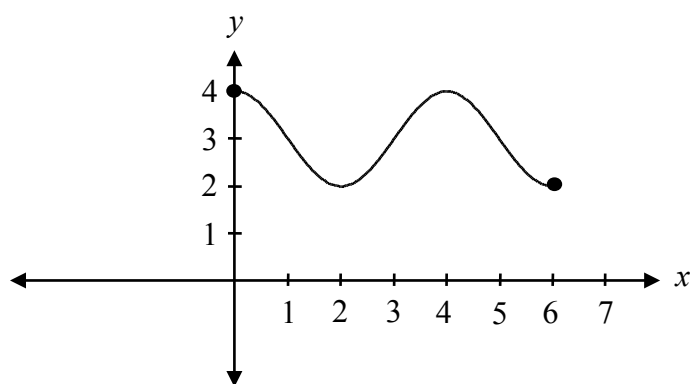
1 mark for justification

2 marks

Sketch the graph of $y = -\sin\left(\frac{\pi}{2}(x-1)\right) + 3$ over the domain $[0, 6]$.

Solution

$$\text{period} = \frac{2\pi}{\frac{\pi}{2}} = 4$$



1 mark for vertical reflection
1 mark for horizontal translation
1 mark for period
1 mark for vertical translation

4 marks

Note(s):

- Deduct $\frac{1}{2}$ mark if the domain $[0, 6]$ is not completely sketched.

When $P(x) = 3x^4 - kx^3 + 5x - 14$ is divided by $(x + 2)$, the remainder is -8 .

Determine the value of k .

Solution

Method 1

$$x = -2$$

$\frac{1}{2}$ mark for $x = -2$

$$-8 = 3(-2)^4 - k(-2)^3 + 5(-2) - 14$$

1 mark for remainder theorem

$$-8 = 48 + 8k - 10 - 14$$

$$-8 = 24 + 8k$$

$$-32 = 8k$$

$$k = -4$$

$\frac{1}{2}$ mark for solving for k

2 marks

Method 2

-2	3	-k	0	5	-14
	↓	-6	$2k + 12$	$-4k - 24$	$8k + 38$
	3	$-k - 6$	$2k + 12$	$-4k - 19$	$8k + 24$

$\frac{1}{2}$ mark for $x = -2$

1 mark for synthetic division
(or any other equivalent strategy)

$$-8 = 8k + 24$$

$$-32 = 8k$$

$$k = -4$$

$\frac{1}{2}$ mark for solving for k

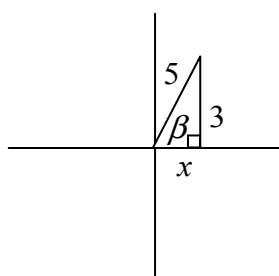
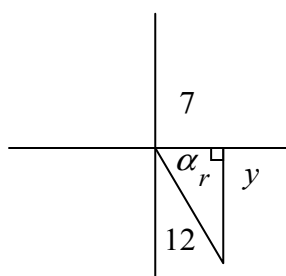
2 marks

Given that $\cos \alpha = \frac{7}{12}$ where α is in quadrant IV, and $\sin \beta = \frac{3}{5}$ where β is in quadrant I, determine the exact value of:

- a) $\sin(\alpha - \beta)$
 b) $\csc(\alpha - \beta)$

Solution

a)



$$12^2 - 7^2 = y^2 \qquad 5^2 - 3^2 = x^2$$

$$95 = y^2 \qquad 16 = x^2$$

$$\pm\sqrt{95} = y \qquad \pm 4 = x$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= \left(\frac{-\sqrt{95}}{12} \right) \left(\frac{4}{5} \right) - \left(\frac{7}{12} \right) \left(\frac{3}{5} \right)$$

$$= \frac{-4\sqrt{95}}{60} - \frac{21}{60}$$

$$= \frac{-4\sqrt{95} - 21}{60}$$

$$\text{b) } \csc(\alpha - \beta) = \frac{60}{-4\sqrt{95} - 21}$$

$\frac{1}{2}$ mark for value of x

$\frac{1}{2}$ mark for value of y

$\frac{1}{2}$ mark for $\sin \alpha$

$\frac{1}{2}$ mark for $\cos \beta$

1 mark for substitution into correct identity

3 marks

1 mark

Note(s):

- accept any of the following values for x : $x = \pm 4$, or $x = 4$
- accept any of the following values for y : $y = \pm\sqrt{95}$, $y = \sqrt{95}$, or $y = -\sqrt{95}$

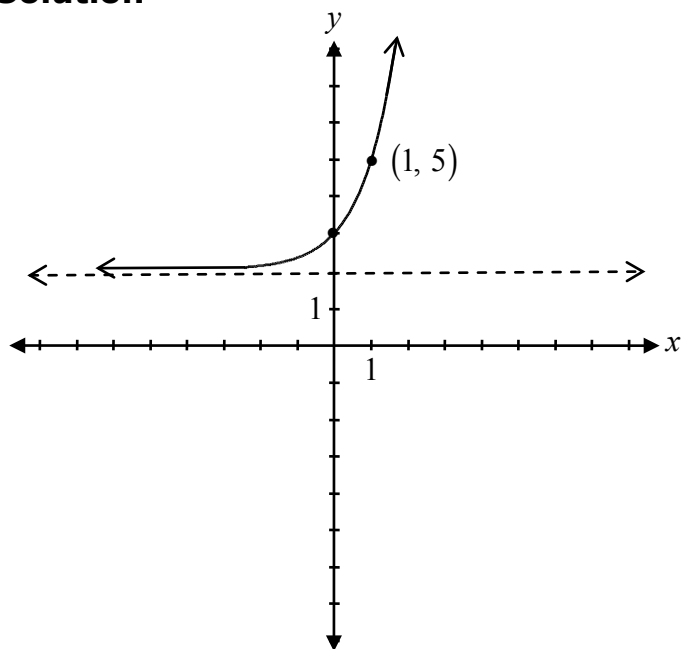
Describe the difference between the graph of $f(x) = \frac{7(x+2)}{x+2}$ and the graph of $g(x) = \frac{7(x-2)}{x+2}$ at $x = -2$.

Solution

The graph of $f(x) = \frac{7(x+2)}{x+2}$ has a point of discontinuity and the graph of $g(x) = \frac{7(x-2)}{x+2}$ has an asymptote.

1 mark

Sketch the graph of $f(x) = 3^x + 2$.

Solution

1 mark for increasing exponential function
1 mark for asymptotic behaviour at $y = 2$

2 marks

Solve algebraically:

$${}_nC_3 = n - 2$$

Solution

$$\frac{n!}{(n-3)!3!} = n - 2$$

½ mark for substitution

$$\frac{n(n-1)\cancel{(n-2)}\cancel{(n-3)}!}{\cancel{(n-3)}!3!} = \cancel{n-2}$$

1 mark for factorial expansion

$$n(n-1) = 6$$

½ mark for simplification of factorials

$$n^2 - n - 6 = 0$$

$$(n-3)(n+2) = 0$$

$$n = 3 \quad \cancel{n = -2}$$

½ mark for rejecting extraneous root

½ mark for the value of n

3 marks

Describe the error that was made when solving the following equation:

$$\sin^2 \theta + \sin \theta - 2 = 1$$

$$\sin^2 \theta + \sin \theta = 3$$

$$\sin \theta (\sin \theta + 1) = 3$$

$$\sin \theta = 3 \quad \sin \theta + 1 = 3$$

$$\sin \theta = 2$$

$\therefore$ No solution

$\therefore$ No solution

Solution

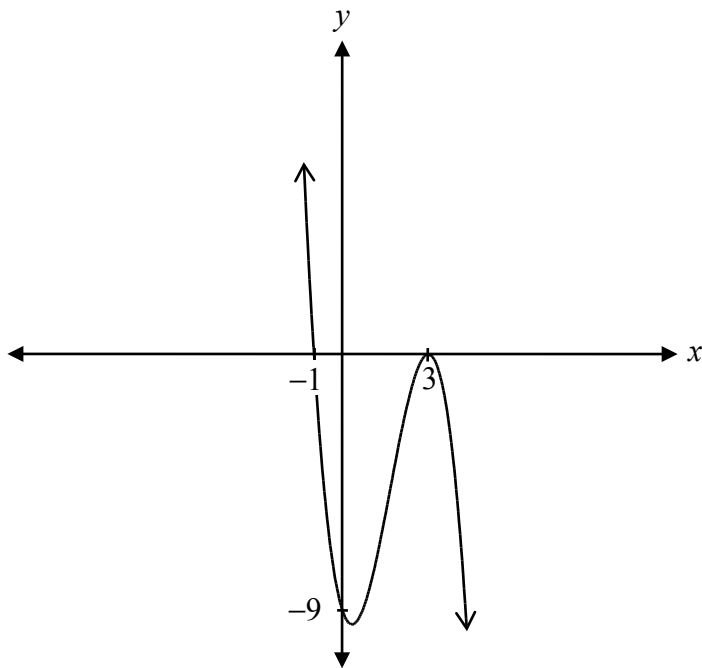
The student did not apply the zero product principle before factoring.

1 mark

Sketch the graph of the polynomial function with the following characteristics.

- a y -intercept of -9
- zeroes at -1 and 3
- the zero at -1 has a multiplicity of 1 and the zero at 3 has a multiplicity of 2

Solution



1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

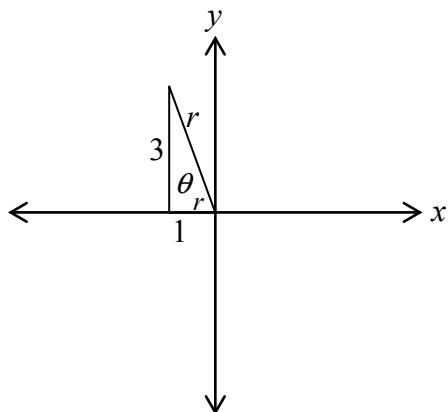
1 mark for multiplicity ($\frac{1}{2}$ mark for multiplicity at $x = 3$, $\frac{1}{2}$ mark for multiplicity at $x = -1$)

$\frac{1}{2}$ mark for shape of a cubic function

3 marks

Given $\cot \theta = -\frac{1}{3}$, where θ is in quadrant II, determine the exact value of $\sin \theta$.

Solution



$$\cot \theta = \frac{x}{y}$$

$$r^2 = x^2 + y^2$$

$$r^2 = (-1)^2 + (3)^2 \quad \frac{1}{2} \text{ mark for substitution}$$

$$r^2 = 10$$

$$r = \pm\sqrt{10} \quad \frac{1}{2} \text{ mark for solving for } r$$

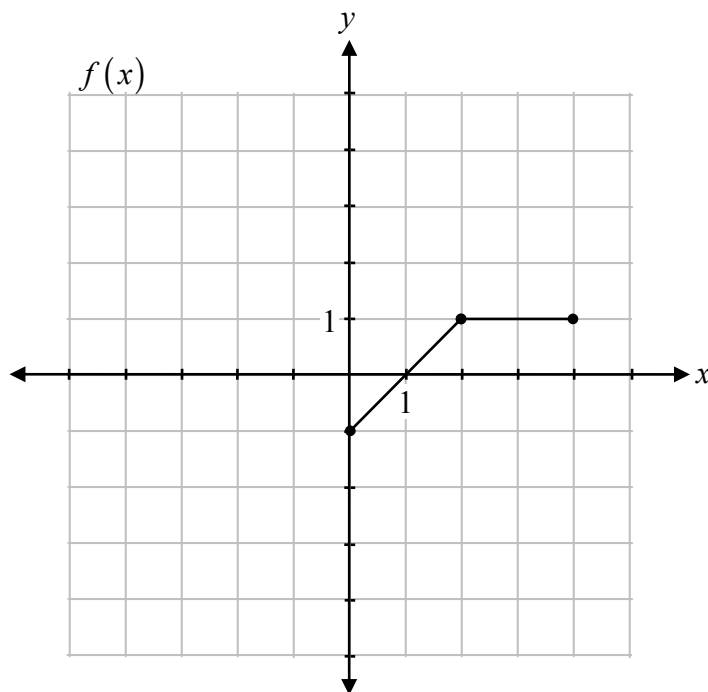
$$\sin \theta = \frac{3}{\sqrt{10}} \quad 1 \text{ mark for } \sin \theta \text{ (}\frac{1}{2} \text{ mark for the quadrant, } \frac{1}{2} \text{ mark for the value)}$$

2 marks

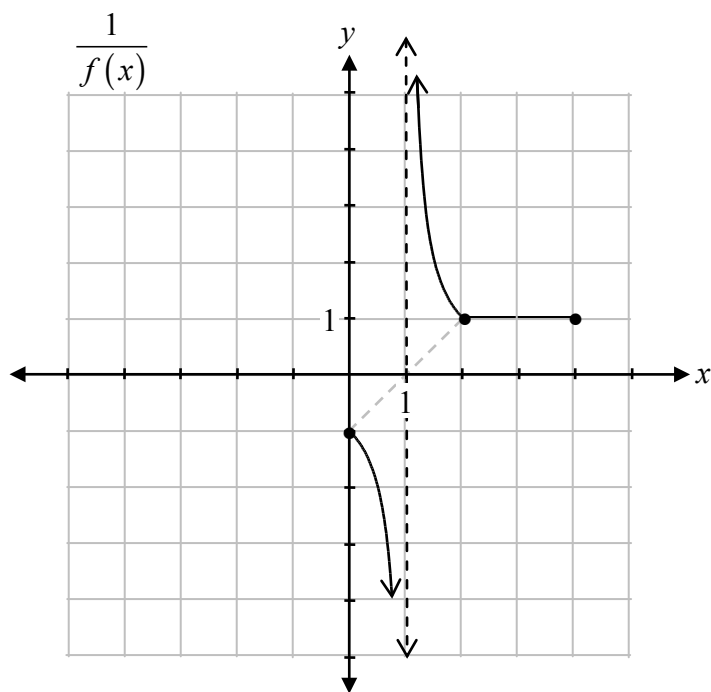
Note(s):

- accept any of the following values for r : $r = \pm\sqrt{10}$, $r = \sqrt{10}$

Given the function $f(x)$, sketch the graph of the reciprocal, $\frac{1}{f(x)}$.



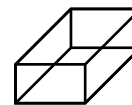
Solution



1 mark for asymptotic behaviour at $x = 1$
 $\frac{1}{2}$ mark for graph left of vertical asymptote at $x = 1$
 $\frac{1}{2}$ mark for graph right of vertical asymptote at $x = 1$

2 marks

The volume of a planter, in the shape of a rectangular prism, can be modelled by the polynomial function $V(x) = x^3 + 3x^2 - 34x + 48$.



Determine the factors of the function, $V(x)$, which represent possible dimensions of this planter.

Solution

$$2^3 + 3(2)^2 - 34(2) + 48 = 0$$

$\therefore x - 2$ is a factor

2	1	3	-34	48
	↓	2	10	-48
	1	5	-24	0

1 mark for identifying one possible value of x

1 mark for synthetic division (or any other equivalent strategy)

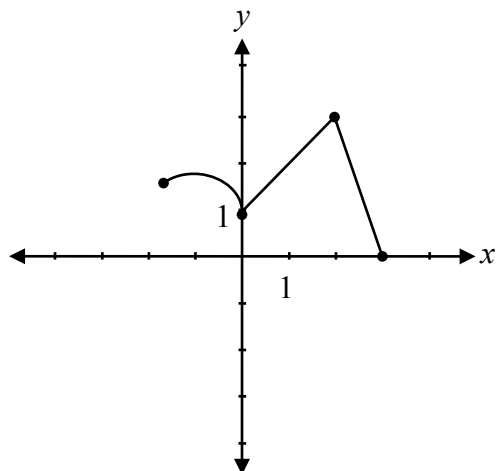
$$V(x) = (x - 2)(x^2 + 5x - 24)$$

$$V(x) = \underline{(x - 2)(x + 8)(x - 3)}$$

1 mark for identifying all factors

3 marks

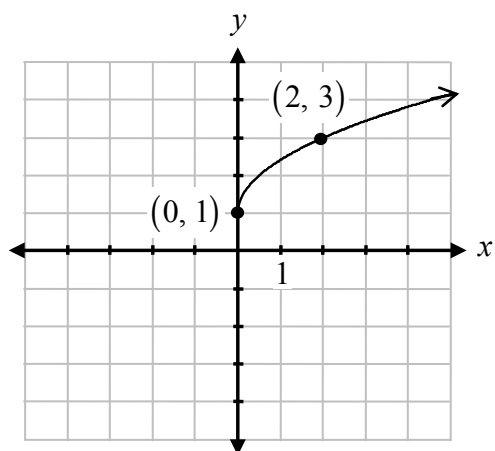
Describe how to determine the range of the inverse of the following graph.

**Solution**

The domain of the graph becomes the range of the inverse.

1 mark

Sketch the graph of the function $y = \sqrt{2x} + 1$.

Solution

1 mark for shape of a radical function

1 mark for vertical translation

1 mark for horizontal compression

3 marks

Given the following characteristics of a sinusoidal function:

- an amplitude of 2
- a vertical translation down 3 units
- a period of $\frac{\pi}{4}$

- a) Determine an equation of this sinusoidal function in the form $y = a \sin b(x - c) + d$.
- b) Determine the range of this function.

Solution

a) $b = \frac{2\pi}{\frac{\pi}{4}}$

$$b = 8$$

$$y = 2 \sin(8x) - 3$$

1 mark for the value of b

$\frac{1}{2}$ mark for amplitude

$\frac{1}{2}$ mark for vertical translation

2 marks

b) Range: $\{y \mid y \in \mathbb{R}, -5 \leq y \leq -1\}$

or

1 mark

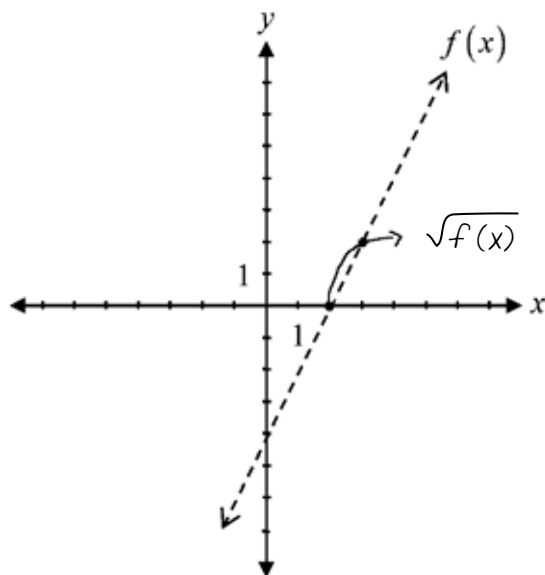
Range: $[-5, -1]$

Note(s):

- Other answers are possible for a).

Suah was given the graph of $f(x)$ and asked to graph $y = \sqrt{f(x)}$.

Her solution is given on the graph below.



Describe the error Suah made when sketching the graph of $y = \sqrt{f(x)}$.

Solution

Suah's graph did not cross the invariant point at $y = 1$.

1 mark

Solve:

$$9^{2x+1} = 27^x$$

Solution

$$3^{2(2x+1)} = 3^{3x}$$

1 mark for changing to a common base

$$3^{4x+2} = 3^{3x}$$

1 mark for exponent law (½ mark for each side)

$$4x + 2 = 3x$$

½ mark for equating exponents

$$x = -2$$

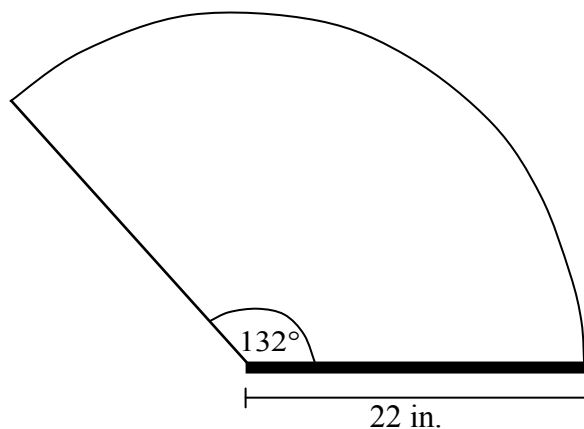
½ mark for solving for x

3 marks

Question 1

T1

A section of a car windshield is cleaned by a wiper as shown in the diagram below. The arm of the wiper is 22 inches, and it rotates through a central angle of 132° . Determine the length of the arc that is created by the tip of the wiper.



Solution

$$\theta = 132 \times \frac{\pi}{180}$$

1 mark for conversion

$$= \frac{132\pi}{180} \text{ or } \frac{11\pi}{15}$$

$$s = \theta r$$

$$s = \frac{11\pi}{15} (22)$$

1 mark for substitution

$$s = \frac{242\pi}{15} \text{ inches}$$

2 marks

or

$$s = 50.684 \text{ inches}$$

There are 20 boys and 11 girls who can be selected to be on a team.

Determine the number of ways that 7 boys and 5 girls can be selected for this team.

Solution

$${}_{20}C_7 \cdot {}_{11}C_5 = 35\,814\,240$$

½ mark for ${}_{20}C_7$

½ mark for ${}_{11}C_5$

1 mark for the product of combinations

2 marks

Question 3

R10

A water filtration system which removes impurities from a sample of water can be modelled by

$$P = 0.25(0.55)^n, \text{ where:}$$

P = the percentage of impurities remaining, in decimal form

n = the number of filters

Determine, algebraically, how many filters are required so that less than 1% of the impurities remain in the water sample. Express your answer as a whole number.

Solution

$$0.01 = 0.25(0.55)^n \quad \frac{1}{2} \text{ mark for substitution}$$

$$0.04 = (0.55)^n$$

$$\log(0.04) = n \log(0.55) \quad \frac{1}{2} \text{ mark for applying logarithms}$$

$$\frac{\log(0.04)}{\log(0.55)} = n \quad \frac{1}{2} \text{ mark for power law}$$

$$n = 5.384 \ 203 \quad \frac{1}{2} \text{ mark for solving for } n$$

$\therefore$ 6 filters are needed

2 marks

In the binomial expansion of $\left(x^2 - \frac{2}{y}\right)^8$, determine the middle term in simplified form.

Solution

$$t_5 = {}_8C_4 \left(x^2\right)^{8-4} \left(-\frac{2}{y}\right)^4 \quad 2 \text{ marks (1 mark for } {}_8C_4, \frac{1}{2} \text{ mark for each consistent factor)}$$

$$t_5 = 70x^8 \left(\frac{16}{y^4}\right)$$

$$t_5 = \frac{1120x^8}{y^4}$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark for exponents)

3 marks

Solve the following equation algebraically over the interval $[0, 2\pi]$.

$$6\sin^2 \theta + \sin \theta - 1 = 0$$

Solution

$$(3\sin \theta - 1)(2\sin \theta + 1) = 0$$

$$3\sin \theta - 1 = 0$$

$$2\sin \theta + 1 = 0$$

$$\sin \theta = \frac{1}{3}$$

$$\sin \theta = -\frac{1}{2}$$

1 mark for solving for $\sin \theta$ ($\frac{1}{2}$ mark for each branch)

$$\theta_r = 0.339836$$

$$\theta = 0.340$$

$$\theta = \frac{7\pi}{6}$$

$$\theta = 2.802$$

$$\theta = \frac{11\pi}{6}$$

2 marks ($\frac{1}{2}$ mark for each value of θ)

or

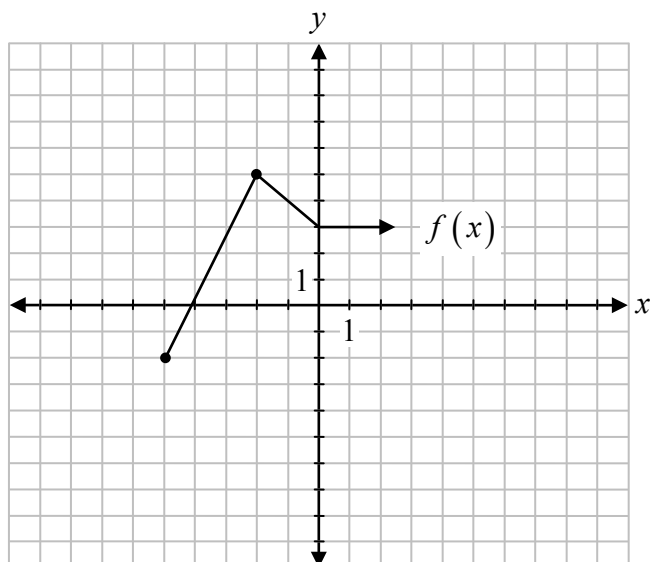
$$\theta = 0.340, 2.802, 3.665, 5.760$$

3 marks

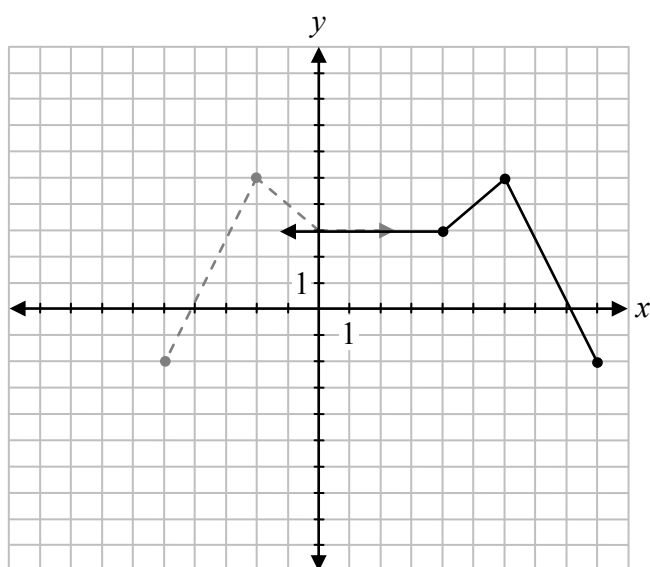
Question 6

R4, R5

Given the graph of $y = f(x)$, sketch the graph of $y = f(-x + 4)$.



Solution



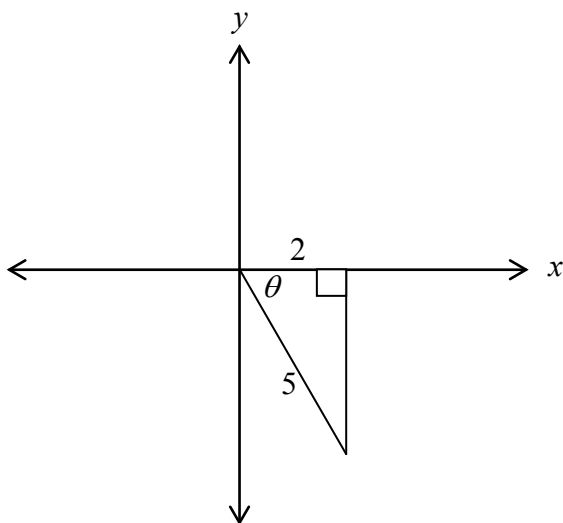
1 mark for reflection over the y -axis
1 mark for horizontal translation

2 marks

Question 7

T2, T3

Given the following triangle, determine $\csc \theta$.



Solution

$$x^2 + y^2 = r^2$$

$$2^2 + y^2 = 5^2$$

½ mark for substitution

$$y^2 = 21$$

$$y = \pm\sqrt{21}$$

½ mark for solving for y

$$\csc \theta = -\frac{5}{\sqrt{21}}$$

1 mark for $\csc \theta$ (½ mark for quadrant, ½ mark for value)

2 marks

Note(s):

- Accept any of the following values for y : $y = \pm\sqrt{21}$, $y = \sqrt{21}$, or $y = -\sqrt{21}$.

Solve algebraically.

$${}_nP_2 = 9n$$

Solution

$$\frac{n!}{(n-2)!} = 9n$$

½ mark for substitution into equation

$$\frac{n(n-1)\cancel{(n-2)!}}{\cancel{(n-2)!}} = 9n$$

1 mark for factorial expansion

$$n(n-1) = 9n$$

½ mark for simplification of factorials

$$n^2 - n = 9n$$

$$n^2 - 10n = 0$$

$$n(n-10) = 0$$

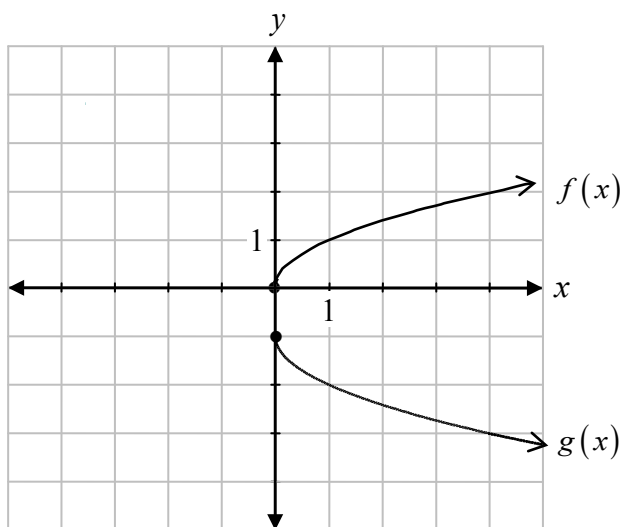
$$\cancel{n=0} \quad n = 10$$

½ mark for rejecting the extraneous root

½ mark for the value of n

3 marks

Describe the transformations applied to the graph of $f(x)$ to obtain the graph of $g(x)$.



Solution

A reflection over the x -axis then a vertical translation of one unit down.

or

A vertical translation of one unit up then a reflection over the x -axis.

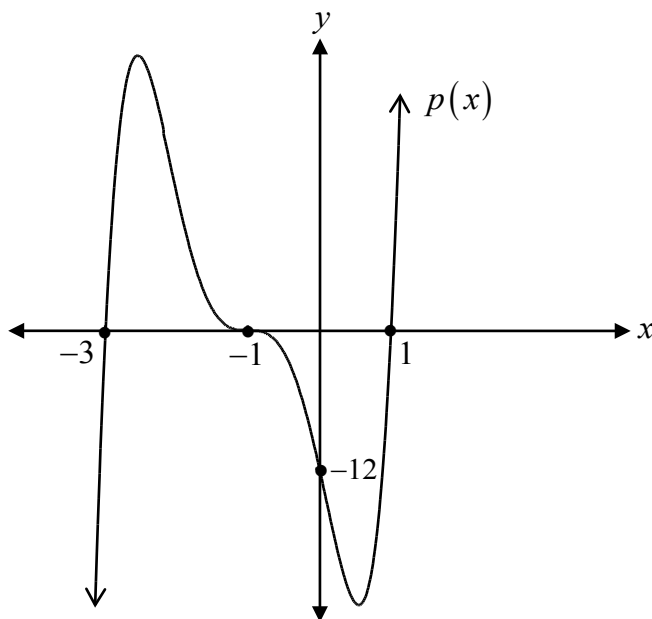
1 mark for reflection over the x -axis
1 mark for vertical translation

2 marks

Note(s):

- Award a maximum of 1 mark if correct transformations are given in the incorrect order.

Determine, algebraically, the value of the leading coefficient of the graph of the polynomial function, $p(x)$.

**Solution**

$$p(x) = a(x+3)(x+1)^3(x-1)$$

$$-12 = a(3)(1)^3(-1)$$

$$-12 = -3a$$

$$a = 4$$

$\frac{1}{2}$ mark for factors of $p(x)$

$\frac{1}{2}$ mark for odd multiplicity for $(x+1)$ greater than 1

$\frac{1}{2}$ mark for substituting $p(0) = -12$

$\frac{1}{2}$ mark for value of a

2 marks

Question 11

P1

Frank, Liam, Chan, and Thao are going to a movie.

Determine the number of ways they can sit in a row of four chairs, if Frank and Chan must sit beside each other.

Solution

$3!2!$

$\frac{1}{2}$ mark for $3!$

$\frac{1}{2}$ mark for $2!$ as a product of factorials

12

1 mark

Determine, algebraically, if $f(x) = \frac{1}{x+5}$ and $g(x) = \frac{1}{x-5}$ are inverses of each other.

Justify your answer.

Solution

Method 1

Let $f(x) = y$

$$y = \frac{1}{x+5}$$

$$x = \frac{1}{y+5}$$

1 mark for switching x and y values

$$y+5 = \frac{1}{x}$$

$$y = \frac{1}{x} - 5$$

$\frac{1}{2}$ mark for solving for y

$$f^{-1}(x) = \frac{1}{x} - 5$$

$$\therefore f^{-1}(x) \neq g(x)$$

$\frac{1}{2}$ mark for justification

2 marks

Method 2

$$f(g(x)) = \frac{1}{\left(\frac{1}{x-5}\right) + 5}$$

1 mark for $f(g(x))$ or $g(f(x))$

$$= \frac{1}{\frac{1+5x-25}{x-5}}$$

$$= \frac{1}{\frac{5x-24}{x-5}}$$

$$= \frac{x-5}{5x-24}$$

$\frac{1}{2}$ mark for simplification

$$g(f(x)) = \frac{1}{\left(\frac{1}{x+5}\right) - 5}$$

$$= \frac{1}{\frac{1-5x-25}{x+5}}$$

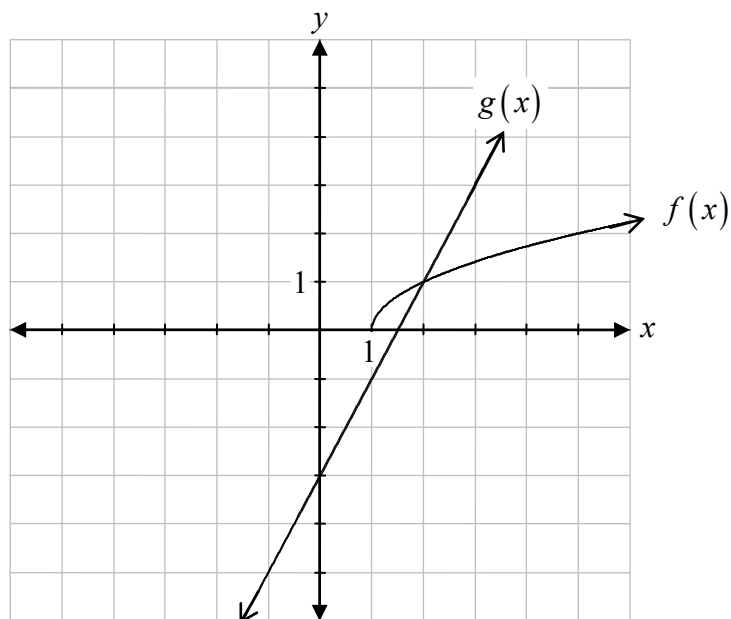
$$= \frac{1}{\frac{-5x-24}{x+5}}$$

$$= \frac{x+5}{-5x-24}$$

$\therefore$ They are not inverses because $f(g(x)) \neq x$ or $g(f(x)) \neq x$. $\frac{1}{2}$ mark for justification

2 marks

Using the graphs of $y = f(x)$ and $y = g(x)$, solve $f(x) = g(x)$.



Solution

$$x = 2$$

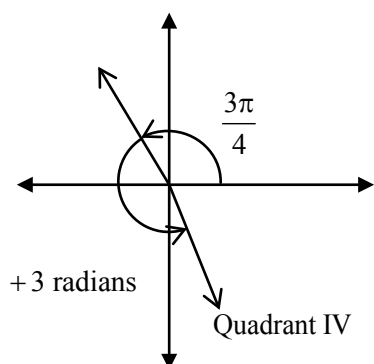
1 mark

An angle in standard position measures $\frac{3\pi}{4}$.

Determine in which quadrant the terminal arm of this angle is located after a rotation of 3 radians.

Justify your answer.

Solution



or

The angle $\frac{3\pi}{4}$ terminates in quadrant II. A rotation of 3 radians is almost a half rotation; therefore, the terminal arm is located in quadrant IV.

1 mark for justification

1 mark

Prove the following identity for all permissible values of θ .

$$\frac{\sin 2\theta}{1 - \cos 2\theta} = \cot \theta$$

Solution

Left-Hand Side	Right-Hand Side
$\frac{\sin 2\theta}{1 - \cos 2\theta}$ $\frac{2 \sin \theta \cos \theta}{1 - (1 - 2 \sin^2 \theta)}$ $\frac{2 \sin \theta \cos \theta}{2 \sin^2 \theta}$ $\frac{\cos \theta}{\sin \theta}$ $\cot \theta$	$\cot \theta$ 1 mark for correct substitution of identities 1 mark for appropriate algebraic strategies 1 mark for logical process to prove the identity 3 marks

If the range of $y = f(x)$ is $-3 \leq y \leq 6$, determine the range of $y = 2f(3x)$.

Solution

$[-6, 12]$

or

$-6 \leq y \leq 12$

1 mark

Maurice incorrectly solved the equation, $\sin \theta + 1 = 0$, over the interval $[0^\circ, 360^\circ]$.

$$\begin{aligned}\sin \theta + 1 &= 0 \\ \sin \theta &= -1 \\ \sin \theta &= 270^\circ\end{aligned}$$

Describe his error.

Solution

Maurice should have written that θ is equal to 270° not $\sin \theta = 270^\circ$.

1 mark

Match the following radical functions with their graphs.

Solution

Place the appropriate letter in this column.

$$f(x) = 2\sqrt{-(x+3)} \quad \underline{\text{B}}$$

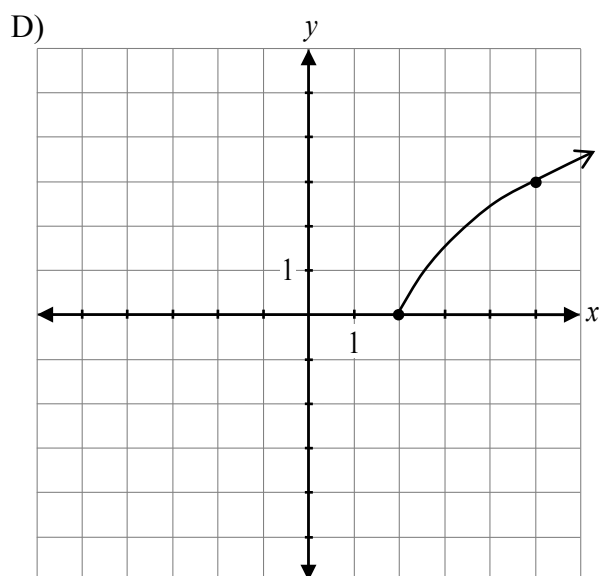
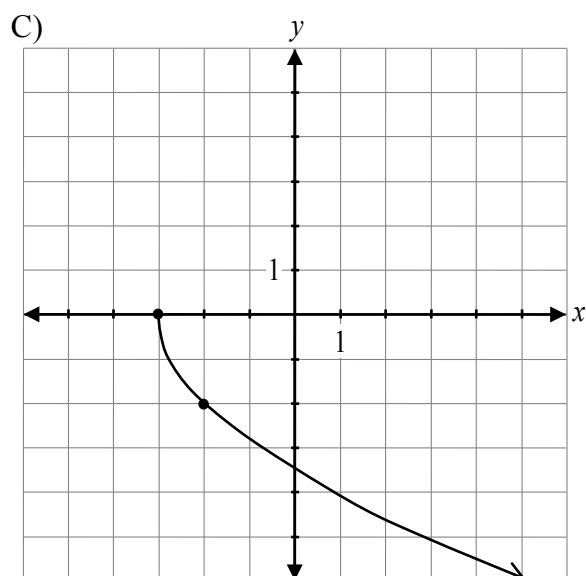
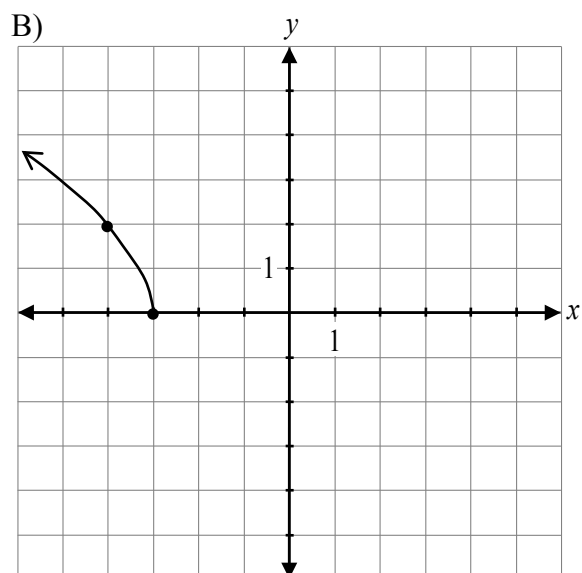
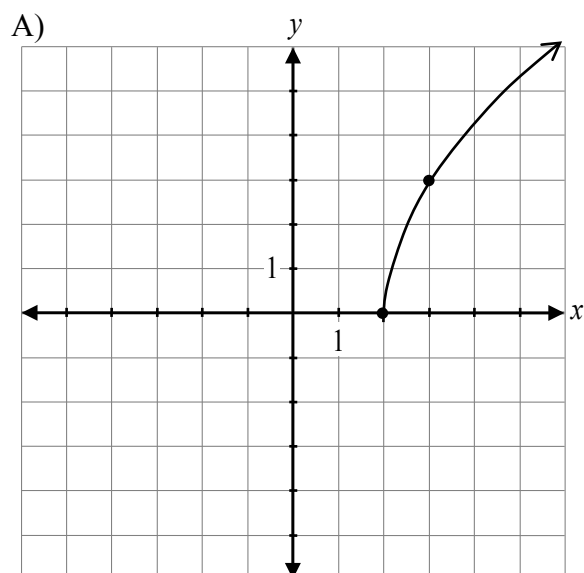
$$g(x) = -2\sqrt{(x+3)} \quad \underline{\text{C}}$$

$$h(x) = 3\sqrt{(x-2)} \quad \underline{\text{A}}$$

$$k(x) = \sqrt{3(x-2)} \quad \underline{\text{D}}$$

½ mark for each correct answer

2 marks



Express $p(x) = x^3 - 2x^2 - 4x + 8$ as a product of factors.

Solution

$$\begin{aligned} p(2) &= 2^3 - 2(2)^2 - 4(2) + 8 \\ &= 8 - 8 - 8 + 8 \\ &= 0 \end{aligned}$$

1 mark for identifying one possible value of x

$$\begin{array}{r|rrrrr} 2 & 1 & -2 & -4 & 8 \\ & \downarrow & 2 & 0 & -8 \\ \hline & 1 & 0 & -4 & 0 \end{array}$$

1 mark for synthetic division (or any equivalent strategy)

$$p(x) = (x - 2)(x^2 - 4)$$

1 mark for product of factors

or

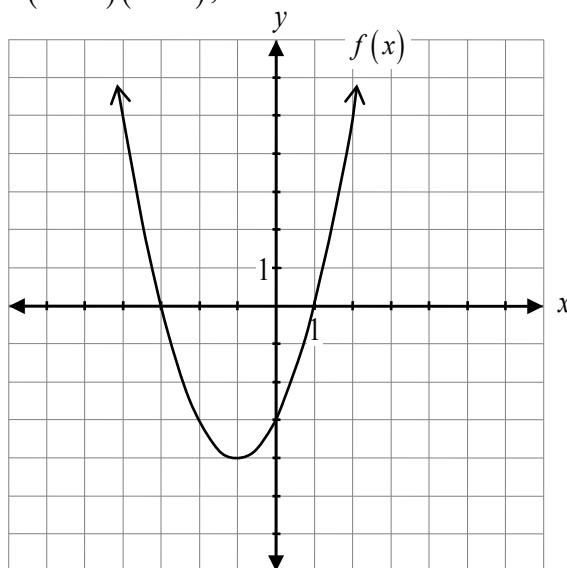
$$p(x) = (x - 2)(x - 2)(x + 2)$$

or

$$p(x) = (x - 2)^2(x + 2)$$

3 marks

Given the graph of $f(x) = (x+3)(x-1)$,

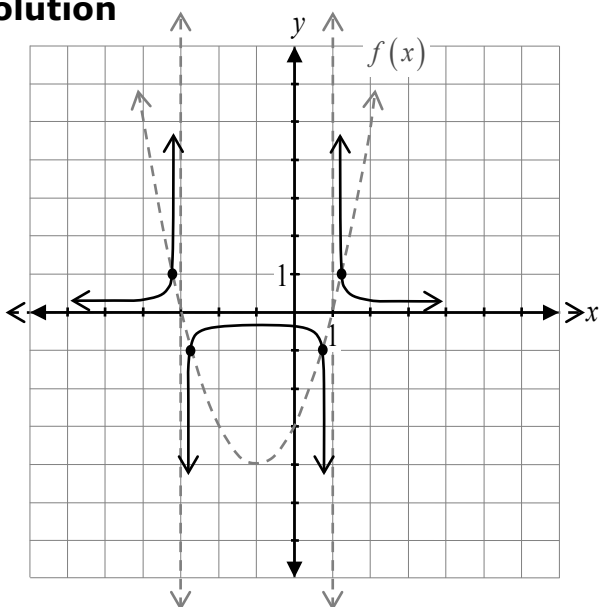


a) sketch the graph of $g(x) = \frac{1}{f(x)}$.

b) describe how to sketch the graph of $h(x) = |f(x)|$.

Solution

a)



1 mark for asymptotic behaviour at $x = 1$ and $x = -3$

$\frac{1}{2}$ mark for asymptotic behaviour at $y = 0$

$\frac{1}{2}$ mark for graph left of $x = -3$

$\frac{1}{2}$ mark for graph between $x = -3$ and $x = 1$

$\frac{1}{2}$ mark for graph right of $x = 1$

3 marks

b) Change all negative y-values to positive values. The positive y-values do not change.

1 mark

Note(s):

- Award 1 mark if only vertical asymptotes at $x = 1$ and $x = -3$ are drawn.

Describe how the value of m in the equation $y = \log_3(x - m)$, $m \in \mathbb{R}$, affects the asymptote on the graph of $y = \log_3 x$.

Solution

The vertical asymptote is translated horizontally left or right m units from the y -axis.

1 mark

Solve algebraically.

$$25^x = \left(\frac{1}{5}\right)^{-3x+1}$$

Solution

$$\left(5^2\right)^x = \left(5^{-1}\right)^{-3x+1}$$

$$5^{2x} = 5^{3x-1}$$

$$2x = 3x - 1$$

$$x = 1$$

1 mark for changing to a common base

½ mark for exponent law

½ mark for equating exponents

2 marks

Solve $\cos 2\theta = 0$, where $\theta \in \mathbb{R}$.

Solution

Method 1

$$1 - 2\sin^2 \theta = 0$$

1 mark for double-angle identity

$$\sin^2 \theta = \frac{1}{2}$$

$$\sin \theta = \frac{\sqrt{2}}{2}$$

$$\sin \theta = \frac{-\sqrt{2}}{2}$$

1 mark for solving for $\sin \theta$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$\theta = \frac{5\pi}{4}, \frac{7\pi}{4}$$

1 mark for values of θ
(½ mark for each branch)

$$\left. \begin{aligned} \theta &= \frac{\pi}{4} + \pi k \\ &= \frac{3\pi}{4} + \pi k \end{aligned} \right\} \text{where } k \in \mathbb{Z} \text{ or } \theta = \frac{\pi}{4} + \frac{\pi k}{2}, \text{ where } k \in \mathbb{Z}$$

1 mark for general solution

4 marks

Method 2

$$2\cos^2 \theta - 1 = 0$$

1 mark for double-angle identity

$$\cos^2 \theta = \frac{1}{2}$$

$$\cos \theta = \frac{\sqrt{2}}{2}$$

$$\cos \theta = \frac{-\sqrt{2}}{2}$$

1 mark for solving for $\cos \theta$

$$\theta = \frac{\pi}{4}, \frac{7\pi}{4}$$

$$\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$$

1 mark for values of θ
(½ mark for each branch)

$$\left. \begin{aligned} \theta &= \frac{\pi}{4} + \pi k \\ &= \frac{3\pi}{4} + \pi k \end{aligned} \right\} \text{where } k \in \mathbb{Z} \text{ or } \theta = \frac{\pi}{4} + \frac{\pi k}{2}, \text{ where } k \in \mathbb{Z}$$

1 mark for general solution

4 marks

Note(s):

- Deduct a maximum of 1 mark if student omits second branch when taking the square root.

Method 3

$$\cos 2\theta = 0$$

$$2\theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

2 marks for values of 2θ (1 mark for each value)

$$\left. \begin{array}{l} 2\theta = \frac{\pi}{2} + 2\pi k \\ 2\theta = \frac{3\pi}{2} + 2\pi k \end{array} \right\} \text{where } k \in \mathbb{Z}$$

1 mark for general solution

$$\theta = \frac{\pi}{4} + \pi k$$

1 mark for values of θ

$$\theta = \frac{3\pi}{4} + \pi k$$

4 marks

Describe a difference between the graphs of $y = f(x)$ and $y = g(x)$.

$$f(x) = -2(x+1)^2(x+3)$$

$$g(x) = 2(x+1)^2(x+3)$$

Solution

The end behaviour is different, because the graph of $f(x)$ falls to the right and the graph of $g(x)$ rises to the right.

or

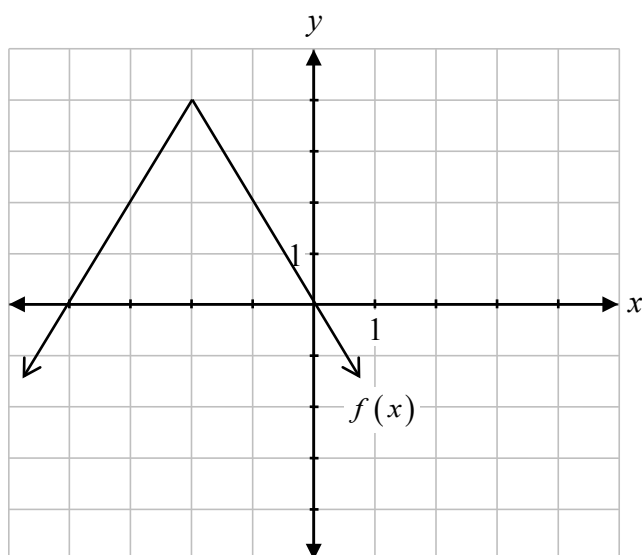
The y -intercept of $f(x)$ is negative while the y -intercept of $g(x)$ is positive.

or

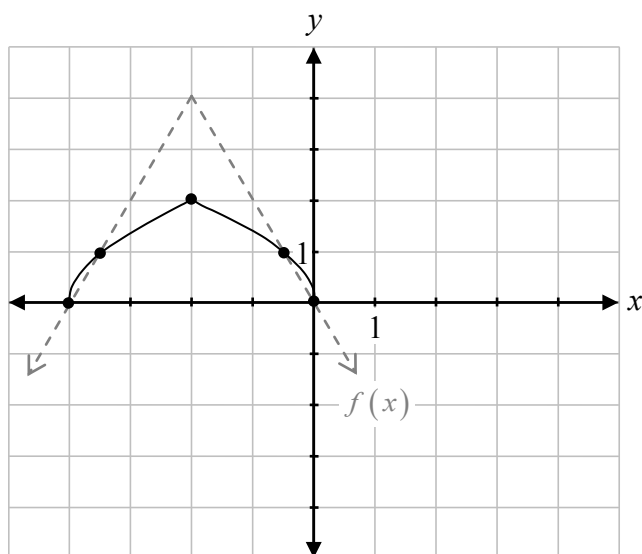
One graph is a reflection over the x -axis of the other graph.

1 mark

Given the graph of $y = f(x)$, sketch the graph of $\sqrt{f(x)}$.



Solution



1 mark for restricted domain

$\frac{1}{2}$ mark for shape between both invariant points,
 $\{0 \leq y \leq 1\}$

$\frac{1}{2}$ mark for shape above invariant points, $\{1 \leq y \leq 2\}$

2 marks

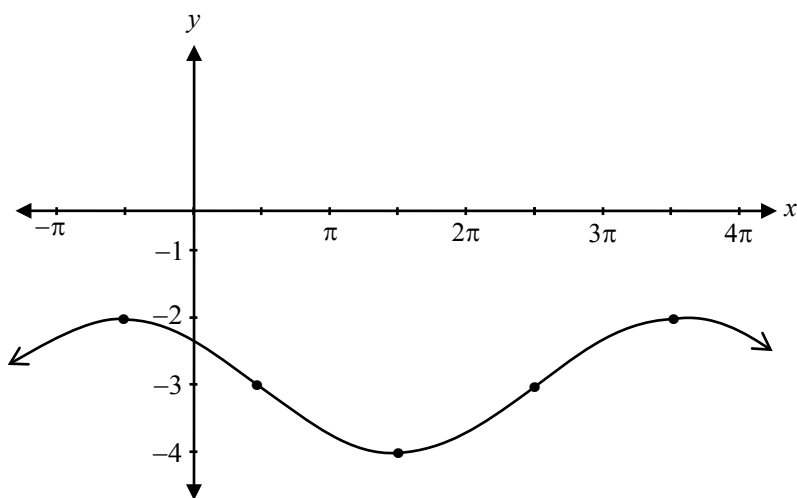
Describe the relationship between the zeros of the function $f(x) = (2x - 1)(x + 3)^2$, the roots of the equation $(2x - 1)(x + 3)^2 = 0$, and the x -intercepts of the graph of $y = f(x)$.

Solution

The zeros, roots, and x -intercepts all have the same values.

1 mark

Sketch a graph of at least one period of the function $f(x) = \cos\left[\frac{1}{2}\left(x + \frac{\pi}{2}\right)\right] - 3$.

Solution

1 mark for period
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Verify that $\theta = \frac{4\pi}{3}$ is a solution of the equation $4 \cos^2 \theta - 1 = 0$.

Solution

$$\text{Left-hand side} = 4 \cos^2 \left(\frac{4\pi}{3} \right) - 1$$

$$= 4 \left(\frac{-1}{2} \right)^2 - 1$$

½ mark for value of $\cos \left(\frac{4\pi}{3} \right)$

$$= 4 \left(\frac{1}{4} \right) - 1$$

$$= 0$$

½ mark for verification

$$= \text{Right-hand side}$$

1 mark

Describe how to determine the equation of the horizontal asymptote of a rational function when the degree of the polynomial in the numerator and the degree of the polynomial in the denominator are equal.

Solution

The horizontal asymptote is $y = \frac{a}{b}$ where a is the leading coefficient in the numerator and b is the leading coefficient in the denominator.

or

Use polynomial division to divide the numerator by the denominator. The equation of the horizontal asymptote will have the same value as the quotient.

1 mark

Evaluate.

$$\frac{\cot\left(-\frac{5\pi}{6}\right)}{\sin\left(\frac{17\pi}{3}\right)}$$

Solution

$$\frac{\frac{\sqrt{3}}{-\frac{\sqrt{3}}{2}}}{\sin\left(\frac{17\pi}{3}\right)}$$

$$(\sqrt{3})\left(-\frac{2}{\sqrt{3}}\right)$$

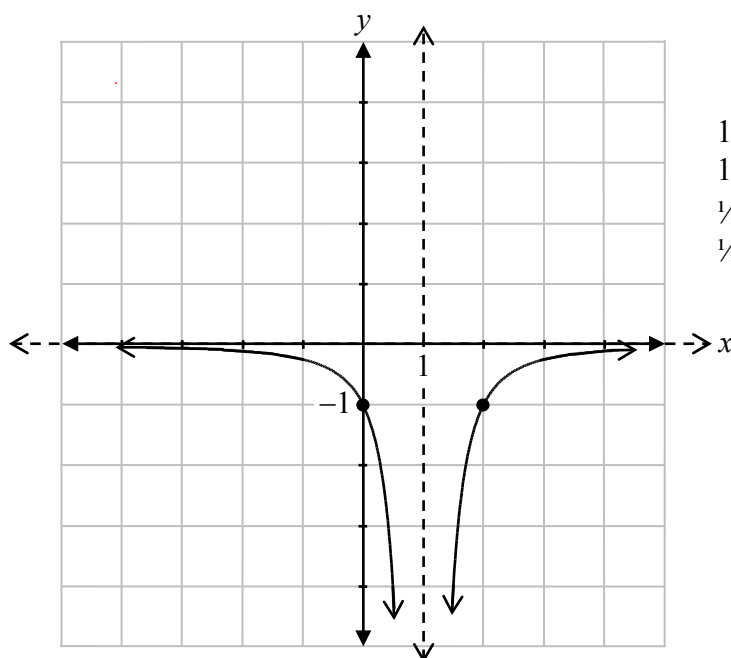
$$-2$$

1 mark for $\cot\left(-\frac{5\pi}{6}\right)$ ($\frac{1}{2}$ mark for value, $\frac{1}{2}$ mark for quadrant)

1 mark for $\sin\left(\frac{17\pi}{3}\right)$ ($\frac{1}{2}$ mark for value, $\frac{1}{2}$ mark for quadrant)

2 marks

Sketch the graph of the function $f(x) = \frac{-1}{(x-1)^2}$ and determine the range.

Solution

1 mark for asymptotic behaviour at $x = 1$
1 mark for asymptotic behaviour at $y = 0$
 $\frac{1}{2}$ mark for graph left of $x = 1$
 $\frac{1}{2}$ mark for graph right of $x = 1$

Range: $\{y \in \mathbb{R} \mid y < 0\}$

or

Range: $(-\infty, 0)$

1 mark for range (consistent with graph)

4 marks

Given $f(x) = \sqrt{x-2}$ and $g(x) = x^2 + 1$,

- a) determine $g(f(x))$.
- b) explain why the domain of $g(f(x))$ is restricted.

Solution

a) $g(f(x)) = (\sqrt{x-2})^2 + 1$ 1 mark for composition
 $= x - 2 + 1, x \geq 2$
 $= x - 1, x \geq 2$

1 mark

- b) The domain of $g(f(x))$ must be restricted because the domain of $f(x)$ is restricted.

or

The value of the radicand must be positive.

1 mark

Solve algebraically.

$$2\log_a 3 + \log_a 4 = 2, \text{ where } a > 0$$

Solution

$$\log_a (3^2 \cdot 4) = 2$$

$$\log_a 36 = 2$$

$$a^2 = 36$$

$$a = 6$$

1 mark for power law

1 mark for product law

1 mark for exponential form

3 marks

Solve $\sec \theta + 2 = 0$ over the interval $[0, 2\pi]$.

Solution

$$\sec \theta + 2 = 0$$

$$\sec \theta = -2$$

$$\cos \theta = -\frac{1}{2}$$

$$\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$$

1 mark for reciprocal

1 mark for values of θ ($\frac{1}{2}$ mark for each value)

2 marks

Determine the x -intercept of the graph of $f(x) = e^x - 1$.

Solution**Method 1**

$$0 = e^x - 1$$

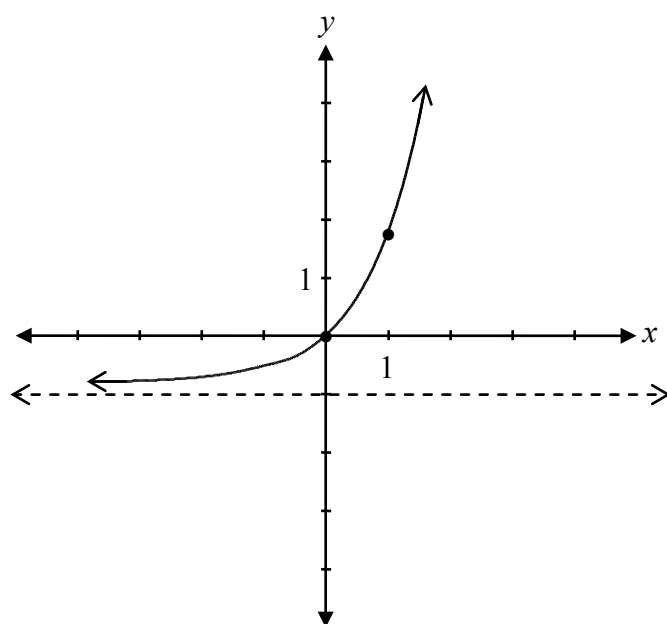
$\frac{1}{2}$ mark for substitution

$$1 = e^x$$

$$x = 0$$

$\frac{1}{2}$ mark for solving for x

1 mark

Method 2

$$\therefore x = 0$$

1 mark

Question 45

P4

Given the 5th row of Pascal's triangle, determine the values of the next row.

1 4 6 4 1

Solution

1 5 10 10 5 1

1 mark

Evaluate.

$$\log_2 80 - \log_2 10$$

Solution

$$\log_2 \left(\frac{80}{10} \right)$$

1 mark for quotient law

$$\log_2 8$$

$$3$$

1 mark for evaluating the logarithm

2 marks

State the amplitude of $f(x) = -2 \sin(x - \pi) - 1$.

Solution

2

1 mark

Determine the exact value of $\cos 15^\circ$.

Solution

$$\cos(15^\circ) = \cos(60^\circ - 45^\circ)$$

$$= \cos 60^\circ \cos 45^\circ + \sin 60^\circ \sin 45^\circ$$

1 mark for substitution into correct identity

$$= \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

2 marks for values ($\frac{1}{2}$ mark for each exact value)

$$= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4}$$

$$= \frac{\sqrt{2} + \sqrt{6}}{4}$$

3 marks

Note(s):

- Other combinations are possible.

Question 49

R1

Given $f(x) = x^2 + 5x + 6$, $g(x) = x + 3$, and $h(x) = f(x) - g(x)$,

a) determine $h(x)$.

b) sketch the graph of $y = h(x)$.

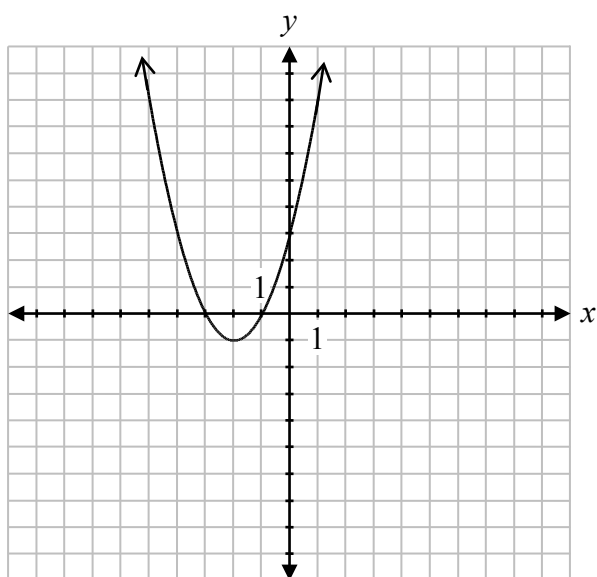
Solution

a)

$$\begin{aligned} f(x) - g(x) &= x^2 + 5x + 6 - (x + 3) \\ &= x^2 + 4x + 3 \end{aligned}$$

1 mark

b)



1 mark for graph consistent with a)

1 mark

Question 1

T1

Pierre pushes his car into a garage. The radius of a tire on his car is 22 cm. Determine the distance travelled by his car if the tire rotated a total of 1000° .

Solution

$$\theta = (1000)\left(\frac{\pi}{180}\right)$$

1 mark for conversion

$$= \frac{50\pi}{9}$$

$$s = \theta r$$

$$s = \left(\frac{50\pi}{9}\right)(22)$$

1 mark for substitution

$$= \frac{1100\pi}{9} \text{ cm}$$

2 marks

or

$$s = 383.972 \text{ cm}$$

Solve, algebraically.

$$7^{\frac{x}{2}} = 85$$

Solution

$$\log 7^{\frac{x}{2}} = \log 85$$

½ mark for applying logarithms

$$\frac{x}{2} \log 7 = \log 85$$

1 mark for power law

$$x \log 7 = 2 \log 85$$

$$x = \frac{2 \log 85}{\log 7}$$

$$x = 4.566\ 142$$

½ mark for evaluating quotient of logarithms

$$x = 4.566$$

2 marks

Solve, algebraically, over the interval $[0, 2\pi)$.

$$\sin x (\sec x + 3) = 0$$

Solution

$$\sin x = 0$$

$$\sec x = -3$$

$\frac{1}{2}$ mark for solving for $\sin x$

$\frac{1}{2}$ mark for solving for $\sec x$

$$\cos x = -\frac{1}{3}$$

1 mark for reciprocal

$$x_r = 1.230\,959$$

$$x = 0, \pi$$

$$x = 1.911, 4.373$$

2 marks for values of x (1 mark for each branch)

4 marks

Question 4

R10

Brahim invests \$2500 at an annual interest rate of 6.75% compounded monthly. Determine, algebraically, how many years it will take for his investment to reach an amount of \$10 500.

Use the formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

where A = the amount of the investment after t years

P = the principal of the investment

r = the annual interest rate (as a decimal)

n = the number of compounding periods per year

t = the length of the investment in years

Solution

$$10\,500 = 2500 \left(1 + \frac{0.0675}{12} \right)^{12t} \quad \frac{1}{2} \text{ mark for substitution}$$

$$4.2 = (1.005\,625)^{12t} \quad \frac{1}{2} \text{ mark for simplification}$$

$$\log 4.2 = 12t \log 1.005\,625 \quad \frac{1}{2} \text{ mark for applying logarithms}$$

$$\frac{\log 4.2}{12 \log 1.005\,625} = t \quad 1 \text{ mark for power law}$$

$$21.320\,250 = t$$

$$21.320 \text{ years} = t$$

$\frac{1}{2}$ mark for evaluating quotient of logarithms

3 marks

There are 13 adults and 18 children who can be selected to go on a trip. Determine the number of ways 4 adults and 7 children can be selected if Sandra, one of the adults, must be selected.

Solution

$${}_1C_1 \cdot {}_{12}C_3 \cdot {}_{18}C_7$$

1 mark for ${}_{12}C_3$

$\frac{1}{2}$ mark for ${}_{18}C_7$

7 001 280

$\frac{1}{2}$ mark for product of combinations

2 marks

Note:

- ${}_1C_1$ does not need to be shown.

In the binomial expansion of $\left(\frac{2}{x^2} - x^3\right)^9$, determine and simplify the 6th term.

Solution

$$t_6 = {}_9C_5 \left(\frac{2}{x^2}\right)^4 (-x^3)^5 \quad 2 \text{ marks (1 mark for } {}_9C_5; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$t_6 = 126 \left(\frac{16}{x^8}\right) (-x^{15})$$

$$t_6 = -2016x^7 \quad 1 \text{ mark for simplification (}\frac{1}{2}\text{ mark for coefficient; } \frac{1}{2}\text{ mark for exponent)}$$

3 marks

Given that $f(x) = \{(-1, 0), (0, 2), (1, -3), (2, 4)\}$, evaluate $f(f(0))$.

Solution

$$f(f(0))$$

$$f(2)$$

$$4$$

$\frac{1}{2}$ mark for $f(0) = 2$

$\frac{1}{2}$ mark for $f(f(0))$

1 mark

The point $\left(-\frac{5}{6}, b\right)$ is on the unit circle and is in quadrant III.

Determine the exact value of b .

Solution

$$\left(-\frac{5}{6}\right)^2 + b^2 = 1 \quad \frac{1}{2} \text{ mark for substitution}$$

$$b^2 = 1 - \frac{25}{36}$$

$$b^2 = \frac{11}{36}$$

$$b = \pm \frac{\sqrt{11}}{6}$$

$$b = -\frac{\sqrt{11}}{6}$$

$\frac{1}{2}$ mark for the exact value of b

1 mark

Question 9

P4

Given the following row of Pascal's Triangle, determine the values of the next row.

1 6 15 20 15 6 1

Solution

1 7 21 35 35 21 7 1

1 mark

Question 10

R3, R5

The following transformations are applied to $f(x)$, resulting in a new function, $g(x)$.

- reflection over the x -axis
- vertical stretch by a factor of 3
- horizontal stretch by a factor of 4

State the equation of $g(x)$ in terms of $f(x)$.

Solution

$$\underline{g(x) = -3f\left(\frac{1}{4}x\right)}$$

1 mark for vertical reflection

1 mark for vertical stretch

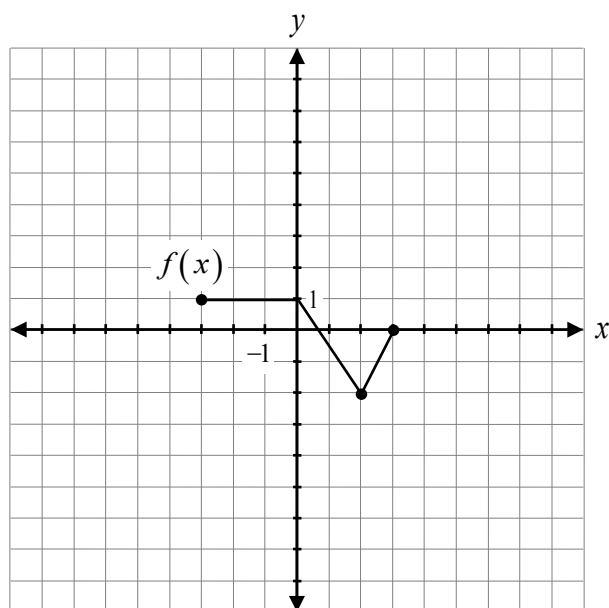
1 mark for horizontal stretch

3 marks

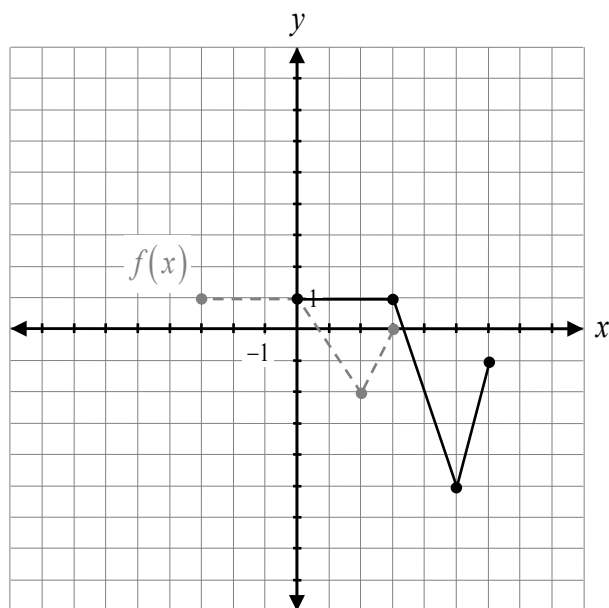
Question 11

R4

Given the graph of $f(x)$, sketch the graph of $y + 1 = 2f(x - 3)$.



Solution



1 mark for vertical stretch
1 mark for horizontal translation
1 mark for vertical translation

3 marks

State the equation of the horizontal asymptote of $f(x) = \frac{2x^2 - 3x + 5}{4x^2 + 2x - 7}$.

Solution

$$y = \frac{1}{2}$$

1 mark

Question 13

R11

Given that $(x + 4)$ is one of the factors of $p(x) = x^3 + 6x^2 - 32$, express $p(x)$ in completely factored form.

Solution

$$\begin{array}{r|rrrrr} -4 & 1 & 6 & 0 & -32 & \\ & \downarrow & -4 & -8 & 32 & \\ \hline & 1 & 2 & -8 & 0 & \end{array}$$

$\frac{1}{2}$ mark for $x = -4$

1 mark for synthetic division (or any equivalent strategy)

$$p(x) = (x + 4)(x^2 + 2x - 8)$$

$$p(x) = (x + 4)(x + 4)(x - 2)$$

or

$\frac{1}{2}$ mark for consistent product of factors

2 marks

$$p(x) = (x + 4)^2(x - 2)$$

Prove the identity for all permissible values of θ .

$$\frac{2 \cos^2 \theta}{1 - \cot \theta} = \frac{\sin 2\theta}{\tan \theta - 1}$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{2 \cos^2 \theta}{1 - \frac{\cos \theta}{\sin \theta}}$	$\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta}{\cos \theta} - 1}$
$\frac{2 \cos^2 \theta}{\frac{\sin \theta - \cos \theta}{\sin \theta}}$	$\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta - \cos \theta}{\cos \theta}}$
$\frac{2 \cos^2 \theta \sin \theta}{\sin \theta - \cos \theta}$	$\frac{2 \sin \theta \cos^2 \theta}{\sin \theta - \cos \theta}$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Solution**Method 2**

Left-Hand Side	Right-Hand Side
$\frac{2 \cos^2 \theta}{1 - \cot \theta}$	$\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta}{\cos \theta} - 1}$ $\frac{2 \sin \theta \cos \theta}{\frac{\sin \theta - \cos \theta}{\cos \theta}}$ $\frac{2 \sin \theta \cos^2 \theta}{\sin \theta - \cos \theta}$ $\frac{\sin \theta (2 \cos^2 \theta)}{\sin \theta \left(1 - \frac{\cos \theta}{\sin \theta}\right)}$ $\frac{2 \cos^2 \theta}{1 - \cot \theta}$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Solution**Method 3**

Left-Hand Side	Right-Hand Side
$\frac{2 \cos^2 \theta}{1 - \cot \theta}$	$\frac{\sin 2\theta}{\tan \theta - 1}$
$\frac{2 \cos^2 \theta}{1 - \frac{1}{\tan \theta}}$	
$\frac{2 \cos^2 \theta}{\frac{\tan \theta - 1}{\tan \theta}}$	
$\frac{2 \cos^2 \theta}{\tan \theta - 1} (\tan \theta)$	
$\frac{2 \cos^2 \theta \left(\frac{\sin \theta}{\cos \theta} \right)}{\tan \theta - 1}$	
$\frac{2 \cos \theta \sin \theta}{\tan \theta - 1}$	
$\frac{\sin 2\theta}{\tan \theta - 1}$	

1 mark for algebraic strategies

1 mark for logical process to prove the identity

1 mark for correct substitution of identities

3 marks

Restaurant A has 5 types of hamburgers, 2 types of french fries, and 10 types of drinks.

Restaurant B has 4 types of hamburgers, 5 types of french fries, and 6 types of drinks.

If a meal is made up of a hamburger, french fries, and a drink, justify which restaurant offers a greater variety of meals.

Solution

Restaurant A: $5 \cdot 2 \cdot 10 = 100$ meals

Restaurant B: $4 \cdot 5 \cdot 6 = 120$ meals

Restaurant B offers a greater variety of meals.

1 mark

Express $\log_7(2x - 5) + 2\log_7 3$ as a single logarithm.

Solution

$$\log_7(2x - 5) + \log_7 3^2 \quad 1 \text{ mark for power law}$$

$$\log_7(2x - 5) + \log_7 9$$

$$\log_7(9(2x - 5)) \quad 1 \text{ mark for product law}$$

or

2 marks

$$\log_7(18x - 45)$$

Explain why $11!$ is not the total number of 11-letter arrangements that can be made from the word CELEBRATION.

Solution

The total number of possible arrangements is half of $11!$ because switching the two Es creates duplicate arrangements.

1 mark

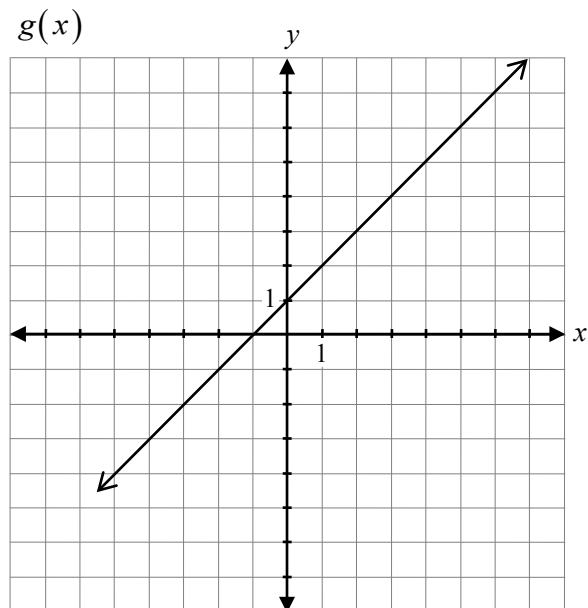
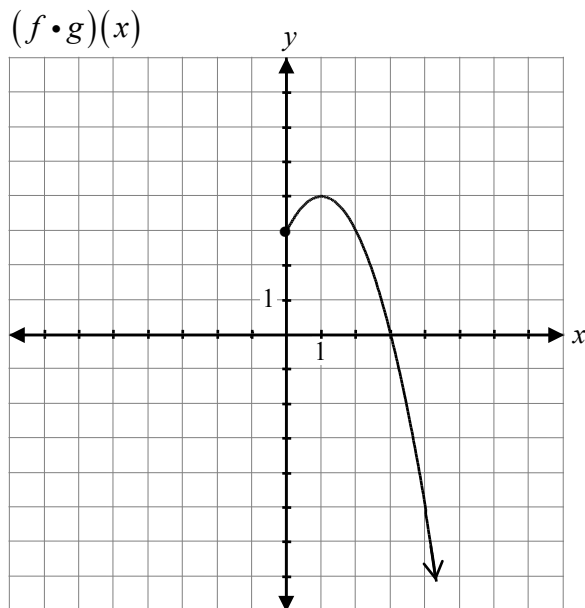
Given $f(x) = \frac{(2x+1)(x-8)}{(x-8)(x+4)}$, state the equation(s) of the vertical asymptote(s).

Solution

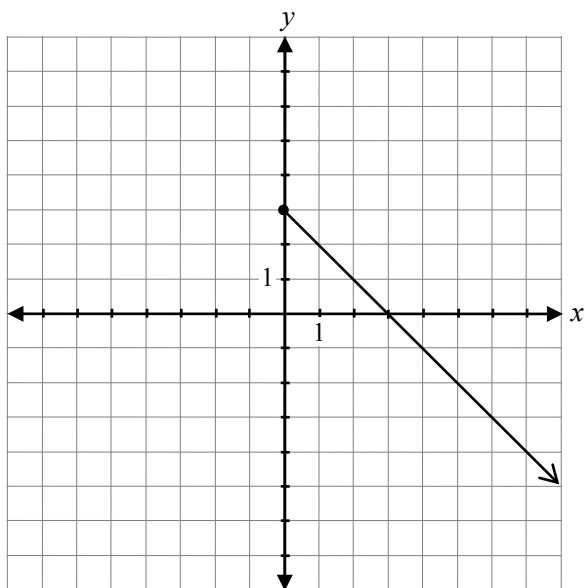
$$x = -4$$

1 mark

Given the graphs of $(f \cdot g)(x)$ and $g(x)$, sketch the graph of $f(x)$.



Solution



1 mark for operation of division
1 mark for restricted domain

2 marks

Brian was asked to state the zeros of the polynomial $p(x) = (x + 2)(x - 5)(x - 1)$.

Brian's response:

$$\text{Zeros: } (x+2), (x-5), (x-1)$$

Explain why his response is incorrect.

Solution

Brian stated the factors instead of the zeros.

1 mark

Simplify ${}_{n+3}C_2$.

Solution

$$\frac{(n+3)!}{(n+3-2)!2!}$$

½ mark for substitution

$$\frac{(n+3)!}{(n+1)!2!}$$

$$\frac{(n+3)(n+2)\cancel{(n+1)!}}{\cancel{(n+1)!}2}$$

1 mark for factorial expansion

$$\frac{(n+3)(n+2)}{2}$$

½ mark for simplification of factorials

or

2 marks

$$\frac{n^2 + 5n + 6}{2}$$

Verify that the equation $2 \cos^2 x = \sin x + 1$ is true for $x = \frac{\pi}{6}$.

Solution

Left-Hand Side	Right-Hand Side	
$2 \cos^2 \left(\frac{\pi}{6} \right)$	$\sin \left(\frac{\pi}{6} \right) + 1$	$\frac{1}{2}$ mark for substitution
$2 \left(\frac{\sqrt{3}}{2} \right)^2$	$\frac{1}{2} + 1$	1 mark for exact values ($\frac{1}{2}$ mark for each)
$2 \left(\frac{3}{4} \right)$		
$\frac{3}{2}$	$\frac{3}{2}$	$\frac{1}{2}$ mark for simplification
$\therefore \text{LHS} = \text{RHS}$		2 marks

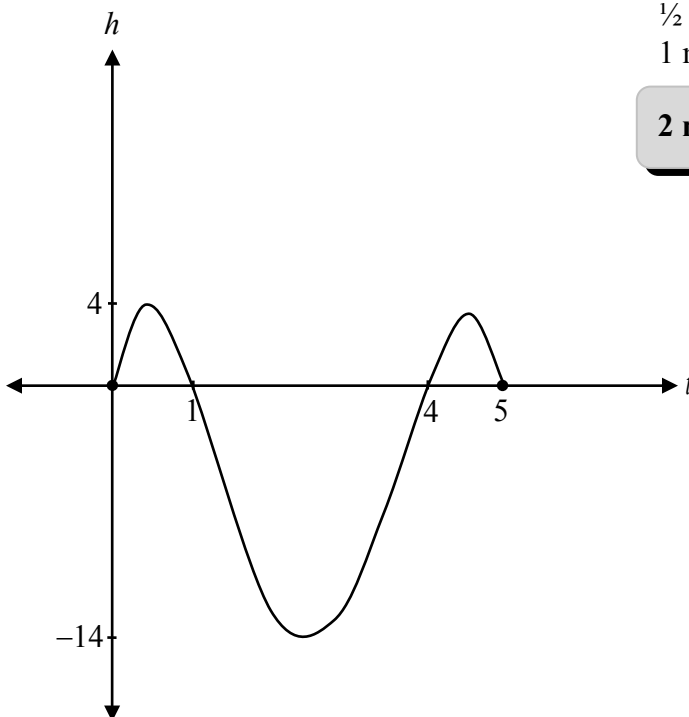
The height of a fish jumping out of the water can be modelled by the function

$h(t) = -t(t-1)(t-4)(t-5)$ where $h(t)$ is the height of the fish above or below the water in cm, and t is the time in seconds, $t \geq 0$.

- a) Sketch a graph representing the height of the fish with respect to time over the interval $[0, 5]$.
- b) State, from the graph in a), the total number of seconds that the fish is above the water.

Solution

a)



½ mark for h -intercept

½ mark for other t -intercepts

1 mark for vertical reflection of a quartic function

2 marks

b) 2 seconds

1 mark for time consistent with graph in a)

1 mark

Note:

- Scale values on h -axis are not required.

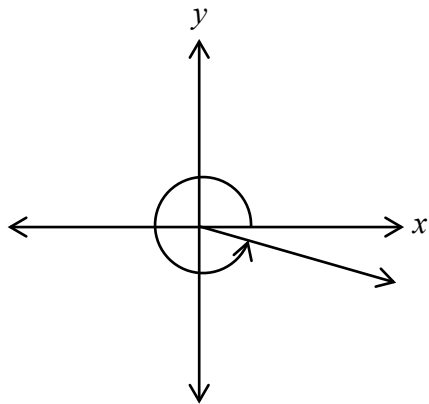
Describe the behaviour of the graph of $y = 5^x + 4$ as it approaches $y = 4$.

Solution

There is a horizontal asymptote at $y = 4$, so the graph approaches $y = 4$ without ever touching it.

1 mark

Sketch the angle of 6 radians in standard position.

Solution

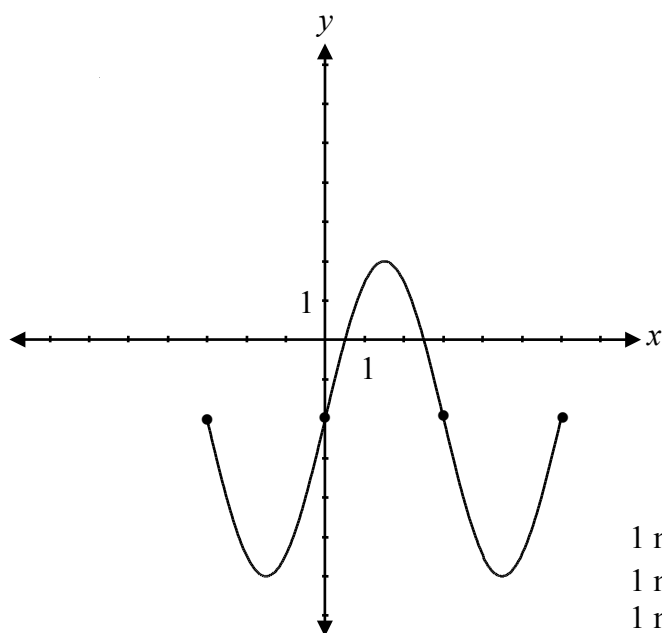
$\frac{1}{2}$ mark for an angle in quadrant IV
 $\frac{1}{2}$ mark for an appropriate angle

1 mark

Note:

- If the directional arrow is not indicated, deduct an E1 error (final answer not stated).

Sketch the graph of the function $y = 4 \sin\left(\frac{\pi}{3}x\right) - 2$ over the domain $[-3, 6]$.

Solution

1 mark for shape of $y = \sin x$
1 mark for amplitude
1 mark for period
1 mark for vertical translation

4 marks

Note:

- Deduct $\frac{1}{2}$ mark for procedural error for not completing the domain of $[-3, 6]$.
- If period mark not awarded, do not deduct for E9 error (scale value on x -axis not indicated).

Question 36

R1, R14

Given $f(x) = 3x - 12$ and $g(x) = x - 4$,

a) determine the equation of $h(x) = \left(\frac{f}{g}\right)(x)$.

b) describe what the non-permissible value represents on the graph of $h(x)$.

Solution

$$\text{a) } h(x) = \frac{3x - 12}{x - 4}$$

1 mark for division of $\left(\frac{f}{g}\right)(x)$

$$h(x) = \frac{3(\cancel{x-4})}{(\cancel{x-4})}$$

1 mark

$$h(x) = \underline{3, x \neq 4}$$

b) The non-permissible value is a point of discontinuity (hole).

1 mark

Determine an equation of a radical function, $f(x)$, with a domain of $x \geq 5$ and a range of $y \geq -2$.

Solution

$$f(x) = \sqrt{x-5} - 2$$

1 mark for horizontal translation

1 mark for vertical translation

2 marks

Note:

- Other equations are possible.

Determine the exact value of $\cos\left(\frac{17\pi}{12}\right)$.

Solution

$$\begin{aligned}\cos\left(\frac{17\pi}{12}\right) &= \cos\left(\frac{3\pi}{4} + \frac{2\pi}{3}\right) \\ &= \cos\frac{3\pi}{4}\cos\frac{2\pi}{3} - \sin\frac{3\pi}{4}\sin\frac{2\pi}{3}\end{aligned}$$

1 mark for substitution into correct identity

$$\begin{aligned}&= \left(-\frac{\sqrt{2}}{2}\right)\left(-\frac{1}{2}\right) - \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) \\ &= \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4} \\ &= \frac{\sqrt{2} - \sqrt{6}}{4}\end{aligned}$$

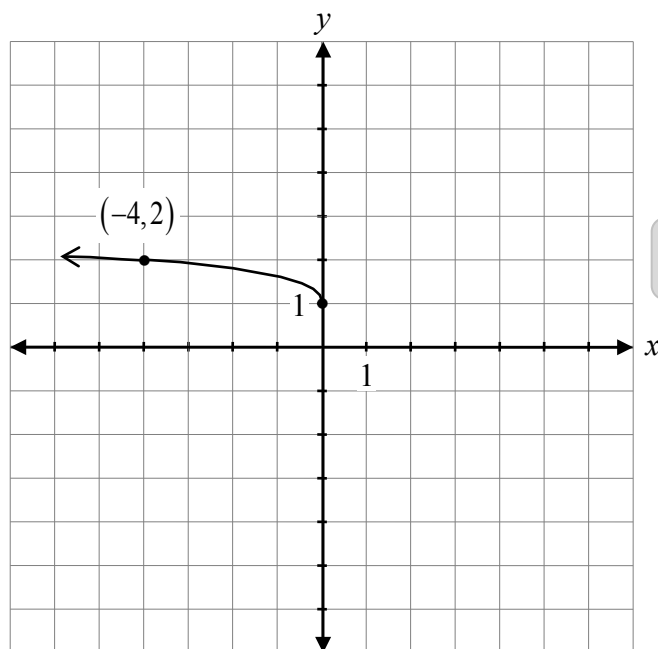
2 marks for exact values ($\frac{1}{2}$ mark for each)

3 marks

Note:

- Other combinations are possible.
- Deduct a maximum $\frac{1}{2}$ mark for arithmetic errors in simplification.

Sketch the graph of $y = \frac{1}{2}\sqrt{-x} + 1$.

Solution

1 mark for shape of a radical function
1 mark for vertical compression
1 mark for horizontal reflection
1 mark for vertical translation

4 marks

Given $f(x) = \frac{3x}{4} + 9$, determine the equation of $f^{-1}(x)$.

Solution

Let $y = f(x)$

$$y = \frac{3x}{4} + 9$$

$$x = \frac{3y}{4} + 9$$

1 mark for switching x and y values

$$4x = 3y + 36$$

$$3y = 4x - 36$$

$$y = \frac{4}{3}x - 12$$

$\frac{1}{2}$ mark for solving for y

$$f^{-1}(x) = \frac{4x}{3} - 12$$

$\frac{1}{2}$ mark for writing equation of $f^{-1}(x)$

2 marks

Describe the end behaviour of the polynomial function $p(x) = -(x - 2)(x + 3)^2$.

Solution

The graph rises as x approaches negative infinity and falls as x approaches positive infinity.

1 mark

Given $\csc \theta = -4$ and θ is in quadrant IV,

a) determine the exact value of $\cos \theta$.

b) determine the exact value of $\cot \theta$.

Solution

a) $\sin \theta = -\frac{1}{4}$ 1 mark for reciprocal

$$1^2 - \left(-\frac{1}{4}\right)^2 = \cos^2 \theta$$

$$\frac{15}{16} = \cos^2 \theta$$

$$\pm \frac{\sqrt{15}}{4} = \cos \theta$$

$$\cos \theta = \frac{\sqrt{15}}{4}$$
 1 mark for $\cos \theta$ ($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

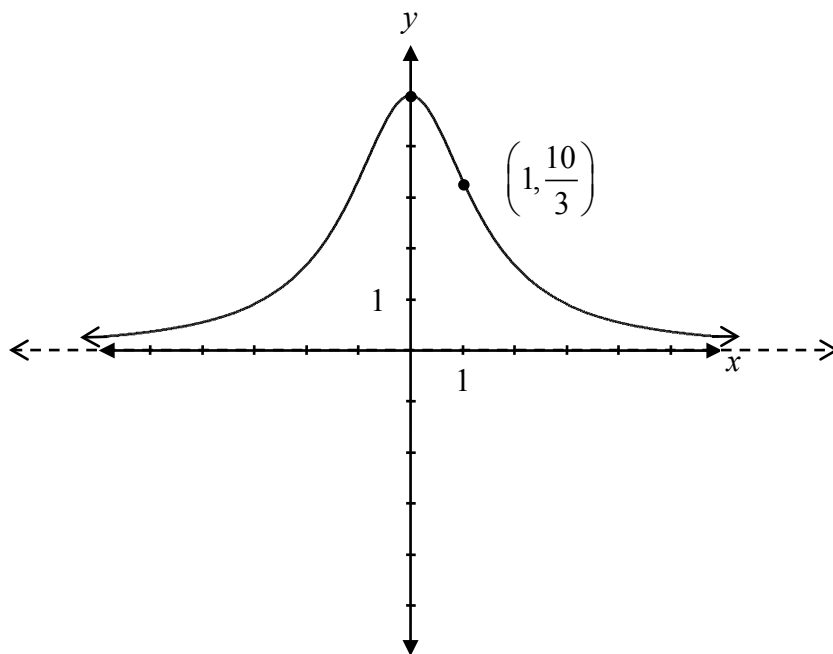
2 marks

b) $\cot \theta = \frac{\frac{\sqrt{15}}{4}}{-\frac{1}{4}}$
 $= -\sqrt{15}$

1 mark for $\cot \theta$ consistent with answer in a)
($\frac{1}{2}$ mark for quadrant; $\frac{1}{2}$ mark for value)

1 mark

Sketch the graph of the function $f(x) = \frac{10}{x^2 + 2}$.

Solution

1 mark for asymptotic behaviour approaching $y = 0$

1 mark for shape ($\frac{1}{2}$ mark for graph left of y -axis; $\frac{1}{2}$ mark for graph right of y -axis)

2 marks

Explain why only one of the following equations could be solved algebraically without using logarithms.

$$3^{5x} = 6^{2x-1} \quad \text{or} \quad 16^{2x+3} = \left(\frac{1}{2}\right)^{4x-5}$$

Solution

The equation $16^{2x+3} = \left(\frac{1}{2}\right)^{4x-5}$ can be solved without the use of logarithms because 16 and $\frac{1}{2}$ can be changed to a common base of 2.

1 mark

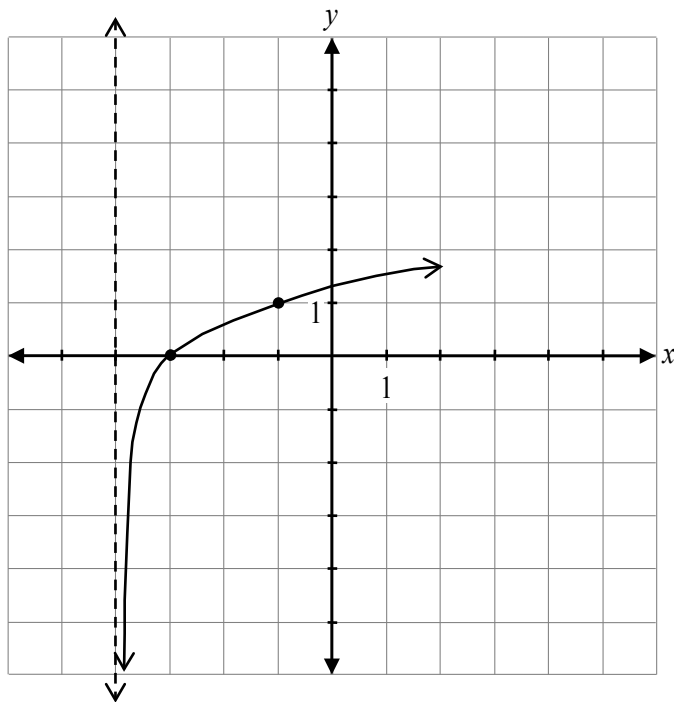
Given a graph of $y = f(x)$, describe how to sketch the graph of $y = |f(x)|$.

Solution

Reflect all the points with negative y -values over the x -axis.

1 mark

Sketch the graph of $y = \log_3(x + 4)$.

Solution

1 mark for increasing logarithmic function
1 mark for asymptotic behaviour
approaching $x = -4$

2 marks

Solve, algebraically.

$$\log x + \log 4 - \log(x - 2) = \log 5$$

Solution

Method 1

$$\log\left(\frac{4x}{x-2}\right) = \log 5$$

$$\frac{4x}{x-2} = 5$$

$$4x = 5(x - 2)$$

$$4x = 5x - 10$$

$$10 = x$$

1 mark for product law

1 mark for quotient law

½ mark for equating arguments

½ mark for solving for x

3 marks

Method 2

$$\log x + \log 4 - \log(x - 2) - \log 5 = 0$$

$$\log\left(\frac{4x}{5(x-2)}\right) = 0$$

$$10^0 = \frac{4x}{5(x-2)}$$

$$5(x - 2) = 4x$$

$$5x - 10 = 4x$$

$$x = 10$$

1 mark for product law

1 mark for quotient law

½ mark for converting to exponential form

½ mark for solving for x

3 marks

Given $\sin \theta = \frac{1}{2}$, determine all possible values of θ over the interval $[-2\pi, 2\pi]$.

Solution

$$\theta = -\frac{11\pi}{6}, -\frac{7\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$$

1 mark for positive values of θ ($\frac{1}{2}$ mark for each)

1 mark for negative values of θ ($\frac{1}{2}$ mark for each)

2 marks

Question 1

P2

Avery has 4 adventure books, 5 mystery books, and 1 comic book.

Determine the number of ways he can arrange all of the books on his shelf if each type of book must be grouped together.

Solution

$$\frac{3!}{\text{types of books}} \cdot \frac{4!}{\text{adventure books}} \cdot \frac{5!}{\text{mystery books}} \cdot \frac{1!}{\text{comic book}} = 17280 \text{ ways}$$

1 mark for arrangement of types of books
1 mark for arrangement of adventure, mystery, and comic books

2 marks

Note:

- 1! does not need to be shown.

Question 2

T5, T6

Solve the following equation, algebraically, over the interval $[0, 2\pi]$.

$$5 \cos^2 \theta - \cos \theta - \sin^2 \theta = 0$$

Solution

$$5 \cos^2 \theta - \cos \theta - (1 - \cos^2 \theta) = 0$$

1 mark for substitution of an appropriate identity

$$5 \cos^2 \theta - \cos \theta - 1 + \cos^2 \theta = 0$$

$$6 \cos^2 \theta - \cos \theta - 1 = 0$$

$$(3 \cos \theta + 1)(2 \cos \theta - 1) = 0$$

$$\cos \theta = -\frac{1}{3}$$

$$\cos \theta = \frac{1}{2}$$

1 mark for solving for $\cos \theta$ ($\frac{1}{2}$ mark for each branch)

$$\theta_r = 1.230\ 959$$

$$\theta = 1.911, 4.373$$

$$\theta = \frac{\pi}{3}, \frac{5\pi}{3}$$

2 marks for solving for θ ($\frac{1}{2}$ mark for each value of θ)

or

$$\theta = 1.911, 4.373$$

$$\theta = 1.047, 5.236$$

4 marks

Given that $-6048x^2y^5$ is the sixth term in the expansion of $(3x - m)^7$, determine m .

Solution

$$-6048x^2y^5 = {}_7C_5(3x)^2(-m)^5 \quad 2 \text{ marks (1 mark for } {}_7C_5; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$-6048x^2y^5 = 21(9x^2)(-m^5)$$

$$-6048x^2y^5 = -189x^2m^5$$

$$32y^5 = m^5$$

$\frac{1}{2}$ mark for simplification

$$2y = m$$

$\frac{1}{2}$ mark for m

3 marks

Question 4

R10

A series of blood tests measures the concentration of a prescribed drug. This concentration decreases according to the formula $A = Pe^{rt}$ where:

A is the concentration at time t

P is the initial concentration

r is the rate of change

t is the time, in hours, after the first blood test

The initial concentration is 3.8900 units/mL. Three hours later, the concentration is 1.7505 units/mL.

- a) Determine the rate of change, r , algebraically.
- b) Determine the concentration of the prescribed drug four hours after the initial concentration of 3.8900 units/mL was measured. Express the answer correct to 4 decimal places.

Solution

a) $1.7505 = 3.8900e^{r(3)}$ $\frac{1}{2}$ mark for substitution

$$0.45 = e^{3r}$$

$$\ln 0.45 = 3r \ln e$$

$$\frac{\ln 0.45}{3} = r$$

or

$$-0.266\ 169\dots = r$$

2 marks

b) $A = 3.8900e^{-0.266\ 169(4)}$

$$A = 1.341\ 424\dots$$

$$A = 1.3414 \text{ units/mL}$$

1 mark for answer consistent with a)

1 mark

Question 5

T1

Ariane uses the formula $s = \theta r$ to determine the arc length of a circle that has a central angle of 20° and a radius of 15 cm.

Below is Ariane's work:

$$\begin{aligned} s &= \theta r \\ s &= (20)(15) \\ s &= 300 \text{ cm} \end{aligned}$$

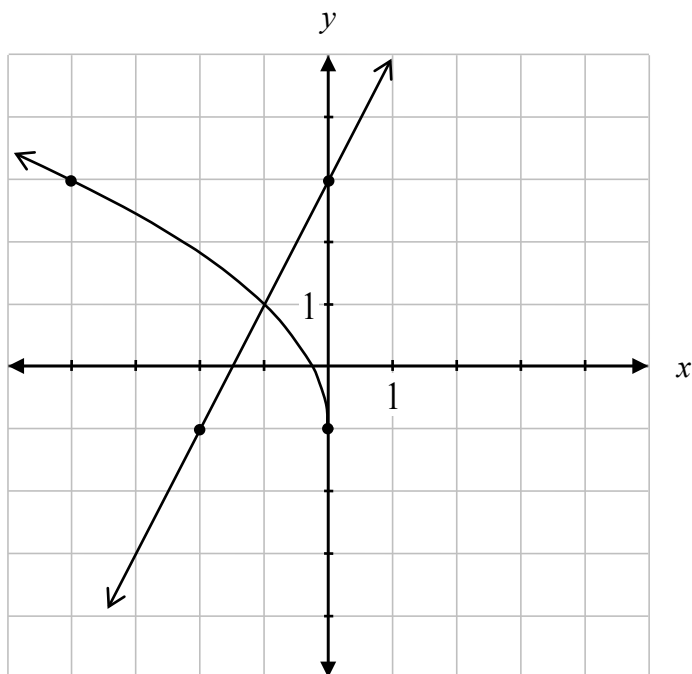
Describe her error.

Solution

When using the formula $s = \theta r$, the angle must be in radians. Ariane did not convert the central angle from degrees to radians.

1 mark

Using the graphs below, state the solution of the equation $2x + 3 = 2\sqrt{-x} - 1$.



Solution

$$x = -1$$

1 mark

Solve, algebraically.

$$\log_2(x+3) = 5 - \log_2(x-1)$$

Solution

Method 1

$$\log_2(x+3) + \log_2(x-1) = 5$$

$$\log_2[(x+3)(x-1)] = 5$$

1 mark for product law

$$x^2 + 2x - 3 = 2^5$$

1 mark for exponential form

$$x^2 + 2x - 3 = 32$$

$$x^2 + 2x - 35 = 0$$

$$(x+7)(x-5) = 0$$

$$\cancel{x = -7} \quad x = 5$$

½ mark for solving for the permissible value of x

½ mark for showing the rejection of the extraneous root

3 marks

Method 2

$$\log_2(x+3) + \log_2(x-1) = 5$$

$$\log_2[(x+3)(x-1)] = \log_2 2^5$$

1 mark for product law

$$x^2 + 2x - 3 = 32$$

1 mark for equating arguments

$$x^2 + 2x - 35 = 0$$

$$(x+7)(x-5) = 0$$

$$\cancel{x = -7} \quad x = 5$$

½ mark for the permissible value of x

½ mark for showing the rejection of the extraneous root

3 marks

Explain why the value of n must be greater than or equal to the value of r , when using ${}_nC_r$.

Solution

The number of objects to select, r , cannot be greater than the total number of objects, n .

1 mark

Question 9

R2, R3, R5

Given that $y = |x|$, determine the equation of the resulting function, $g(x)$, after the following transformations:

- reflection in the x -axis
- vertical translation 5 units down
- horizontal stretch by a factor of 3

Solution

$$g(x) = -\left|\frac{1}{3}x\right| - 5$$

1 mark for vertical reflection
1 mark for vertical translation
1 mark for horizontal stretch

3 marks

Explain why the graph of $y = \frac{x-1}{x^2+x-6}$ has a horizontal asymptote at $y = 0$.

Solution

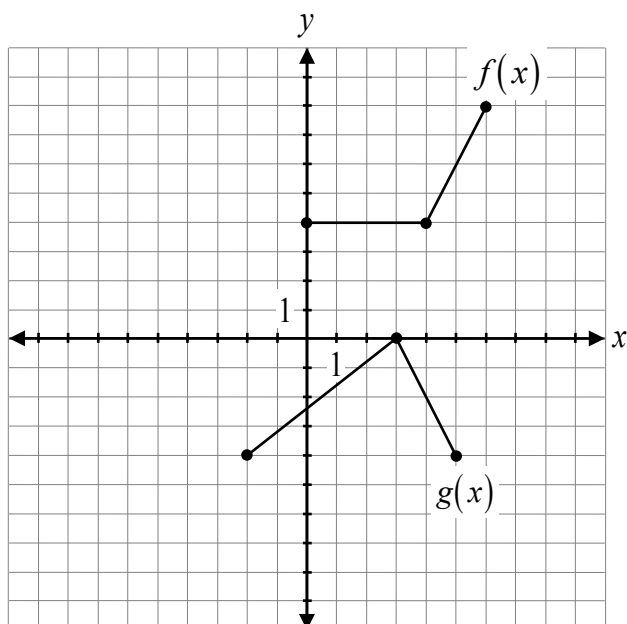
As x approaches positive or negative infinity, y approaches zero.

1 mark

or

The degree of the numerator is less than the degree of the denominator.

Given the graphs of $f(x)$ and $g(x)$, evaluate $g(f(2))$.

**Solution**

$$f(2) = 4 \quad \frac{1}{2} \text{ mark for the value of } f(2)$$

$$g(4) = -2 \quad \frac{1}{2} \text{ mark for consistent value of } g(f(2))$$

1 mark

Kennedy was asked to solve the equation $\tan \theta = 1$ over all real numbers.

Below is Kennedy's solution:

$$\tan \theta = 1$$
$$\theta = \frac{\pi}{4}, \frac{5\pi}{4}$$

Describe her error.

Solution

Kennedy did not include the general solution in her answer.

1 mark

Question 13

P2, P3

Solve, algebraically.

$${}_nC_3 = 3({}_nP_2)$$

Solution

$$\frac{n!}{(n-3)!3!} = \frac{3n!}{(n-2)!}$$

1 mark for substitution into equation (½ mark for each side)

$$\frac{n!(n-2)!}{(n-3)!3!} = 3n!$$

$$\frac{n!(n-2)\cancel{(n-3)!}}{\cancel{(n-3)!}} = 3!(3n!)$$

1 mark for factorial expansion

$$n-2 = 3!(3)$$

½ mark for simplification of factorials

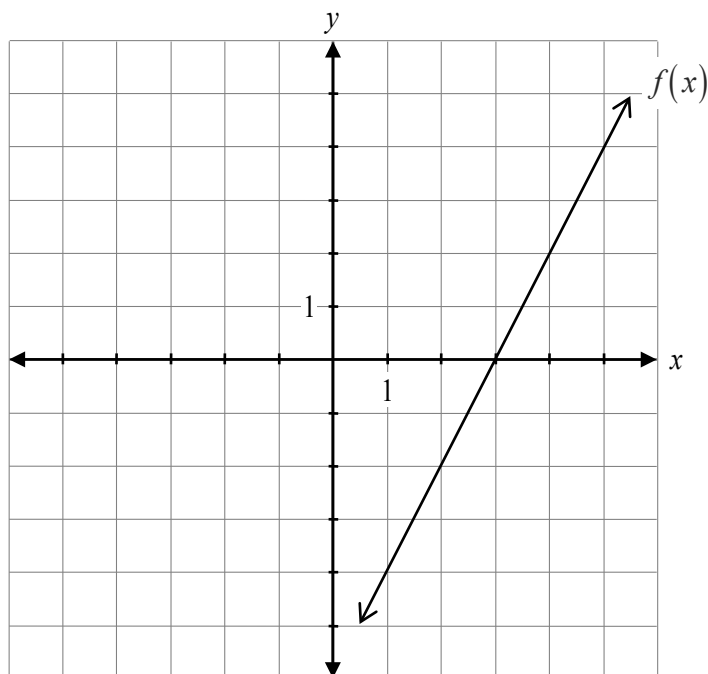
$$n-2 = 18$$

$$n = 20$$

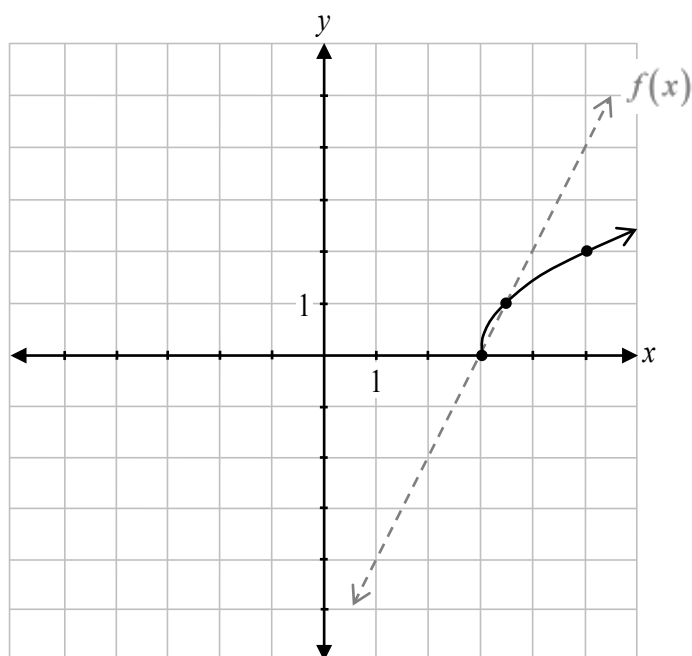
½ mark for solving for n

3 marks

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



Solution



1 mark for restricting domain
 $\frac{1}{2}$ mark for shape between invariant points
 $\frac{1}{2}$ mark for shape to the right of invariant points

2 marks

Prove the identity for all permissible values of θ .

$$\frac{\sec \theta - \tan \theta \sin \theta}{\tan \theta \sin \theta} = \csc^2 \theta - 1$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta} \cdot \sin \theta}{\frac{\sin \theta}{\cos \theta} \cdot \sin \theta}$	$\csc^2 \theta - 1$
$\frac{\frac{1 - \sin^2 \theta}{\cancel{\cos \theta}}}{\frac{\sin^2 \theta}{\cancel{\cos \theta}}}$	
$\frac{1 - \sin^2 \theta}{\sin^2 \theta}$	
$\frac{1}{\sin^2 \theta} - \frac{\sin^2 \theta}{\sin^2 \theta}$	
$\csc^2 \theta - 1$	

1 mark for substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Method 2

Left-Hand Side	Right-Hand Side
$\frac{\sec \theta}{\tan \theta \sin \theta} - 1$	$\csc^2 \theta - 1$
$\frac{\frac{1}{\cos \theta}}{\frac{\sin \theta}{\cos \theta}} - 1$	
$\frac{1}{\cos \theta} \cdot \frac{\cos \theta}{\sin^2 \theta} - 1$	
$\frac{1}{\sin^2 \theta} - 1$	
$\csc^2 \theta - 1$	

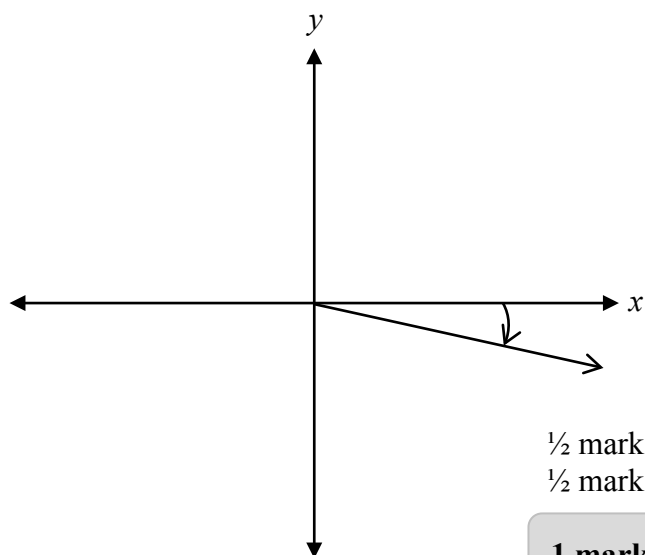
1 mark for substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Sketch the angle of $-\frac{\pi}{12}$ radians in standard position.

Solution

$\frac{1}{2}$ mark for an appropriate angle in quadrant IV
 $\frac{1}{2}$ mark for correct direction

1 mark

Note:

- If the directional arrow is not indicated, deduct an E1 error (final answer not stated)

Given that $h(x) = 2x^2 - 7x - 15$, determine possible equations of the functions $f(x)$ and $g(x)$ if $h(x) = f(x) \cdot g(x)$.

Solution

$$f(x) = \underline{2x + 3}$$

$$g(x) = \underline{x - 5}$$

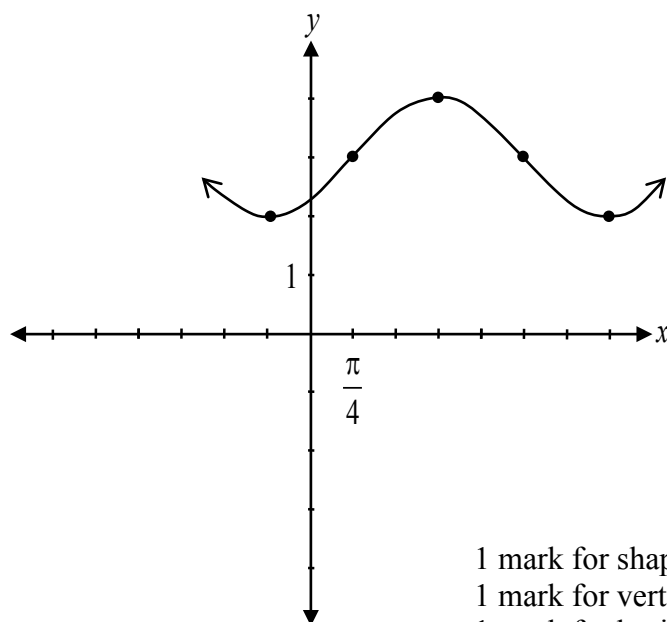
1 mark for two correct factors of $h(x)$

1 mark

Note:

- Other answers are possible.

Sketch the graph of at least one period of the function $y = -\cos\left(x + \frac{\pi}{4}\right) + 3$.

Solution

- 1 mark for shape of a sinusoidal function with correct period
- 1 mark for vertical reflection
- 1 mark for horizontal translation
- 1 mark for vertical translation

4 marks

Justify that $(x - 5)$ is not a possible factor of the function $P(x) = x^3 - 3x^2 - 4x + 12$.

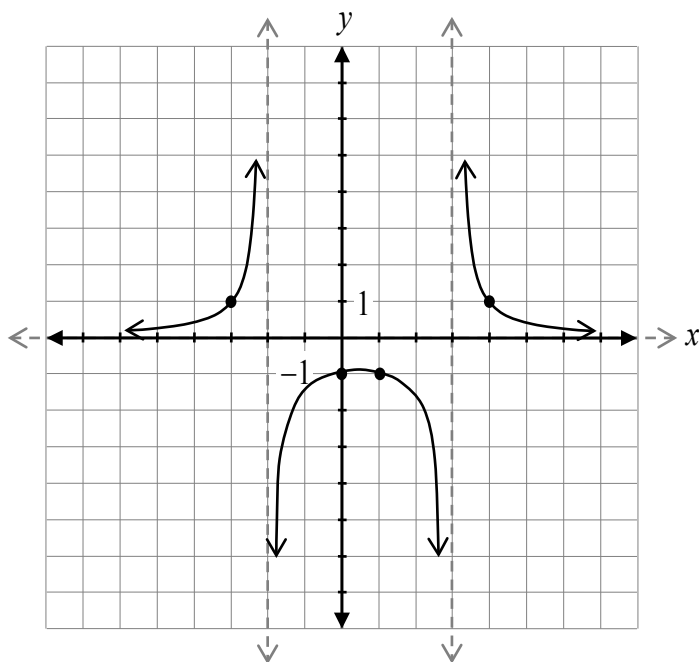
Solution

When $x = 5$ is substituted into $P(x)$, $P(5)$ does not equal 0.

1 mark

Sketch the graph of $f(x) = \frac{6}{(x+2)(x-3)}$ and state the y -intercept.

Solution



y -intercept: -1

1 mark for vertical asymptotic behaviour

($\frac{1}{2}$ mark for behaviour approaching $x = -2$; $\frac{1}{2}$ mark for behaviour approaching $x = 3$)

1 mark for horizontal asymptotic behaviour approaching $y = 0$

$1\frac{1}{2}$ marks for shape ($\frac{1}{2}$ mark for shape in each section)

$\frac{1}{2}$ mark for y -intercept

4 marks

Question 30

P2

Determine how many 3-digit odd numbers less than 300 are possible using the digits 1, 2, 3, 4, 5, 6 if repetition is not allowed.

Solution

$$\text{case 1: } \frac{1}{1} \cdot \frac{4}{-} \cdot \frac{2}{\text{odd}} = 8$$

$\frac{1}{2}$ mark for case 1

$$\text{case 2: } \frac{1}{2} \cdot \frac{4}{-} \cdot \frac{3}{\text{odd}} = 12$$

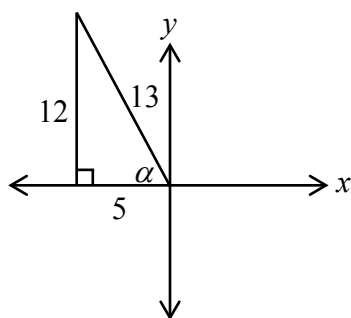
$\frac{1}{2}$ mark for case 2

$$8 + 12 = 20 \text{ numbers}$$

1 mark for addition of cases

2 marks

Given that $\cos \alpha = -\frac{5}{13}$ and $\sin \beta = \frac{2}{3}$, where α and β terminate in the same quadrant, determine the exact value of $\cos(\alpha - \beta)$.

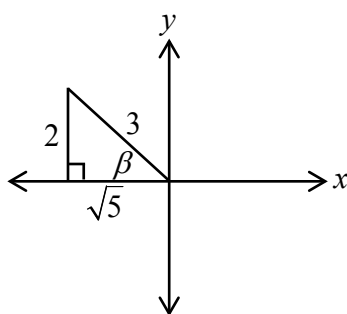
Solution

$$x^2 + y^2 = r^2$$

$$25 + y^2 = 169$$

$$y^2 = 144$$

$$y = \pm 12$$



$$x^2 + y^2 = r^2$$

$$x^2 + 4 = 9$$

$$x^2 = 5$$

$$x = \pm\sqrt{5}$$

$\frac{1}{2}$ mark for value of x
 $\frac{1}{2}$ mark for value of y

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$= \left(-\frac{5}{13}\right)\left(-\frac{\sqrt{5}}{3}\right) + \left(\frac{12}{13}\right)\left(\frac{2}{3}\right)$$

$$= \frac{5\sqrt{5}}{39} + \frac{24}{39}$$

$$= \frac{5\sqrt{5} + 24}{39}$$

$\frac{1}{2}$ mark for $\cos \beta$

$\frac{1}{2}$ mark for $\sin \alpha$

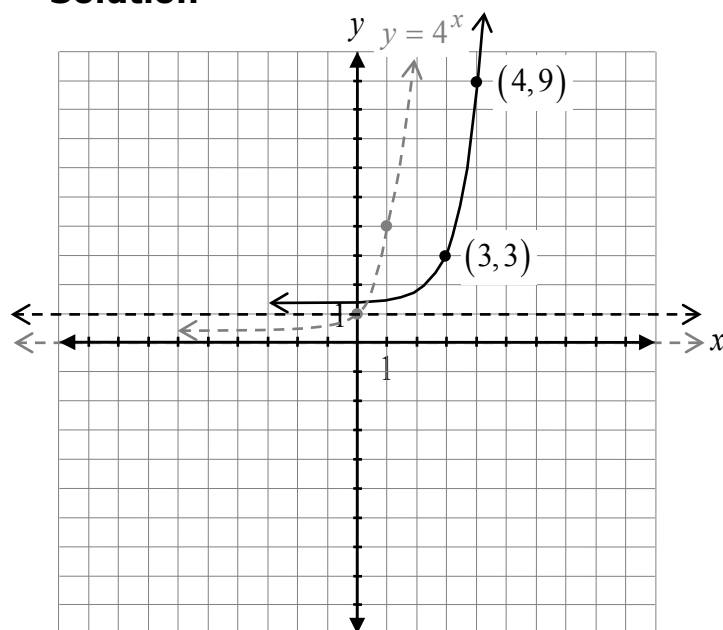
1 mark for substitution into correct identity

3 marks

Notes:

- Accept any of the following values for x : $x = \pm\sqrt{5}$, $x = \sqrt{5}$, or $x = -\sqrt{5}$.
- Accept any of the following values for y : $y = \pm 12$ or $y = 12$.

Given the graph of $y = 4^x$, sketch the graph of $y = 2(4)^{x-3} + 1$.

Solution

1 mark for vertical stretch
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Determine the coterminal angle of $\frac{\pi}{5}$ over the interval $[-2\pi, 0]$.

Solution

$$-\frac{9\pi}{5}$$

1 mark

State the domain of the graph of $y = \log(x - 4) - 8$.

Solution

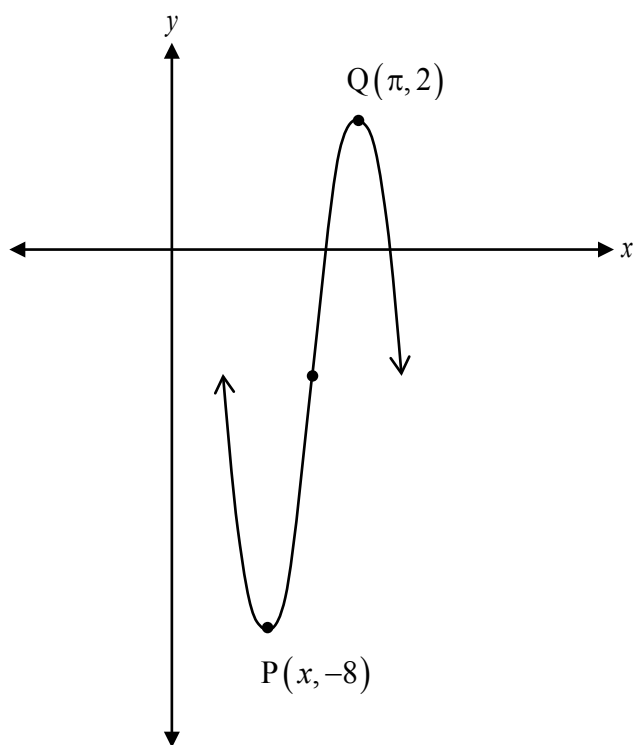
$$x > 4$$

or

$$(4, \infty)$$

1 mark

Given the graph of $y = 5 \sin \left[2 \left(x + \frac{\pi}{4} \right) \right] - 3$, determine the exact value of the x -coordinate in the point P.

**Solution**

$$x = \frac{\pi}{2}$$

1 mark

Verify that the following equation is true for $x = \frac{5\pi}{6}$.

$$\frac{\cos x}{1 - \sin x} = \frac{1 + \sin x}{\cos x}$$

Solution

Left-Hand Side	Right-Hand Side
$\frac{\cos \frac{5\pi}{6}}{1 - \sin \frac{5\pi}{6}}$	$\frac{1 + \sin \frac{5\pi}{6}}{\cos \frac{5\pi}{6}}$
$\frac{-\frac{\sqrt{3}}{2}}{1 - \frac{1}{2}}$	$\frac{1 + \frac{1}{2}}{-\frac{\sqrt{3}}{2}}$ 1 mark for exact values ($\frac{1}{2}$ mark for $\cos \frac{5\pi}{6}$; $\frac{1}{2}$ mark for $\sin \frac{5\pi}{6}$)
$\frac{-\frac{\sqrt{3}}{2}}{\frac{1}{2}}$	$\frac{\frac{3}{2}}{-\frac{\sqrt{3}}{2}}$
$-\frac{\sqrt{3}}{1}$	$-\frac{3}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$ 1 mark for simplification ($\frac{1}{2}$ mark for LHS; $\frac{1}{2}$ mark for RHS)
$-\sqrt{3}$	$-\sqrt{3}$

2 marks

Given that $(x + 1)$ is one of the factors of $P(x) = x^3 - x^2 + kx - 8$, determine the value of k .

Solution

Method 1

$$x = -1$$

$\frac{1}{2}$ mark for $x = -1$

$$0 = (-1)^3 - (-1)^2 + k(-1) - 8$$

1 mark for remainder theorem

$$0 = -1 - 1 - k - 8$$

$$k = -10$$

$\frac{1}{2}$ mark for solving for k

Method 2

$$x = -1$$

$\frac{1}{2}$ mark for $x = -1$

$$\begin{array}{r|rrrr} -1 & 1 & -1 & k & -8 \\ & & -1 & 2 & -k-2 \\ \hline & 1 & -2 & k+2 & -k-10 \end{array}$$

1 mark for synthetic division

$$-k - 10 = 0$$

$$k = -10$$

$\frac{1}{2}$ mark for solving for k

2 marks

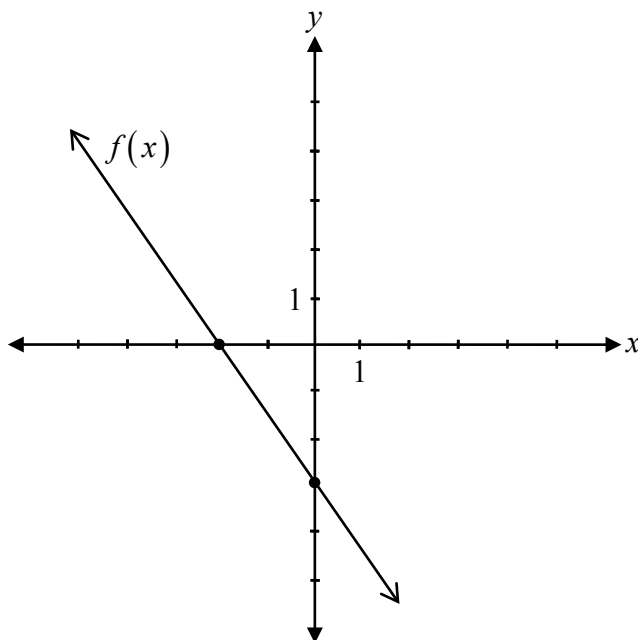
Given the function $f(x) = \sqrt{x}$, describe how to use transformations to determine the domain of the function $g(x) = f(x + 2) + 1$.

Solution

The graph of $g(x)$ is a horizontal translation 2 units to the left of the graph of $f(x)$, which changes the domain from $x \geq 0$ to $x \geq -2$.

1 mark

Given the graph of $y = f(x)$, state the equation of the vertical asymptote of $y = \frac{1}{f(x)}$.



Solution

$$x = -2$$

1 mark

Solve, algebraically.

$$16^x = 64^{2x-1}$$

Solution

$$4^{2x} = 4^{3(2x-1)}$$

1 mark for changing to a common base ($\frac{1}{2}$ mark for each)

$$4^{2x} = 4^{6x-3}$$

$\frac{1}{2}$ mark for exponent law

$$2x = 6x - 3$$

$\frac{1}{2}$ mark for equating exponents

$$3 = 4x$$

2 marks

$$\frac{3}{4} = x$$

Question 41

R11

Given that one of the factors of $P(x) = x^3 + 2x^2 - 5x - 6$ is $(x + 3)$, express $P(x)$ in completely factored form.

Solution

$$\begin{array}{r|rrrr} -3 & 1 & 2 & -5 & -6 \\ & \downarrow & -3 & 3 & 6 \\ \hline & 1 & -1 & -2 & 0 \end{array}$$

$\frac{1}{2}$ mark for $x = -3$

1 mark for synthetic division (or any equivalent strategy)

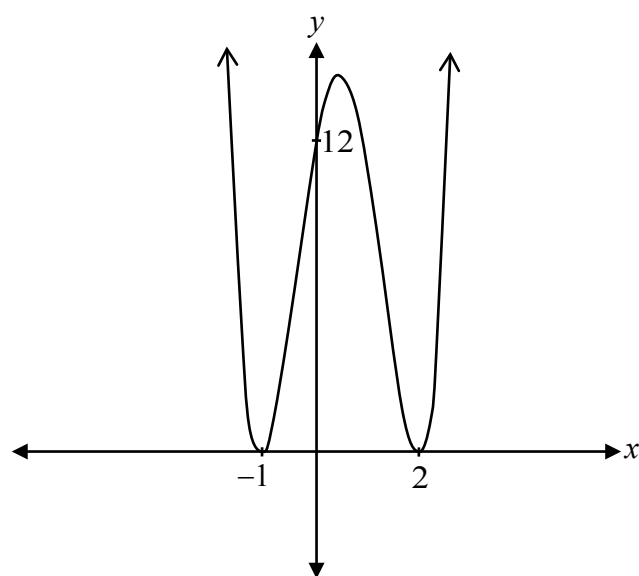
$$P(x) = (x + 3)(x^2 - x - 2)$$

$$P(x) = \underline{(x + 3)(x - 2)(x + 1)}$$

$\frac{1}{2}$ mark for consistent product of factors

2 marks

Sketch the graph of $p(x) = 3(x+1)^2(x-2)^2$.

Solution

1 mark for x -intercepts

$\frac{1}{2}$ mark for y -intercept

1 mark for multiplicity of 2 at $x = -1$ and at $x = 2$
($\frac{1}{2}$ mark for each)

$\frac{1}{2}$ mark for end behaviour

3 marks

Given that $f(x) = x^2 - 4$ and $g(x) = \sqrt{x}$, determine $f(g(x))$ and state its domain.

Solution

$$f(g(x)) = (\sqrt{x})^2 - 4$$

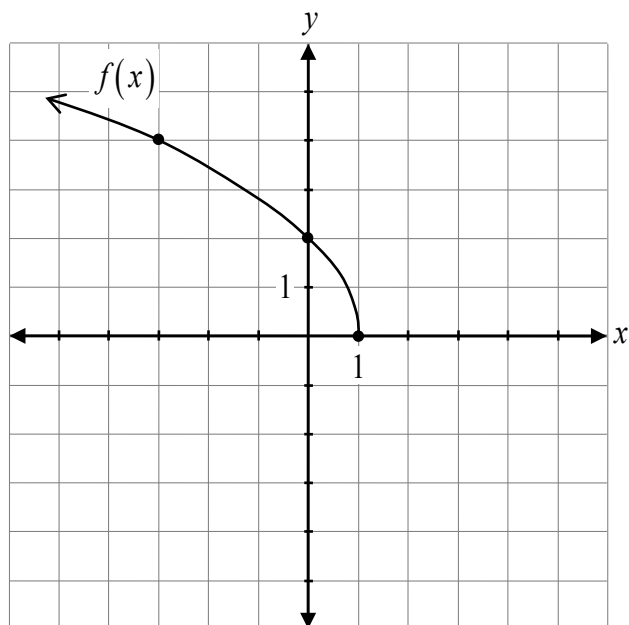
$$f(g(x)) = \underline{x - 4, x \geq 0}$$

1 mark for composite function

1 mark for domain consistent with composite function

2 marks

Determine a possible equation of the function $f(x)$.

**Solution**

$$f(x) = 2\sqrt{-(x-1)}$$

1 mark for vertical stretch
1 mark for horizontal reflection
1 mark for horizontal translation

3 marks

or

$$f(x) = \sqrt{-4(x-1)}$$

1 mark for horizontal compression
1 mark for horizontal reflection
1 mark for horizontal translation

3 marks

Explain why the graph of $y = \log_2 x$ does not have a y -intercept.

Solution

The domain of the graph is $x > 0$.

1 mark

or

There is a vertical asymptote at $x = 0$.

Evaluate.

$$\sin^2\left(-\frac{\pi}{3}\right) + \cos\left(\frac{17\pi}{6}\right)\sec\left(\frac{\pi}{6}\right)$$

Solution

$$\begin{aligned} \left(-\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{2}{\sqrt{3}}\right) & \quad 1 \text{ mark for } \sin\left(-\frac{\pi}{3}\right) \text{ (}\frac{1}{2}\text{ mark for quadrant; } \frac{1}{2}\text{ mark for value)} \\ \frac{3}{4} - 1 & \quad 1 \text{ mark for } \cos\left(\frac{17\pi}{6}\right) \text{ (}\frac{1}{2}\text{ mark for quadrant; } \frac{1}{2}\text{ mark for value)} \\ -\frac{1}{4} & \quad 1 \text{ mark for } \sec\left(\frac{\pi}{6}\right) \text{ (}\frac{1}{2}\text{ mark for quadrant; } \frac{1}{2}\text{ mark for value)} \end{aligned}$$

3 marks

Determine the coordinates of the point of discontinuity (hole) on the graph of $y = \frac{x^2 - 3x}{x}$.

Solution

$(0, -3)$ 1 mark for point of discontinuity (hole) at $x = 0$

1 mark

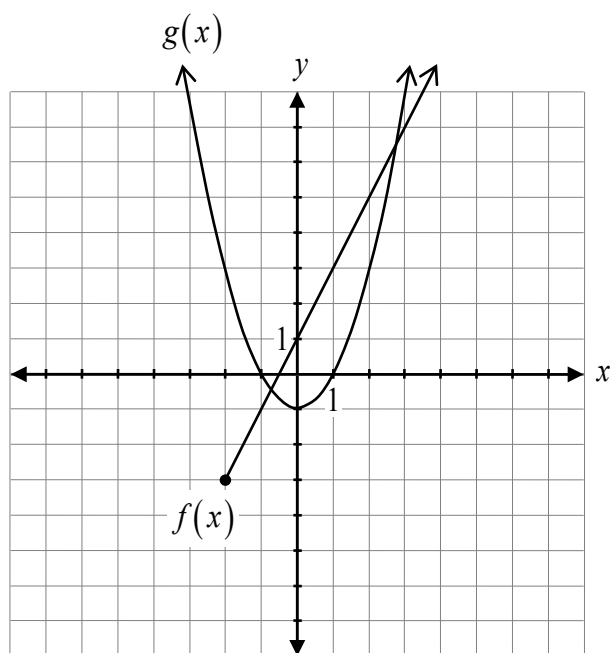
Note:

- Deduct $\frac{1}{2}$ mark for procedural error of incorrect y -value.

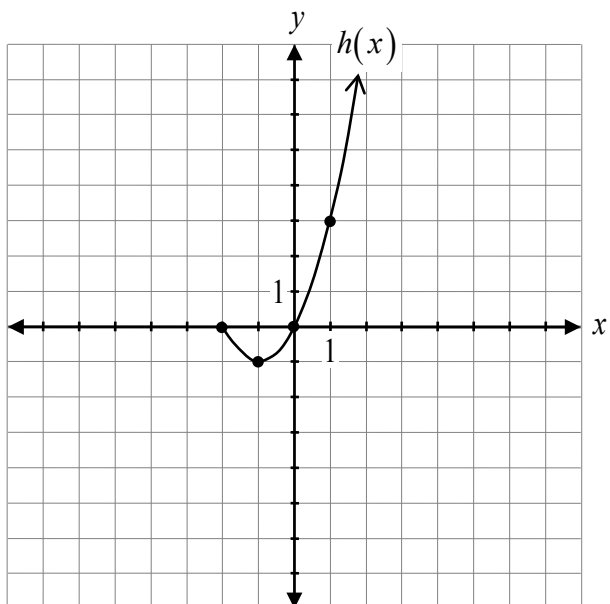
Question 48

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) + g(x)$.



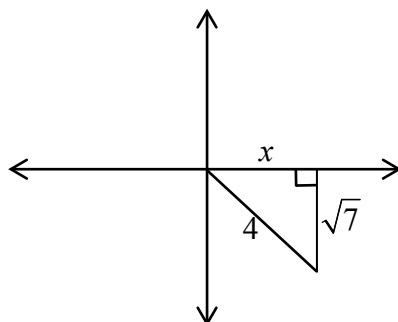
Solution



1 mark for operation of addition
1 mark for restricted domain

2 marks

Given that $\csc \theta = -\frac{4}{\sqrt{7}}$ and $\cos \theta > 0$, determine the exact value of $\tan \theta$.

Solution

$$\sin \theta = -\frac{\sqrt{7}}{4}$$

$$x^2 + (\sqrt{7})^2 = (4)^2$$

½ mark for substitution

$$x^2 = 16 - 7$$

$$x^2 = 9$$

$$x = \pm 3$$

½ mark for solving for x

$$\tan \theta = -\frac{\sqrt{7}}{3}$$

1 mark for $\tan \theta$ (½ mark for quadrant; ½ mark for value)

2 marks

Note:

- Accept any of the following values for x : $x = \pm 3$, $x = 3$, or $x = -3$.

Question 1

P1

A soccer team consists of 14 players. Determine the number of ways a captain and an assistant captain can be selected.

Solution

$${}_{14}P_2 = 182$$

or

$$\underline{14} \cdot \underline{13} = 182$$

1 mark

Question 2

R10

The pH of a person's blood can be found using the formula

$$\text{pH} = 6.1 + \log\left(\frac{B}{C}\right)$$

where B represents the concentration of bicarbonate in the blood, in mEq/L

C represents the concentration of carbonic acid in the blood, in mEq/L

Kansas has a blood pH of 7.41. In her blood sample, the concentration of carbonic acid is 1.41 mEq/L. Determine the concentration of bicarbonate in her blood.

Express the answer correct to one decimal place.

Solution

Method 1

$$7.41 = 6.1 + \log\left(\frac{B}{1.41}\right) \quad \frac{1}{2} \text{ mark for substitution}$$

$$1.31 = \log\left(\frac{B}{1.41}\right)$$

$$10^{1.31} = \frac{B}{1.41} \quad 1 \text{ mark for exponential form}$$

$$B = 28.788 \ 505\dots$$

$$B = 28.8 \text{ mEq/L} \quad \frac{1}{2} \text{ mark for solving for } B$$

2 marks

Method 2

$$7.41 = 6.1 + \log\left(\frac{B}{1.41}\right) \quad \frac{1}{2} \text{ mark for substitution}$$

$$1.31 = \log\left(\frac{B}{1.41}\right)$$

$$1.31 = \log B - \log 1.41 \quad 1 \text{ mark for quotient law}$$

$$1.31 + \log 1.41 = \log B$$

$$1.4592\dots = \log B$$

$$B = 28.788 \ 505\dots$$

$$B = 28.8 \text{ mEq/L} \quad \frac{1}{2} \text{ mark for solving for } B$$

2 marks

Solve $(2 \tan x - 1)(\tan x + 1) = 0$ where $x \in \mathbb{R}$.

Solution

Method 1

$$2 \tan x - 1 = 0$$

$$\tan x + 1 = 0$$

$$\tan x = \frac{1}{2}$$

$$\tan x = -1$$

$$x_r = 0.463\,647\dots$$

$$x = 0.464$$

$$x = \frac{3\pi}{4}$$

$$x = 3.605$$

$$x = \frac{7\pi}{4}$$

2 marks for solving for x ($\frac{1}{2}$ mark for each value)

$$\left. \begin{array}{l} x = 0.464 + 2k\pi \\ x = 3.605 + 2k\pi \\ x = \frac{3\pi}{4} + 2k\pi \\ x = \frac{7\pi}{4} + 2k\pi \end{array} \right\} k \in \mathbb{Z}$$

1 mark for general solution

3 marks

Method 2

$$\tan x = \frac{1}{2}$$

$$\tan x = -1$$

$$x_r = 0.463\,647\dots$$

$$x = \frac{3\pi}{4}$$

2 marks for solving for x (1 mark for each branch)

$$x = 0.464$$

$$\left. \begin{array}{l} x = 0.464 + k\pi \\ x = \frac{3\pi}{4} + k\pi \end{array} \right\} k \in \mathbb{Z}$$

1 mark for general solution

3 marks

Determine and simplify the 7th term in the binomial expansion of $\left(\frac{2}{x^2} + 3x\right)^9$.

Solution

$$t_7 = {}_9C_6 \left(\frac{2}{x^2}\right)^3 (3x)^6 \quad 2 \text{ marks (1 mark for } {}_9C_6; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$= 84 \left(\frac{8}{x^6}\right) (729x^6)$$

$$= 489\,888$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient; $\frac{1}{2}$ mark for exponent)

3 marks

Question 5

T1

A winch, with a diameter of 14 cm, is used to pull a boat out of the water. As the winch rotates, it pulls in a cable. Determine the length of cable that is pulled in if the winch rotates 526° .



Solution

$$\theta = (526) \left(\frac{\pi}{180} \right)$$

1 mark for conversion

$$\theta = \frac{263\pi}{90}$$

$$\theta = 9.180\,431\dots$$

$$s = \theta r$$

$$s = \left(\frac{263\pi}{90} \right) (7)$$

1 mark for substitution

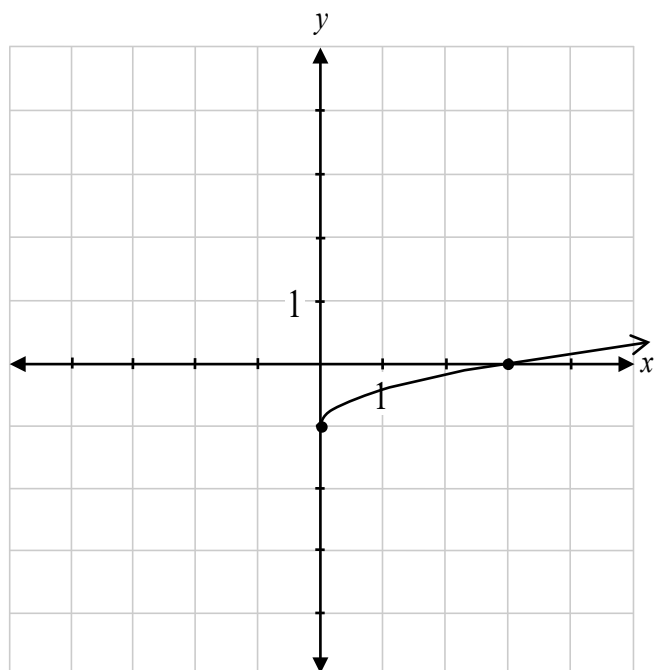
$$s = \frac{1841\pi}{90} \text{ cm}$$

2 marks

or

$$s = 64.263 \text{ cm}$$

Sketch the graph of $f(x) = \sqrt{\frac{1}{3}x} - 1$.

Solution

1 mark for shape of a radical function
1 mark for horizontal stretch
1 mark for vertical translation

3 marks

Question 7

R2, R3, R5

Given $y = f(x)$, determine the equation of the resulting function, $g(x)$, after the following transformations:

- reflection over the y -axis
- vertical stretch by a factor of 3
- horizontal translation 2 units left

Solution

$$g(x) = \underline{3f(-(x+2))}$$

1 mark for horizontal reflection

1 mark for vertical stretch

1 mark for horizontal translation

3 marks

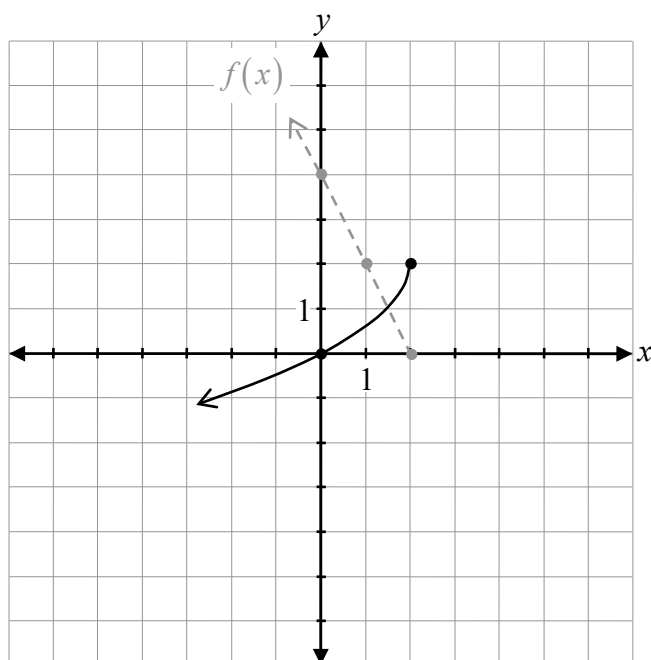
Explain why the graph of $y = \frac{6x+7}{3x+5}$ has a horizontal asymptote at $y = 2$.

Solution

When the degree of the numerator and the denominator are the same, the horizontal asymptote is $y = \frac{a}{b}$ where a is the leading coefficient in the numerator and b is the leading coefficient in the denominator.

1 mark

Given the graph of $y = f(x)$, sketch the graph of $y = -\sqrt{f(x)} + 2$.

Solution

1 mark for shape of $\sqrt{f(x)}$

1 mark for vertical reflection

1 mark for vertical translation

3 marks

Using the laws of logarithms, completely expand the given expression.

$$\log\left(\frac{A}{\sqrt[3]{B} \cdot C^4}\right)$$

Solution

$$\log A - \frac{1}{3}\log B - 4\log C$$

1 mark for product law

1 mark for power law ($\frac{1}{2}$ mark for each)

1 mark for quotient law

3 marks

Prove the following identity for all permissible values of θ .

$$\cot \theta - \tan \theta = \frac{2 \cos 2\theta}{\sin 2\theta}$$

Solution

Left-Hand Side	Right-Hand Side
$\cot \theta - \tan \theta$ $\frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta}$ $\frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta}$ $\frac{\cos 2\theta}{\sin \theta \cos \theta}$	$\frac{2 \cos 2\theta}{\sin 2\theta}$ $\frac{2 \cos 2\theta}{2 \sin \theta \cos \theta}$ $\frac{\cos 2\theta}{\sin \theta \cos \theta}$

1 mark for substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 12

R11

Express $p(x) = -2x^3 + x^2 + 13x + 6$ in completely factored form.

Solution

$$p(-2) = -2(-2)^3 + (-2)^2 + 13(-2) + 6$$

$$p(-2) = -2(-8) + 4 - 26 + 6$$

$$p(-2) = 0$$

1 mark for identifying one zero of $p(x)$

$\therefore (x + 2)$ is a factor.

$$\begin{array}{r|rrrr} -2 & -2 & 1 & 13 & 6 \\ & \downarrow & 4 & -10 & -6 \\ \hline & -2 & 5 & 3 & 0 \end{array}$$

1 mark for synthetic division (or any equivalent strategy)

$$p(x) = (x + 2)(-2x^2 + 5x + 3)$$

$$p(x) = -(x + 2)(2x^2 - 5x - 3)$$

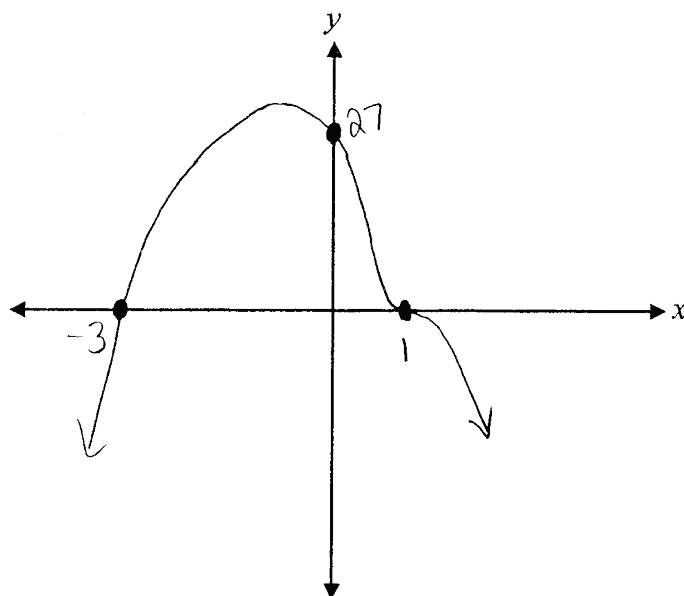
$$p(x) = \underline{-(x + 2)(2x + 1)(x - 3)}$$

1 mark for consistent product of factors

3 marks

Kaitlyn was asked to sketch the graph of the polynomial function $p(x) = -(x-1)(x+3)^3$.

Kaitlyn's graph:



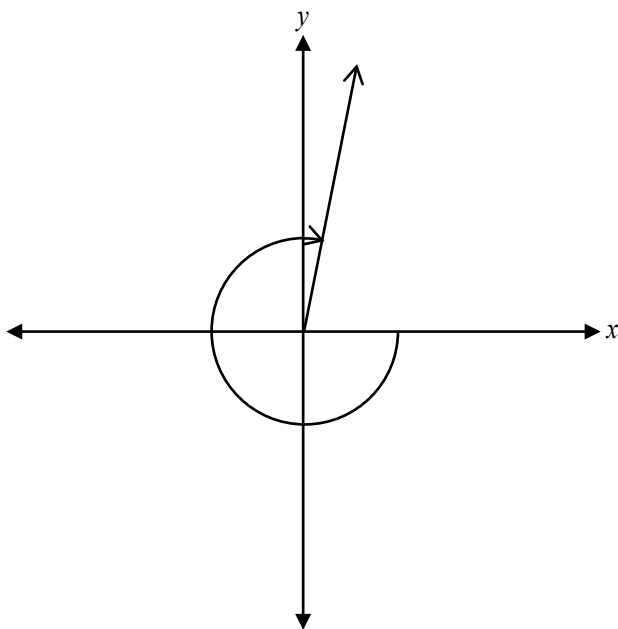
Describe an error on her graph.

Solution

Kaitlyn's graph shows a multiplicity of 3 at $x = 1$ instead of at $x = -3$.

1 mark

Sketch the angle of -5 radians in standard position.

Solution

$\frac{1}{2}$ mark for an appropriate angle in quadrant I

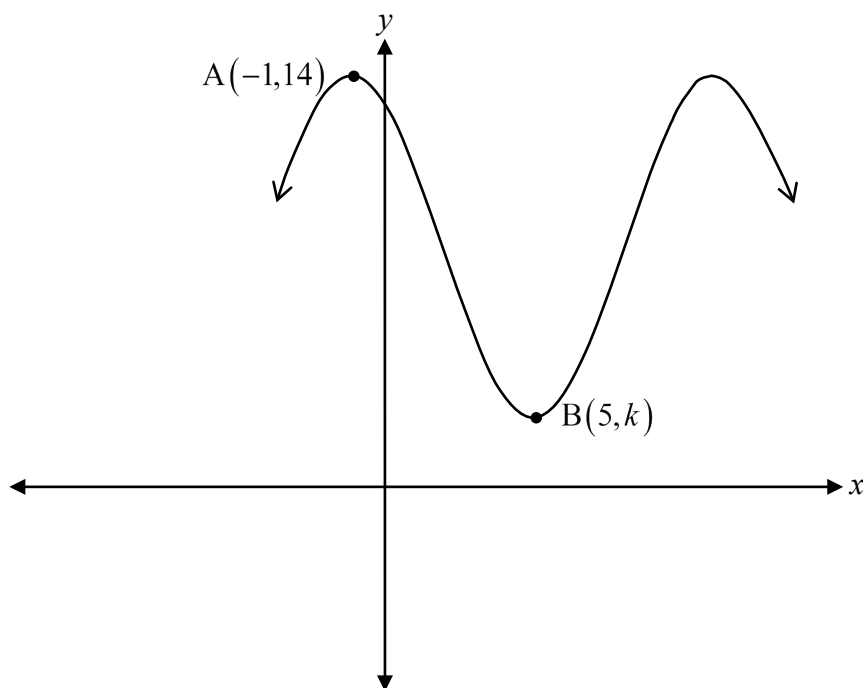
$\frac{1}{2}$ mark for the correct direction of an angle in standard position

1 mark

Question 15

T4

The graph of a sinusoidal function is sketched below. The point A is a maximum and the point B is a minimum. State the value of k if the amplitude of the function is 6.

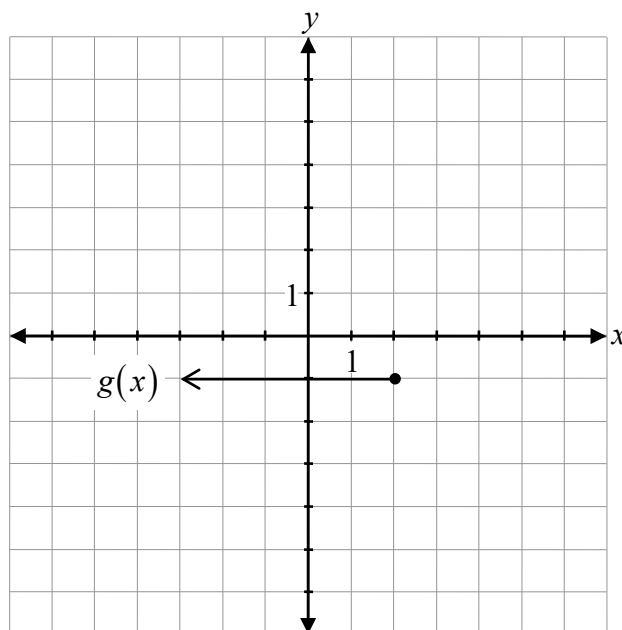
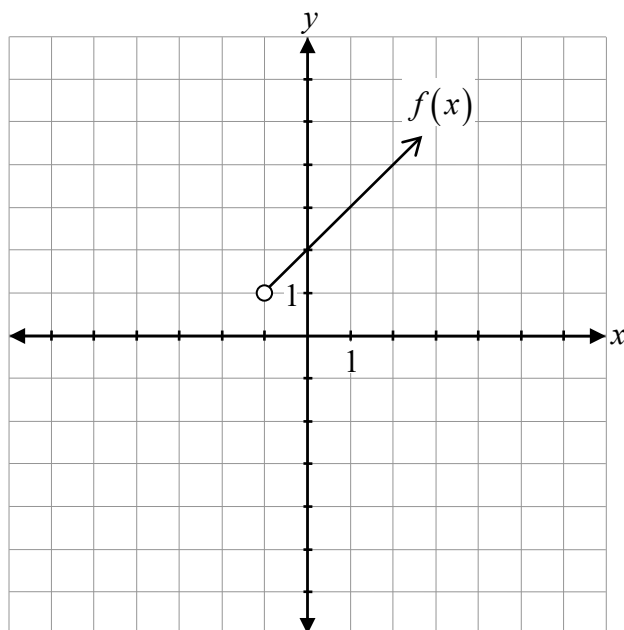


Solution

$$k = 2$$

1 mark

Given the graphs of $f(x)$ and $g(x)$, state the domain of $(f + g)(x)$.

**Solution**

Domain: $\{x \mid -1 < x \leq 2, x \in \mathbb{R}\}$

1 mark

or

Domain: $(-1, 2]$

Question 20

T6

Identify a non-permissible value of θ in the identity, $\frac{\cot \theta}{\csc \theta} = \cos \theta$.

a) π

b) $\frac{\pi}{4}$

c) $\frac{3\pi}{2}$

d) $\frac{\pi}{2}$

Question 21

R10

Identify the solution of the equation, $e^{2x} = 16$.

a) $x = \ln 8$

b) $x = 8$

c) $x = \frac{\ln 16}{2}$

d) $x = \frac{\log 16}{2}$

Question 22

P3

A committee of 6 people is to be selected from 90 students. Identify an expression that represents the number of possible committees if two of the students, Evelyn and Wayne, must be on the committee.

a) ${}_{90}C_4$

b) ${}_{90}C_6$

c) ${}_{88}C_6$

d) ${}_{88}C_4$

Match the following functions with their correct graphs.

Place the appropriate letter in this column.

Solution

$$f(x) = \frac{1}{x-2}$$

C

$$g(x) = \frac{-x}{x-2}$$

D

$$h(x) = \frac{2x}{x-1}$$

A

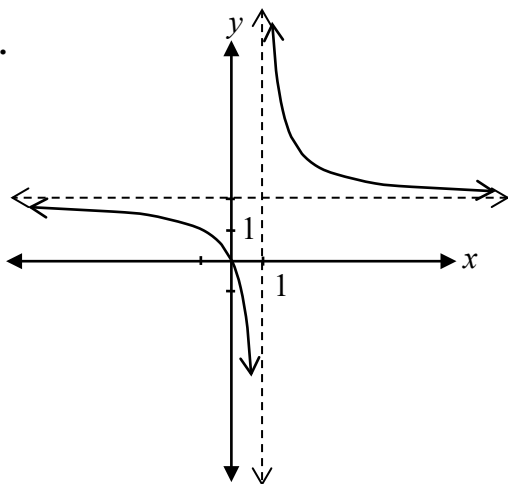
$$k(x) = \frac{2x}{x+1}$$

B

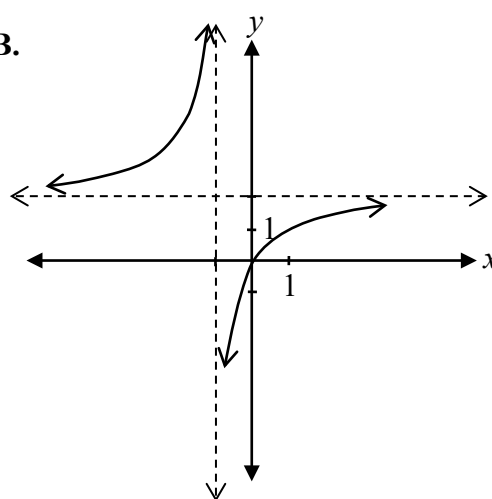
½ mark for each answer

2 marks

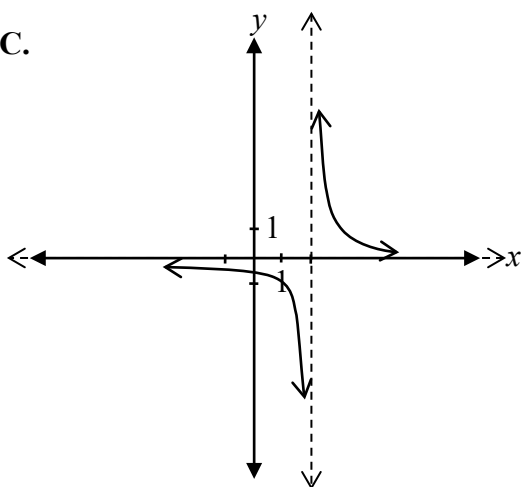
A.



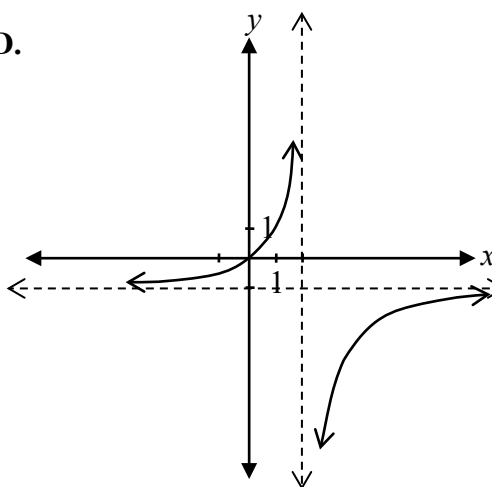
B.



C.



D.



Evaluate.

$$\sin^2\left(\frac{11\pi}{3}\right)\cot\left(\frac{\pi}{4}\right) + \csc\left(\frac{11\pi}{6}\right)$$

Solution

$$\left(\frac{-\sqrt{3}}{2}\right)^2 (1) + (-2)$$

1 mark for $\sin\left(\frac{11\pi}{3}\right)$ (½ mark for quadrant; ½ mark for value)

$$\left(\frac{3}{4}\right)(1) - 2$$

1 mark for $\cot\left(\frac{\pi}{4}\right)$ (½ mark for quadrant; ½ mark for value)

$$\frac{3}{4} - 2$$

1 mark for $\csc\left(\frac{11\pi}{6}\right)$ (½ mark for quadrant; ½ mark for value)

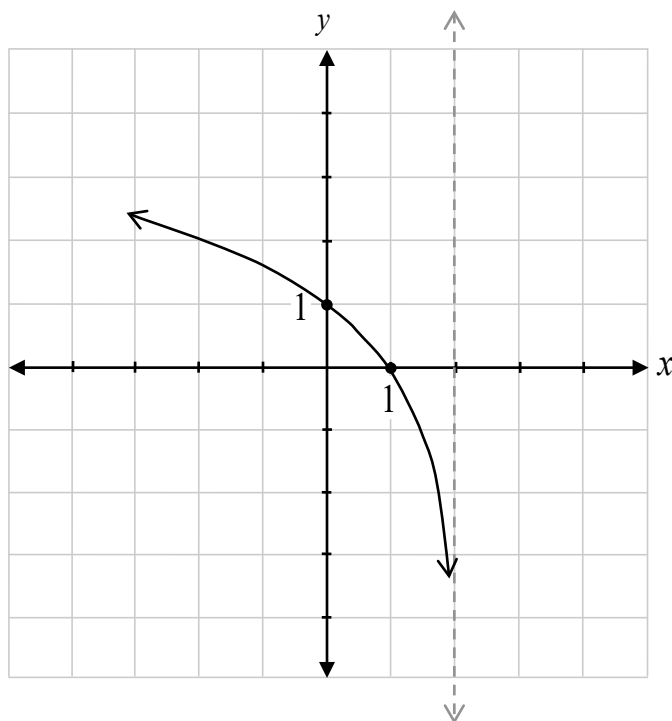
$$-\frac{5}{4}$$

3 marks

or

$$-1\frac{1}{4}$$

Sketch the graph of $y = \log_2[-(x-2)]$.

Solution

1 mark for shape of a logarithmic function

1 mark for horizontal reflection

1 mark for horizontal translation

3 marks

Justify that ${}_{12}C_7 = {}_{12}C_5$.

Solution

$$\begin{aligned} {}_{12}C_7 &= \frac{12!}{(12-7)!7!} & {}_{12}C_5 &= \frac{12!}{(12-5)!5!} \\ &= \frac{12!}{5!7!} & &= \frac{12!}{7!5!} \end{aligned}$$

$$\frac{12!}{5!7!} = \frac{12!}{7!5!}$$

1 mark

Solve, algebraically.

$$\left(\frac{1}{4}\right)^{3x} = 8^{x-3}$$

Solution

$$\left(2^{-2}\right)^{3x} = \left(2^3\right)^{x-3}$$

$$2^{-6x} = 2^{3x-9}$$

$$-6x = 3x - 9$$

$$-9x = -9$$

$$x = 1$$

1 mark for changing to a common base (½ mark for each side)

½ mark for exponent law

½ mark for equating exponents

2 marks

State the domain and range of the function $y = -\sqrt{x-5} + 4$.

Solution

Domain: $\{x|x \geq 5, x \in \mathbb{R}\}$ 1 mark for domain

Range: $\{y|y \leq 4, y \in \mathbb{R}\}$ 1 mark for range

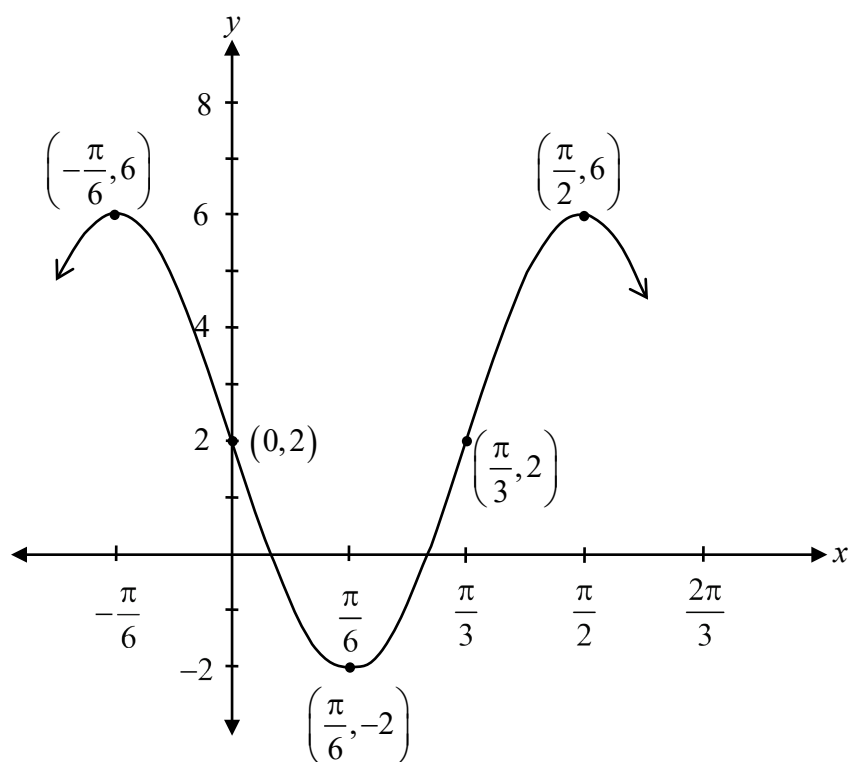
or

Domain: $[5, \infty)$ 1 mark for domain

Range: $(-\infty, 4]$ 1 mark for range

2 marks

Sketch at least one period of the graph of $y = 4 \cos \left[3 \left(x + \frac{\pi}{6} \right) \right] + 2$.

Solution

1 mark for shape of a sinusoidal function with correct amplitude

1 mark for period

1 mark for horizontal translation

1 mark for vertical translation

4 marks

Determine, algebraically, the equation of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 4
- $p(x)$ has zeros at -5 and 1
- all zeros of $p(x)$ have a multiplicity of 2
- the graph of $p(x)$ has a y -intercept at 75

Solution

$$p(x) = a(x+5)^2(x-1)^2$$

1 mark for multiplicity of 2 at $x = -5$ and $x = 1$

$$75 = a(5)^2(-1)^2$$

$$75 = 25a$$

$$a = 3$$

1 mark for consistent value of a

$$p(x) = \underline{3(x+5)^2(x-1)^2}$$

2 marks

Vitaly was asked to solve the exponential equation $5^{2x+7} = 3 \cdot 7^x$.

Vitaly's work:

$$\begin{aligned}\log(5^{2x+7}) &= \log(3 \cdot 7^x) \\ (2x+7)\log 5 &= \log(21^x) \\ 2x\log 5 + 7\log 5 &= x\log 21 \\ 7\log 5 &= x\log 21 - 2x\log 5 \\ 7\log 5 &= x(\log 21 - 2\log 5) \\ x &= \frac{7\log 5}{\log 21 - 2\log 5}\end{aligned}$$

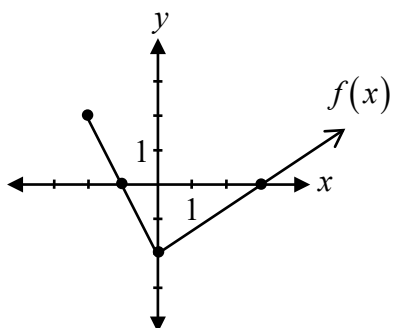
Describe his error.

Solution

In step 2, Vitaly incorrectly simplified $3 \cdot 7^x$ to 21^x .
He should have applied the product law of logarithms.

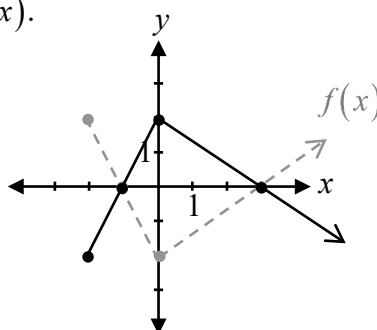
1 mark

Given the graph of $f(x)$,



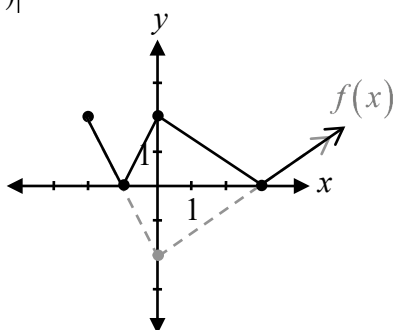
Solution

a) sketch the graph of $g(x) = -f(x)$.



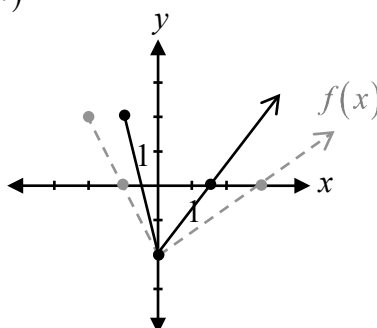
1 mark

b) sketch the graph of $k(x) = |f(x)|$.



1 mark

c) sketch the graph of $r(x) = f(2x)$.



1 mark

Determine the exact value of $\sin\left(\frac{17\pi}{12}\right)$.

Solution

$$\sin\left(\frac{17\pi}{12}\right) = \sin\left(\frac{\pi}{6} + \frac{5\pi}{4}\right)$$

$$= \sin\left(\frac{\pi}{6}\right)\cos\left(\frac{5\pi}{4}\right) + \cos\left(\frac{\pi}{6}\right)\sin\left(\frac{5\pi}{4}\right)$$

1 mark for substitution into correct identity

$$= \left(\frac{1}{2}\right)\left(\frac{-\sqrt{2}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{-\sqrt{2}}{2}\right)$$

2 marks for exact values ($\frac{1}{2}$ mark for each)

$$= \frac{-\sqrt{2} - \sqrt{6}}{4}$$

3 marks

Note:

- Other combinations are possible.

Question 37

R1

Given $f(x) = \sqrt{x+3}$ and $g(x) = x-8$,

a) determine the value of $g(f(1))$.

b) justify that the value of $(f-g)(-5)$ is undefined.

Solution

a) $f(1) = 2$ $\frac{1}{2}$ mark for the value of $f(1)$

$$g(2) = -6$$

$g(f(1)) = -6$ $\frac{1}{2}$ mark for consistent value of $g(f(1))$

1 mark

b) The domain of $f(x)$ is $x \geq -3$; $x = -5$ is outside the domain.

or

$$\begin{aligned} f(-5) &= \sqrt{-5+3} \\ &= \sqrt{-2} \end{aligned}$$

It is not possible to evaluate the square root of a negative radicand, therefore $f(-5)$ is undefined.

1 mark

Solve, algebraically.

$$48({}_nC_4) = {}_{(n+1)}P_4$$

Solution

$$48\left(\frac{n!}{(n-4)!4!}\right) = \frac{(n+1)!}{((n+1)-4)!}$$

1 mark for substitution into equation (½ mark for each side)

$$48\left(\frac{n!}{(n-4)!4!}\right) = \frac{(n+1)!}{(n-3)!}$$

$$\frac{48}{4!} = \frac{(n-4)!(n+1)!}{(n-3)!n!}$$

$$\frac{48}{24} = \frac{\cancel{(n-4)!} (n+1) \cancel{(n)!}}{(n-3) \cancel{(n-4)!} \cancel{(n)!}}$$

1 mark for factorial expansion
½ mark for simplification of factorials

$$2 = \frac{n+1}{n-3}$$

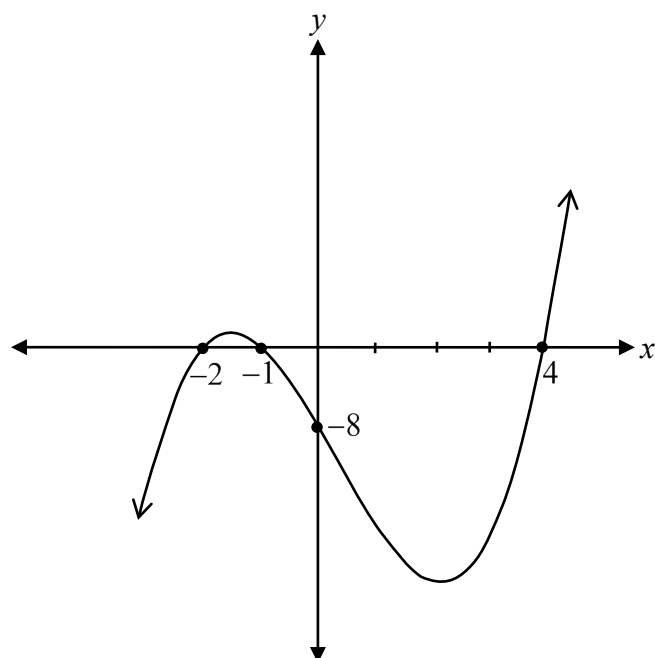
$$2n - 6 = n + 1$$

$$n = 7$$

½ mark for solving for n

3 marks

Sketch the graph of $p(x) = (x+1)(x+2)(x-4)$.

Solution

1 mark for x -intercepts
 $\frac{1}{2}$ mark for y -intercept
 $\frac{1}{2}$ mark for end behaviour

2 marks

Evaluate.

$$\log_5(\log_2 32)$$

Solution

$$\log_5 5$$

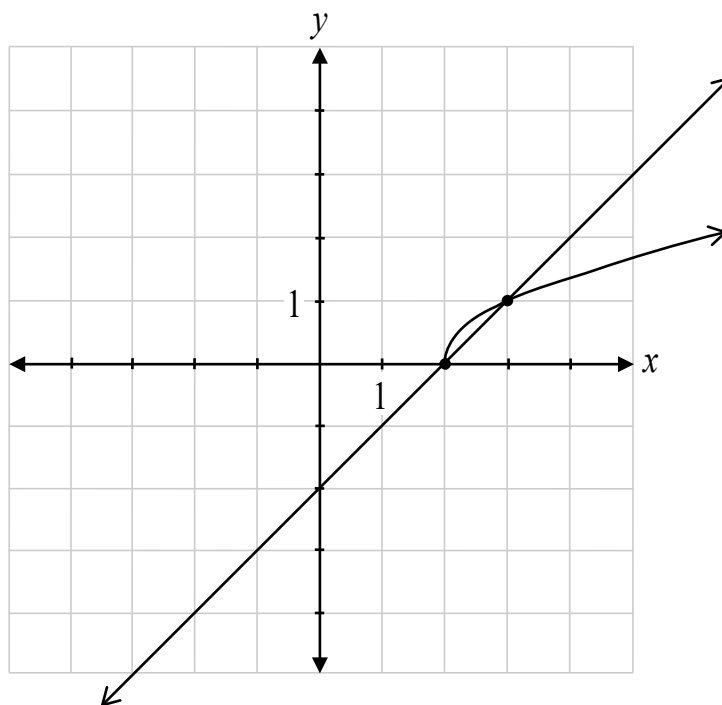
$\frac{1}{2}$ mark for evaluating $\log_2 32$

$$1$$

$\frac{1}{2}$ mark for consistent value

1 mark

Describe how to use the graph to solve the equation $\sqrt{x-2} = x-2$.

**Solution**

State the x -coordinates of the points of intersection.

1 mark

Verify, by substitution, that the equation $\frac{\sec \theta}{\tan \theta + \cot \theta} = \sin 2\theta$ is true for $\theta = \frac{\pi}{3}$.

Solution

Left-Hand Side	Right-Hand Side
$\frac{\sec \frac{\pi}{3}}{\tan \frac{\pi}{3} + \cot \frac{\pi}{3}}$ $\frac{2}{\sqrt{3} + \frac{1}{\sqrt{3}}}$ $\frac{2}{\frac{3+1}{\sqrt{3}}}$ $\frac{2\sqrt{3}}{4}$ $\frac{\sqrt{3}}{2}$	$\sin \frac{2\pi}{3}$ $\frac{\sqrt{3}}{2}$
	½ mark for substitution 2 marks for exact values (½ mark for each) ½ mark for simplification
LHS = RHS	

3 marks

Describe the transformation(s) to the graph of $y = 3^x$ which results in a graph with a y -intercept of -2 .

Solution

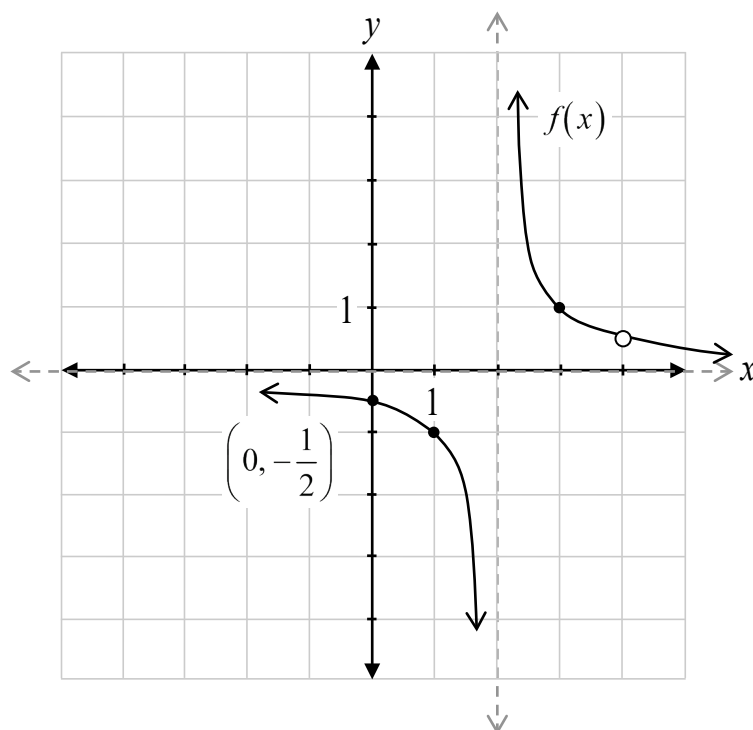
A vertical translation down 3 units.

1 mark

Note:

- Other answers are possible.

Determine the equation of the rational function represented by the graph.

**Solution**

$$f(x) = \frac{x-4}{(x-2)(x-4)}$$

1 mark for factor of $(x-2)$ in denominator

1 mark for factor of $(x-4)$ in both numerator and denominator

2 marks

Justify that $f(x) = 3x + 2$ and $g(x) = \frac{x-3}{2}$ are not inverses of each other.

Solution**Method 1**

$$\text{Let } y = f(x)$$

$$y = 3x + 2$$

$$\text{Let } y = f^{-1}(x)$$

$$x = 3y + 2$$

1 mark for switching x and y

$$\frac{x-2}{3} = y$$

$$\frac{x-2}{3} = f^{-1}(x)$$

$$g(x) \neq f^{-1}(x)$$

1 mark for justification

2 marks**Method 2**

If they are inverses, then $g(f(x)) = x$, and $f(g(x)) = x$.

$$\begin{aligned} g(3x+2) &= \frac{3x+2-3}{2} \\ &= \frac{3x-1}{2} \end{aligned}$$

1 mark for $g(f(x))$ or $f(g(x))$

Since $g(f(x)) \neq x$, they are not inverses of each other.

1 mark for justification

2 marks

The terminal arm of an angle, θ , in standard position passes through the point $(-3, 6)$ and intersects the unit circle at the point $P(\theta)$. Determine the coordinates of the point $P(\theta)$.

Solution

$$x^2 + y^2 = r^2$$

$$(-3)^2 + (6)^2 = r^2 \quad \frac{1}{2} \text{ mark for substitution}$$

$$9 + 36 = r^2$$

$$45 = r^2$$

$$\sqrt{45} = r \quad \frac{1}{2} \text{ mark for solving for } r$$

$$P(\theta) = \left(\frac{-3}{\sqrt{45}}, \frac{6}{\sqrt{45}} \right) \quad \begin{array}{l} 1 \text{ mark for the coordinates of } P(\theta) \\ (\frac{1}{2} \text{ mark for } x\text{-coordinate; } \frac{1}{2} \text{ mark for } y\text{-coordinate}) \end{array}$$

or

$$P(\theta) = \left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right)$$

or

$$P(\theta) = \left(\frac{-\sqrt{5}}{5}, \frac{2\sqrt{5}}{5} \right)$$

2 marks

Determine the exact value of $2\sin\theta\cos\theta$ if $\theta = \frac{\pi}{12}$.

Solution

$$2\sin\theta\cos\theta = \sin 2\theta \quad 1 \text{ mark for substitution into correct identity}$$

$$= \sin\left(2\left(\frac{\pi}{12}\right)\right)$$

$$= \sin\left(\frac{\pi}{6}\right)$$

$$= \frac{1}{2}$$

1 mark for consistent value

2 marks

Question 1

P1

Manitoba postal codes consist of three letters and three digits. Determine the total number of possible postal codes in Manitoba if the following characteristics must be met:

- The postal code must begin with the letter R.
- Only 18 letters of the alphabet can be used.
- The letters and digits must alternate.
- The letters and digits can be repeated.

Solution

$$\frac{1 \cdot 10 \cdot 18 \cdot 10 \cdot 18 \cdot 10}{R} = 324\,000$$

1 mark

Note:

The restriction of the first letter does not need to be shown.

Question 2

R10

The percent of people who click on a link, in response to seeing an advertisement on YouTube, can be modeled by the formula:

$$R(t) = -(0.8)^{0.2t} + 0.6$$

where $R(t)$ is the percentage, in decimal form, of people who click on the link

t is the time in days.

Determine, algebraically, the number of days it will take for 45% of people who see the advertisement to click on the link.

Solution

$$0.45 = -(0.8)^{0.2t} + 0.6$$

$$-0.15 = -(0.8)^{0.2t}$$

$$0.15 = (0.8)^{0.2t}$$

$$\log(0.15) = \log(0.8)^{0.2t}$$

$$\log(0.15) = 0.2t \cdot \log(0.8)$$

$$\frac{\log(0.15)}{0.2 \cdot \log(0.8)} = t$$

$$t = 42.508958\dots$$

$$t = 42.509 \text{ days}$$

½ mark for applying logarithms

1 mark for power law

½ mark for evaluating quotient of logarithms

2 marks

Question 3

P4

Determine and simplify the middle term in the binomial expansion of $\left(\frac{5}{x} + 4x^3\right)^6$.

Solution

$$t_4 = {}_6C_3 \left(\frac{5}{x}\right)^3 (4x^3)^3 \quad 2 \text{ marks (1 mark for } {}_6C_3; \frac{1}{2} \text{ mark for each consistent factor)}$$

$$= 20 \left(\frac{125}{x^3}\right) (64x^9)$$

$$= 160\,000x^6 \quad 1 \text{ mark for simplification (}\frac{1}{2}\text{ mark for coefficient; } \frac{1}{2}\text{ mark for exponent)}$$

3 marks

Question 4

T5

Solve $3\tan^2 x + 5\tan x - 6 = 0$, over the interval $[0, 2\pi]$.

Solution

$$\tan x = \frac{-5 \pm \sqrt{5^2 - 4(3)(-6)}}{2(3)}$$

$$\tan x = \frac{-5 \pm \sqrt{97}}{6}$$

1 mark for solving for $\tan x$

$$\tan x = 0.808\,142\dots$$

$$\tan x = -2.474\,809\dots$$

$$x_r = 0.679\,686\dots$$

$$x_r = 1.186\,784\dots$$

$$x = 0.680, 3.821$$

$$x = 1.955, 5.096$$

2 marks for solving for x ($\frac{1}{2}$ mark for each value)

3 marks

Question 5

T1

Given $\theta = -80^\circ$,

a) determine the measure of θ , in radians.

b) state all the coterminal angles of θ .

Solution

a) $\theta = -80^\circ \left(\frac{\pi}{180^\circ} \right)$

1 mark

$$\theta = -\frac{4\pi}{9}$$

or

$$\theta = -1.396$$

b) $\theta = -80^\circ + 360^\circ k, k \in \mathbb{Z}$

or

$$\theta = -\frac{4\pi}{9} + 2k\pi, k \in \mathbb{Z}$$

or

$$\theta = -1.396 + 2k\pi, k \in \mathbb{Z}$$

1 mark

Question 6

R12

State a possible value of n if the polynomial function $p(x) = (x+1)^n(x-4)^2$ has a range of $(-\infty, \infty)$.

Solution

$$n = 1$$

or

$$n = 3$$

1 mark

Note:

Accept any odd whole number for n .

Question 7

R2, R3, R5

Describe the transformations used to obtain the graph of the function $y = -2f(x + 8) - 5$ from the graph of $y = f(x)$.

Solution

The graph of $y = f(x)$ is reflected over the x -axis, vertically stretched by a factor of 2, translated left 8 units and down 5 units.

- 1 mark for vertical reflection
- 1 mark for vertical stretch
- 1 mark for horizontal translation
- 1 mark for vertical translation

4 marks

Note:

Deduct a maximum of 1 mark for concept error of incorrect order of vertical transformations.

Question 8

R14

State the domain and range of the function $f(x) = \frac{-2}{x^2}$.

Solution

Domain: $\{x|x \neq 0, x \in \mathbb{R}\}$ or $(-\infty, 0) \cup (0, \infty)$

1 mark for domain

Range: $\{y|y < 0, y \in \mathbb{R}\}$ or $(-\infty, 0)$

1 mark for range

2 marks

Question 9

P4

Given the following row of Pascal's Triangle, state the values in the next row.

1 5 10 10 5 1

Solution

1 6 15 20 15 6 1

1 mark

Question 10

R6

Given $f(x) = 3 - 2x$, determine the equation of $f^{-1}(x)$.

Solution

$$\begin{aligned}\text{Let } y &= f(x) \\ y &= 3 - 2x\end{aligned}$$

To determine the inverse of $f(x)$, switch x and y .

$$\begin{aligned}x &= 3 - 2y && \text{1 mark for switching } x \text{ and } y \\ 2y &= 3 - x \\ y &= \frac{3 - x}{2} && \frac{1}{2} \text{ mark for solving for } y \\ f^{-1}(x) &= \frac{3 - x}{2} && \frac{1}{2} \text{ mark for writing equation in terms of } f^{-1}(x)\end{aligned}$$

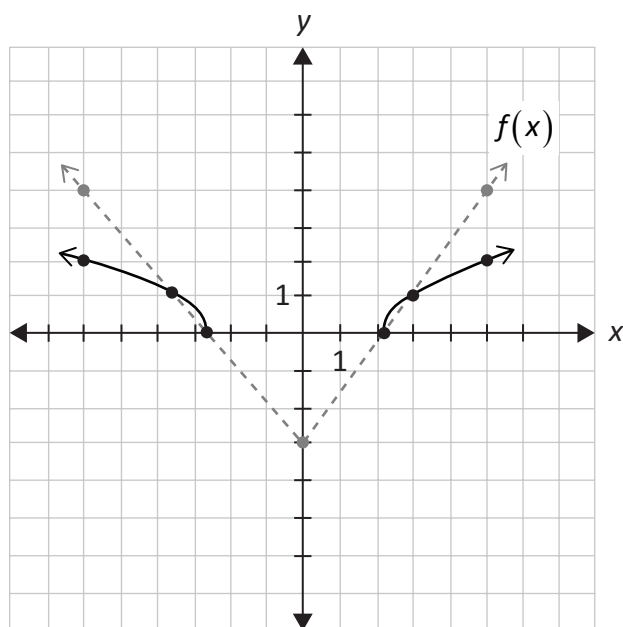
2 marks

Question 11

R13

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.

Solution



1 mark for restricting domain

$\frac{1}{2}$ mark for shape between invariant points, $\{0 \leq y \leq 1\}$

$\frac{1}{2}$ mark for shape above invariant points, $\{y \geq 1\}$

2 marks

Question 12

T6

Prove the identity for all permissible values of x .

$$\frac{\csc^2 x + \sec^2 x}{\tan x + \cot x} = \csc x \sec x$$

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$	$\csc x \sec x$
$\frac{\frac{\cos^2 x + \sin^2 x}{\sin^2 x \cos^2 x}}{\frac{\cos^2 x + \sin^2 x}{\cos x \sin x}}$	
$\frac{\cancel{\cos x} \cancel{\sin x}}{\sin^{\cancel{2}} x \cos^{\cancel{2}} x}$	
$\frac{1}{\sin x \cos x}$	
$\csc x \sec x$	

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 12

T6

Method 2

Left-Hand Side	Right-Hand Side
$\frac{1}{\sin^2 x} + \frac{1}{\cos^2 x}$ $\frac{\tan x}{1} + \frac{1}{\tan x}$ $\frac{\cos^2 x + \sin^2 x}{\sin^2 x \cos^2 x}$ $\frac{\tan^2 x + 1}{\tan x}$ $\frac{1}{\sin^2 x \cos^2 x}$ $\frac{\sec^2 x}{\tan x}$ $\frac{\tan x}{\sin^2 x \cancel{\cos^2 x} \cancel{\sec^2 x}}$ $\frac{\cancel{\sin x}}{\cos x}$ $\frac{\cos x}{\sin^2 x}$ $\frac{1}{1}$ $\frac{1}{\sin x \cos x}$ $\csc x \sec x$	$\csc x \sec x$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 13

R9

Justify that the graph of $f(x) = e^{x+1} + 5$ does not have an x-intercept.

Solution

$$0 = e^{x+1} + 5$$

$$-5 = e^{x+1}$$

$e^{x+1} > 0$, therefore there is no solution.

or

The range of the graph is $\{y \mid y > 5, y \in \mathbb{R}\}$.

or

The horizontal asymptote is $y = 5$ and it is an increasing exponential function above the asymptote.

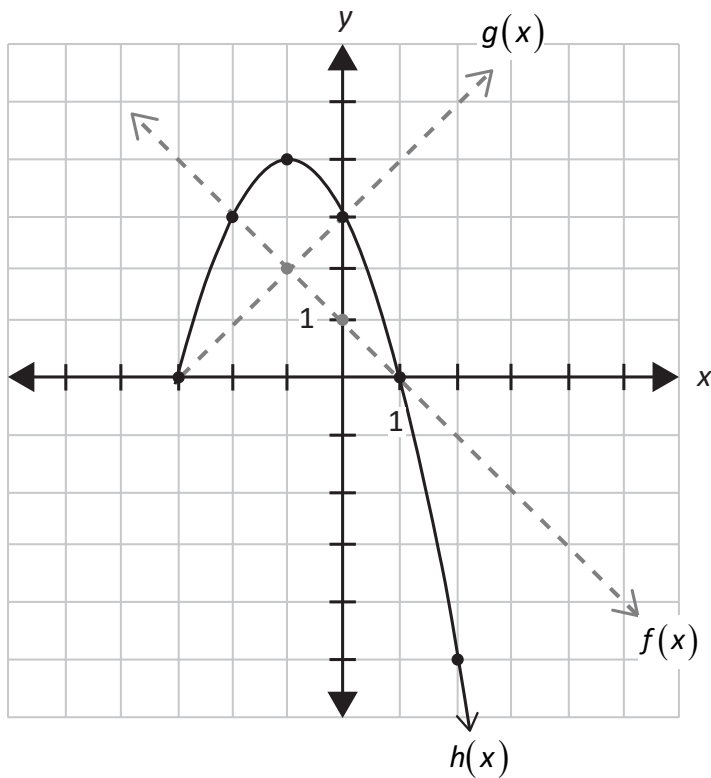
1 mark

Question 14

R1

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) \cdot g(x)$.

Solution



1 mark for operation of multiplication
1 mark for restricted domain

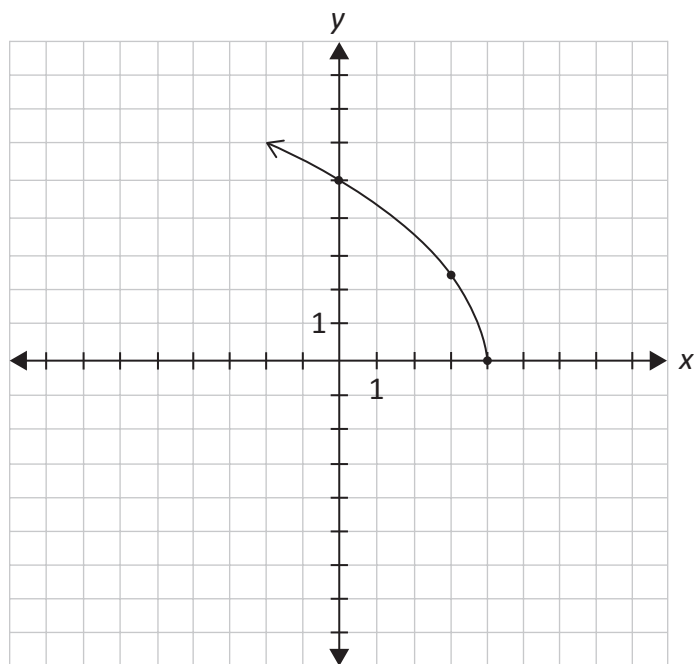
2 marks

Question 15

R13

Sketch the graph of $f(x) = \frac{5}{2}\sqrt{-(x-4)}$.

Solution



- 1 mark for shape of a radical function
- 1 mark for vertical stretch
- 1 mark for horizontal reflection
- 1 mark for horizontal translation

4 marks

Question 16**R11**

Identify the number of x -intercepts on the graph of the polynomial function,

$$p(x) = 3x(x - 8)(x^2 + 5).$$

a. 1

b. 2

c. 3

d. 4

Question 17**R5**

Identify the equation that represents the graph of $y = x^2 - 3$ after a reflection over the x -axis.

a. $y = -x^2 - 3$

b. $y = x^2 - 3$

c. $y = x^2 + 3$

d. $y = -x^2 + 3$

Question 18**R7**

Given $2^{\log_2 3} = x$, identify the value of x .

a. 1

b. 3

c. 6

d. 8

Question 19

R14

Identify the y-intercept on the graph of $y = \frac{x-3}{x^2-5x+6}$.

a. -3

b. -2

c. $-\frac{1}{2}$

d. 3

Question 20

P2

Identify the total number of arrangements for 10 students and 2 teachers to sit in a row if the teachers must sit together.

a. $10!2!$

b. $11!$

c. $11!2!$

d. $12!$

Question 21

R11

Identify which expression represents the remainder when the polynomial $P(x)$ is divided by $(x-4)$.

a. $P(-4)$

b. $P(4)$

c. $P(x-4)$

d. $P(x+4)$

Question 22

T4

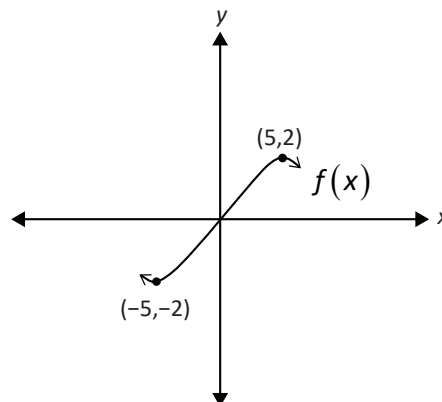
Identify the period of the graph of the sinusoidal function.

a. 10π

b. 10

c. 20π

d. 20



Question 23

R8

Identify an equivalent expression for $\log_3 9 + \log_3 5$.

a. $\log_3 14$

b. $2\log_3 5$

c. $\log_3 45$

d. $\log_3 \left(\frac{9}{5} \right)$

Question 24

T1

Identify the coterminal angle(s) of $\theta = -\frac{11\pi}{10}$ over the interval $[-3\pi, \pi]$.

a. $\theta = \frac{9\pi}{10}$

b. $\theta = -\frac{31\pi}{10}, -\frac{21\pi}{10}$

c. $\theta = -\frac{31\pi}{10}, \frac{9\pi}{10}$

d. $\theta = -\frac{21\pi}{10}, \frac{9\pi}{10}$

Question 25

R11

Express $p(x) = x^3 - 13x - 12$ in completely factored form.

Solution

$$p(-1) = (-1)^3 - 13(-1) - 12$$

$$p(-1) = 0$$

$\therefore (x + 1)$ is a factor

1 mark for identifying one possible zero of $P(x)$

-1		1	0	-13	-12
			-1	1	12
		1	-1	-12	0

1 mark for synthetic division (or equivalent strategy)

$$p(x) = (x + 1)(x^2 - x - 12)$$

$$p(x) = \underline{(x + 1)(x - 4)(x + 3)}$$

1 mark for consistent product of factors

3 marks

Question 26

T5

Verify, by substitution, that the equation $\frac{\cos^2 \theta}{\sin^2 \theta} = \frac{2 \csc \theta}{\sec^2 \theta}$ is true for $\theta = \frac{\pi}{6}$.

Solution

Left-Hand Side	Right-Hand Side	
$\frac{\cos^2\left(\frac{\pi}{6}\right)}{\sin^2\left(\frac{\pi}{6}\right)}$	$\frac{2 \csc\left(\frac{\pi}{6}\right)}{\sec^2\left(\frac{\pi}{6}\right)}$	
$\frac{\left(\frac{\sqrt{3}}{2}\right)^2}{\left(\frac{1}{2}\right)^2}$	$\frac{2(2)}{\left(\frac{2}{\sqrt{3}}\right)^2}$	2 marks for exact values (½ mark for each)
$\frac{\frac{3}{4}}{\frac{1}{4}}$	$\frac{2(2)}{\left(\frac{4}{3}\right)}$	
	$\frac{12}{4}$	
3	3	1 mark for simplification (½ mark for LHS; ½ mark for RHS)

3 marks

Question 27

R9

Given $f(x) = \log_3 x + 2$,

a) determine the x-intercept of $f(x) = \log_3 x + 2$.

b) sketch the graph of $f(x) = \log_3 x + 2$.

Solution

a)

$$0 = \log_3 x + 2$$

$$-2 = \log_3 x$$

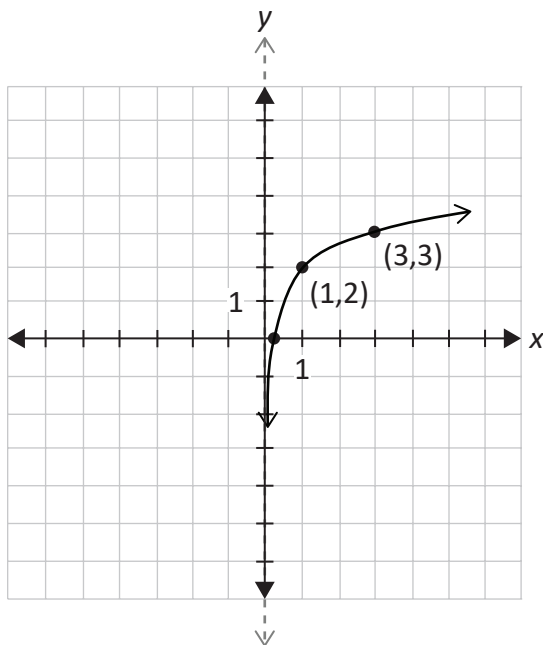
$$3^{-2} = x$$

$$\frac{1}{3^2} = x$$

$$\frac{1}{9} = x$$

1 mark

b)



1 mark for shape of a logarithmic function
1 mark for vertical translation

2 marks

Question 28

T3

Evaluate.

$$\sec^2\left(\frac{5\pi}{4}\right) \cdot \tan^2\left(-\frac{2\pi}{3}\right)$$

Solution

$$\left(-\frac{2}{\sqrt{2}}\right)^2 \left(\frac{\sqrt{3}}{1}\right)^2$$

1 mark for $\sec\left(\frac{5\pi}{4}\right)$ (½ mark for quadrant; ½ mark for value)

$$\left(\frac{4}{2}\right)\left(\frac{3}{1}\right)$$

1 mark for $\tan\left(-\frac{2\pi}{3}\right)$ (½ mark for quadrant; ½ mark for value)

6

2 marks

Question 29

R1

Given the functions $f(x) = x - 3$ and $g(x) = \sqrt{x} + 2$, state the domain of $g(f(x))$.

Solution

$$\{x \mid x \geq 3, x \in \mathbb{R}\}$$

1 mark

or

$$[3, \infty)$$

Question 30

T6

Evaluate.

$$\sin 70^\circ \cos 25^\circ - \cos 70^\circ \sin 25^\circ$$

Solution

$$\sin 70^\circ \cos 25^\circ - \cos 70^\circ \sin 25^\circ = \sin(70^\circ - 25^\circ) \quad \frac{1}{2} \text{ mark for substitution into correct identity}$$

$$= \sin 45^\circ$$

$$= \frac{\sqrt{2}}{2}$$

$\frac{1}{2}$ mark for exact value

1 mark

Question 31

R10

Solve, algebraically.

$$\log_{15}(x^2 - 1) = 1$$

Solution

Method 1

$$15^1 = x^2 - 1$$

$$0 = x^2 - 16$$

$$0 = (x + 4)(x - 4)$$

$$x = -4, x = 4$$

1 mark for exponential form

1 mark for solving for x ($\frac{1}{2}$ mark for each value)

2 marks

Method 2

$$\log_{15}(x^2 - 1) = \log_{15} 15$$

$$x^2 - 1 = 15$$

$$x^2 = 16$$

$$x = \pm 4$$

$\frac{1}{2}$ mark for logarithmic form

$\frac{1}{2}$ mark for equating arguments

1 mark for solving for x ($\frac{1}{2}$ mark for each value)

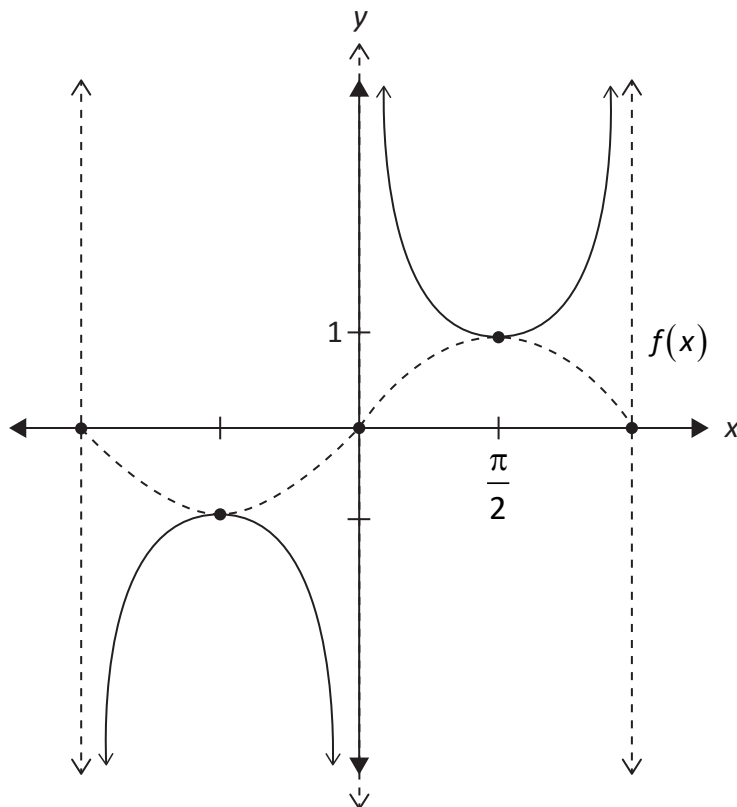
2 marks

Question 32

R1

Given the graph of $y = f(x)$, sketch the graph of $y = \frac{1}{f(x)}$.

Solution



1 mark for asymptotic behaviour approaching $x = -\pi$, $x = 0$, and $x = \pi$

1 mark for invariant points ($\frac{1}{2}$ mark for $y = -1$; $\frac{1}{2}$ mark for $y = 1$)

2 marks

Question 33

T4

A light is attached to the end of a blade of a wind turbine. The height of the light, with respect to the ground, follows the sinusoidal equation:

$$h(t) = 80 \cos\left(\frac{\pi}{2}t\right) + 240$$

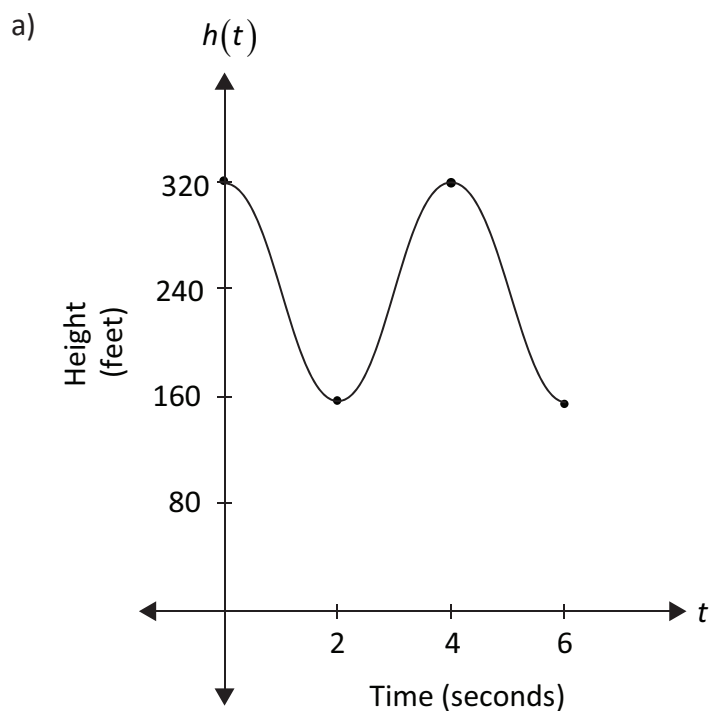
where $h(t)$ is the height of the light with respect to the ground, measured in feet and,

t is the time, measured in seconds.



- Sketch the graph of the height of the light with respect to the ground, over an interval of 6 seconds.
- If the turbine begins to rotate faster, describe the resulting effect on the period of the graph of $h(t)$.

Solution



1 mark for shape of a sinusoidal function with correct amplitude
1 mark for period
1 mark for vertical translation

3 marks

1 mark

- The period would decrease.

Question 34

P2

Elodie was asked to determine the total number of arrangements of the letters in the word EXCELLENCE.

Her solution:

$$10! = 3628\ 800$$

Describe her error.

Solution

She did not consider the repeating letters.

1 mark

Question 35

T6

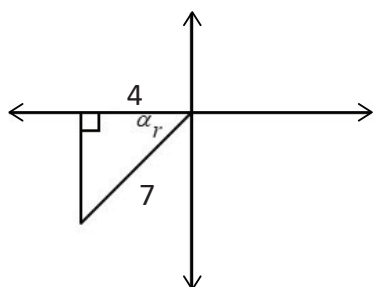
Given that $\cos \alpha = -\frac{4}{7}$ where α is in quadrant III and $\sin \beta = \frac{5}{13}$ where β is in quadrant II, determine the exact value of:

a) $\cos(\alpha + \beta)$

b) $\sec(\alpha + \beta)$

Solution

a)

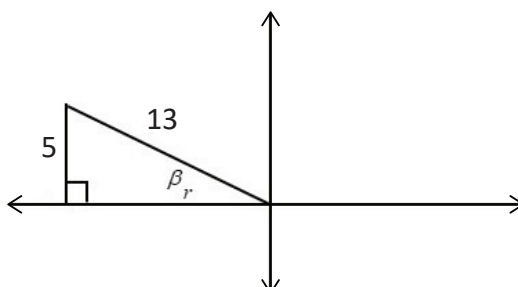


$$x^2 + y^2 = r^2$$

$$16 + y^2 = 49$$

$$y^2 = 33$$

$$y = \pm\sqrt{33}$$



$$x^2 + y^2 = r^2$$

$$x^2 + 25 = 169$$

$$x^2 = 144$$

$$x = \pm 12$$

$\frac{1}{2}$ mark for value of x

$\frac{1}{2}$ mark for value of y

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$= \left(-\frac{4}{7}\right)\left(-\frac{12}{13}\right) - \left(-\frac{\sqrt{33}}{7}\right)\left(\frac{5}{13}\right)$$

$$= \frac{48 + 5\sqrt{33}}{91}$$

$\frac{1}{2}$ mark for consistent value of $\cos \beta$

$\frac{1}{2}$ mark for consistent value of $\sin \alpha$

1 mark for substitution into correct identity

3 marks

b) $\sec(\alpha + \beta) = \frac{91}{48 + 5\sqrt{33}}$

1 mark for consistent value of $\sec(\alpha + \beta)$

1 mark

Note:

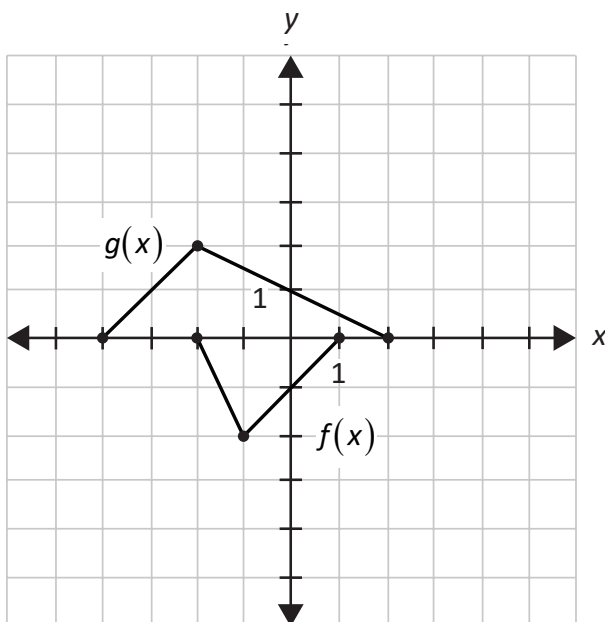
Accept any of the following values for x : $x = \pm 12$, $x = -12$, or $x = 12$.

Accept any of the following values for y : $y = \pm\sqrt{33}$, $y = -\sqrt{33}$, or $y = \sqrt{33}$.

Question 36

R4, R5

Determine the equation for $g(x)$, in terms of $f(x)$.



Solution

$$g(x) = -f\left(\frac{x}{2}\right)$$

1 mark for vertical reflection
1 mark for horizontal stretch

2 marks

Question 37

R7

State possible values for m and n that satisfy the equation, $\log_m n = 5$.

Solution

Some possible solutions are:

$$m = 2 \quad n = 32$$

or

$$m = 3 \quad n = 243$$

or

$$m = 4 \quad n = 1024$$

or

$$m = 5 \quad n = 3125$$

1 mark

Notes:

Other values for m and n are possible.

Question 38

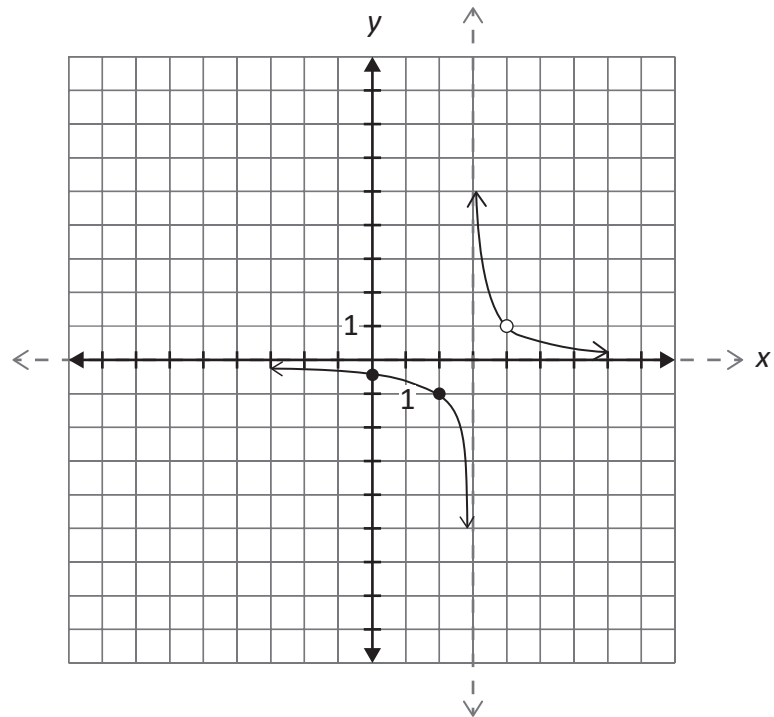
R14

Sketch the graph of $f(x) = \frac{x-4}{x^2-7x+12}$.

Solution

$$f(x) = \frac{\cancel{(x-4)}}{\cancel{(x-4)}(x-3)} \\ = \frac{1}{x-3}, \quad x \neq 4$$

$\therefore$ there is a point of discontinuity (hole) at (4,1)



1 mark for asymptotic behaviour approaching $x = 3$

1 mark for asymptotic behaviour approaching $y = 0$

1 mark for point of discontinuity (hole) at $x = 4$

1 mark for shape of rational function ($\frac{1}{2}$ mark for graph left of $x = 3$, $\frac{1}{2}$ mark for graph right of $x = 3$)

4 marks

Notes:

Deduct $\frac{1}{2}$ mark for procedural error (incorrect y -value for point of discontinuity (hole)).

Question 39

P2

Justify that the value of ${}_3P_4$ does not exist.

Solution

Method 1

When using ${}_nP_r$, $n \geq r$.

Method 2

$${}_3P_4 = \frac{3!}{(3-4)!}$$

$${}_3P_4 = \frac{3!}{(-1)!}$$

$(-1)!$ does not exist, therefore ${}_3P_4$ does not exist.

1 mark

Question 40

T4

State the range of the sinusoidal function that has the following characteristics:

- an amplitude of 3
- a maximum at $\left(\frac{\pi}{2}, 1\right)$

Solution

$$\{y \mid -5 \leq y \leq 1, y \in \mathbb{R}\}$$

or

$$[-5, 1]$$

1 mark

Question 41

P3

Solve, algebraically.

$${}_nC_2 = 15$$

Solution

$$\frac{n!}{2!(n-2)!} = 15$$

½ mark for substitution

$$\frac{n(n-1)\cancel{(n-2)!}}{2!\cancel{(n-2)!}} = 15$$

1 mark for factorial expansion

½ mark for simplification of factorials

$$n(n-1) = 30$$

$$n^2 - n - 30 = 0$$

$$(n-6)(n+5) = 0$$

$$n = 6 \quad \cancel{n = -5}$$

½ mark for the permissible value of n

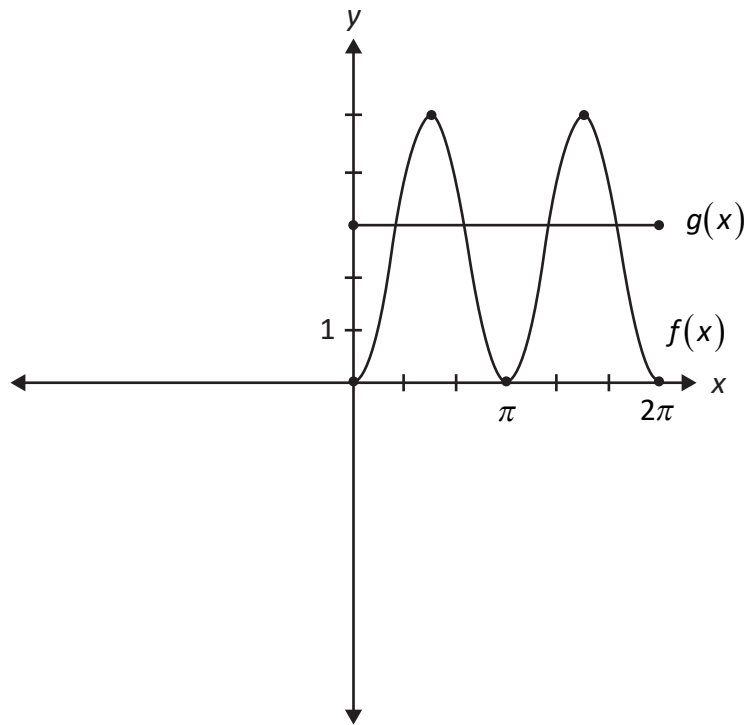
½ mark for showing the rejection of the extraneous root

3 marks

Question 42

T5

Describe how to use the graphs of $f(x) = 5\sin^2 x$ and $g(x) = 3$ to solve the equation, $5\sin^2 x = 3$, over the interval $[0, 2\pi]$.



Solution

The solution will be the x -values where the two graphs intersect.

1 mark

Question 43

R8

Solve, algebraically.

$$2\log_3 5 - \frac{1}{3}\log_3 125 = \log_3 a$$

Solution

Method 1

$$\log_3 5^2 - \log_3 125^{\frac{1}{3}} = \log_3 a$$

1 mark for power law (½ mark for each)

$$\log_3 25 - \log_3 5 = \log_3 a$$

$$\log_3 \left(\frac{25}{5} \right) = \log_3 a$$

1 mark for quotient law

$$\log_3 (5) = \log_3 a$$

$$a = 5$$

1 mark for equating arguments

3 marks

Method 2

$$\log_3 5^2 - \log_3 125^{\frac{1}{3}} - \log_3 a = 0$$

1 mark for power law (½ mark for each)

$$\log_3 \left(\frac{25}{5a} \right) = 0$$

1 mark for quotient law

$$3^0 = \frac{25}{5a}$$

1 mark for exponential form

$$5a = 25$$

$$a = 5$$

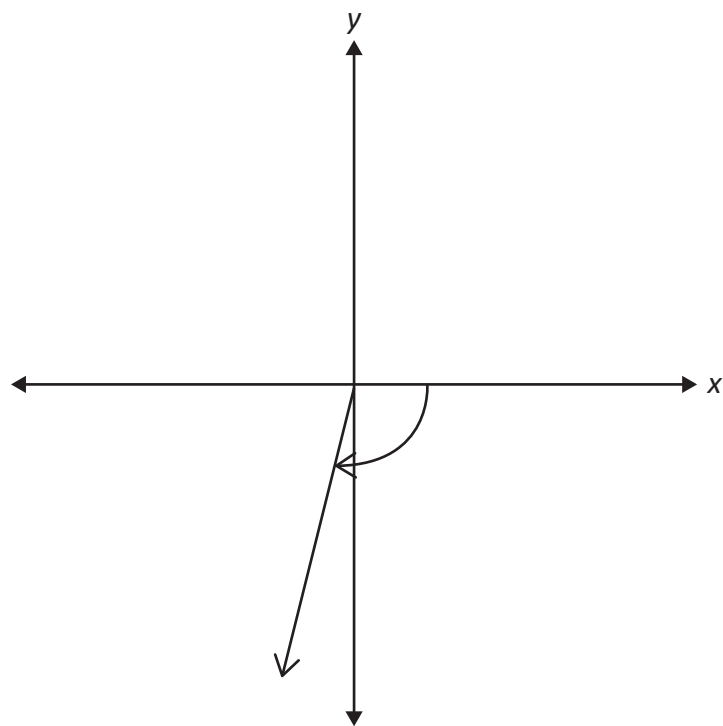
3 marks

Question 44

T1

Sketch the angle, $\frac{-7\pi}{12}$, in standard position.

Solution



½ mark for an appropriate angle
in quadrant III
½ mark for correct direction

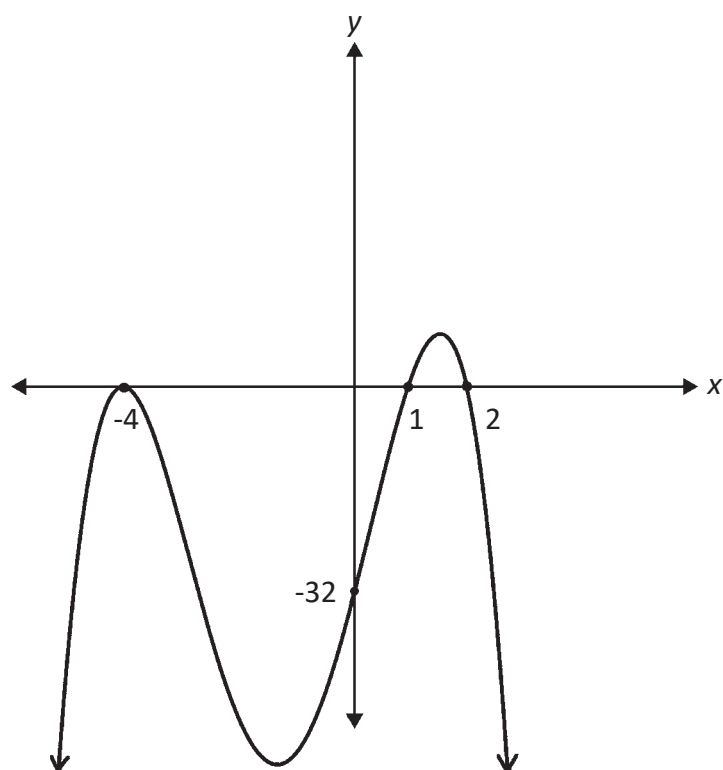
1 mark

Question 45

R12

Sketch the graph of $p(x) = -(x-1)(x-2)(x+4)^2$.

Solution



1 mark for x-intercepts
½ mark for y-intercept
1 mark for multiplicity of 2 at $x = -4$
½ mark for end behaviour

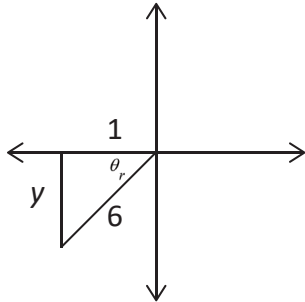
3 marks

Question 46

T3

The point, $P(\theta) = \left(-\frac{1}{6}, y\right)$, lies on the unit circle and is located in Quadrant III. Determine the exact value of $\csc \theta$.

Solution



$$(-1)^2 + y^2 = (6)^2$$

½ mark for substitution

$$1 + y^2 = 36$$

$$y^2 = 35$$

$$y = \pm\sqrt{35}$$

½ mark for solving for y

$$\csc \theta = -\frac{6}{\sqrt{35}}$$

1 mark for consistent value of $\csc \theta$

(½ mark for quadrant; ½ mark for value)

or

2 marks

$$\csc \theta = -\frac{6\sqrt{35}}{35}$$

Notes:

Accept any of the following values for y : $y = \pm\sqrt{35}$, $y = \sqrt{35}$, or $y = -\sqrt{35}$.